
**Search for the Inert Doublet Model in a
Final State with Two Leptons at the
CMS Experiment**

Edward Curtis

IMPERIAL COLLEGE LONDON

DEPARTMENT OF PHYSICS

Declaration of Originality

I, Edward Curtis, declare that the work in this thesis is my own. Any figures or results that were not made by myself, or that were taken from other sources, have been appropriately referenced. Chapters 2 and 3, and the first two sections of chapter 4, do not contain original work by me. However, these provide the necessary background information to understand the rest of the thesis. The remaining sections of chapter 4 were developed with other CMS members, but was entirely implemented by myself. Similarly, the IDM analysis chapter at CMS (chapter 5) was developed in conjunction with other CMS members, but all technical implementations were performed solely by myself. The results in chapter 6, on the IDM search at the FCC, were made public in Reference [1]. I did not produce the simulations for the analysis, however I was responsible for everything after that, including implementation of the neural network and the analysis of the results. For the whole thesis, a full list of references is given in the bibliography.

Copyright

The copyright of this thesis rests with the author. Unless otherwise indicated, its contents are licensed under a Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International Licence (CC BY NC-SA).

Under this licence, you may copy and redistribute the material in any medium or format. You may also create and distribute modified versions of the work. This is on the condition that; you credit the author, do not use it for commercial purposes and share any derivative works under the same licence.

When reusing or sharing this work, ensure you make the licence terms clear to others by naming the licence and linking to the licence text. Where a work has been adapted, you should indicate that the work has been changed and describe those changes.

Please seek permission from the copyright holder for uses of this work that are not included in this licence or permitted under UK Copyright Law.

Abstract

This thesis describes two searches for neutral scalars of the Inert Doublet Model (IDM) in the dilepton final state. The first search is performed using proton-proton collision data collected from 2016-2022 with the CMS detector at centre of mass energies of $\sqrt{s} = 13.0$ TeV and $\sqrt{s} = 13.6$ TeV, corresponding to a total integrated luminosity of 172.2 fb^{-1} . No significant excess above the SM prediction is found. 95% CL upper limits are placed on the masses of the two neutral IDM scalars, H and A , resulting in a maximum exclusion up to $M_H = 108$ GeV at a mass splitting of approximately $M_A - M_H = 80$ GeV.

The second search is a projection study looking at the sensitivity to the IDM at the Future Circular Collider with electron-positron collisions at $\sqrt{s} = 240$ GeV and $\sqrt{s} = 365$ GeV, corresponding to 10.8 ab^{-1} and 2.7 ab^{-1} of integrated luminosity, respectively. Using solely simulated events and a simplified detector model, expected limits are derived for the two neutral IDM scalars, resulting in an expected exclusion of nearly the entire kinematically-allowed parameter space.

In addition, a study is presented on the performance of graph neural networks (GNNs) for electron and photon identification versus pileup, carried out in the context of the High-Luminosity LHC CMS endcap upgrade. Utilising only the raw energy deposits in the detector, the GNN is capable of automatically extracting meaningful features from the data, and is shown to outperform current methods based on boosted decision trees.

Acknowledgements

I would like to thank the Imperial College HEP group and the Science and Technology Facilities Council, for giving me the opportunity to carry out this research. In addition, I must thank my supervisors, Anne-Marie and Nick, for their invaluable insight into particle physics. Without either of you guiding me through this process, this thesis would not have been possible. This is extended to Alexandre Nikitenko, Olga Kodolova and Tania Robens, who laid much of the groundwork for the main analysis. I would also like to thank all my colleagues at Imperial for not only providing indispensable input, but for creating a supportive atmosphere throughout my studies. Additionally, I feel I owe a debt of gratitude to all my colleagues outside of Imperial, especially those I was lucky enough to work with during my time at CERN. Finally, thank you to my friends and family for helping me relax and escape from the stresses of the PhD. Your love and support have enriched my life beyond measure; I am profoundly grateful to know every one of you.

Contents

1	Introduction	8
2	Theory and Motivation	10
2.1	The Standard Model of Particle Physics	10
2.1.1	Fields and Lagrangians	12
2.1.2	Gauge Theory	15
2.1.3	The Standard Model Lagrangian	18
2.2	The Dark Matter Problem	23
2.3	The Inert Doublet Model	26
3	The LHC and the CMS Experiment	30
3.1	The Large Hadron Collider	30
3.2	The CMS Detector	33
3.2.1	Tracker	34
3.2.2	Electromagnetic Calorimeter	35
3.2.3	Hadronic Calorimeter	36
3.2.4	Muon Chambers	38
3.2.5	Trigger	40
3.3	Object Reconstruction	40
3.3.1	Particle Flow	40
3.3.2	Muons	43
3.3.3	Electrons	44
3.3.4	Jets	46
3.3.5	Bottom Jets	48
3.3.6	Missing Transverse Energy	49
3.4	Event Simulation	52
4	The High Granularity Calorimeter	55
4.1	Introduction	55
4.2	High Granularity Calorimeter	57
4.3	Electron and Photon Identification	60
4.3.1	Boosted Decision Trees	60
4.3.2	Graph Neural Networks	64

4.3.3	BDT and GNN Performance	66
5	Search for Inert Scalars at CMS in the Dilepton Final State	70
5.1	Introduction	70
5.2	Samples	72
5.3	Base Selections	74
5.4	Event Selection	77
5.4.1	Input Features	78
5.4.2	Training and Performance	83
5.4.3	Binning Optimisation	89
5.4.4	pNN Interpolation	93
5.5	Statistical Model	97
5.6	Modelling Overview	98
5.7	Background Modelling	102
5.7.1	WW/tt Control Region	102
5.7.2	WZ Control Region	104
5.7.3	MisId Control Region	106
5.7.4	ZZ Control Region	113
5.7.5	Drell-Yan	116
5.8	MC Corrections and Uncertainties	121
5.9	Results	124
6	Search for Inert Scalars at the FCC in the Dilepton Final State	133
6.1	Introduction	133
6.2	Samples	135
6.3	Object Selections	136
6.4	Event Selection	137
6.5	Expected Limits	142
7	Conclusion	146
	Bibliography	148

List of Figures

2.1	Particle content of the SM	13
2.2	Orbital velocity of stars and the bullet cluster observation	25
2.3	Leading IDM Feynman diagrams for dilepton production	29
3.1	Schematic diagram of the CERN accelerator complex	32
3.2	Total integrated luminosity delivered by the LHC to CMS for each year	32
3.3	The CMS pileup rates for each year	33
3.4	Perspective view of the CMS detector	34
3.5	Schematic diagram of the CMS tracker	35
3.6	Schematic diagram of the CMS ECAL	37
3.7	Schematic diagram of the CMS HCAL	38
3.8	Schematic diagram of the CMS Muon Chambers	39
3.9	Example of particle detection at CMS.	41
3.10	Distributions of electron identification variables	46
3.11	Jet energy resolution for CHS and PUPPI jets.	48
3.12	Performance of the DeepJet algorithm.	49
3.13	Illustration of the event kinematics of a Z boson with hadronic recoil.	51
3.14	Resolution performance of PF- $\vec{E}_T^{\text{miss}}$ and PUPPI- $\vec{E}_T^{\text{miss}}$	51
3.15	Schematic of a full CMS event simulation.	54
4.1	Schematic diagram of the HGCAL	58
4.2	Before and after HGCAL timing cut	59
4.3	Simulation of clusters in the HGCAL	60
4.4	Distributions of HGCAL L1T BDT features	63
4.5	Example of an undirected graph	65
4.6	EdgeConv layer and GNN architecture	66
4.7	HGCAL L1T BDT and GNN output scores	68
4.8	HGCAL L1T BDT and GNN ROC curves	68
4.9	HGCAL L1T BDT and GNN efficiencies versus p_T and η	69
5.1	Parameters of simulated IDM samples	73
5.2	Generator-level plots of IDM samples	74
5.3	Distributions after the base selections	84
5.4	Example of iterative method to find input features	85

5.5	ROC curves for different IDM mass points	85
5.6	Comparison between BDTs trained at individual mass points and the pNN	87
5.7	Distribution of the full pNN output for $M_H = 70, M_A = 120$ GeV	88
5.8	Distribution of the full pNN output for $M_H = 70, M_A = 160$ GeV	88
5.9	Comparison between individual binning optimisation and simultaneous optimisation	90
5.10	Comparison of bin boundaries for highest pNN scores	91
5.11	Blinded signal region distributions for the $\mu^- \mu^+$ channel	91
5.12	Blinded signal region distributions for the $e^- e^+$ channel	92
5.13	Example of spline interpolation	95
5.14	Comparison between spline interpolation and actual signal distributions	96
5.15	Schematic overview of the different control regions	101
5.16	Distributions for the $W^- W^+ / t\bar{t}$ control region with no b-jets	103
5.17	Distributions for the $W^- W^+ / t\bar{t}$ control region with at least 1 b-jet	103
5.18	Distributions for the opposite-sign WZ control region	105
5.19	Distributions for the same-sign WZ control region	105
5.20	Fake rates for electrons and muons for 2016.	107
5.21	Fake rates for electrons and muons for 2017.	107
5.22	Fake rates for electrons and muons for 2018.	108
5.23	Fake rates for electrons and muons for 2022.	108
5.24	Distributions for the $e^\pm e^\pm$ MisId control region	111
5.25	Distributions for the $\mu^\pm \mu^\pm$ MisId control region	112
5.26	Distributions for the $e^\pm \mu^\pm$ MisId control region	112
5.27	ZZ fit values with a selection of $\text{pNN}(M_H, M_A) > 0.9$	114
5.28	ZZ fit values with a selection of $\text{pNN}(M_H, M_A) > 0.8$	114
5.29	Distributions for ZZ control region	115
5.30	Differences between NLO and LO Drell-Yan samples before the k-factors corrections	117
5.31	Drell-Yan NLO/LO k-factors for all years	118
5.32	Differences between NLO and LO Drell-Yan samples after the k-factors corrections	119
5.33	Distributions of the validation region for $Z/\gamma^* + \text{jets}$	120
5.34	Example distributions of systematic uncertainties	123
5.35	Post-fit signal region distributions for $M_H = 70$ and $M_A = 110, 120$ GeV	126
5.36	Post-fit signal region distributions for $M_H = 70$ and $M_A = 130, 160$ GeV	127
5.37	Post-fit distributions for $M_H = 70$ and $M_A = 120$ GeV	128
5.38	Post-fit distributions for $M_H = 70$ and $M_A = 120$ GeV	129
5.39	Post-fit distributions for $M_H = 70$ and $M_A = 160$ GeV	130
5.40	Post-fit distributions for $M_H = 70$ and $M_A = 160$ GeV	131
5.41	95% CL exclusion limits in the $M_H - \Delta(M_A, M_H)$ plane	132
6.1	Distributions of the invariant mass and dilepton momentum after base selections	138

6.2	Dimuon energy and mass distributions for $\sqrt{s} = 240$ GeV	138
6.3	Dimuon energy and mass distributions for $\sqrt{s} = 365$ GeV	139
6.4	pNN output distributions for $\sqrt{s} = 240$ GeV	143
6.5	pNN output distributions for $\sqrt{s} = 365$ GeV	144
6.6	IDM exclusion limits and significance for $\sqrt{s} = 240$ GeV	145
6.7	IDM exclusion limits and significance for $\sqrt{s} = 365$ GeV	145

List of Tables

2.1	Couplings for the IDM particles	29
4.1	Features for the HGCAL L1T e/γ BDT	62
5.1	Definitions for the electron and muon selections	75
5.2	Base selections to reduce SM backgrounds.	76
5.3	Initial set of input features for the pNN feature optimisation procedure . . .	81
5.4	Final set of input features to the pNN	82
5.5	Hyperparameters for the pNN architecture grid search	86
5.6	Dominant systematic uncertainties	125
6.1	Base selections to reduce SM backgrounds	138
6.2	Background and signal yields at $\sqrt{s} = 240$ GeV	139
6.3	Background and signal yields at $\sqrt{s} = 365$ GeV	140

Chapter 1

Introduction

The principal goal of particle physics is to identify the base constituents of matter that make up the universe, and to understand the fundamental interactions between them. Our current most successful theory is the *Standard Model* (SM). Established in the 1970s, the SM describes how all the known elementary particles interact via the strong, weak and electromagnetic force [2–6]. Over the 5 decades since its formulation, the SM has undergone intense experimental scrutiny across a vast range of energies, with many measurements performed at unprecedented levels of precision [7]. The final component of the SM to be experimentally verified, the Higgs Boson, was discovered by the ATLAS and CMS experiments in 2012, marking a vital milestone for the Standard Model and a monumental achievement for particle physics [8, 9].

Despite the success of the SM, evidence points to it being incomplete. Notably, the SM lacks a description of gravity, with the model being incompatible with General Relativity. Additionally, cosmological models predict the existence of dark matter, which conflicts with the SM as it contains no viable particle candidate to match observations [10, 11]. Furthermore, the SM cannot describe neutrino oscillations [12], nor the matter-antimatter asymmetry of the universe [13]. This tension, therefore, necessitates the search for a more complete theory of Nature. Such theories are collectively referred to as *Beyond the Standard Model* (BSM) theories.

This thesis begins with an outline of the Standard Model of particle physics in Chapter 2, introducing the fundamental constituents of matter and detailing the interaction forces between them. This is followed by an overview of the experimental evidence for dark matter and the incompatibility of such observations with the SM. Using this tension as motivation, a BSM theory is described that contains a viable dark matter candidate: the Inert Doublet Model (IDM). By extending the currently-known Higgs sector of the Standard Model, the IDM introduces four new fundamental particles: H , A and $H^\pm$. The lightest of which, H , is a stable, weakly-interacting, massive particle that forms the dark matter candidate. This chapter details the interactions of these new particles, before outlining the possible signatures of the IDM at a collider. Notably, this focuses on the dilepton final state, as

initial studies have shown this to be the most promising channel for an IDM search.

Chapter 3 outlines the experimental apparatus and methods utilised in the search for the Inert Doublet Model. This includes an overview of the Compact Muon Solenoid (CMS) detector, covering the design and motivation for each of the sub-detector components, as well as the methods utilised to identify and construct different physics objects. Following this, Chapter 4 details the upgrade of the CMS end-cap calorimeters to the High-Granularity Calorimeter (HGCAL) in preparation for the future operation of the LHC: the High-Luminosity LHC. Using the raw information from the HGCAL, a new technique involving graph neural networks is presented for identifying electrons and photons versus background sources.

Chapter 5 describes the main analysis undertaken by the author, searching for the Inert Doublet Model in the dilepton final state, using proton-proton collision data collected at CMS from 2016 to 2022. This starts with the definitions of the required physics objects, followed by a set of base selections to remove easily-reducible SM backgrounds. Subsequently, the main signal extraction method is described, consisting of a deep neural network, and the signal region is defined using the output of this network. Following this, various methods are presented for validating the modelling of the SM backgrounds in the signal region. Finally, the systematic uncertainties are discussed. These are included in a maximum likelihood fit to set limits on the masses of the two neutral IDM particles, H and A .

This thesis concludes with Chapter 6, which presents a prospective sensitivity study for the Future Circular Collider (FCC) when it is operating in its initial stage colliding electron-positron pairs (FCC-ee). Specifically, this chapter studies the sensitivity to the Inert Doublet Model in the dilepton final state when operating at a centre of mass energy of $\sqrt{s} = 240$ GeV and $\sqrt{s} = 365$ GeV. Using purely simulation and a simplified detector model, the analysis applies similar techniques to those used in Chapter 5, to estimate expected limits on the masses of the two neutral IDM particles, H and A . Ultimately, this study highlights the benefits of low-energy precision colliders when searching for the direct production of BSM particles.

Chapter 2

Theory and Motivation

Contents

2.1	The Standard Model of Particle Physics	10
2.1.1	Fields and Lagrangians	12
2.1.2	Gauge Theory	15
2.1.3	The Standard Model Lagrangian	18
2.2	The Dark Matter Problem	23
2.3	The Inert Doublet Model	26

This chapter begins with an overview of the Standard Model of particle physics, including the particle content and the associated symmetry structure. This is followed by an outline on its construction, first introducing the concept of the Lagrangian and gauge theories, before moving onto the different sections of the SM. Next, the problem of dark matter is described, with a brief overview of the experimental observations and the tension of these observations with the SM. Finally, the last section presents a BSM theory, called the Inert Doublet Model (IDM), which is a minimal extension to the SM to include an additional Higgs doublet, leading to a viable dark matter candidate.

2.1 The Standard Model of Particle Physics

The Standard Model of particle physics is a quantum field theory (QFT) in which the excitations of fields give rise to the observed particles. The set of particles described by the SM is presented in figure 2.1. These are split into two major groups: the spin- $\frac{1}{2}$ fermions that constitute normal matter; and the spin-integer bosons that are the force carriers, with all bosons being spin-1 except the Higgs boson which is spin-0.

The spin- $\frac{1}{2}$ fermions can be further divided into *quarks* and *leptons*. Quarks interact with the strong force and, therefore, possess *colour charge* which comes in three types: red, green and blue. The mediator of the strong force, the gluon, also possess colour charge, which allows gluons to couple to themselves. Quarks also carry an *electric charge*¹ of $+\frac{2}{3}$

¹Given in the SI units of elementary charge, e .

or $-\frac{1}{3}$, which enables them to couple to the spin-1 mediator of the electromagnetic force, the photon (γ). Additionally, quarks hold weak isospin and/or weak hypercharge, allowing them to couple to the weak force carriers, the $W^\pm$ and the Z .

The leptons, on the other hand, do not possess colour charge, and as such do not couple to the strong force. Leptons are split based on their electric charge into those that have a charge of -1 (ℓ), and the *neutrinos* (ν) that are electrically neutral (charge of 0). However, all leptons do interact via the weak force.

All fermions are grouped into three generations, with each generation containing the same properties except for their masses. Each includes two quarks, one with charge $+\frac{2}{3}$ and one with $-\frac{1}{3}$, and two leptons, one with charge -1 and a neutrino. The first generation contains the lightest particles, which are stable and include the primary constituents of visible matter. Finally, for each fermion, there exists a corresponding anti-fermion that has the same mass, but opposite charge and parity.

Central to the construction of the SM is Noether's Theorem [14]. Noether's theorem states that if the Lagrangian is invariant under a continuous transformation, then there is a corresponding conserved quantity. Each invariant transformation is called a symmetry of the Lagrangian. Conversely, if there is a conserved quantity in Nature, then there is an expected symmetry of the Lagrangian. In the case of the SM, the underlying symmetries are *gauge symmetries*, where the Lagrangian is invariant to local phase transformations that change the phase of fields depending on the location in space-time. Such theories are referred to as *Gauge Theories*. The Standard Model is a gauge theory with gauge structure

$$\text{SU}(3)_c \times \text{SU}(2)_L \times \text{U}(1)_Y, \quad (2.1)$$

where the subscripts c , L and Y denote colour, left-chirality and hypercharge, respectively. The gauge structure can be decomposed into two main parts: the Quantum Chromodynamics (QCD) sector, corresponding to the $\text{SU}(3)_c$ gauge theory; and the Electroweak (EW) sector, corresponding to $\text{SU}(2)_L \times \text{U}(1)_Y$.

The QCD sector is concerned with the kinematics and couplings of the strong force, which is mediated by eight massless gluon fields that couple to quarks. This coupling is then dependent on a coupling strength parameter, g_s . Moreover, as the group $\text{SU}(3)_c$ is non-Abelian, this consequently leads to gluon self-interactions. Finally, QCD is not a chiral theory, and as such it makes no distinction between left and right-handed fermions.

The Electroweak sector contains four gauge bosons $W_\mu^i (i = 1, 2, 3)$ and B_μ which have a coupling of g and g' for the $\text{SU}(2)_L$ and the $\text{U}(1)_Y$ groups, respectively. Unlike the strong force, the electroweak sector is a chiral theory, and therefore acts differently on left and right-handed fermions. This is realised by having only left-handed particles transform under $\text{SU}(2)_L$, whereas both left and right-handed particles transform under $\text{U}(1)_Y$. As a result, the chiral nature introduces parity violation in the weak force.

The gauge structure 2.1 does not produce the observed particles as it alone cannot permit mass terms that are invariant under the gauge transformations, clearly contradicting the observed masses of the gauge bosons and fermions. This is resolved, however, through the Brout-Englert-Higgs (BEH) mechanism [5, 6, 15], which introduces mass terms from spontaneous symmetry breaking in the EW sector:

$$\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y \rightarrow \mathrm{SU}(3)_c \times \mathrm{U}(1)_Q. \quad (2.2)$$

After symmetry breaking, a massive scalar boson, the Higgs boson, appears which introduces the necessary mass terms to the $W^\pm$, Z and fermions.

Overall, the Standard Model Lagrangian can be broken down into four main sections:

$$\mathcal{L}_{\mathrm{SM}} = \mathcal{L}_{\mathrm{gauge}} + \mathcal{L}_{\mathrm{fermion}} + \mathcal{L}_{\mathrm{Higgs}} + \mathcal{L}_{\mathrm{Yukawa}}. \quad (2.3)$$

To fully understand the SM, a review of each of its components is given, starting with the relevant theory on fields and Lagrangians in section 2.1.1, followed by an overview of Gauge theories in the context of complex scalar fields in section 2.1.2. Finally, each component of $\mathcal{L}_{\mathrm{SM}}$ is discussed in section 2.1.3.

2.1.1 Fields and Lagrangians

The Standard Model is described by a Lagrangian density², $\mathcal{L}_{\mathrm{SM}}$, which is a real function of a set of fields and their derivatives. This is used to define the *action*: $S = \int_{\mathbb{R}^4} \mathcal{L}_{\mathrm{SM}} d^4x$. Requiring that the action remains stationary under variations of the fields leads to the Euler-Lagrange equations, which in turn yield the equations of motion for the fields.

An example of this is the Dirac equation which is the equation of motion for all spin- $\frac{1}{2}$ fermions in the Standard Model:

$$\mathcal{L}_{\mathrm{Dirac}} = \bar{\psi}(i\gamma^\mu \partial_\mu - m)\psi \xrightarrow{\text{Euler-Lagrange}} (i\gamma^\mu \partial_\mu - m)\psi = 0, \quad (2.4)$$

where m is the mass of the particle and γ^μ ($\mu \in 0, 1, 2, 3$) are the 4×4 gamma matrices. The gamma matrices are defined such that $\{\gamma^\mu, \gamma^\nu\} = \gamma^\mu \gamma^\nu + \gamma^\nu \gamma^\mu = 2g^{\mu\nu}$ where $g^{\mu\nu}$ is the standard Minkowski metric $g^{\mu\nu} = \mathrm{diag}(1, -1, -1, -1)$. Finally, ψ is the particle field, which is a four-component *spinor*, and its adjoint $\bar{\psi} = \psi^\dagger \gamma^0$. As previously mentioned, the Weak force acts differently on left-handed particles than on right-handed particles, which motivates decomposing spinors into left and right-handed chiral components. This is accomplished by first defining a new gamma matrix

$$\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3, \quad (2.5)$$

²The Lagrangian and Lagrangian Density will be used interchangeably throughout this thesis.

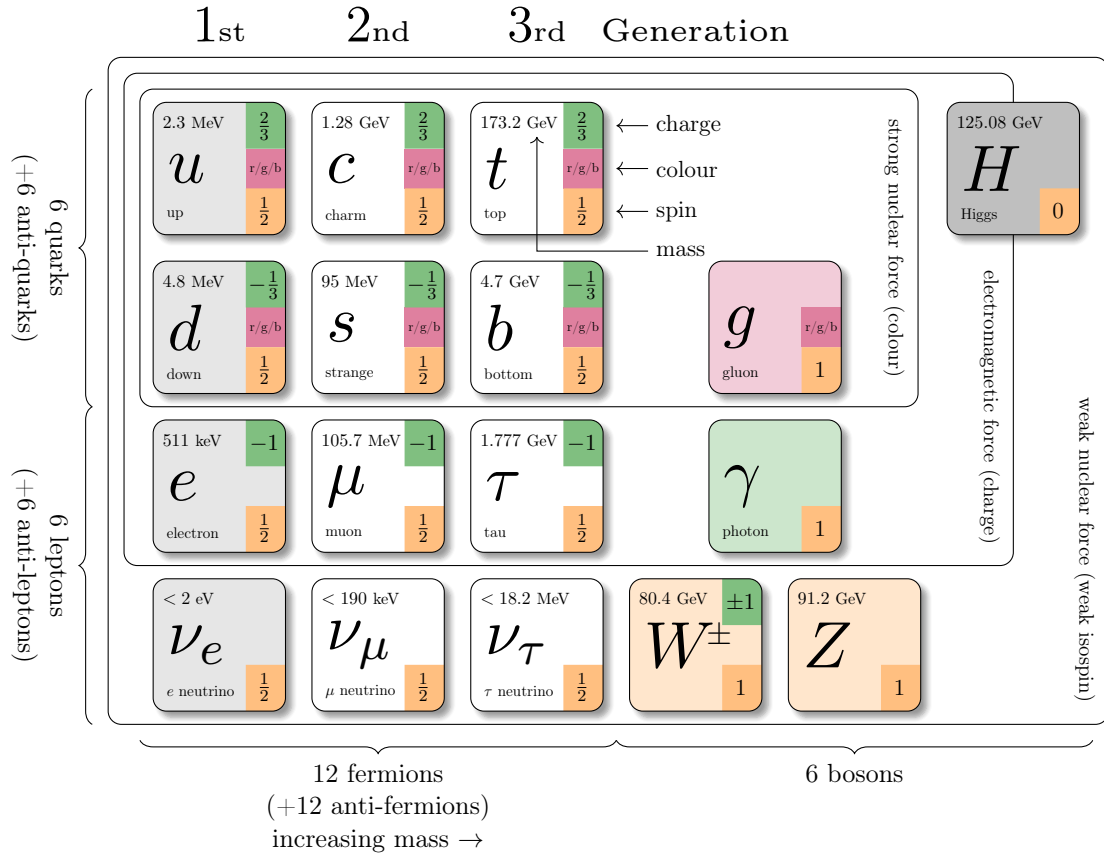


Figure 2.1: The particle content of the Standard Model, with the mass, electric charge, colour charge and spin for each particle. All the masses of the particles are taken from Reference [16], except for the Higgs boson mass which is taken from Reference [17].

which is then used to define projection operators

$$P_L = \frac{1}{2}(1 + \gamma^5), \quad P_R = \frac{1}{2}(1 - \gamma^5). \quad (2.6)$$

The left and right-handed components of a spinor field are then isolated by applying the projection operators:

$$\psi_{L/R}(x) = P_{L/R}\psi(x). \quad (2.7)$$

By separating the field into both components, the Lagrangian decomposes into the following form

$$\mathcal{L}_{\text{Dirac}} = \bar{\psi}_L i \not{\partial} \psi_L + \bar{\psi}_R i \not{\partial} \psi_R - m(\bar{\psi}_R \psi_L + \bar{\psi}_L \psi_R), \quad (2.8)$$

where for convenience we have used the notation $a_\mu \gamma^\mu = \not{a}$ for an operator a . Therefore, if the particle has mass, then left and right-handed states are coupled through the final term, leading to mixing. If, however, the particle is massless, then these entirely decouple, allowing for a consistent theory with only one chiral state.

As the weak force is a parity violating theory, it is instructive to demonstrate the parity properties of these states. Under the Parity operation, P , which reverses space coordinates $\vec{x} \rightarrow -\vec{x}$, the newly introduced γ^5 matrix transforms as

$$P : \gamma^5 \rightarrow -\gamma^5. \quad (2.9)$$

As γ^5 is in the definition of the projection operators (equation 2.6), under parity it exchanges P_L and P_R , leading chiral states to transform as:

$$P : \quad \psi_{L/R}(\vec{x}, t) \rightarrow \psi_{R/L}(-\vec{x}, t), \quad (2.10)$$

thus changing a left-handed spinor into a right-handed spinor and vice versa. As a consequence, by having different interactions for the two chiral states, under a parity transformation these interactions are swapped, leading to a parity-violating theory. This decomposition into left and right-handed states — and their associated parity properties — provides the foundation for constructing physical particle states. The details of this construction are presented in section 2.1.3.

2.1.2 Gauge Theory

Abelian Gauge Theory

Using a complex scalar field, $\psi(x)$, and its complex conjugate, $\psi^*(x)$, the following Lagrangian can be defined:

$$\mathcal{L} = \partial_\mu \psi(x) \partial^\mu \psi^*(x) - V(\psi(x) \psi^*(x)), \quad (2.11)$$

which is the product of pairs of $\psi(x)$, $\psi^*(x)$ and their derivatives, such that the Lagrangian remains real. This is invariant under the phase transformation:

$$\psi(x) \rightarrow \psi'(x) = e^{ig\alpha} \psi(x), \quad (2.12)$$

where α and g are arbitrary real numbers. Transformations of this type are referred to as *global* transformations, as they change the phase of the field equally everywhere [18]. Specifically, this transformation is called a U(1) phase transformation. Moreover, because different values of α_i commute, $[\alpha_i, \alpha_j] = 0$, successive transformations of this type commute³, $e^{ig\alpha_1} e^{ig\alpha_2} = e^{ig\alpha_2} e^{ig\alpha_1}$. This is called an *Abelian* transformation.

If, however, it is demanded that α be a *local* transformation dependent on spacetime, $\alpha = \alpha(x)$, then issues arise from the derivative terms as

$$\begin{aligned} \partial_\mu \psi'(x) &= \partial_\mu (e^{ig\alpha(x)} \psi(x)) = e^{ig\alpha(x)} \partial_\mu \psi(x) + ig\psi(x) [\partial_\mu \alpha(x)] e^{ig\alpha(x)} \\ &\neq e^{ig\alpha(x)} \partial_\mu \psi(x). \end{aligned} \quad (2.13)$$

Therefore, the Lagrangian picks up additional terms and is not invariant. To overcome this, an additional vector field is introduced to form the *covariant derivative*:

$$D_\mu \equiv \partial_\mu + igA_\mu(x). \quad (2.14)$$

If we then require $A_\mu(x)$ to transform under the gauge symmetry as:

$$A'_\mu(x) = A_\mu(x) - \frac{1}{g} \partial_\mu \alpha(x) \quad (2.15)$$

then the covariant derivative term transforms as required under the local U(1) transformation:

³Here, g remains constant, and as it is just a real number it also commutes.

$$\begin{aligned}
D'_\mu \psi'(x) &= [\partial_\mu + igA'_\mu(x)]\psi'(x) \\
&= [\partial_\mu + ig(A_\mu(x) - \frac{1}{g}\partial_\mu\alpha(x))](e^{ig\alpha}\psi(x)) \\
&= e^{ig\alpha(x)}\partial_\mu\psi(x) + igA_\mu(x)e^{ig\alpha(x)}\psi(x) \\
&= e^{ig\alpha(x)}[\partial_\mu + igA_\mu(x)]\psi(x) \\
&= e^{ig\alpha(x)}D_\mu\psi(x).
\end{aligned} \tag{2.16}$$

However, because the field $A_\mu(x)$ is dynamical — it varies in space and time — we have to also include a gauge-invariant kinetic term for it in the Lagrangian:

$$\mathcal{L}_{\text{gauge}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu}, \tag{2.17}$$

where $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ is the *field tensor*. If the field was massive, it would require an additional mass term of the form $-\frac{1}{2}m^2 A_\mu A^\mu$, however this is only gauge invariant for $m = 0$, forcing the field to be massless.

Combining all the elements together results in the final Lagrangian for a gauge-invariant complex scalar theory coupled to a vector field:

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + D_\mu\psi(x)D^\mu\psi^*(x) - V(\psi(x)\psi^*(x)). \tag{2.18}$$

In summary, by requiring local gauge invariance, a vector field $A_\mu(x)$ had to be introduced with the constraint that it is massless. An Abelian gauge theory similar to the one outlined here will form the basis of Quantum Electrodynamics (QED) which describes the interactions between the fermions and the massless photon field.

Non-Abelian Gauge Theory

The Abelian procedure can be generalised to the non-Abelian case where the fields enter in multiplets:

$$\Psi(x) = \begin{pmatrix} \psi_1(x) \\ \psi_2(x) \\ \vdots \\ \psi_N(x) \end{pmatrix} \tag{2.19}$$

within the Lagrangian

$$\mathcal{L} = \partial_\mu \bar{\Psi}(x)\partial^\mu \Psi(x) - V(\bar{\Psi}(x)\Psi(x)), \tag{2.20}$$

where $\bar{\Psi}(x)$ is the hermitian conjugate of $\Psi(x)$. Similar to the Abelian case, we now want a local gauge transformation that acts on the multiplets:

$$\begin{aligned}\Psi(x) &\rightarrow \Psi(x)' = e^{ig\alpha(x)\cdot\mathbf{T}}\Psi(x) \\ &= U(\alpha)\Psi(x),\end{aligned}\tag{2.21}$$

where $\alpha_i(x)$ ($i = (1, \dots, N)$) are arbitrary functions of spacetime, and T_i are $N \times N$ matrices which are a representation of the group generators, and g is a coupling constant [18]. As T_i are matrices, however, they do not generally commute and instead form a *non-Abelian* group:

$$[T_i, T_j] = if_{ijk}T_k,\tag{2.22}$$

where f_{ijk} are called the *structure constants* of the group. As in the Abelian case, the derivative terms are not gauge invariant. Therefore, a similar gauge-invariant covariant derivative is defined as

$$D_\mu \equiv \partial_\mu + ig\mathbf{T} \cdot \mathbf{W}_\mu(x),\tag{2.23}$$

where $\mathbf{W}_\mu(x) = (W_\mu^1(x), W_\mu^2(x), \dots, W_\mu^N(x))$ are N gauge fields. By demanding that the new fields transform under the gauge transformation as⁴:

$$\mathbf{T} \cdot \mathbf{W}'_\mu(x) = U(\alpha)[\mathbf{T} \cdot \mathbf{W}_\mu(x) - \frac{i}{g}U^{-1}(\alpha)\partial_\mu U(\alpha)]U^{-1}(\alpha)\tag{2.24}$$

then the covariant derivative transforms as required:

$$D_\mu \Psi(x) \rightarrow D'_\mu \Psi'(x) = U(\alpha)D_\mu \Psi(x),\tag{2.25}$$

Finally, we need to introduce the kinetic terms for each of the vector fields, however the form used previously (equation 2.17) would not be gauge invariant due to the symmetry generators being non-Abelian. Instead, a gauge-invariant kinetic energy term is formed using the non-Abelian field tensor:

$$G_{\mu\nu}^j = \partial_\mu W_\nu^j - \partial_\nu W_\mu^j + gf^{jkl}W_\mu^k W_\nu^l,\tag{2.26}$$

which, if $f^{jkl} = 0$, reduces to the previous Abelian case. Whilst $G_{\mu\nu}^j$ for different fields could be combined to form Lorentz invariant terms, such as $G_{\mu\nu}^0 G^{1,\mu\nu}$, they are not in

⁴This is identical to the transformation in equation 2.15 if the generators of $U(\alpha)$ are Abelian.

general gauge invariant. As such, the only way to form the gauge invariant kinetic energy term is as follows:

$$\mathcal{L}_{\text{kinetic}} = -\frac{1}{4}G_{\mu\nu}^j G^{j,\mu\nu}, \quad (2.27)$$

where $j = (1, \dots, N)$. Unlike the Abelian case, the non-Abelian fields $\mathbf{W}_\mu(x)$ are self-coupled through this additional term in $G_{\mu\nu}^j G^{j,\mu\nu}$. Likewise, however, no mass term is permissible for the gauge fields as terms of the form $\mathbf{W}_\mu(x)\mathbf{W}^\mu(x)$ would violate gauge invariance.

Overall, the final Lagrangian for this non-Abelian gauge theory is given by

$$\mathcal{L} = -\frac{1}{4}G_{\mu\nu}^j G^{j,\mu\nu} + D_\mu \bar{\Psi}(x) D^\mu \Psi(x) - V(\bar{\Psi}(x)\Psi(x)). \quad (2.28)$$

In summary, using a multiplet of fields and requiring the Lagrangian to be invariant under non-Abelian transformations, we were forced to include N massless gauge fields that not only couple to the scalar multiplets, but also couple to themselves.

2.1.3 The Standard Model Lagrangian

The Standard Model Lagrangian can be broken down into four main sections:

$$\mathcal{L}_{\text{SM}} = \mathcal{L}_{\text{gauge}} + \mathcal{L}_{\text{fermion}} + \mathcal{L}_{\text{Higgs}} + \mathcal{L}_{\text{Yukawa}}. \quad (2.29)$$

Each of these will be studied individually in the following sections.

The Gauge Sector

The SM symmetry structure in equation 2.1 introduces gauge fields whose dynamics are described by the gauge sector. The Lagrangian for this is given by

$$\mathcal{L}_{\text{gauge}} = -\frac{1}{4}B_{\mu\nu}B^{\mu\nu} - \frac{1}{4}W_{\mu\nu}^j W^{j,\mu\nu} - \frac{1}{4}G_{\mu\nu}^a G^{a,\mu\nu}, \quad (2.30)$$

where $B_{\mu\nu}$ is the field tensor for the Abelian group $U(1)_Y$, and $W_{\mu\nu}^j$ and $G_{\mu\nu}^a$ are the field tensors for the Non-Abelian groups $SU(2)_L$ and $SU(3)_c$, respectively [18]. The field tensors take the following form:

$$\begin{aligned} B_{\mu\nu} &= \partial_\mu B_\nu - \partial_\nu B_\mu \\ W_{\mu\nu}^j &= \partial_\mu W_\nu^j - \partial_\nu W_\mu^j + g\epsilon^{jkl}W_\mu^k W_\nu^l & j, k, l = 1, 2, 3 \\ G_{\mu\nu}^a &= \partial_\mu G_\nu^a - \partial_\nu G_\mu^a + g_s f^{abc}G_\mu^b G_\nu^c & a, b, c = 1, \dots, 8, \end{aligned} \quad (2.31)$$

where f^{abc} and ϵ^{jkl} are the $SU(3)_c$ and the $SU(2)_L$ structure functions. Following section 2.1.2, the non-Abelian gauge fields, G_μ^a and W_μ^j , pick up three and four-point self-interacting terms from the field tensors, respectively. Whereas, the Abelian field, B_μ , does not. Notably, however, no mass terms for the gauge fields can be included as this would violate the required gauge invariance.

The Fermionic Sector

The fermionic sector describes the dynamics and interactions of the spin- $\frac{1}{2}$ matter fields. Both the quark and lepton fields couple to the gauge fields through the covariant derivative

$$D_\mu = \partial_\mu + ig' B_\mu Y + ig W_\mu^j I^j + ig_s G_\mu^a T^a, \quad (2.32)$$

where Y , I^j and T^a are the generators for the $U(1)_Y$, $SU(2)_L$ and $SU(3)_c$ symmetry groups, respectively [18]. The strength of each of the couplings is dependent on the coupling strengths g' , g and g_s .

The form of the generators depends on the fields they act upon. For the strong force, the gluons couple only to quarks, for which the generators are equal to $T^a = \frac{\lambda^a}{2}$, where λ^a ($a = (1, \dots, 8)$) are the eight 3×3 generators of the $SU(3)$ group, referred to as the Gell-Mann matrices [18]. The quark fields, therefore, transform as $SU(3)_c$ triplets. For example, for the up quark:

$$u_L = \begin{pmatrix} u_L^r \\ u_L^g \\ u_L^b \end{pmatrix}, \quad (2.33)$$

where r, g, b are the conserved colour charges from Noether's Theorem. For fields that do not couple to gluons, the $SU(3)_c$ generators instead take the form $T^a = 0$.

As previously mentioned, the weak force is a chiral theory, acting only on left-handed particles and not right-handed particles. Therefore, when acting on left-handed fields the generators $I^j = \frac{\sigma^j}{2}$ where σ^j are the Pauli matrices, and the associated charge is called *isospin*, $t^3 = \pm \frac{1}{2}$. Consequently, the left-handed fields transform as $SU(2)_L$ doublets, such as quark doublets

$$q_L^i = \begin{pmatrix} u_L \\ d_L \end{pmatrix}, \begin{pmatrix} c_L \\ s_L \end{pmatrix}, \begin{pmatrix} t_L \\ b_L \end{pmatrix} \quad (2.34)$$

and lepton doublets

$$\ell_L^i = \begin{pmatrix} \nu_{e,L} \\ e_L \end{pmatrix}, \begin{pmatrix} \nu_{\mu,L} \\ \mu_L \end{pmatrix}, \begin{pmatrix} \nu_{\tau,L} \\ \tau_L \end{pmatrix} \quad (2.35)$$

On the other hand, when acting on right-handed fermion fields, the generators $I^j = 0$, resulting in right-handed fields transforming as singlets under $SU(2)_L$, with isospin charge $t^3 = 0$. These singlets are grouped as such:

$$\begin{aligned} u_R^i &= u_R, c_R, t_R \\ d_R^i &= d_R, s_R, b_R \\ e_R^i &= e_R, \mu_R, \tau_R. \end{aligned} \quad (2.36)$$

Note that within the SM there are no right-handed neutrino fields as they do not interact with the weak, strong or electromagnetic forces.

Finally, for $U(1)_Y$, the generators act on any fields with hypercharge charge y , which is related to the electric charge through the relation

$$Q = t^3 + \frac{1}{2}y, \quad (2.37)$$

where t^3 is isospin.

Using the covariant derivative defined in equation 2.32, and the fields defined above, the fermionic Lagrangian can be constructed:

$$\begin{aligned} \mathcal{L}_{\text{fermion}} = & \bar{q}_L^j i \not{D} q_L^j + \bar{u}_R^j i \not{D} u_R^j + \bar{d}_R^j i \not{D} d_R^j \\ & + \bar{l}_L^j i \not{D} l_L^j + \bar{e}_R^j i \not{D} e_R^j. \end{aligned} \quad (2.38)$$

Notably, however, this does not contain mass terms. This is due to the fact that mass terms enter through combinations of left and right-handed fields: $m(\bar{\psi}_L \psi_R + \bar{\psi}_R \psi_L)$. As the left-handed fields transform differently to right-handed fields, mass terms of this form are not gauge invariant unless $m = 0$.

The Higgs Sector

The Lagrangian described thus far has many of the required components for a theory of nature. It, however, misses one major component to comply with experimental observations: the particle masses. Not only are mass terms for the gauge fields forbidden — a necessity to describe the massive W and Z bosons that we observe — but also the fermionic mass terms needed to describe the quark and lepton masses. In the Standard Model, mass

terms can be reintroduced into the Lagrangian in a gauge invariant way using the Brout-Englert-Higgs (BEH) Mechanism, which is accomplished via a process called *spontaneous symmetry breaking* (SSB) [5, 6, 19].

Spontaneous symmetry breaking arises when the lowest energy state of a system does not obey the symmetry of the Lagrangian, which in turn generates mass terms for the symmetry generators that are broken. In the case of the SM, the symmetry breaking occurs to the $SU(2)_L \times U(1)_Y$ groups, leading to a single $U(1)_Q$ symmetry that changes the gauge symmetry as follows:

$$SU(3)_c \times SU(2)_L \times U(1)_Y \rightarrow SU(3)_c \times U(1)_Q. \quad (2.39)$$

The $SU(3)_c$ group remains unbroken, however, leaving the gluons massless.

Spontaneous symmetry breaking is achieved by first introducing the Higgs field, H , which transforms as a doublet under $SU(2)_L$ and is parameterised as

$$\phi = \begin{pmatrix} \phi^+ \\ \phi^0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} \phi_1 + i\phi_2 \\ \phi_3 + i\phi_4 \end{pmatrix}, \quad (2.40)$$

where ϕ_i are real. The Higgs field then enters the Lagrangian via

$$\mathcal{L}_{\text{Higgs}} = (D_\mu \phi)^\dagger (D^\mu \phi) - \mu^2 \phi^\dagger \phi - \frac{1}{4} (\phi^\dagger \phi)^2, \quad (2.41)$$

where the covariant derivate is defined in equation 2.32, with the $SU(3)_c$ generators being 0 [19]. The first term constitutes the kinetic part of the Higgs Lagrangian, whilst the latter terms form the Higgs potential, $V(\phi)$,

$$V(\phi) = \mu^2 \phi^\dagger \phi + \frac{1}{4} (\phi^\dagger \phi)^2. \quad (2.42)$$

If $\mu^2 > 0$, then the potential $V(\phi)$ has a minimum at $\phi = 0$. If, however, $\mu^2 < 0$, then the minimum of $V(\phi)$ is not at 0, but some constant value, given by

$$\phi = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v \end{pmatrix}, \quad (2.43)$$

where v is referred to as the *vacuum expectation value* (VEV) of ϕ . To obtain the physical particle spectrum, perturbations around the minimum are considered in the form

$$\phi = e^{iG^j(x)\frac{\sigma^j}{2}} \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v + h(x) \end{pmatrix}, \quad (2.44)$$

which, under a suitable gauge transformation called the *Unitary Gauge* [19], is equal to

$$\phi = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v + h(x) \end{pmatrix}, \quad (2.45)$$

where $h(x)$ is the Higgs Boson.

Using the unitary gauge, it appears that 3 degrees of freedom have been removed from ϕ and the Lagrangian. In reality, through the coupling of the scalar field to the gauge fields in the covariant derivative, the degrees of freedom reappear as longitudinal components of the gauge bosons, thereby giving the bosons mass. This is shown explicitly by expanding the kinetic term, $(D_\mu \phi)^\dagger (D^\mu \phi)$, using the matrix form of the covariant derivative:

$$D_\mu = \partial_\mu + \frac{i}{2} \begin{pmatrix} g' B_\mu + g W_\mu^3 & g(W_\mu^1 - i W_\mu^2) \\ g(W_\mu^1 + i W_\mu^2) & g' B_\mu - g W_\mu^3 \end{pmatrix} \quad (2.46)$$

and only keeping terms that contain v :

$$\begin{aligned} (D_\mu \phi)^\dagger (D^\mu \phi) &= \frac{v^2}{8} [g^2 (W_\mu^1 - i W_\mu^2)(W^{1,\mu} + i W^{2,\mu}) \\ &\quad + (g' B_\mu - g W_\mu^3)(g' B^\mu - g W^{3,\mu})] + \mathcal{O}(h(x)). \end{aligned} \quad (2.47)$$

The massive gauge boson mass terms are identified as

$$\begin{aligned} W_\mu^\pm &= \frac{1}{\sqrt{2}} (W_\mu^1 \mp i W_\mu^2) & \text{with mass } m_W &= \frac{gv}{2} = 81.4 \text{ GeV} \\ Z_\mu &= \frac{1}{\sqrt{g^2 + g'^2}} (g W_\mu^3 - g' B_\mu) & \text{with mass } m_Z &= \sqrt{g^2 + g'^2} \frac{v}{2} = 91.2 \text{ GeV}, \end{aligned} \quad (2.48)$$

which directly depend on the vacuum expectation value, $v = 246$ GeV, and the EW couplings. The massless photon, which has no associated mass term, is defined as

$$A_\mu = \frac{1}{\sqrt{g^2 + g'^2}} (g W_\mu^3 + g' B_\mu) \quad \text{with mass } m_A = m_\gamma = 0. \quad (2.49)$$

By introducing the Higgs field, ϕ , and after spontaneous symmetry breaking of the EW interaction, the required terms naturally emerge: three massive gauge bosons, $W^\pm$ and Z , and a single massless gauge boson, the photon γ . Expanding the Lagrangian using the field redefinitions in equations 2.48 and 2.49 reveals additional interaction terms

$$\mathcal{L}_{\text{Higgs}} \supset \frac{1}{2} \partial_\mu h \partial^\mu h + [m_W^2 W_\mu^+ W^{-,\mu} + \frac{1}{2} m_Z^2 Z_\mu Z^\mu] \left(1 + \frac{h}{v}\right)^2, \quad (2.50)$$

which indicate that the coupling strength of the Higgs to the gauge bosons is proportional to the gauge boson masses, resulting in no direct coupling of the Higgs to the photon.

Finally, considering the potential terms in $\mathcal{L}_{\text{Higgs}}$ after symmetry breaking, we find

$$V(\phi) = \mu^2 h^2 + \lambda v h^3 + \frac{1}{4} \lambda h^4 + \frac{\mu^4}{4\lambda}, \quad (2.51)$$

where the mass of h is given by $m_h = \sqrt{2}\mu = \sqrt{2}\lambda v$, and the higher order terms indicate triple and quartic self-coupling interactions.

The Yukawa Sector

The Yukawa sector is concerned with the mass generation of the leptons, which originally were not permissible as terms of the form $m(\bar{\psi}_L \psi_R + \bar{\psi}_R \psi_L)$ are not gauge invariant. The fermion masses are instead generated by coupling the leptons to the Higgs field, ϕ ; this is the Yukawa sector of the SM Lagrangian.

$$\mathcal{L}_{\text{Yukawa}} = -\lambda_j \bar{l}_{L,j} \phi e_{R,j} - \lambda_a \bar{Q}_{L,a} \phi d_{R,a} - \lambda_a \bar{Q}_{L,a} \tilde{\phi} u_{R,a} + \text{h.c.}, \quad (2.52)$$

where λ_j are the Yukawa couplings for a fermion of type j , and $\tilde{\phi} = \epsilon_{kl}(\phi^l)^\dagger$ with ϵ_{kl} being the totally anti-symmetry matrix, and h.c. is the hermitian conjugate of the first three terms. After spontaneous symmetry breaking, equation 2.52 decomposes into (for a single lepton field for simplicity):

$$\mathcal{L}_{\text{Yukawa}} \supset -\frac{\lambda_e}{\sqrt{2}} v (\bar{e}_L e_R + \bar{e}_R e_L) - \frac{\lambda_e}{\sqrt{2}} h (\bar{e}_L e_R + \bar{e}_R e_L). \quad (2.53)$$

The first term is the exact mass term required, leading to lepton masses $m_j = \frac{\lambda_j v}{\sqrt{2}}$. The values λ_j are not predicted by the SM, but are instead measured through experiments. The second term introduces interactions between the fermions and the field h , with the coupling strength, similar to the gauge boson case, proportional to the mass of the fermions.

2.2 The Dark Matter Problem

A central question in cosmology is determining the composition of the universe. This is well justified: the amount of matter in the universe directly affects its evolution, playing a critical role in understanding the origin, and fate, of the universe. Even if the exact amount is known, the composition of matter remains of interest. The relative fraction of matter that is stars, dust, cold stellar remnants or some other unknown exotic substance

can provide insights into the universe's evolution since its conception. This field of research is not without surprise, however, with observations suggesting that normal baryonic matter comprises only $\sim 5\%$ of the total budget [20]. Instead, the matter content is dominated by two other groups: a non-baryonic matter referred to as *dark matter* ($\sim 27\%$); and an unknown energy known as *dark energy* ($\sim 68\%$). Dark energy is primarily responsible for the accelerated expansion of the universe and is closely related to the cosmological constant in Einstein's field equations. However, relatively little is known about its nature. The focus of this thesis is instead on dark matter (DM). Whilst there exists no direct detection of dark matter, its existence can be inferred through astrophysical observations of its gravitational effects. A few sources of evidence are presented, but for a more complete review of dark matter, refer to References [10, 11].

One of the earliest pieces of evidence for dark matter came in 1933 from observations of rotating galactic clusters, which are groups of galaxies that are gravitationally bound together [21]. By measuring the orbital velocity of each galaxy in the Coma galaxy cluster (containing roughly 1000 galaxies), and the total area that they occupied, the average mass of the galaxy could be inferred. Surprisingly, however, the calculated mass was nearly 2 orders of magnitude larger than the observed mass from their luminosity, indicating an additional source of matter not radiating light. More recent studies of the Coma cluster include better estimates of the baryonic matter, notably from the inclusion of X-ray imaging to detect the intra-cluster gas [11]. This reduces the initial discrepancy, but still predicts that around 85% of the mass is from dark matter, with only 1% and 14% being stars and intra-cluster gas, respectively.

On a smaller scale than galactic clusters, discrepancies are also seen in the rotation of individual galaxies. In spiral galaxies, gravitationally bound stars and interstellar gas orbit around a galactic centre, with rotational velocity dependent on the mass distribution of the galaxy. Therefore, by measuring the orbital velocity versus radial distance, the distribution of mass can be inferred. If the stars and interstellar gas were the only source of mass in the galaxy, then the orbital velocity is expected to decrease versus radial distance past a critical point [22]. However, in 1980, it was observed that instead of decreasing with radius, the velocity stayed almost completely flat, requiring again an unobserved source of mass [23]. An example galaxy is shown in figure 2.2a, which shows the observed distribution of visible matter (disk) and gas, and the inferred dark matter distribution required to match the observed velocities. Notably, figure 2.2a presents another feature of dark matter: it forms structures further out from the centre of the galaxy, which are referred to as *dark matter halos*.

As a final example, consider the phenomenon of *gravitational lensing*, whereby light is deflected due to the curvature of spacetime around a massive object [10]. On a cosmological scale, galaxies and clusters can deflect the light of background objects, acting as a gravitational lens and leading to distorted images of the object. The deflection depends on the mass distribution of the intermediate object, allowing gravitational lenses to be used to infer the mass distribution of object, including both visible and dark matter. An impor-

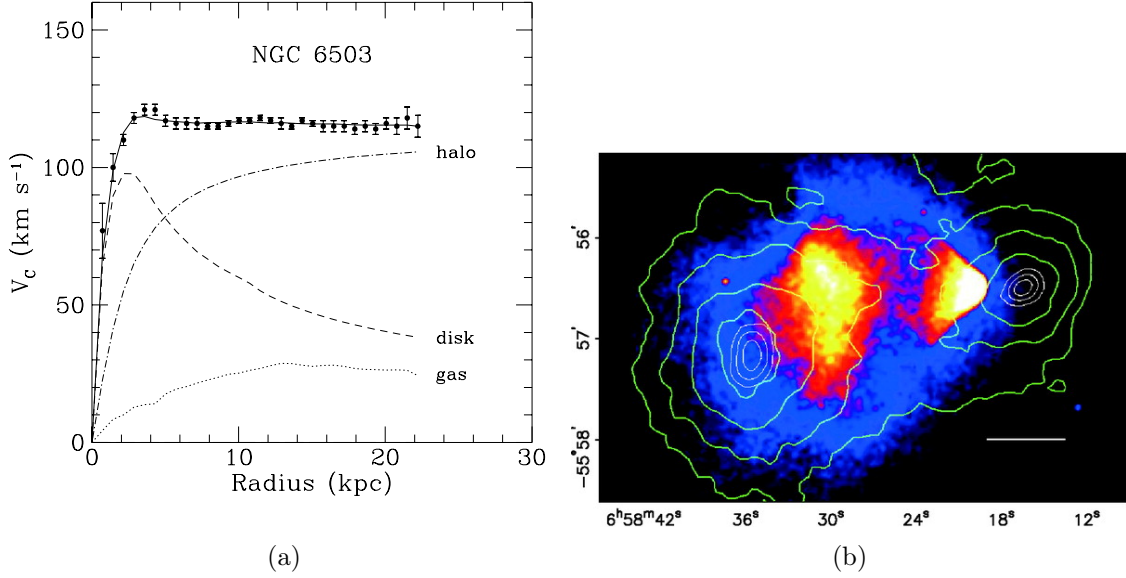


Figure 2.2: (a) Orbital velocity of stars versus radius to centre of the galaxy, taken from [22]. The dashed line indicates the visible component, the dotted indicates the gas, and the dash-dot indicates the inferred dark matter distribution (halo). (b) Bullet cluster observation, taken from [24]. The colours indicate the temperature of the gas, with black corresponding to the coldest and white the hottest. The green lines indicate the reconstructed mass distribution from gravitational lensing.

tant example of this is the Bullet cluster observation of two colliding clusters [24]. As the clusters collide, the stars pass with minimal interactions, whereas the gas feels a drag force, allowing both to be observed separately. Subsequently, gravitational lensing can be used to create a mass-density map of all matter in the cluster, including dark matter, which can be compared to the observed mass distribution. The results are shown in figure 2.2b. Crucially, the gravitational potential does not overlap with the interstellar gas, indicating that a significant amount of the matter passes undisturbed, yet it is not accounted for by the observed stars. Not only does this indicate the majority of the matter is not held in the visible matter, but it also allows for constraints on the self-interactions of the dark matter, as stronger interactions would lead to larger drag forces that are inconsistent with the observations.

Whilst the astrophysical evidence strongly supports a new form of matter, its exact nature remains unknown due to the absence of direct detection. The leading model that best reproduces the observed data is that dark matter takes the form of a weakly interacting massive particle (WIMP). Specifically, a particle that has little to no interactions with photons — hence it is "dark" — and weak interactions with baryonic matter, leading to the observed cosmological structures and dark matter halos. Within the Standard Model, the leading candidates would be the neutrinos. However, their mass is too small to account for the required dark matter density. Consequently, without a viable candidate, focus has shifted to beyond the Standard Model. One such extension that includes a viable dark matter candidate is the Inert Doublet Model.

2.3 The Inert Doublet Model

The Inert Doublet Model (IDM) is a simple extension of the SM to include a second $SU(2)_L$ Higgs doublet, ϕ_D , along with the usual $SU(2)_L$ Higgs doublet, ϕ_S , and an unbroken $\mathbb{Z}_2$ symmetry [25]. The additional doublet, referred to as the *inert doublet*, makes the IDM a Two-Higgs-Doublet-Model (2HDM), but that is separated from other 2HDMs by requiring the $\mathbb{Z}_2$ symmetry to be unbroken even for the vacuum state. Under this symmetry, all standard model fields are even, whilst the inert doublet is odd:

$$\mathbb{Z}_2 : \quad \phi_D \rightarrow -\phi_D, \quad \phi_S \rightarrow \phi_S, \quad SM \rightarrow SM. \quad (2.54)$$

For the Lagrangian, the inert doublet has a standard kinetic term $D_\mu \phi_D D^\mu \phi_D$, with the same covariant derivative defined in equation 2.32. The most generalised, renormalizable, scalar potential can be formed with the two doublets:

$$\begin{aligned} V(\phi_S, \phi_D) = & -\frac{1}{2}[m_{11}^2(\phi_S^\dagger \phi_S) + m_{22}^2(\phi_D^\dagger \phi_D)] + \frac{\lambda_1}{2}(\phi_S^\dagger \phi_S)^2 + \frac{\lambda_2}{2}(\phi_D^\dagger \phi_D)^2 \\ & + \lambda_3(\phi_S^\dagger \phi_S)(\phi_D^\dagger \phi_D) + \lambda_4(\phi_S^\dagger \phi_D)(\phi_D^\dagger \phi_S) + \frac{\lambda_5}{2}[(\lambda_S^\dagger \phi_D)^2 + (\phi_D^\dagger \phi_S)^2], \end{aligned} \quad (2.55)$$

where λ_{1-5} are real parameters. Note that terms with a single $\phi_S^\dagger \phi_D$ are forbidden due to the $\mathbb{Z}_2$ symmetry. Moreover, as the $\mathbb{Z}_2$ symmetry remains unbroken after SSB, the ϕ_D is forced to have a zero VEV, whereas ϕ_S undergoes the same spontaneous symmetry breaking process as in the SM. Consequently, the doublets are written as

$$\phi_S = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v + h(x) \end{pmatrix}, \quad \phi_D = \begin{pmatrix} H^+(x) \\ \frac{1}{\sqrt{2}}(H(x) + iA(x)) \end{pmatrix}, \quad (2.56)$$

where $v = 246$ GeV is the SM VEV, and $h(x)$ is the SM Higgs boson. From the inert doublet, four new particles have now been introduced: two charged scalars $H^\pm$; and two neutral scalars H and A . These new scalars couple to the SM gauge bosons through the usual covariant derivative terms. However, as no Yukawa terms coupling the IDM fields to the fermion fields are $\mathbb{Z}_2$ -symmetric, the symmetry forbids any direct coupling of the IDM particles to the fermionic sector. It should be noted that, contrary to other Two-Higgs-Doublet-Models, H/A do not denote scalar/pseudoscalar particles as ϕ_D has no fermionic interactions. In fact, the exact roles of H and A can be exchanged under the redefinition of $\lambda_5 \rightarrow -\lambda_5$.

One important implication of the $\mathbb{Z}_2$ symmetry is that all interaction vertices must involve an even multiple of IDM particles. Consequently, the lightest of the IDM particles is stable. Following the literature [26–31], the lightest particle is taken to be H , which renders it the dark matter candidate. All relevant IDM couplings are presented in table 2.1, using

the definitions $\lambda_{345} = \lambda_3 + \lambda_4 + \lambda_5$ and $\bar{\lambda}_{345} = \lambda_3 + \lambda_4 - \lambda_5$. In total there are five new free parameters in the IDM: the three masses, $m_H, m_A, m_{H^\pm}$; and two coupling constants λ_2 and λ_{345} . From table 2.1, λ_2 is only involved in the quartic coupling between the IDM scalars, whilst λ_{345} only appears for vertices with the Higgs boson. Furthermore, the couplings of the inert particles to the SM gauge bosons are entirely dependent on SM quantities. Moreover, unlike other 2HDM models, there is no mixing between the different doublets, and therefore no additional mixing parameters are defined.

The IDM parameters are subject to multiple constraints on both theoretical and experimental grounds. Only an overview will be given here, but for more details refer to References [26–31]. On theoretical grounds, the parameters are constrained such that the model constitutes a consistent model. For example, the vacuum state is required to be bounded from below, leading to the constraints $\lambda_{1,2} > 0$. Moreover, the couplings are required to be $\lambda_i < 4\pi$ such that amplitudes can be calculated using perturbation theory. These parameters are further constrained to match the plethora of experimental observations. Notably, to ensure that the widths of the gauge bosons are not affected by the IDM particles, the following is required to kinematically limit the gauge bosons from decaying to the IDM particles:

$$M_{H,A} + M_{H^\pm} \geq M_W, \quad M_H + M_A \geq M_Z, \quad 2M_H^\pm \geq M_Z. \quad (2.57)$$

This is supplemented with constraints from the branching ratio of the Higgs to invisible, which places limits on the IDM-Higgs coupling λ_{345} . Further restrictions arise from direct detection experiments, leading to limits on λ_{345} versus DM mass parameterised as $\lambda_{345} = 0.0018 \times (M_H - 72\text{GeV})$. Finally, additional constraints arise from recasting previous searches, notably SUSY searches at LEP but reinterpreted with the IDM. Considering all theoretical and experimental constraints, a strict mass hierarchy emerges:

$$M_H < M_A < M_{H^\pm}. \quad (2.58)$$

Whilst numerous theoretical and experimental constraints have excluded areas of the IDM parameter space, a large proportion still remains viable. Despite this, there have been no direct searches for production of IDM scalars at a collider. However, numerous studies have been performed comparing the expected sensitivity of different final-state signatures at the LHC [26, 32–35] and at future colliders [1, 36].

Of particular importance to this thesis is Reference [26], which studies the expected cross-sections for the production of IDM particles at the LHC. The production channels considered are all the possible pairings of IDM scalars⁵: $HA, HH^\pm, AH^\pm, H^-H^+$ and AA . Comparing the production rates for a range of benchmark points, Reference [26] provides a rough overview of the accessibility for the different modes. Many benchmark points have

⁵This does not include direct production of HH with no other associated particles as this would not be detectable.

large production cross-sections for $HH^\pm$ production, however the decay channel for this involves a single W boson — from $H^\pm \rightarrow HW^\pm$ — which is difficult to distinguish against the extremely large W boson background at the LHC. Subsequently, the most promising channel is HA since, for a significant number of benchmark points, it provides the largest cross section. In this channel, the scalar A decays via $A \rightarrow ZH$, leading to the general final state $q\bar{q} \rightarrow HA \rightarrow HHZ$. As a result, the optimal IDM signature is the opposite-sign same-flavour (OSSF) dilepton final state from $q\bar{q} \rightarrow HA \rightarrow HHZ \rightarrow HH\ell^-\ell^+$ where $\ell = e, \mu$. The leptons provide a relatively clean event signature against the large hadronic activity of the LHC, whilst also offering more efficient triggering compared to purely hadronic final states. An alternative possibility is from $AH^\pm$ which decays to a trilepton final state: $AH^\pm \rightarrow HH\ell\nu\ell'\ell'$. However, generator-level studies in References [32] and [33] observed stronger sensitivity to the dilepton final state than the trilepton channel.

More generally, the dilepton signature of the IDM is produced through $q\bar{q} \rightarrow HH\ell^+\ell^-$ with a sub-leading contribution from $q\bar{q} \rightarrow HH\ell^+\ell^-\nu_\ell\bar{\nu}_\ell$, where ℓ denotes either an electron or muon, and ν_ℓ is the corresponding neutrino. The leading Feynman diagrams for each of these processes are shown in figures 2.3(a) and 2.3(b). Notably, for the leading diagrams, all vertex couplings are given by SM couplings (see table 2.1). At tree level the parameter λ_2 has no contribution to the dilepton final state⁶. The parameter λ_{345} can enter, as shown in figure 2.3(c), however this is generally sub-dominant for allowed values of λ_{345} . Subsequently, as figure 2.3(a) is the leading diagram, which contains only SM couplings, the kinematics of the dilepton final state is largely dominated by the mass splitting between the inert scalars, $M_A - M_H$, with the cross section driven by the mass scale. Notably, however, the two dark matter particles are often produced back-to-back. As a result, there is a cancellation of their transverse momenta, yielding relatively small missing transverse energy, generally below 100 GeV. Consequently, the IDM signature has reduced overlap with previous DM searches, which typically rely on large missing transverse energy, ultimately making recasts insensitive to the IDM. This, therefore, necessitates a direct search that can fully exploit the IDM topology.

⁶At higher orders than tree-level, λ_2 can have an effect through loop diagrams.

IDM-Gauge Couplings		Higgs Couplings		IDM-IDM Couplings	
Vertex	Coupling	Vertex	Coupling	Vertex	Coupling
$H^- H^+ \gamma$	ie	hHH	$\lambda_{345}v$	$H^+ H^+ H^- H^-$	$2\lambda_2$
$H^- H^+ Z$	$i\frac{g}{2} \frac{\cos(2\theta_W)}{\cos(\theta_W)}$	hAA	$\bar{\lambda}_{345}v$	$HHAA$	λ_2
$HH^\pm W^\mp$	$\pm i\frac{g}{2}$	$hH^+ H^-$	$\lambda_3 v$	$HHHH$	$3\lambda_2$
$AH^\pm W^\mp$	$-\frac{g}{2}$	$hhH^+ H^-$	λ_3	$AAAA$	$3\lambda_2$
$H AZ$	$-\frac{g}{2\cos(\theta_W)}$	$hhHH$	λ_{345}	$H^+ H^- HH$	λ_2
		$hhAA$	$\bar{\lambda}_{345}$	$H^+ H^- AA$	λ_2
		$hhhh$	$3\lambda_1$		

Table 2.1: Coupling constants for the IDM particles. Showing: (left) couplings of IDM particles to the SM gauge bosons, (centre) couplings of IDM particles to the Higgs boson, and (right) quartic couplings between IDM particles.

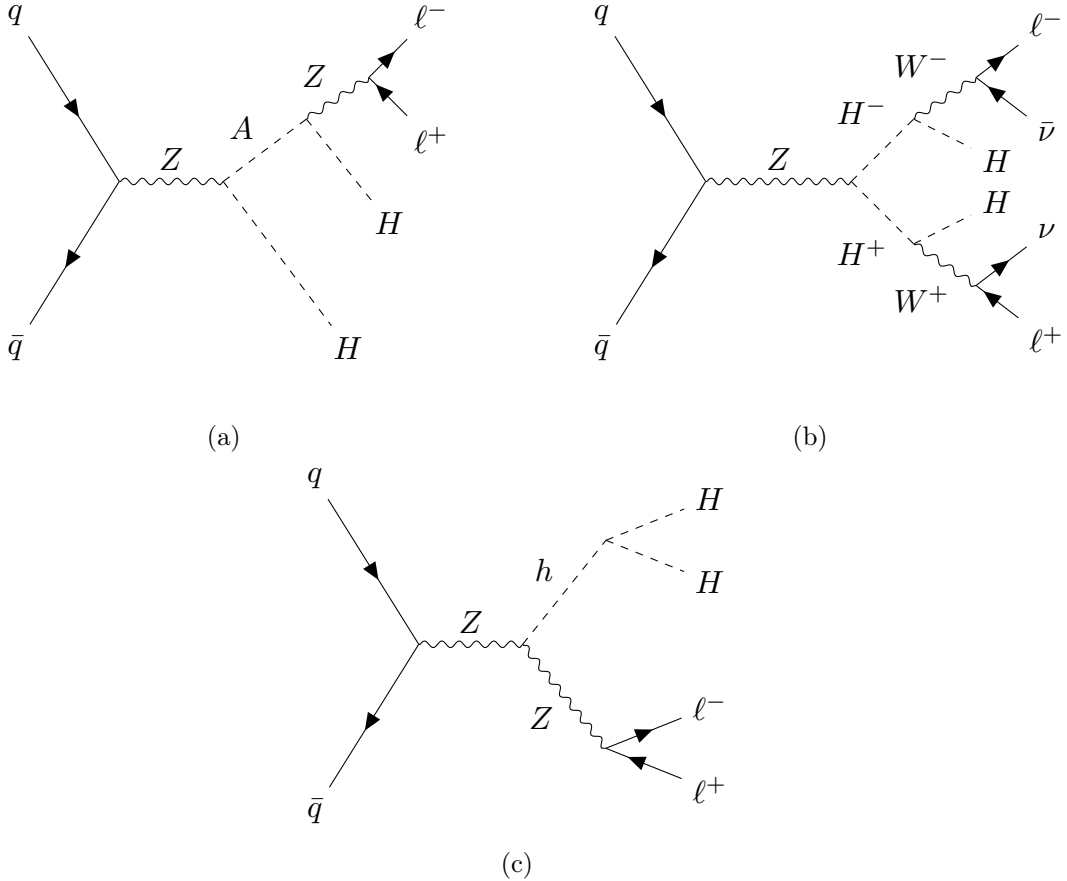


Figure 2.3: Leading Feynman diagrams for the process (a) $q\bar{q} \rightarrow HH\ell^-\ell^+$ and (b) $q\bar{q} \rightarrow HH\ell^-\ell^+\nu\bar{\nu}$ and (c) $q\bar{q} \rightarrow Zh \rightarrow HH\ell^-\ell^+$. Note that for diagrams (a) and (b) all vertex factors are given by SM parameters.

Chapter 3

The LHC and the CMS Experiment

Contents

3.1	The Large Hadron Collider	30
3.2	The CMS Detector	33
3.2.1	Tracker	34
3.2.2	Electromagnetic Calorimeter	35
3.2.3	Hadronic Calorimeter	36
3.2.4	Muon Chambers	38
3.2.5	Trigger	40
3.3	Object Reconstruction	40
3.3.1	Particle Flow	40
3.3.2	Muons	43
3.3.3	Electrons	44
3.3.4	Jets	46
3.3.5	Bottom Jets	48
3.3.6	Missing Transverse Energy	49
3.4	Event Simulation	52

This chapter gives an overview of the Large Hadron Collider (LHC) as well as one of the detectors situated around the collider: The Compact Muon Solenoid (CMS). The basic goals, construction and operation of the LHC are described, with specific focus on data-taking periods relevant to the Inert Doublet Model search presented in chapter 5. A description of the CMS detector follows, covering each sub-detector unit, as well as an overview of the event triggering system. Finally, a section on how specific physics objects are reconstructed and identified is presented.

3.1 The Large Hadron Collider

The Large Hadron Collider (LHC) is a circular collider located near Geneva, Switzerland, with the main aim of studying physics at the TeV scale [37]. Installed in the same 27

km tunnel previously used by the LEP accelerator, the LHC is the world's most powerful particle accelerator, colliding protons at a maximum centre-of-mass energy of $\sqrt{s} = 13.6$ TeV. Collisions happen at four main locations, with a detector placed at each location: ALICE [38], ATLAS [39], CMS [40], and LHCb [41]. A schematic diagram of the CERN accelerator complex is shown in figure 3.1.

Acceleration of particles is performed in 5 stages, with each stage progressively increasing the energy of the particles. First, particles are accelerated to 50 MeV in the Linac4. Next, the Proton Synchrotron Booster (PSB) accelerates them to 1.4 GeV, followed by the Proton Synchrotron (PS) which boosts the energy to 25 GeV. The Super Proton Synchrotron (SPS) then accelerates the particles to 450 GeV before they are finally injected into the LHC, where they are injected as two counter rotating beams. For proton-proton collisions, the beam consists of nearly 3000 bunches, each containing approximately 10^{11} protons, which are separated by 25 ns. All the parameters of the beam at a given time, for example the frequency of bunch crossings and beam size, are summarised by the *instantaneous luminosity*, $\mathcal{L}_{\text{inst}}$ [$\text{b}^{-1}\text{s}^{-1}$], such that the rate of events for a specific process, R , is given by

$$R = \sigma(\sqrt{s})\mathcal{L}_{\text{inst}}, \quad (3.1)$$

where $\sigma(\sqrt{s})$ is the cross-section for that specific process. Therefore, to get the expected number of events, N , for a given process, the rate can be integrated over time:

$$N = \sigma(\sqrt{s}) \int_{\text{time}} \mathcal{L}_{\text{inst}}. \quad (3.2)$$

Whilst an increase in $\mathcal{L}_{\text{inst}}$ results in more events of interest, one disadvantage is the increase in additional scattering processes in each event. These additional collisions, called *pileup* (PU), contaminate the detector, making it harder to effectively reconstruct the particles and processes of interest. Furthermore, the radiation from PU quickens the degradation of the detector, which in turn necessitates the need for more radiation-tolerant materials.

The centre-of-mass energies and pileup conditions for each data-taking year are presented in figures 3.2 and 3.3. It should be noted that the total luminosity delivered by the LHC will not equal the total luminosity measured by the CMS experiment due to detector downtimes. Operation of the LHC began with Run 1 in 2010 with a centre-of-mass energy of $\sqrt{s} = 7$ TeV, increasing to 8 TeV in 2012. This was followed by a planned shutdown, called Long Shutdown 1 (LS1), where upgrades to the detector were performed in preparation for the higher centre-of-mass energies of the subsequent runs. After this shutdown period, in 2015, Run 2 commenced with an operating $\sqrt{s} = 13$ TeV until the end of 2018. During Run 2, the CMS experiment recorded: 36.3 fb^{-1} in 2016 with an average PU of 27; 41.5 fb^{-1} in 2017 with an average PU of 38; and 59.7 fb^{-1} in 2018 with an average PU of 37. This was followed by another period of upgrades (LS2) in preparation for Run 3, which, upon starting in 2022, operated at a $\sqrt{s} = 13.6$ TeV. During this period CMS collected:

38.0 fb^{-1} in 2022 with an average PU of 46; 30.1 fb^{-1} in 2023 with an average PU of 52; and 112.7 fb^{-1} in 2024 with an average PU of 57.

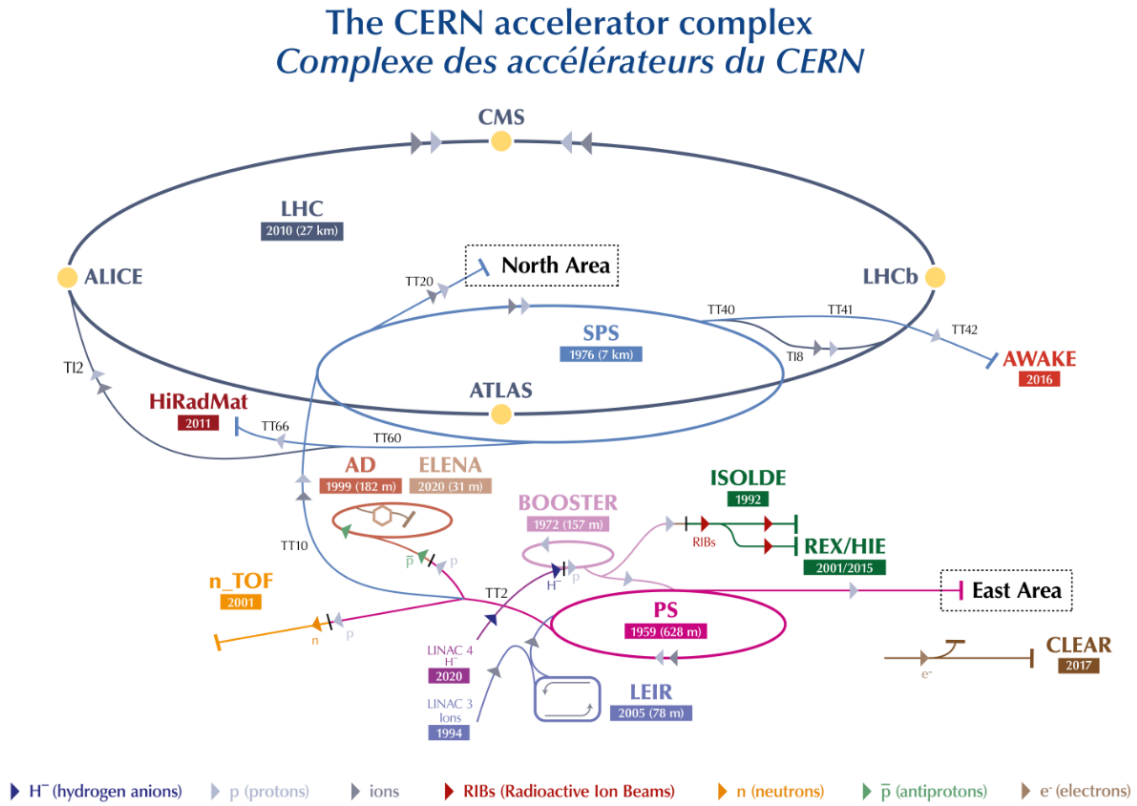


Figure 3.1: Schematic of the CERN accelerator complex and experimental facilities [37].

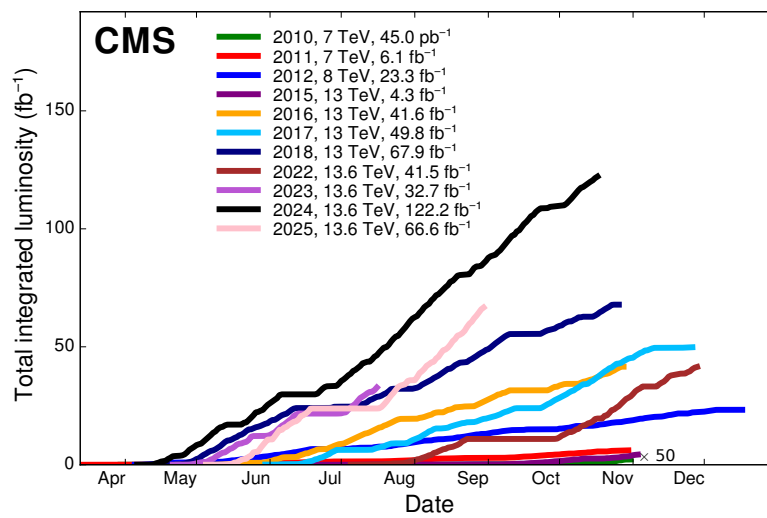


Figure 3.2: The total integrated luminosity delivered to the CMS experiment by the LHC for each data-taking year. Figure taken from [42].

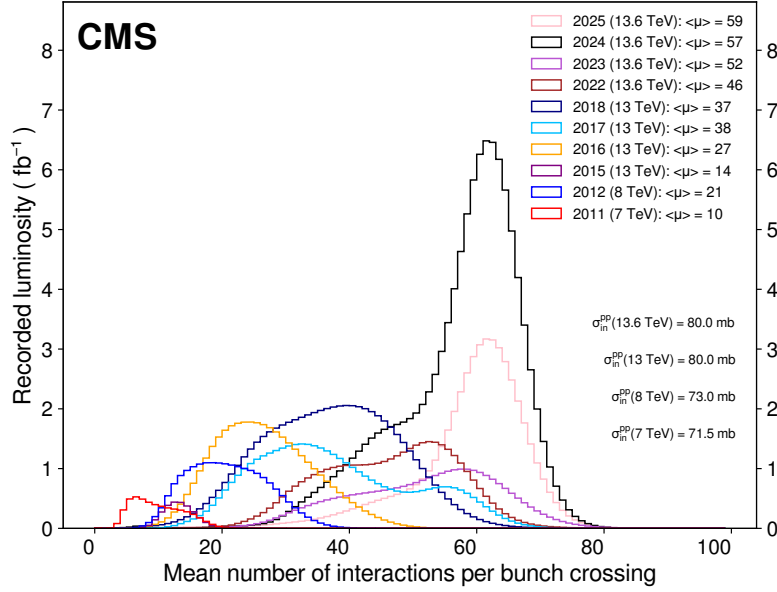


Figure 3.3: The recorded luminosity versus pileup rate for each year. Note that the pileup rate is denoted $\langle \mu \rangle$. Figure taken from [42].

3.2 The CMS Detector

The Compact Muon Solenoid (CMS) is one of the two general-purpose detectors situated on the LHC ring, designed to explore a range of possible physics signatures, most notably to achieve strong sensitivity to the SM Higgs Boson [40]. This is achieved through a series of concentric sub-detectors layered around the beam axis. Combined, these form a near-hermetic coverage, which is essential for fully reconstructing all visible particles produced in collisions—a feature especially important when inferring the production of particles that do not interact with the detector. A schematic of the CMS detector is shown in figure 3.4.

The first of these sub-detectors is the pixel vertex detector and tracker, which provides measurements of charged particle momenta, as well as the location of different particle interactions. Further out from the beam pipe, surrounding the tracker, is the Electromagnetic calorimeter, designed to measure the energy deposited from EM showers from electrons and photons. Following this is the hadronic calorimeter (HCAL), which measures the energy deposited by neutral and charged hadrons through nuclear interactions. This is encased by the superconducting solenoid, providing a magnetic field of 3.8 T symmetrically around the beam pipe, and which enables the precise measurement of charged particles as they bend through the magnetic flux. Finally, this is surrounded by the muon chambers, interleaved between layers of the magnetic return yoke, that is positioned as such due to the fact that muons are generally penetrating enough to pass through all the other sub-detectors.

The CMS detector uses a right-handed coordinate system with the origin placed at the centre of the interaction point, and with the z-axis parallel to the beam pipe, the x-axis pointing towards the centre of the LHC, and the y-axis pointing directly up. The azimuthal

angle ϕ is measured as the angle relative to the x-axis in the x-y plane, and the polar angle η is measured as the angle relative to the z-axis in the z-y plane. From the polar angle, the pseudorapidity angle can be defined as $\eta = -\ln(\tan(\theta/2))$, which has the benefit that, for particles travelling close to the speed of light, differences in pseudorapidity are invariant to lorentz transformations along the beam axis.

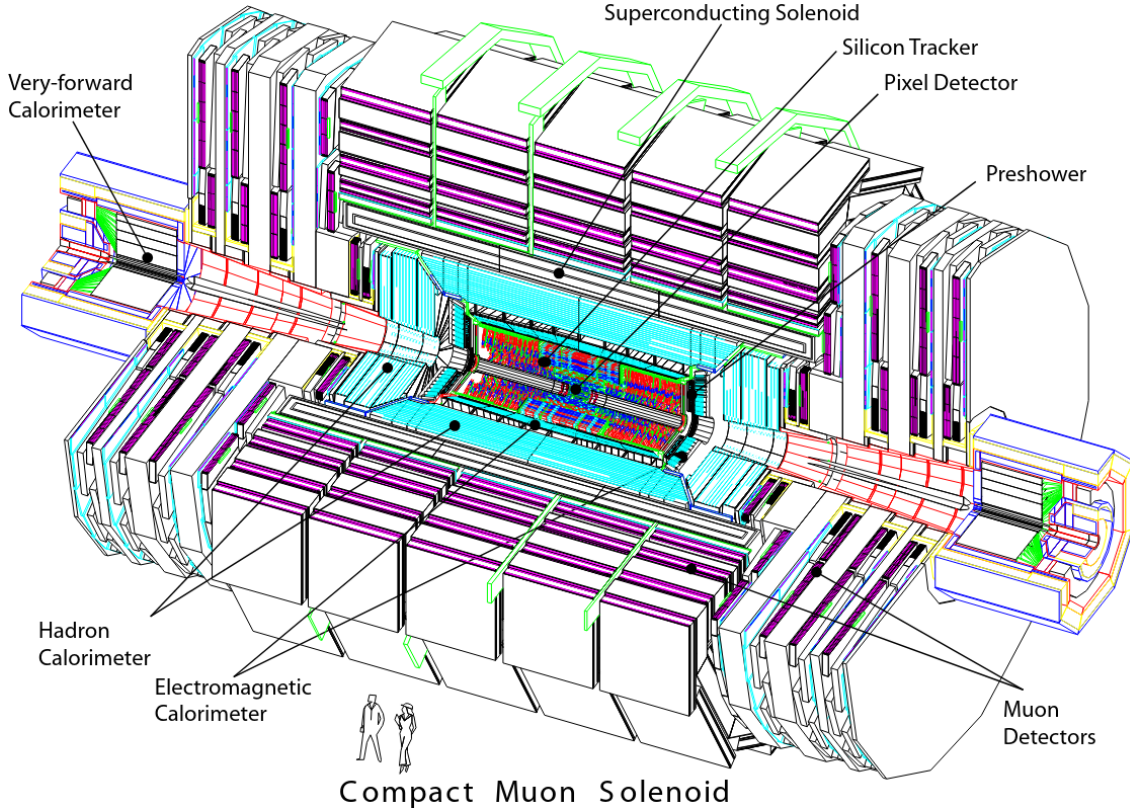


Figure 3.4: Perspective view of the CMS detector [40].

3.2.1 Tracker

The CMS tracker is designed to reconstruct the trajectories of charged particles [40] [43], which in turn allows for precise determination of observables such as particle momenta and the location of their primary vertex (PV). Precise measurements of such observables requires high granularity to ensure enough resolution to separate at small distances, and the 25 ns between proton-proton collisions further requires a fast response time from the tracking elements. Moreover, given the location of the tracker close to the beam pipe, it is required to have high radiation tolerance to withstand the high particle interaction rate. A schematic of the CMS tracker is shown in figure 3.5.

As indicated in figure 3.5, the tracker is split into several sub-detector sections. At the centre of the tracker system, closest to the beam pipe, is the pixel detector which covers a range of $|\eta| < 2.5$. Initially, the pixel detector consisted of three cylindrical barrel layers and two forward disks, however this was upgraded between the 2016 and 2017 runs to cope with the higher luminosity of Run 2 [43]. The upgraded design included four cylindrical layers, located at a radial distance of 30 mm to 160 mm from the interaction point (IP), and

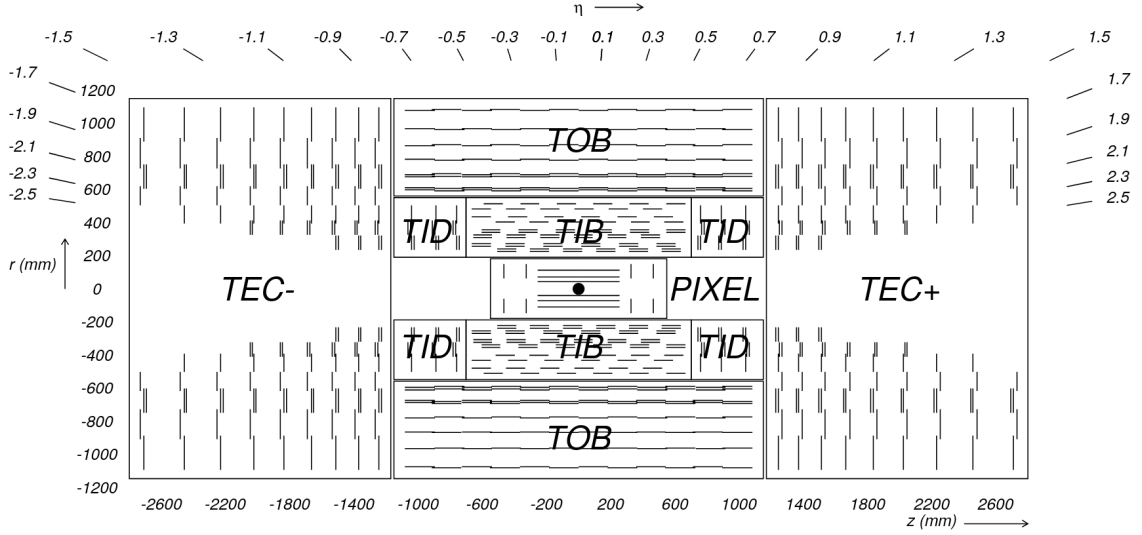


Figure 3.5: Schematic cross-section of the CMS tracker (before the pixel upgrade) in the z - r plane. Single lines indicate a detector module, and double lines represent back-to-back modules [40].

three end-cap disks at 291, 396 and 516 mm from the IP (in both z directions). Together, these contain approximately 124 million individual silicon pixels, each with size $100 \times 150 \mu\text{m}$, and result in a spatial resolution of $10 \mu\text{m}$ and $20 \mu\text{m}$ in the r/ϕ and z direction, respectively.

Further out from the interaction point, the particle flux is reduced, allowing for the use of silicon strip detectors with coarser granularity. These strip detectors are divided into four components: the Tracker Inner Barrel (TIB); the Tracker Outer Barrel (TOB); the Tracker Inner Disks (TID); and the Tracker Endcaps (TEC). The TIB and the TOB contain 4 and 6 cylindrical silicon strip layers, respectively, extending the tracker system to 1200 mm away from the beam pipe, and to ± 1080 mm in the z direction. These are then complimented by two silicon disc units, the TID with 3 circular layers and the TEC with 9 circular layers, and these extend to ± 2820 mm in z . In total the tracking system covers a pseudorapidity range of $|\eta| < 2.4$.

With this design, the tracker ensures that for $|\eta| < 2.4$, there will be at least four hits in the pixel tracker and nine in the strip tracker for each charged particle. This enables precise 3-dimensional reconstruction of charged particle trajectories as they travel helically through the magnetic field, which in turn allows for precise measurements of the particle's momentum. Moreover, the high granularity close to the beam pipe enables precise determination of the primary vertex of the interaction by extrapolating the particle's trajectory back to the origin.

3.2.2 Electromagnetic Calorimeter

Following the tracking system is the Electromagnetic Calorimeter (ECAL). The primary purpose of the ECAL is to reconstruct the energy from electromagnetic showers that are produced by electrons and photons [40]. Given its function, the ECAL is designed to

have a short radiation length (the mean distance for an electron's energy to reduce by a factor e) such that a significant proportion of the electron/photon energy is deposited in the ECAL, as well as a narrow Molière radius (the mean radius that contains 90% of the electron energy) which enables better separation between showers due to less overlap between them. A diagram of the CMS ECAL is shown in figure 3.6.

The ECAL is a hermetic, homogeneous, calorimeter that is composed of lead tungstate (PbWO_4) crystals. Photons and electrons interact with the crystals, producing scintillation light that is collected using photo-sensors to measure the energy. Lead tungstate is used as it has the requirements previously stated: it has a short radiation length of approximately $X_0 = 0.89$ cm; a narrow Molière radius of $r_m = 2.9$ cm; radiation tolerance; and a fast response time within the 25 ns bunch crossings of the LHC collisions.

The ECAL is split into two parts: the barrel (EB) covering a pseudorapidity of $|\eta| < 1.479$ and the endcaps (EC) covering a pseudorapidity of $1.479 < |\eta| < 3.0$. The barrel region is made from 61,200 crystals, arranged with a granularity of 365 crystals in ϕ and (2×85) in η , and are mounted with a quasi-projective geometry such that they overlap to reduce particles passing in-between crystals. Furthermore, at 230 mm in length, corresponding to $25.8 X_0$, showers from electrons and photons with up to 500 GeV of energy are almost entirely ($>99\%$) contained in the ECAL. The endcaps are composed of 7,324 stacked crystals arranged in an approximate circular shape, each with a length of 220 mm, corresponding to $24.7 X_0$. Moreover, immediately before the endcaps are pre-shower sampling calorimeters consisting of two alternating layers of lead and silicon strip detectors. The lead layers initiate EM showers, whilst the silicon layers measure the energy and the transverse shower profiles, and together the additional information is used to help distinguish photons from pions.

When measuring a particle of energy E , the energy resolution of the ECAL, σ , is given by the following equation:

$$\left(\frac{\sigma}{E}\right)^2 = \left(\frac{S}{\sqrt{E}}\right)^2 + \left(\frac{N}{E}\right)^2 + C^2. \quad (3.3)$$

This is decomposed into three components: a stochastic term, $S = 0.28 \text{ GeV}^{\frac{1}{2}}$, that is due to random fluctuations in lateral shower containment; a noise term, $N = 0.12 \text{ GeV}$, due to electronic noise; and a constant term, $C = 0.003$, from non-uniformity of light collection and calibration errors.

3.2.3 Hadronic Calorimeter

Surrounding the ECAL is the Hadronic Calorimeter (HCAL) [40]. The central role of the HCAL is for reconstructing, and measuring the energy of, strongly interacting particles that undergo hadronisation and produce collimated beams of particles called jets.

The HCAL is a sampling calorimeter consisting of plastic scintillators between layers of

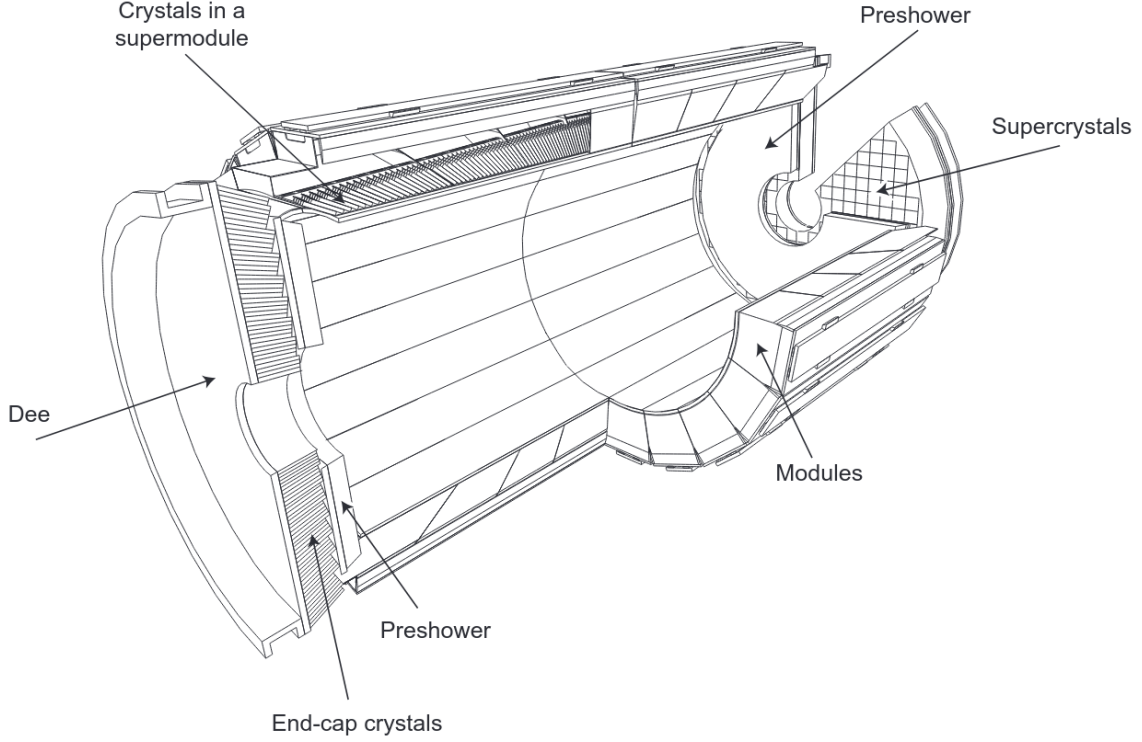


Figure 3.6: Schematic cross-section of the CMS ECAL [40].

dense absorber (brass or steel) that reduce the interaction lengths and allow for a more compact calorimeter. A diagram of the CMS HCAL is shown in figure 3.7. Closest to the interaction point are the Hadron Barrel (HB) and the Hadron Endcap (HE) sub-detectors, which are both housed within the magnetic bore. These are made from alternating layers of brass absorber, chosen for its short interaction length and non-magnetic properties, and plastic scintillator, chosen for their radiation hardness. The HB, which covers a pseudorapidity range of $|\eta| < 1.3$, is positioned 1.77 m away from the interaction point and extends radially to 2.95 m. This corresponds to a depth of 5.8 to 10.6 nuclear interaction lengths, depending on the pseudorapidity. For each direction in z , the HB consists of 18 identical wedges that are split into 4 azimuthal sections and 16 η sections, resulting in a granularity of 72 in ϕ and (2×16) in η or, equivalently, $(\Delta\eta, \Delta\phi) = (0.087, 0.087)$. The HE extends this out to a pseudorapidity of $1.3 < \eta < 3.0$, and provides a granularity of $(\Delta\eta, \Delta\phi) = (0.087, 0.087)$ for $|\eta| < 1.6$ and $(\Delta\eta, \Delta\phi) = (0.17, 0.17)$ for $|\eta| \geq 1.6$ whilst providing a minimum depth of 10 nuclear interaction lengths.

Outside the magnetic bore are the Hadron Outer (HO) and Hadron Forward (HF) sub-detectors. The HO extends the radial distance of the hadronic calorimeter to a pseudorapidity range of $|\eta| < 1.3$. This uses the same plastic scintillator as the HB, but instead uses the magnetic bore itself as the dense absorber, and extends the inner HCAL section to a minimum of 11.8 nuclear radiation lengths. At the very forward region is the HF, which increases the pseudorapidity coverage up to $|\eta| < 5.2$. This region of the detector experiences extremely high levels of particle flux, and as such requires materials that are extremely radiation resistant. Given this, the HF is composed of radiation-hard quartz

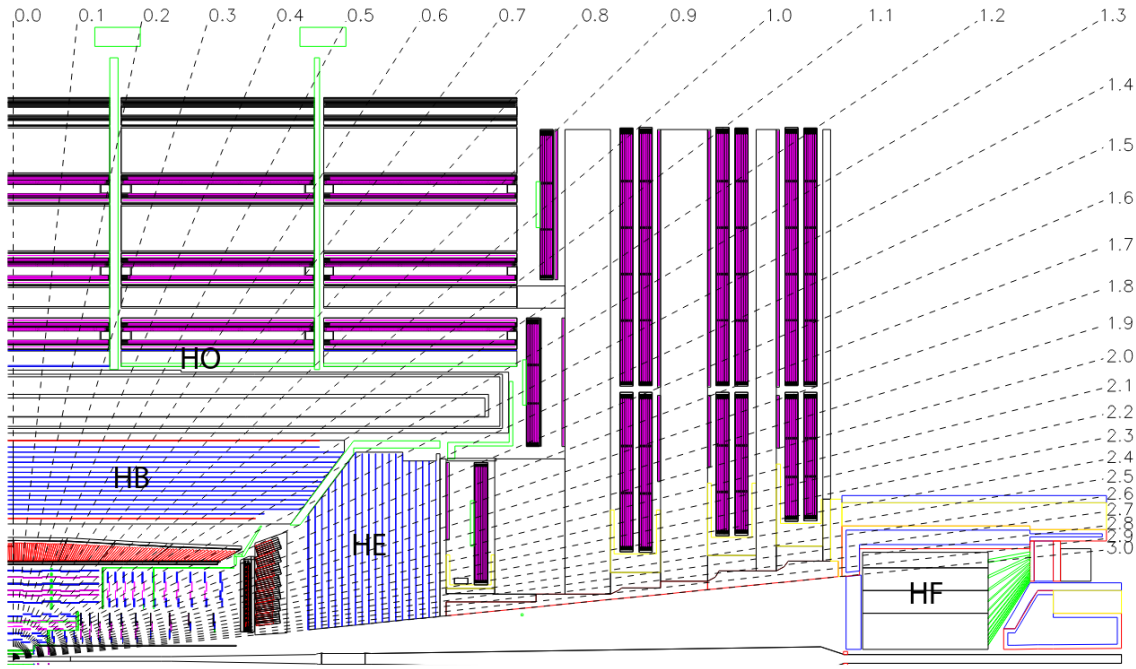


Figure 3.7: Schematic cross-section of the CMS HCAL [40].

fibres sandwiched between steel absorbers. The quartz fibres act as the active material, which measures the energy of charged particles by collecting the Cherenkov radiation they emit as they pass through.

The HCAL energy resolution in the barrel was measured using a beam of isolated electrons and charged hadrons (predominantly charged pions), with particle momenta ranging from 2 GeV/c up to 350 GeV/c [44]. The resolution of the incident particle's energy, E , can be expressed using the following equation:

$$\left(\frac{\sigma}{E}\right)^2 = \left(\frac{S}{\sqrt{E}}\right)^2 + C^2, \quad (3.4)$$

where $S = 0.943 \text{ GeV}^{\frac{1}{2}}$ is the stochastic term and $C = 0.084$ is the constant term.

3.2.4 Muon Chambers

The muon detection systems are located outside the solenoid coil, placed as such due to muons being highly penetrating, and, therefore, often passing through the previous detector systems. With an overall coverage in pseudorapidity of $|\eta| < 2.4$, the muon detectors are split into three subsystems: the Drift Tube Chambers (DT), the Cathode Strip Chambers (CSC), and the Resistive Plate Chambers (RPC) [45]. A schematic diagram is shown in figure 3.8. Each subsystem uses gaseous ionizing detectors, that have the benefit of being cheap yet reliable, which is necessary given the large area they are required to cover. Gas-based detectors work by charged particles entering and ionizing the gas, producing free electrons and ions. An applied electric field inside the detector then causes the ionized electrons to drift towards electrodes, thus producing a measurable electrical signal. Addi-

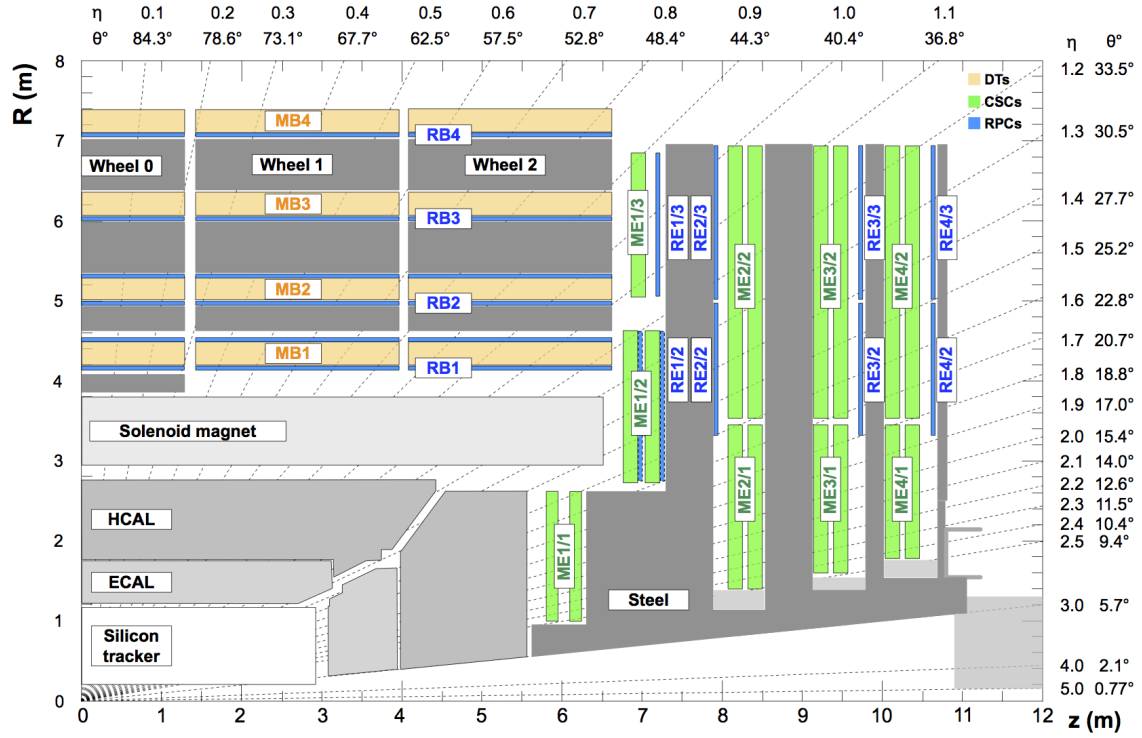


Figure 3.8: Schematic cross-section of a quadrant of the CMS Muon Chambers [45].

tionally, the position of the muon within the cell can be calculated by measuring the time delay between the muon entering the cell to the resultant electrical signal.

The Drift Tube Chambers cover a range of $|\eta| < 1.2$ and are split into four cylindrical stations which are interspersed between the layers of the magnetic return yoke. The first three stations contain 8 chambers in ϕ and 4 chambers in the z direction, whilst the fourth station has only the 8 chambers in ϕ . Each chamber is then subdivided into individual drift cells which are arranged in a staggered, overlapping formation such that there are no dead spots in efficiency. Depending on the exact drift chamber, the spatial resolution is $77\text{--}133\text{ }\mu\text{m}$ in ϕ , and $133\text{--}393\text{ }\mu\text{m}$ in z [45]. The pseudorapidity range is then extended with the Cathode Strip Chambers, which cover a range of $0.9 < |\eta| < 2.4$. In each z direction, these are split into four stations that are, similar to the DT, interspersed between the magnetic return yoke and placed perpendicular to the beam line.

Complimentary to both the DT and CSC subsystems are the Resistive Plate Chambers (RPC) for $|\eta| < 1.6$. There are 6 RPC units within the barrel, with 1 unit on either side of the first two DTs, and 1 before the last two DTs. Furthermore, an additional 4 RPCs are placed within the CSC systems. Unlike the DT and the CSC, the RPC units operate with a far higher electrical field, which results in electron avalanches. This means that, whilst offering far coarser spatial resolution than the DT and CSCs, the RPCs offer significantly faster response times, which allows for precise, dedicated, triggering on muons.

3.2.5 Trigger

When the LHC is running, protons collide at a frequency of 40 MHz. At this rate it is not possible to read out all the information collected by the detector. Moreover, a significant amount of collisions will have no processes of interest. Therefore, to reduce the rate of data collection, and to save only collisions of interest, a triggering system is employed [46].

The CMS Trigger system is a two-tier system, starting with the Level-1 Trigger (L1T), followed by the High-Level Trigger (HLT). The L1T, operating exclusively with custom-programmable hardware called Field Programmable Gate Arrays, reduces the event rate from 40 MHz down to 100 (110) kHz for Run 2 (Run 3) [47]. At this rate, the available information to the L1T is limited by the maximum allowed latency of 4 μ s. As a consequence, only coarse information on the energy deposits from the calorimeters and the muon chambers are available, with no information from the tracker, and these are used by the L1 global trigger to decide whether to keep an event.

This is followed by the second stage: the HLT. For Run 2 this stage was performed entirely on a network of thousands of CPU cores, whereas for Run 3 this was upgraded to a combination of CPUs and GPUs [47]. At this stage, the full detector information is available, including input from the tracker, and more sophisticated algorithms that match the offline processing can be applied. This enables significantly more precise reconstruction of physics objects. Furthermore, with the inclusion of GPUs for Run 3, even more sophisticated algorithms can be deployed, delivering improved tracking despite the higher PU environment. The HLT further reduces the event rate down to around 1 (2.6) kHz for Run 2 (Run 3) whilst, crucially, maintaining a high efficiency for processes of interest [46].

3.3 Object Reconstruction

3.3.1 Particle Flow

Each individual sub-detector is designed to identify specific particles, such as the ECAL being designed to detect electrons and photons. However, by combining the information from each sub-detector, the measurements of particles — and the differentiation between different particles — is greatly improved. The method CMS employs to reconstruct different particles using all the sub-detectors is called the Particle Flow (PF) algorithm [48]. This approach is exemplified in figure 3.9, which shows a simplified slice of the CMS detector and the typical signatures from different particles. Electrons and photons deposit nearly all of their energy in the ECAL in the form of EM showers. While the resultant EM shower is similar for electrons and photons, they can still be differentiated by the presence of electron tracks in the tracker, similarly for the identification of neutral hadrons from charged hadrons in the HCAL. Finally, a typical muon can be identified by their helical tracks that pass through each sub-detector and out to the muon chambers.

The main inputs into the PF algorithm are tracks from the tracker and muon sub-detectors, and energy clusters from the ECAL and HCAL. Tracks are constructed in three stages:

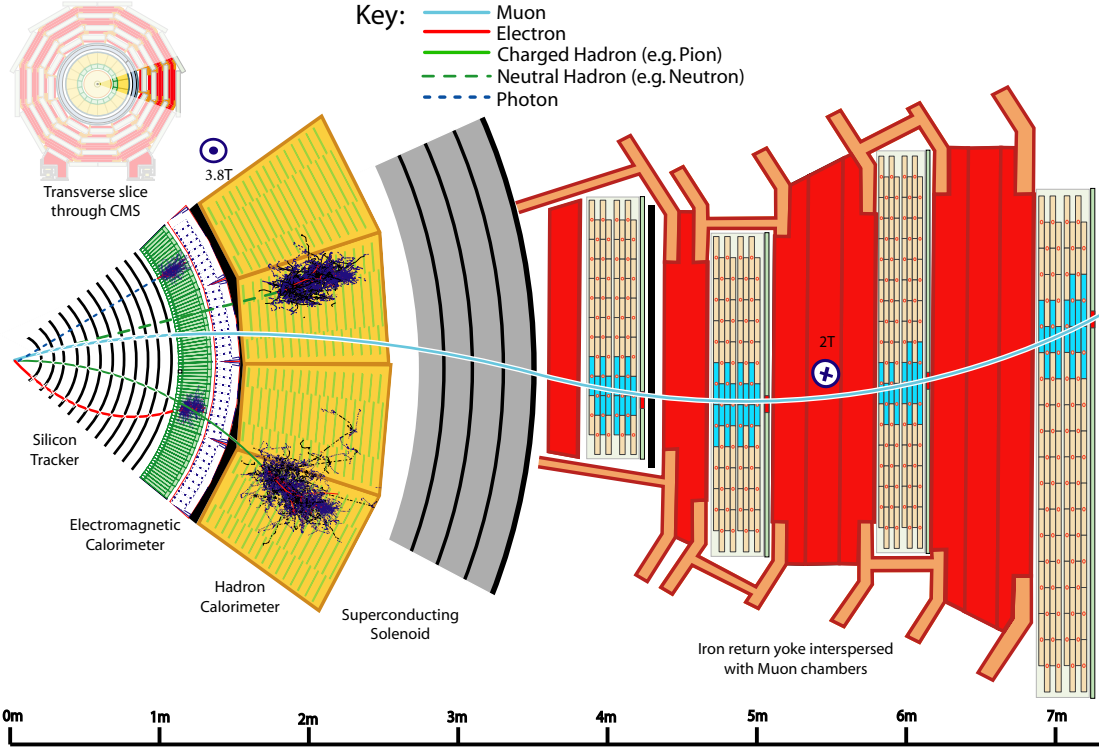


Figure 3.9: A slice of the CMS detector showing the typical signatures for different particles. The electron shown is negatively charged, whilst the muon and the charged pion are both positively charged [48].

1. Potential track candidates are formed by multiple (2-3) hits in the tracker that are compatible with expected charged tracks. These are called *seeds*.
2. Using the seed as an initial estimate, these are extrapolated outwards and connected with additional tracker hits using a Kalman filter (KF) [49].
3. A curve is fit to the set of selected hits, and this fitted curve is used to determine particle properties such as initial vertex, direction, transverse momentum and an overall track goodness-of-fit.

However, given the high granularity of the tracker, it is not possible to use all the hits as potential seeds, as the number of possible combinations exceeds the computing limits. Instead, an iterative approach is used which starts with the easiest, most isolated, seeds and then with each iteration focuses on reconstructing more difficult tracks such as those from lower transverse momentum (p_T) particles, or those from highly displaced vertices. This is repeated for 10 iterations. At each stage, when a hit has been assigned to a track, it is removed before the next iteration. This allows for significantly less computing time whilst maintaining a high efficiency of track reconstruction, as well as a low misidentification rate of independent hits being identified as a single track [48].

The standard track finding procedure is unsuitable for electrons, however, due to large energy losses from radiative processes. Radiative losses cause large changes in momentum

as the track develops, which leads to poor-fitting tracks with fewer hits, and inaccurate estimation of the particle parameters. To account for this, two separate methods are used for selecting electron tracks: an ECAL-based method and a tracker-based method. For the ECAL-based method, instead of creating tracks from seeds that start in the tracker, tracks are seeded by ECAL clusters and extrapolated back to the IP. Whereas, for the tracker-based method, tracks are seeded from the tracker but are extrapolated with a Gaussian-sum filter [50] instead of a KF filter, which allows for sudden changes in trajectories from large energy losses.

For muons, tracks are split into three different types depending on their location. The first type are Standalone Muons, which are muon tracks reconstructed from hits within the DT and CSC sub-detectors. The second type are Global Muons, which are made from Standalone Muons that have a trajectory matched to a track in the tracker. If a match is found, hits from both tracks are combined and a new fit is formed. Notably, for high p_T muons ($p_T > 200$ GeV), this helps improve the momentum resolution. Finally, the third type of muon tracks are Tracker Muons. These are formed by extrapolating tracks from the tracker out to the muons systems if the tracks have $p_T > 0.5$ GeV, and are only kept if the extrapolated trajectory matches hits in the muon system.

Along with the tracks, the other input into the PF algorithm are the clusters: local deposits of energy in the calorimeters. Clusters are formed in three stages. First, initial seeds are identified that have an energy deposit above a location-dependent threshold, and that have more energy than neighbouring cells. Depending on the detector system, the neighbouring cells can either be the four cells that share an edge with the seed, or the eight cells that also includes cells that share a corner with the seed. Second, from the initial seeds Topological Clusters are formed by combining the seeds with the neighbouring cells that have energy deposits larger than twice the noise level of that cell. Finally, as a single topological cluster can have multiple seeds from overlapping particles, a maximisation algorithm is performed to separate the different particle contributions. For each seed the algorithm assumes that the energy deposit is Gaussian, and the maximisation procedure finds the location and energy of the gaussian functions most likely to result in the measured signal. After this procedure, the location and energy of the gaussians are taken as the cluster parameters.

These clusters alone are not suitable for fully reconstructing electrons and photons. As the tracker has a radiation length of 0.4 to 2 X_0 , depending on η [48], electrons and photons likely emit bremsstrahlung or undergo conversion, respectively. For photons that have converted into electron-positron pairs, these pairs will deposit energy into multiple regions of the ECAL. Whereas for electrons, bremsstrahlung results in a considerable amount of energy being radiated before reaching the ECAL, with an average energy loss of 84% at the thickest point, which is mainly spread out in the ϕ direction due to the magnetic field. Therefore, to accurately measure the electron or photon energy, it is vital that all clusters originating from the same initial particle are collected. This is achieved by combining several clusters together — with a larger extension in ϕ — to form what is known as Superclusters (SC) [48].

Using the PF elements presented above (tracks and clusters), a linking algorithm is used to connect elements from different sub-detectors that likely originated from the same particle. These link the different elements together into *PF Blocks* which are then used as input for the particle identification algorithms. Initially, tracks are extrapolated out to the ECAL and the HCAL, and links are formed between tracks that overlap with the clusters. If a track overlaps with multiple clusters then the cluster closest to the track is chosen. A similar procedure is used for matching clusters from the finer, more granular, calorimeters to the coarser calorimeters. For example, a link is formed between an ECAL cluster if it is within the envelope of an HCAL cluster, and if multiple HCAL clusters overlap with the ECAL cluster, then the HCAL cluster that is closest is chosen. Furthermore, ECAL Superclusters are linked to ECAL clusters if they share at least one ECAL cell. Finally, as nuclear interactions can lead to kinks in the track or the production of multiple secondary particles, links are established between any tracks that share a secondary vertex.

Next, using the PF Blocks, particle candidates are formed in the following order: first, potential muon candidates are identified; followed by electrons and photons; then ending with charged and neutral hadrons [48]. At each stage, once a potential candidate has been established, all PF elements associated with that candidate are removed to ensure that no elements are double-counted in the proceeding steps.

3.3.2 Muons

Using variables derived from the PF muon candidates, a set of selections are defined to further separate real, prompt, muons from ones originating from other sources, such as heavy flavour decays or jets faking muon signatures [45]. The variables include: information on the track fit; the number of hits in the tracker and muon systems; the compatibility between tracker tracks and standalone tracks within the muon systems (for global muons); compatibility with the primary vertex; and the existence of kinks along the track. Using these variables, the selections, referred to as *working points*, balance between muon ID efficiency and selection purity. These are labelled Loose, Medium and Tight Identifications (IDs), with the Tight ID offering the highest selection purity.

The Loose ID is optimised to have a high efficiency for selecting prompt muons originating from the primary vertex or from light and heavy flavour decays, as well as efficiently rejecting charged hadrons being identified as muons. This is defined as any muon that passes as a global muon or a tracker muon from the PF algorithm. Whereas, the Tight ID is aimed at identifying prompt muons, whilst reducing contributions from light/heavy flavour decays and from hadronic punch-through (jets which are not fully absorbed by the calorimeters and reach the muon detectors). Tight muons need to pass the Loose ID as well as the following requirements:

- Six or more hits in the inner tracker, and at least one in the pixel tracker.
- It must be both a tracker muon and a global muon.
- The tracker muon must match hits in at least two of the muon stations.

- The global muon fit must have a $\chi^2/\text{DOF} < 10$ along with a hit in the muon system.
- It must be consistent with the primary vertex, and have a transverse displacement with respect to the beam spot of $|dXY| < 0.2$ cm, and a longitudinal displacement of $|dZ| < 0.5$ cm.

Moreover, contamination of prompt muons from heavy-flavour decays can be further reduced with additional cuts on the isolation of the muon. Isolation is measured by comparing the P_T of the muon (μ) with the p_T of photons (γ), neutral hadrons (h^0), charged hadrons from the primary vertex ($h^\pm$) and charged hadrons from PU ($h^{\pm,PU}$) that are within a geometric cone around the muon candidate. The geometric cone is defined by $\Delta R = \sqrt{\Delta\phi^2 + \Delta\eta^2}$, and the isolation variable is defined as:

$$I_{rel} = \frac{1}{p_T^\mu} \left(\sum p_T^{h^\pm} + \max \left(0, \sum p_T^{h^0} + \sum p_T^\gamma - \Delta\beta \sum p_T^{h^{\pm,PU}} \right) \right), \quad (3.5)$$

where the sum is over all particles within the cone, and $\Delta\beta$ is an estimate of the neutral hadron contribution from PU, which is taken as $\Delta\beta = 0.5$ [51]. Using this variable, a set of working points can be defined. These are defined as: Loose Isolation, with $I_{rel} < 0.25$; Medium Isolation, with $I_{rel} < 0.20$; and Tight Isolation, with $I_{rel} < 0.15$. For prompt muons, the Loose Isolation and Tight Isolation working points have an efficiency of roughly 95% and 98%, respectively, measured using $Z \rightarrow \mu^- \mu^+$ events [45].

The momentum of a reconstructed muon is calculated using the *Tune-P* Algorithm [52], which compares the estimated p_T of the muon from different reconstruction fits depending on their goodness-of-fit. The fits include tracker-only, tracker plus first muon detector plane, global muon but with high-occupancy muon chambers ignored, and a global fit with chambers that potentially have hits from showers. The scale and resolution of the p_T estimate is then calibrated to match that of data using $Z/\gamma, J/\Psi, \Upsilon \rightarrow \mu^- \mu^+$ events [53]. For muons with p_T up to 100 GeV, the scale corrections are of the order 0.2-0.3%, with the resolution corrections being approximately 1-3%.

3.3.3 Electrons

Similar to muon identification, electrons have a set of identification selections to separate real electrons from photon conversions, heavy flavour decays, and misidentified jets [54]. Likewise, they are given as separate IDs — referred to as Loose, Medium and Tight — that balance efficiency and purity. These have been tuned separately for each year of data-taking to correspond to approximately 90% efficiency for the Loose ID, 80% for the Medium ID, and 70% for the Tight ID when identifying prompt electrons.

Unlike the muon IDs, however, the isolation requirements are included in the definition of the electron IDs [55]. In the case of electrons from heavy flavour decays or misidentified jets, the electron candidates will have significantly more energy around their trajectories. As such, requiring electrons to be isolated significantly reduces these backgrounds. The

isolation is defined as

$$I_{PF} = \frac{1}{p_T^e} \left(\sum p_T^{h^\pm} + \max \left(0, \sum p_T^{h^0} + \sum p_T^\gamma - \rho A_{\text{eff}} \right) \right), \quad (3.6)$$

where the sum is over the charged hadrons, neutral hadrons and photons within a chosen ΔR around the electron. Furthermore, the charged hadrons are required to originate from the same PV as the electron. Finally, ρA_{eff} is a correction accounting for the contribution from PU in the event, with ρ being the average energy density in η - ϕ of the event — which is dependent on the number of reconstructed interaction vertices — and A_{eff} is the effective area of the electron cone, which is dependent on the choice of ΔR .

The variables used in the electron selection are the following:

- H/E : Ratio of energy deposited in the HCAL and to the ECAL. Electrons deposit significantly more energy in the ECAL compared to hadrons.
- $\sigma_{i\eta i\eta}$: Lateral shower extent in η . Electron showers are predominantly narrower than hadron clusters.
- $|\eta_{SC} - \eta_{track}|$: Difference between the supercluster (SC) seed pseudorapidity, η_{SC} , and the pseudorapidity of the track from the inner tracker extrapolated to the point of the SC, η_{track} . For electrons these are likely to be similar, whereas for jets these are likely to be different as the track is not extrapolated with the Kalman filter.
- $|\phi_{SC} - \phi_{track}|$: Difference between the SC seed azimuthal angle, ϕ_{SC} , and the azimuthal angle of the track from the inner tracker extrapolated to the point of the SC, ϕ_{track} . For electrons these are likely to be similar, whereas for jets these are likely to be different as the track is not extrapolated with the Kalman filter.
- $|\frac{1}{E_{SC}} - \frac{1}{p_{track}}|$: One over the energy of the SC minus one over the momentum of the particle at the point closest the SC. Given the light mass of the electron these are likely (within measurement uncertainties) to be similar.
- I_{PF} : Electron isolation to separate prompt electrons from ones originating from jets.
- Missing number of hits in tracker, N_{miss} . Electrons from photon conversions are more likely to have missing hits in the tracker.

Examples of some of these variables are shown in figure 3.10. The distributions of the prompt electrons are from a sample dominated by $Z \rightarrow e^- e^+$ by requiring two opposite-sign electrons that have an invariant of $|M_{ee} - M_Z| < 7.5$ GeV [54]. For the distribution of misidentified electrons, these are from a sample of data events dominated by Z +jets events which similarly require a pair of opposite-sign leptons (e or μ) with an invariant mass compatible to a Z boson, but with an additional third electron. Furthermore, to suppress real three lepton events from WZ , which also contain neutrinos, the transverse missing energy is required to be below 25 GeV. Finally, the distributions of the third lepton,

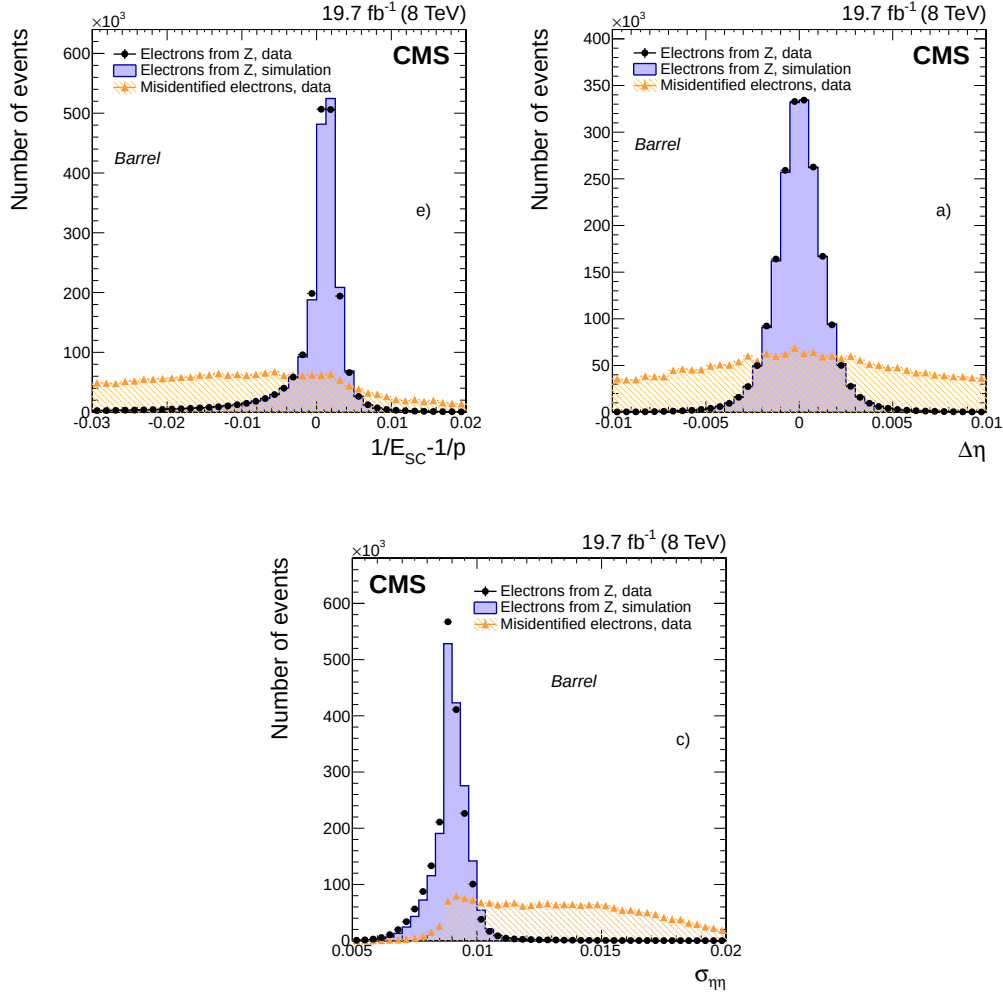


Figure 3.10: Example distributions of the identification variables for prompt electrons versus for misidentified electrons. Plots taken from [54].

that likely originates from a jet, are defined as the "Misidentified Electrons".

The momentum of an electron is calculated using a combination of tracker and ECAL measurements [54], with the calculation of low-energy electrons being dominated by the tracker estimate, and with the ECAL measurement dominating for high-energy electrons. Subsequently, the scale and resolution of the p_T estimate is then calibrated to match that of data using $Z/\gamma, J/\Psi, \Upsilon \rightarrow e^-e^+$ events.

3.3.4 Jets

Quarks and gluons undergo hadronization which leads to a collimated spray of particles called *jets*. As the resultant particles extend out in η and ϕ , the first stage is to recombine the different particles originating from the same quark or gluon into a single jet object. This is referred to as *clustering*. The clustering algorithm used for this thesis is the anti- k_T algorithm [56]. By defining a distance measure representing the distance between particles in the event, the anti- k_T algorithm combines particles into a single object if they are close.

This process is then repeated until no more particles can be combined. Specifically, the distance measures for the anti- k_T algorithm are

$$\begin{aligned} d_{ij} &= \min(p_{T,i}^{-2}, p_{T,j}^{-2}) \frac{\Delta R_{ij}^2}{R^2} \\ d_{iB} &= p_{T,i}^{-2}, \end{aligned} \tag{3.7}$$

where ΔR_{ij}^2 is the angular separation of object i from object j , and R is a parameter that controls the size of the cones for each jet. For this thesis, the parameter R is set to a value of 0.4 for all jets. d_{ij} then gives a measure of how close particles are to each other, and d_{iB} gives a measure of how close particle i is to the beam-line. The jets are then produced with the following method:

1. Compute d_{ij} and d_{iB} for all particles, then find the minimum value.
2. If the minimum value corresponds to a d_{ij} , merge object i with j .
3. If the minimum corresponds to d_{iB} , define object i as a jet and remove from the list of objects.
4. Repeat steps 1-3 until all objects have been clustered into jets.

However, using this method alone will result in many PF candidates from PU not only being incorrectly combined with non-PU particles into jets, but also being clustered into jets themselves. This in turn leads to contamination of the jets of interest, and a degradation of the jet observables. To mitigate these effects, CMS employs two different approaches depending on the year the data is taken. The first approach, used for data collected in 2016-2018, is the Charged Hadron Subtraction (CHS) method [57]. With this method, a primary interaction vertex is defined as the vertex with the highest sum of p_T over its tracks, and then any charged hadron track that is not associated with this primary vertex is treated as PU and removed. Following this, the anti- k_T algorithm clusters the PF candidates into jets, and then a correction based on the jet area is applied to remove contributions from neutral PU. Finally, PU particles that have been clustered to form jets are identified using a Boosted Decision Tree (BDT) discriminator that uses the tracking and jet-shape information as inputs [57].

One issue with the CHS algorithm is that the correction for the neutral PU contribution only applies to the jet four-momentum and, therefore, does not correct for the neutral PU contribution to the jet shape or jet substructure variables. To overcome this limitation, a second approach was used for data collected in 2022-2024 called the Pileup Per Particle Identification (PUPPI) method [58]. This improves on the CHS method by relying on only the PF candidates and corrects for both the charged and neutral PU contributions to the jet four-momentum and substructure variables. Using the tracking information, information on the shape of the clusters, and the event pileup properties, the PUPPI method derives a

weight for each PF candidate which encodes the probability of the particle originating from PU. This weight is then used to scale the particle four-momentum which, ideally, should reweight PU particles with a weight of 0, essentially removing them from the event, whilst having a weight of 1 for non-PU particles. Following this, the anti- k_T algorithm clusters the PF candidates into jets.

To quantify the performance of the CHS algorithm versus the PUPPI algorithm, simulated QCD events are used where the reconstructed jet four-momentum (using either method) is compared to the four-momentum of the jets clustered without PU and before any detector simulation [59]. This is presented in figure 3.11, which shows the jet energy resolution as a function of the number of interactions in the event. For events with more than roughly 20 interactions the PUPPI algorithm generally displays better resolution, especially for lower p_T jets, highlighting the benefit of switching to PUPPI jets for the higher PU environment of Run 3 (2022-24).

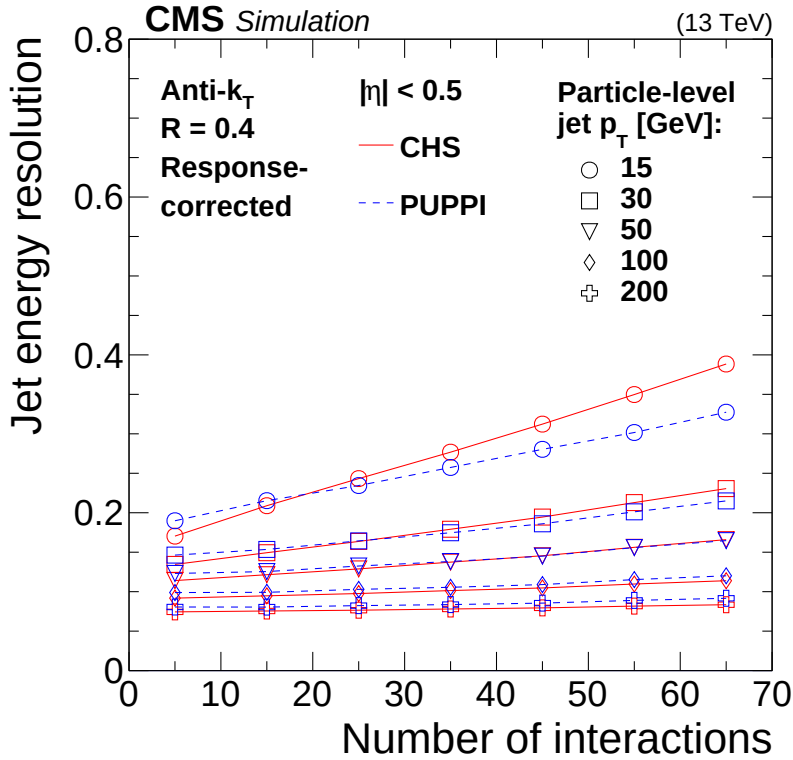


Figure 3.11: Jet energy resolution for CHS and PUPPI jets as a function of the number of interactions in the event [59].

3.3.5 Bottom Jets

Jets initiated by different particles, in general, exhibit different characteristics. For example, jets initiated by gluons tend to have higher particle multiplicities within the jet than those initiated by quarks [60]. Additionally, jets from heavy flavour decays can feature unique multi-prong structures from cascade decays that are not seen in jets from lighter quarks or gluons. An example is the decay of a bottom quark which, due to the longer lifetime of the bottom quark, results in a secondary vertex displaced from the primary

vertex up to a few millimetres. It is, therefore, possible to leverage these differences to tag jets as initiating from a certain source. This is particularly important when trying to target, or limit, certain processes, such as those involving top quarks.

One such algorithm used for tagging jets initiated by bottom quarks (b jets) is the DeepJet algorithm [61]. This uses both low-level and high-level features which can be separated into four groups: global features, such as jet p_T and the number of charged and neutral PF candidates in the jet; charged candidate features, including the candidate's p_T and η ; neutral candidate features; and secondary vertex features, such as the longitudinal displacement of the vertex. In total, approximately 650 input features are used, which are passed through multiple convolutional and fully-connected layers. This results in a multidimensional output that describes the probability of the jet being a b-jet, c-jet, or from a light quark or gluon. The performance of the DeepJet algorithm is shown in figure 3.12, which was obtained using simulated $t\bar{t}$ events with both b-jets decaying hadronically. This shows that for a b-jet efficiency of roughly 68% there is a misidentification rate of around 0.1% when tagging b-jets versus light-jets.

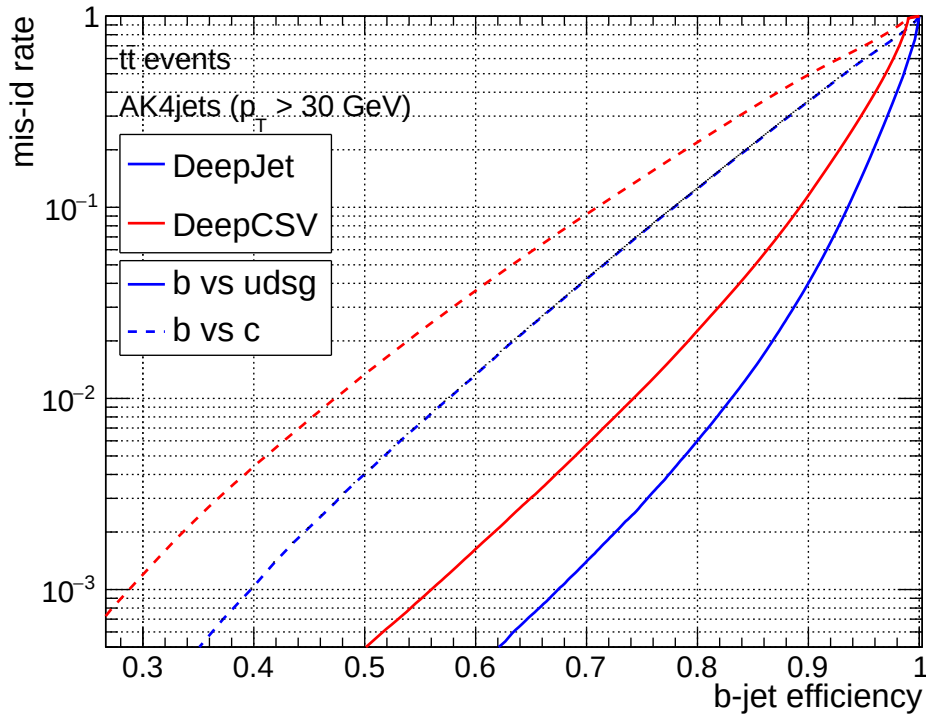


Figure 3.12: Performance of the DeepJet algorithm versus the older tagging algorithm, DeepCSV, using simulated $t\bar{t}$ events. The solid line shows the performance for tagging b jets versus light jets, and the dashed line shows the performance of tagging b jets versus c jets [61].

3.3.6 Missing Transverse Energy

Many processes of interest include particles that do not interact with the detector, and as such leave no direct signature. This includes SM processes involving neutrinos, such as

$W^\pm \rightarrow l^\pm \nu$, or those involving hypothetical BSM particles such as dark matter candidates. However, whilst direct detection of such particles is not possible, their existence can be inferred by utilising the conservation of momentum, which should be zero in the transverse plane. The imbalance measured in this plane is called the *missing transverse energy*, $\vec{E}_T^{\text{miss}}$, and is given by the following equation:

$$\vec{E}_T^{\text{miss}} = - \sum_i \vec{p}_T^i, \quad (3.8)$$

where the sum is over all PF candidates within the events [62]. The magnitude of this is then given by $E_T^{\text{miss}} = |\vec{E}_T^{\text{miss}}|$. The missing transverse energy in this form is referred to as PF- $\vec{E}_T^{\text{miss}}$ as it uses the PF jets. CHS jets are not suitable for this as it only applies to jets with $|\eta| < 2.5$ (the tracker extent) and so PU jets bordering this region are only partially removed which leads to a loss of E_T^{miss} resolution. This is not an issue for PUPPI jets, and as such a corresponding quantity can be derived, which is referred to as PUPPI- $\vec{E}_T^{\text{miss}}$.

With perfect reconstruction, the missing transverse energy would equal the vectorial sum of all invisible particle momenta in the event. However, in reality, detector inefficiencies degrade this. This includes, but is not limited to, the minimum energy thresholds of the calorimeters, the non-linear response of the calorimeters which leads to overestimates or underestimates of the momentum, and inefficiencies in the tracking. To reduce this bias, any corrections made to the measured jets are propagated to the missing energy calculation.

The performance of the reconstructed E_T^{miss} (PF- $\vec{E}_T^{\text{miss}}$ or PUPPI- $\vec{E}_T^{\text{miss}}$) can be studied using Z+jets events where the momentum of the Z is compared to the momentum of the recoiling jets (hadronic recoil) [59]. As no neutrinos are directly produced from the hard scattering process, these two observables should be equal and opposite. However, given the limited resolution when measuring particle momenta — predominantly from the mismeasurements of jets — E_T^{miss} is expected to be smeared. As shown in figure 3.13, the Z boson decays to two leptons, $Z \rightarrow \ell^- \ell^+$ that have momenta $\vec{p}_T(\ell^\pm)$ and which sum to the momentum of the Z: $\vec{q}_T(Z) = \vec{p}_T(\ell^-) + \vec{p}_T(\ell^+)$. The hadronic recoil, $\vec{u}_T$ is then defined as the negative sum of all particle momenta except the Z leptons. The hadronic recoil can then be further decomposed into the component parallel, $u_{\parallel}$, and perpendicular, $u_{\perp}$, to $\vec{q}_T(Z)$. Here, $u_{\parallel}$ is sensitive to the energy resolution of the jets over a range of p_T , whilst $u_{\perp}$ is sensitive to PU contributions. The resolution of both quantities versus the number of interaction vertices is presented in figure 3.14, taken from $Z \rightarrow e^- e^+$ events. Clearly, for events with more interaction vertices, PUPPI- $\vec{E}_T^{\text{miss}}$ outperforms PF- $\vec{E}_T^{\text{miss}}$, which is a result of the PUPPI algorithm more effectively removing contributions from PU.

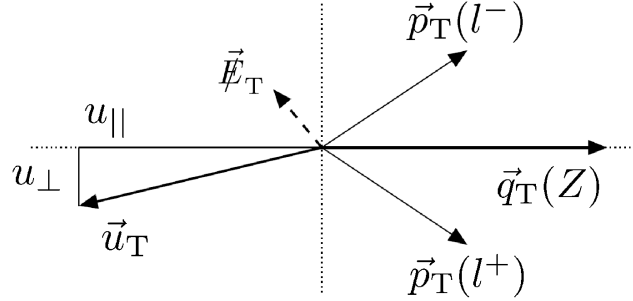


Figure 3.13: Illustration of a Z boson decaying to two leptons in the transverse plane. $\vec{u}_T$ refers to the sum of all PF candidates in the event except the Z leptons [62].

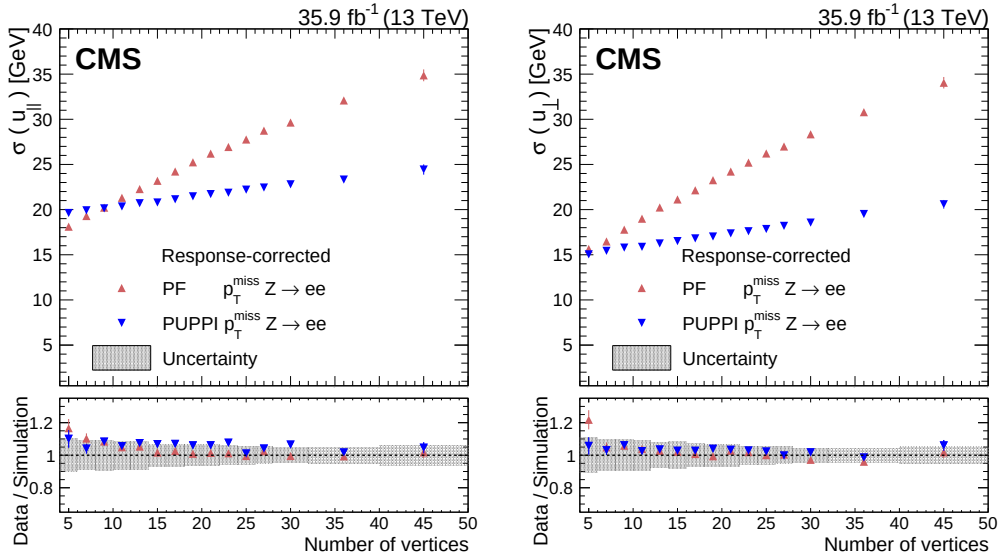


Figure 3.14: Resolution of the parallel (left) and perpendicular (right) component of the hadronic recoil system versus number of vertices for PF-\$\vec{E}_T^{\text{miss}}\$ and PUPPI-\$\vec{E}_T^{\text{miss}}\$ using $Z \rightarrow e^-e^+$ events. Lower plots show data to simulation ratio, with the uncertainty given by the uncertainties in jet energy scale and resolution [59].

3.4 Event Simulation

The goal of the CMS experiment is to infer the properties of the parameters in a model's Lagrangian, $\vec{\alpha}$, using the events and objects defined in the previous section. Ultimately, this relies on the likelihood function [63] that describes the probability of observing an event given the model parameters, $\vec{\alpha}$:

$$L(x|\vec{\alpha}) = \int dz_d \int dz_s \int dz_p p(x|z_d)p(z_d|z_s)p(z_s|z_p)p(z_p|\vec{\alpha}), \quad (3.9)$$

where x is the vector of all event observables such as lepton momenta. z_i are the latent space variables, which are defined as follows:

- z_p are the kinematic variables of all out-going partons in the hard-scattering event, for example their four-momenta and spin. From this, the term $p(z_p|\vec{\alpha})$ defines the probability of observing z_p given the underlying model parameters $\vec{\alpha}$, which is related to the matrix-element calculation, $|\mathcal{M}|^2$, that also includes information on the incoming partons. This term is referred to as the parton-level event.
- z_s describes the showering and hadronization of the out-going partons from the hard-scattering event. Essentially, $p(z_s|z_p)$ encodes the probability of observing z_s given the out-going partons z_p .
- z_d encodes the possible interactions of the out-going particles with the detector. Here, $p(z_d|z_s)$ describes the probability of observing the detector signatures given the particles z_s .

Critically, the integral in equation 3.9, which is also illustrated in figure 3.15, is entirely intractable. Consequently, calculating the likelihood analytically is impossible, which necessitates the need to approximate equation 3.9 using Monte-Carlo (MC) simulation.

The first step in the MC simulation is to calculate the hard-scattering process. For two colliding partons, a and b , which lead to the outgoing particles f , the hard-scattering process is calculated according to

$$\frac{d\sigma}{d\Phi_n} = \sum_{a,b} \int_0^1 dx_a dx_b f_a(x_a) f_b(x_b) \frac{1}{s} |\mathcal{M}_{a,b \rightarrow f}|^2(\Phi_n; \vec{\alpha}), \quad (3.10)$$

where Φ_n is the phase space of the final state, $\mathcal{M}_{a,b \rightarrow f}$ is the matrix element, $\frac{1}{s}$ is one over the squared centre of mass energy, and $f_i(x_i)$ are the parton distribution functions (PDFs). For high momentum transfer events, such as at the LHC, the strong coupling constant is small, allowing for perturbative expansion of $\mathcal{M}_{a,b \rightarrow f}$. This is diagrammatically represented using Feynman diagrams, whereby the leading order (LO) term corresponds to the first term in the expansion, and the subsequent terms are called higher-order terms. Such examples of higher-order terms are Next-to-LO (NLO), Next-to-Next-to-LO (NNLO) and so on. Ultimately, however, it is not possible to calculate all terms in the expansion,

with generally only the first few orders included. As such, truncating the expansion leads to uncertainties on the calculation, which are reduced by including more terms. The PDFs, $f_i(x_i)$, encode the number density of each type of parton to have a fraction x of the total proton momentum, with their form calculated empirically using lepton-proton and proton-proton collision data [64]. However, calculations of such functions are complicated, which introduces uncertainties to not only the predicted event yields, but also their kinematics, ultimately limiting the precision of many measurements. As a result, the PDF functions are often updated to incorporate new experimental data and theoretical improvements. Conversely, for electron-positron collisions, such as in chapter 6, the initial state particles are elementary leptons, and therefore do not require PDFs.

Following the hard-scattering calculation is the parton showering and hadronisation step, which describes the fragmentation from high-energy outgoing partons to low-energy final-state hadrons. This is performed by simulating the QCD radiation off all coloured partons using perturbative QCD calculations. This includes radiated gluons, which can go on to radiate additional gluons, or decay into quark-antiquark pairs, which are generally soft and collinear with the initial parton. The process is repeated for all radiated particles until the momentum transfer is comparable to that of hadron-formation energies, at which point the process is stopped. Parameters are included in the simulations, which can be tuned to match real data, with different sets of values being referred to as parton-shower tunes [65]. Following parton showers is the hadronisation step, which simulates the confinement of coloured partons into colourless hadrons. However, at this low-energy scale, QCD is no longer weak, and perturbative calculations are insufficient. Consequently, hadronisation models are utilised, which are generally theoretically motivated but rely on tunable parameters that are fit to data.

The final-state particle-level objects are subsequently propagated through a simulation of the CMS detector using the GEANT4 package [66]. The package includes a detailed description of the geometry of all CMS subdetectors, and models how the particles traverse through the magnetic field and eventually interact with the detector materials. The final event is then a set of energy deposits, referred to as hits, in each of the detector units. Following this, identical algorithms that are used on real data are used on the event simulation, resulting in the same objects as defined in the previous section.

Simulating each step is performed using a variety of tools, each suited to a specific task in the simulation chain. For this thesis, the parton-level events are simulated with MADGRAPH5_aMC@NLO [67] and POWHEG [68–70], which include the matrix-element calculations performed up to a specific order in the perturbative expansion. All parton showering and hadronisation is then performed using the PYTHIA package [71, 72]. Finally, the response of the full CMS detector is simulated using the GEANT4 software [66] for the CMS analysis in section 5. Whereas, for the FCC-ee analysis in chapter 6, the IDEA detector is simulated using a simplified simulation software called DELPHES [73].

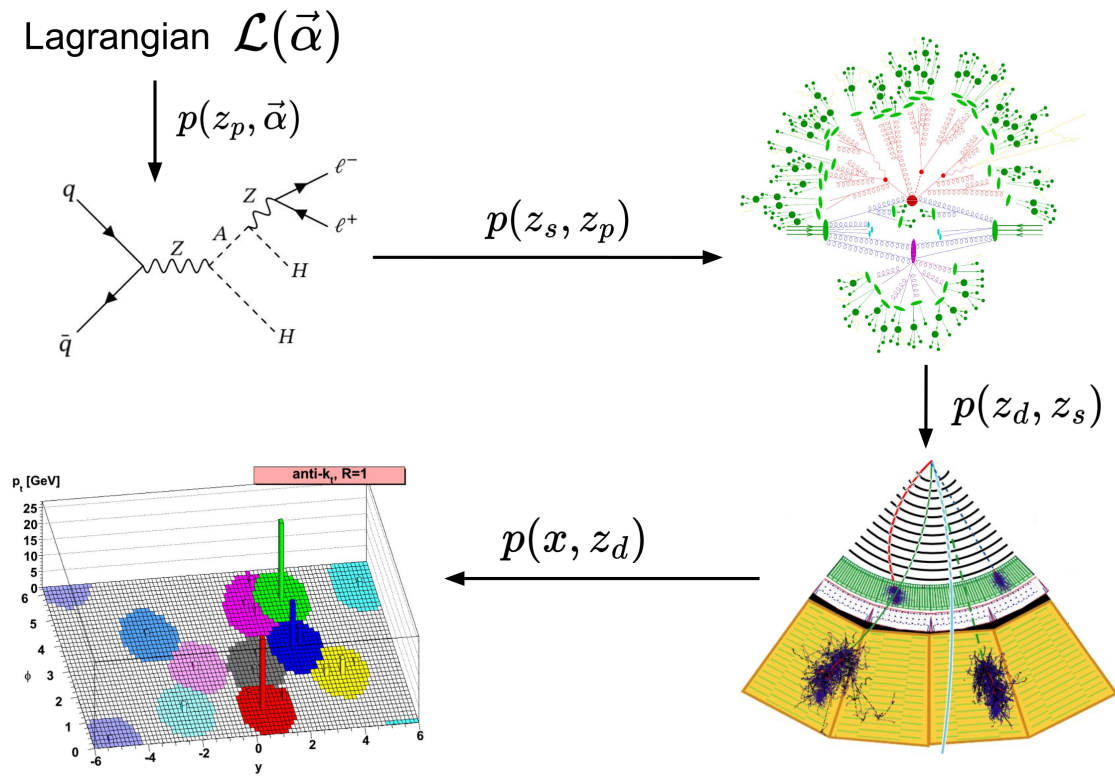


Figure 3.15: An illustration showing the individual elements that go into the intractable likelihood function. The matrix element (top left) is the IDM process $q\bar{q} \rightarrow HH\ell^+\ell^-$. The parton shower (top right), the CMS schematic (bottom right), and the anti- k_T clustering illustration (bottom left) are taken from References [74], [48] and [56], respectively.

Chapter 4

The High Granularity Calorimeter

Contents

4.1	Introduction	55
4.2	High Granularity Calorimeter	57
4.3	Electron and Photon Identification	60
4.3.1	Boosted Decision Trees	60
4.3.2	Graph Neural Networks	64
4.3.3	BDT and GNN Performance	66

4.1 Introduction

One way to improve the physics potential of the LHC is to take more data: more data reduces the statistical uncertainty of measurements, as well as opens up the reach to rarer processes. However, using the current beam conditions of Run 3, and considering that the statistical uncertainty scales as $1/\sqrt{\mathcal{L}}$, the LHC would have to run for another decade to reduce the statistical uncertainty by a factor of two relative to the current dataset. As such, running with the current beam conditions is unsuitable given the marginal improvements. Instead, the LHC will seek to maximise the physics potential by pushing the instantaneous luminosity. Following the completion of Run 3 in mid-2026, the LHC will go into a period of upgrades, referred to as Long Shutdown 3 (LS3), in preparation for the next stage of operation: the High-Luminosity LHC (HL-LHC) [75]. The HL-LHC will operate at an instantaneous luminosity of $5 \times 10^{34} \text{cm}^{-2} \text{s}^{-1}$ — a factor of 2.5 over the current capabilities — rising to $7.5 \times 10^{34} \text{cm}^{-2} \text{s}^{-1}$ towards the end of operation, at a centre-of-mass energy of $\sqrt{s} = 14.0 \text{ TeV}$. At this rate, an expected 250 fb^{-1} *per year* will be delivered by the HL-LHC, roughly double what was delivered from the whole of Run 2. With the scheduled 10-15 years of running, therefore, a total of 3000 fb^{-1} is expected, a factor 10 over the current data set [75]. Achieving this instantaneous luminosity, however, does not come without substantial effort; to operate at this level, significant upgrades must be made to the LHC.

At this instantaneous luminosity, the experiments also face significant challenges. The HL-LHC will operate with an average pileup (PU) rate as high as 200 [75], a significant increase over the current highest rate of around 60, as shown in figure 3.3. This poses problems on many fronts. First, this results in considerably higher radiation levels that would far exceed the limit for many parts of the current detector, leading to a significant degradation of physics performance. Furthermore, in a high PU environment, triggering and reconstructing processes of interest becomes more challenging given the increased occupancy in the detector. To address these challenges, the experiments will also require upgrades. For CMS, the upgrades will be performed during LS3, improving not only the radiation hardness required for the HL-LHC, but also the physics capabilities to deal with the higher PU. The most notable of these are the upgrades to the Level-1 Trigger (L1T) and the endcap calorimeters. Operation of the CMS experiment during the HL-LHC is referred to as CMS Phase-2 [76].

The L1T Phase-2 upgrade is designed to not only maintain the efficiency seen during Run 3, but also significantly improve upon the performance by enabling triggering on more complex physics signatures [77]. This is made possible by upgrading the Data Acquisition System (DAQ), allowing for an increase in the event selection latency to 12.5 μ s, as well as increasing the overall L1T rate to 750 kHz. This measures a large increase over Run 3, which had a maximum latency of 4 μ s, and an overall L1T rate of 110 kHz. As a result, the L1T will have access to the tracker information, allowing for particle-flow level reconstruction at the L1T stage, as well as the capability to implement state-of-the-art offline ML techniques for object identification. Moreover, the triggering is further enhanced by the higher granularity from the calorimeters, and the gain in pseudorapidity coverage from inclusion of the CMS endcap calorimeters at the L1T stage. It should be noted, however, that whilst the L1T latency has increased to 12.5 μ s, 5 μ s are allotted for just creating the HGCal L1T clusters, therefore limiting any ML techniques to at most, but likely less than, 7.5 μ s [78].

The current CMS endcap calorimeters are split into two parts, the ECAL endcap and the HCAL endcap, covering a pseudorapidity of $1.479 < |\eta| < 3.0$ and $1.3 < |\eta| < 3.0$, respectively. In the high-pseudorapidity region, the endcaps experience a significant amount of radiation from the high particle flux. As such, by the end of Run 3, the endcaps will have significantly degraded, and will be unsuitable for further use. Furthermore, the granularity of the current endcaps is not suitable for the high PU environment of the HL-LHC, where better resolution is required to discern genuine prompt objects from PU. Given these issues, for Phase-2, the CMS endcaps are being replaced with a single endcap calorimeter system called the High Granularity Calorimeter (HGCal). The HGCal will include an electromagnetic and a hadronic component with fine granularity in both the transverse and longitudinal plane, as well as timing information, allowing for precise 4-dimensional reconstruction.

This chapter explores the use of graph neural networks (GNNs) for electron/photon identification in the context of the HL-LHC Level-1 trigger, specifically, identifying electrons and

photons versus pileup using only the information available to the trigger from the HGCal. The performance of the GNN is then compared to current methods that rely on boosted decision trees. Notably, however, whilst this study is conducted in the context of the L1T, for actual use at the trigger level, additional studies on computational efficiency would have to be performed, but this is beyond the scope of this thesis. Instead, this work serves as a proof of concept for the effectiveness of GNNs in processing low-level, irregular-shaped data compared to traditional methods that require manual feature extraction. Therefore, this approach is not limited to the trigger level, but is also applicable to a broad range of particle physics applications — many of which involve variable-sized data — both at current and future colliders when more performant algorithms are required.

4.2 High Granularity Calorimeter

The High Granularity Calorimeter (HGCal) is designed to replace the current CMS end-caps that will be nearing their radiation limit by the end of Run 3 [79]. Positioned at high pseudorapidity, and operating under extreme PU conditions during the HL-LHC, the HGCal is required to be highly radiation tolerant to ensure precise energy resolution over the lifetime of the HL-LHC. Furthermore, the high PU environment places additional requirements on the HGCal to maintain a good efficiency for reconstructing and identifying prompt particles from background. This includes high transverse and longitudinal granularity, providing precise information on the development of showers, and enabling discrimination against PU events which often have different shower properties. To ensure full shower containment for both electromagnetic and hadronic particles, a short radiation length and nuclear interaction length are essential, as well as compact shower radii to reduce contamination between showers. Moreover, precise timing information is required to further separate prompt processes of interest from PU, which also aids in the reconstruction of interaction vertices. Finally, to increase the pseudorapidity acceptance of the trigger, and to enable triggering on more complex processes, the HGCal is required to contribute to the L1T.

The HGCal will cover a pseudorapidity range of $1.3 < |\eta| < 3.0$, and is split into two sections: an electromagnetic section (CE-E) followed by a hadronic section (CE-H) [79]. Both sections are sampling calorimeters. The CE-E uses entirely finely-segmented silicon cells as the active material, whilst the CE-H uses silicon cells closer to the interaction point and plastic scintillator further away where particle flux is lower, and the area covered is larger. A schematic diagram of the HGCal is shown in figure 4.1. The silicon cells are highly segmented with an area of $\approx 0.5\text{-}1\text{ cm}^2$ — required due to the higher occupancy closer to the interaction point — and for the plastic scintillators each cell is $\approx 4\text{-}30\text{ cm}^2$. The active material is then interleaved between layers of lead (CE-E) and steel (CE-H) absorbers, increasing the effective depth of the calorimeter and leading to radially-compact showers. The CE-E consists of 28 longitudinal samplings, corresponding to 26 radiation lengths (X_0) and 1.7 nuclear interaction lengths (λ_I), and with the inclusion of the CE-H, gives a total calorimeter thickness for the HGCal of $10.7\lambda_I$. Finally, a significant source of

the particle flux experienced in the HGCal is fast neutrons from PU. To reduce this source of radiation, and help maintain detector performance, a polythene neutron moderator is placed in front of the CE-E.

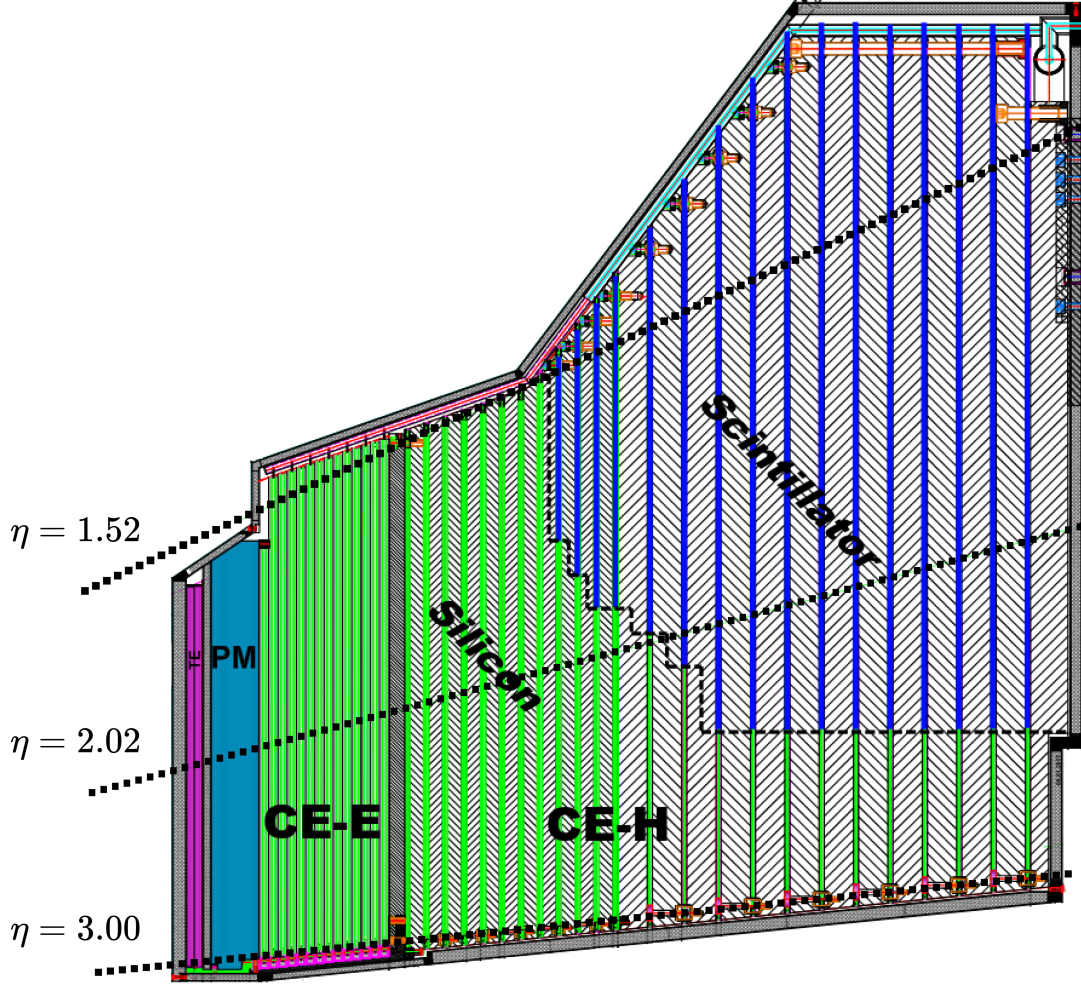


Figure 4.1: Longitudinal diagram of the HGCal. This is split into two main parts, the electromagnetic compartment (CE-E) and the hadronic compartment (CE-H), with the CE-E made from silicon (green) and the CE-H made from silicon and plastic scintillator (blue). In front of the CE-E is a polythene neutron moderator, denoted as PM, and the timing module is denoted by TE (purple). Image modified from [79].

As previously stated, one major upgrade to the calorimeter is the addition of precise timing, adding vital information for rejecting PU, and reducing ambiguity from overlapping trajectories. This is achieved with a specific endcap timing layer preceding the CE-E, similarly made out of silicon cells, that allows for a precise timestamp for an event with a resolution of ≈ 25 ps [79]. The effectiveness of the timing information to remove PU is shown in figure 4.2 which shows the simulation of hits in the HGCal for VBF $H \rightarrow \gamma\gamma$ events before and after the removal of hits with $\Delta t > 90$ ps compared to the primary interaction. After the timing cut, a substantial number of PU hits are removed, leaving a cleaner signature of a jet and photon, and thus allowing for reconstruction less

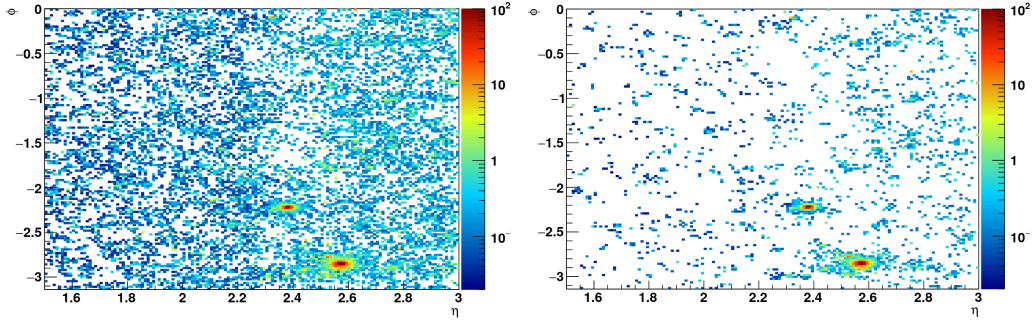


Figure 4.2: Simulation of hits in the HGCal projected onto the front face for VBF $H \rightarrow \gamma\gamma$ events, before (left) timing requirements and after (right) removal of hits with $\Delta t > 90$ ps. The two red clusters are from a jet and a photon. Image taken from [79].

affected by PU. Quantities derived from these are also improved, as preliminary studies found up to a 20% improvement on the missing energy resolution when incorporating the timing information [80]. Moreover, timing information aids in the reconstruction of primary vertices, mitigating the incorrect association of charged PU with the PV. This in turn leads to improvements in lepton isolation, which depends on tracks being associated with the primary vertex, with simulations suggesting isolation efficiency comparable to that of the current Run 3 beam conditions [80].

Including the HGCal at the L1T stage extends the pseudorapidity range of the trigger, allowing for a larger acceptance of high- η processes. Furthermore, advances in computing technology, such as improvements to FPGAs, and the increased L1T latency to 12.5 μ s, allows for more modern machine learning (ML) techniques to be implemented [77]. To trigger on such states, however, requires efficient identification of prompt particles and significant rejection of pileup to operate within the event rate constraints. Moreover, even with the increased L1T latency, the high granularity of the HGCal renders full readout of all cell information impossible. As such, to reduce the data throughput at the L1T stage, only alternate layers in the CE-E are used and all timing information is dropped. Furthermore, neighbouring silicon cells are summed together to form coarser Trigger Cells, effectively increasing their size to 4 cm^2 . The same process is also done for the scintillator tiles, increasing their effective size to 16-100 cm^2 . Using the coarser information, a clustering algorithm is employed (similar to Section 3.3) that identifies cluster seeds in each layer, and from them expands out to form 2D clusters. Following this, 2D clusters are combined to form 3D topological clusters called *HGCal Trigger Primitives*. Using the trigger primitives alone, e/γ candidates are selected if they pass an identification selection. Those that pass are sent to the next level of the L1T for track matching, isolation calculations, and further selections to reduce pileup contamination [77]. The subsequent sections cover two different methods for e/γ identification, one using Boosted Decision Trees (BDT), which is the current baseline method outlined in the Reference [77], and the other using Graph Neural Networks (GNN).

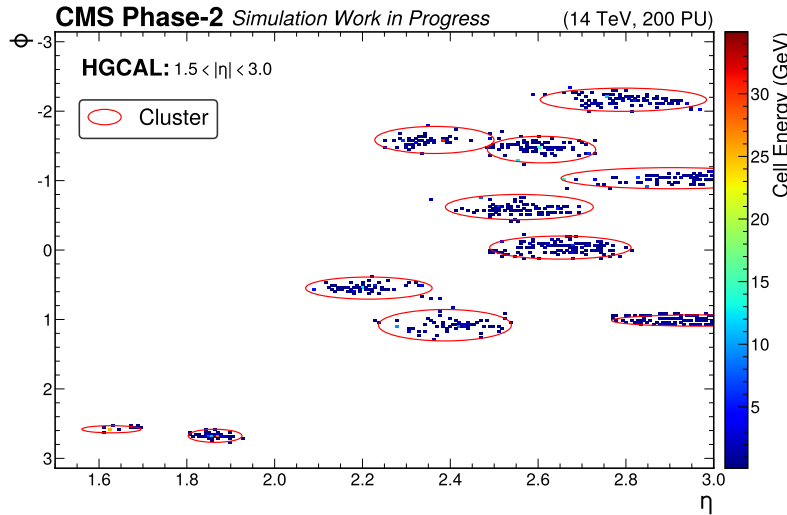


Figure 4.3: Simulation of trigger primitive clusters in the HGCAL for $\eta > 1.5$. The red circles indicate the rough clustering of the trigger cells.

4.3 Electron and Photon Identification

The precise development of showers, provided by the high granularity of the HGCAL, enables discrimination between electrons/photons and pileup, even in the harsh environment of the HL-LHC. This is true despite the reduced granularity at the L1T, as described in the previous section. No differentiation between electrons and photons is made, however, due to their similar signatures in the calorimeters, and the fact that no tracking information is used. Furthermore, no derived quantities are utilised, such as candidate p_T , only the trigger cell location and energy deposits. An example of the clusters in a single simulated event projected onto the face of the HGCAL is shown in figure 4.3.

This section covers the use of different ML techniques for e/γ identification, using only the Trigger Primitive clusters available to the HGCAL L1T. These studies are performed on Monte-Carlo samples, simulated at a centre-of-mass energy of 14 TeV, with an average PU of 200. The simulated clusters are taken as the training samples, with the signal being clusters with $p_T > 20$ GeV matching truth-level electrons, where matching refers to the requirement $\Delta R < 0.2$. The background sample is then defined as clusters with $p_T > 20$ GeV that are not matched to a truth-level electron. In total, there are approximately 200,000 signal clusters and 200,000 background clusters, and these are split into train/validation/test sets corresponding to 70%/10%/20% of the total. The samples used in training/validating/testing the BDT are identical to those used for the GNN.

4.3.1 Boosted Decision Trees

Boosted Decision Trees are a form of supervised machine learning applicable to a variety of problems such as classification or regression [81][82]. By learning a set of simple decision rules from an input data set, a model is formed that can predict a target value $\hat{y}(x)$ from an input $\mathbf{x} = (x_1, x_2, \dots, x_N) \in \mathbb{R}^F$, where each element in $\mathbf{x}$ corresponds to some observable. For classification — which is the use case for this chapter — the data set used

for learning is formed with samples of known classes, $\{(\hat{y}_1, \mathbf{x}_1), (\hat{y}_2, \mathbf{x}_2) \dots (\hat{y}_N, \mathbf{x}_N)\}$, where $\hat{y}_i \in \{0, 1, 2, \dots, K\}$ for $K+1$ classes. For binary classification $y_i \in \{0, 1\}$.

A BDT is an ensemble ML method, where instead of having a single classifier, a set of classifiers are used with their individual predictions combined to give a final, improved, prediction. The basic classifier used for a BDT is a Decision Tree, which for classifying a data set as background (0) or signal (1), is created with the following method:

- The input feature space is partitioned according to a selection on an input feature, x_i . The selection is determined by optimising a purity measure that defines how pure each partition is, with perfect purity being when each partition is only composed of a single class. The feature selection that produces the highest purity partitions is chosen.
- This process is repeated for each of the subsequent partitions.
- For a single partition, if a stopping criterion is met, no more selections are made (but other partitions might still require more selections). Examples of a stopping criterion could be a limit on the number of selections (max depth), or if the number of samples after a split falls below a certain amount.
- The final set of partitions that have no further selections are called leaves. The classification for a leaf is given by $\frac{s}{s+b}$ to give an output probability, with 1 being signal-like and 0 being background-like.

This can, however, be improved using ensemble methods such as boosting [81][82]. Boosting involves training multiple DTs in succession, with each DT aiming to improve the prediction of the previous tree. Specifically, if the initial tree has output $F(x_i)$, then a second tree, $H(x_i)$, is trained with the aim of having $F(x_i) + H(x_i) = y_i$. In reality a single additional tree is not enough, and instead multiple DTs are successively trained, and the final prediction from the ensemble is given by the sum of all DT predictions $\hat{y}(x_i) = \sum_j^{\text{N Trees}} f_j(x_i)$. The number of additional trees is then set as a hyperparameter, with a balance needed as too many trees can be a source of overfitting.

Following the baseline outlined in Reference [77], the XGBoost software package is used for training two BDTs, one for the range $1.5 < |\eta| < 2.7$ and one for $2.7 < |\eta| < 3$ [81]. Training two BDTs is found to be beneficial, especially for the higher η region, as the input features were found to change considerably as a function of η . In total 9 input features are used that describe the longitudinal and transverse shower-shape. These are listed in table 4.1. For a cluster cell with coordinate p — such as cluster cell η — the energy-weighted RMS feature is calculated as:

$$\text{Weighted RMS}(p) = \sqrt{\frac{1}{E_{\text{tot}}} \sum_i^{N_{\text{tcs}}} (p_i - \bar{p})^2}, \quad (4.1)$$

Table 4.1: Input features to the BDT for e/γ identification at the HGAL L1T stage.

Longitudinal shower shape variables	
First layer	First layer in the HGAL that the cluster deposited energy in
Max energy layer	The layer in the HGAL that had the maximum cluster energy deposit
Shower length	Total length of the cluster, given by the difference between the first and last layer
Core shower length	Maximum number of consecutive HGAL layers the cluster has energy deposits in
Weighted RMS(z)	Energy-weighted RMS of trigger cell z coordinates
Transverse shower shape variables	
Weighted RMS(r/z)	Energy-weighted RMS of trigger cell r/z coordinates. The ratio is taken to account for the shower spreading out as it develops in the HGAL. This is a measure of the radial spread of the shower.
Mean of weighted RMS(r/z) per layer	Energy-weighted RMS of r/z coordinate for each HGAL layer
Weighted RMS(η)	Energy-weighted RMS of trigger cell η coordinates. Measure of the shower-spread in η
Weighted RMS(ϕ)	Energy-weighted RMS of trigger cell ϕ coordinates. Measure of the shower-spread in ϕ

where the sum is over the trigger cells in the cluster, each with coordinate p_i and energy E_i , up to the total number of cells in the cluster N_{tcs} . The mean value of the coordinate is given by $\bar{p}$, and the sum of all trigger cell energies is given by $E_{\text{tot}} = \sum_i^{N_{\text{tcs}}} E_i$.

The distributions of the input feature are shown in figure 4.4. Good discrimination is shown between signal and background for both the low- η and high- η case. Notably, the Max Energy Layer distribution demonstrates that nearly all e/γ events deposit their maximum energy in the CE-E section of the HGAL, whereas pileup events are significantly more likely to deposit their maximum energy in the CE-H section.

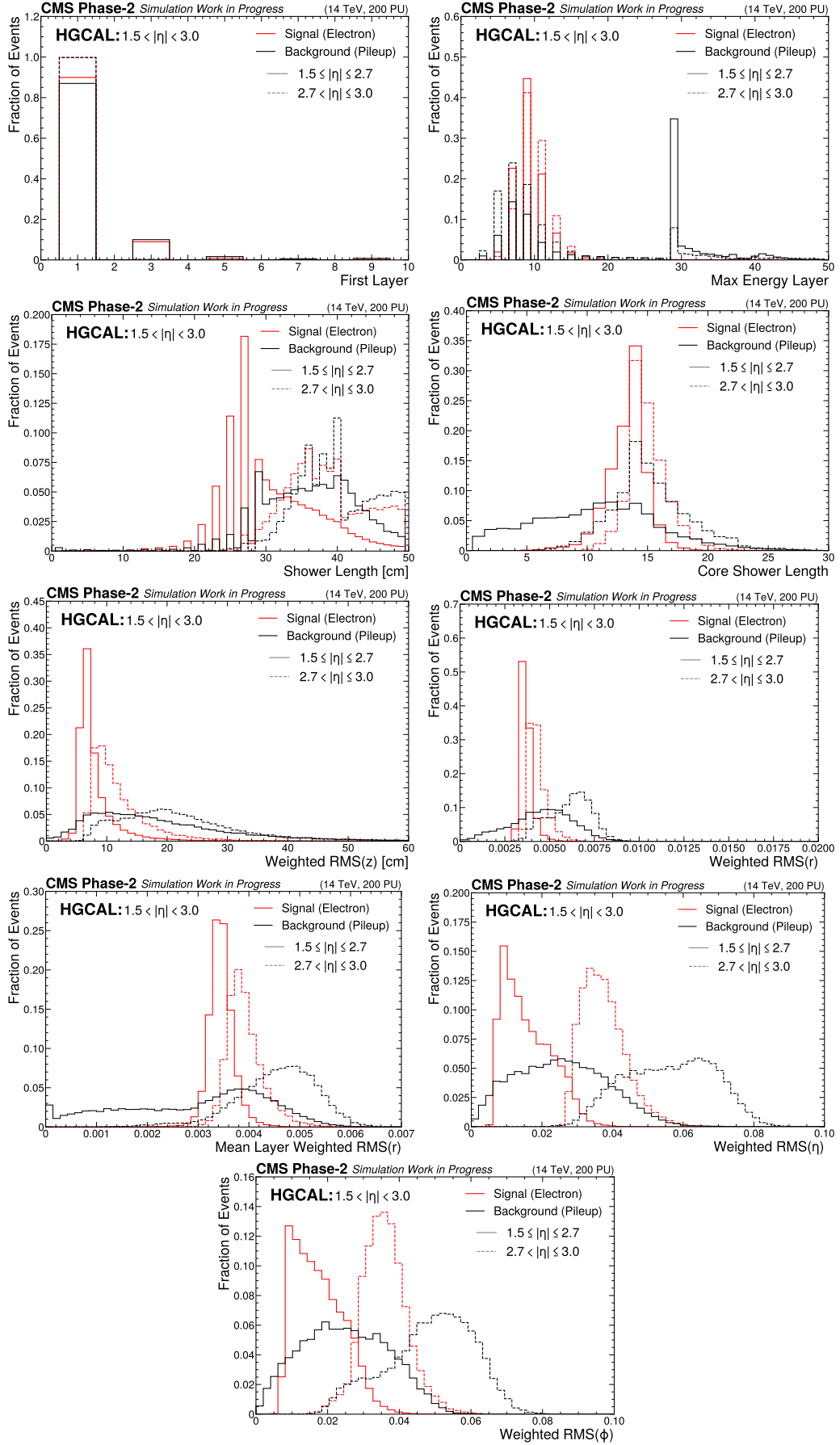


Figure 4.4: Distributions of the input features to the BDT for the HGCAL L1T for e/γ identification. Following Reference [77], a separate BDT is trained on each of the two regions, $1.5 < |\eta| < 2.7$ and $2.7 < |\eta| < 3.0$.

4.3.2 Graph Neural Networks

A major drawback with BDTs for e/γ identification is that they cannot be applied directly to the raw, variable-sized cluster information, as they require fixed-length input vectors. As a result, they rely on features that are manually extracted from the low-level data. If the features selected do not fully encapsulate all the information in the cluster, however, then a degradation in performance is expected. Therefore, to fully utilise the information contained in the trigger primitive clusters, it is desirable to have a model that can be applied directly to the raw information, with feature extraction learnt during training. One such method capable of handling variable-length data is Graph Neural Networks (GNNs).

Graphs are a basic data structure that can represent data that has no fixed size and with arbitrary relations between its components. A graph, $\mathcal{G}(\mathcal{V}, \mathcal{E})$, is formed by a set of nodes, $\mathcal{V}$, and the relationship between these nodes, $\mathcal{E}$. Each node, $v \in \mathcal{V}$, can possess a set of features¹, labelled $\vec{x}_v = (x_v^1, x_v^2, \dots, x_v^F) \in \mathbb{R}^F$, and the edge connecting node v_i and v_j is given by $e_{ij} \in \mathcal{E}$ and these likewise can possess their own feature vector $\vec{e}_{ij} = (e_{ij}^1, e_{ij}^2, \dots, e_{ij}^E) \in \mathbb{R}^E$. Edges between nodes can be undirected, where $e_{ij} = e_{ji}$, or directed where, $e_{ij} \neq e_{ji}$. Furthermore, a graph can also possess a graph-level feature that describe features of the graph as a whole, $\vec{g} = (g^1, g^2, \dots, g^T) \in \mathbb{R}^T$. An example of an undirected graph is shown in figure 4.5. Evidently, these can be composed of an arbitrary number of nodes, with arbitrary connections between them.

In general there are three graph based prediction tasks: node-level, edge-level and graph-level prediction. Node-level prediction is the task of predicting a node's feature vector given its edges to other nodes. Edge-level prediction is the task of predicting the edges of a node given its feature vector. Finally, for graph-level tasks, the aim is to use the entire graph as input to predict a feature of the graph. The focus of this section is graph-level prediction, with the task being to read in the raw cluster information as a graph, and to predict whether it originated from an electron/photon or from a pileup jet. Specifically, the graph nodes are constructed from the trigger cells in the cluster, with the location and energy of the trigger cell being the feature vector. Edges are then constructed using a k-nearest neighbour method, and these edges carry no feature vectors.

The graph neural network architecture is based on the ParticleNet model [83], which is built upon the EdgeConv layer [84]. A schematic of this is presented in figure 4.6a. The EdgeConv layer is split into two parts which are combined at the end: a feedforward network that takes in the feature vector of node x_i and its neighbours x_j ; and a skip connection which is a copy of the input feature vector x_i . For a node i with feature vector x_i , the EdgeConv layer first finds its k-nearest neighbours with feature vectors x_j . Then, for each neighbour it creates the vector $(x_i, x_j - x_i)$, which encodes information of the point itself and the differences to the neighbour, and passes this through a three-layer Multilayer Perceptron (MLP). The number of hidden units in the three layers is denoted $C = (C_1, C_2, C_3)$, and each layer consists of a linear transformation, followed

¹Only the case where each node (edge) has the same feature dimension as every other node (edge) is considered.

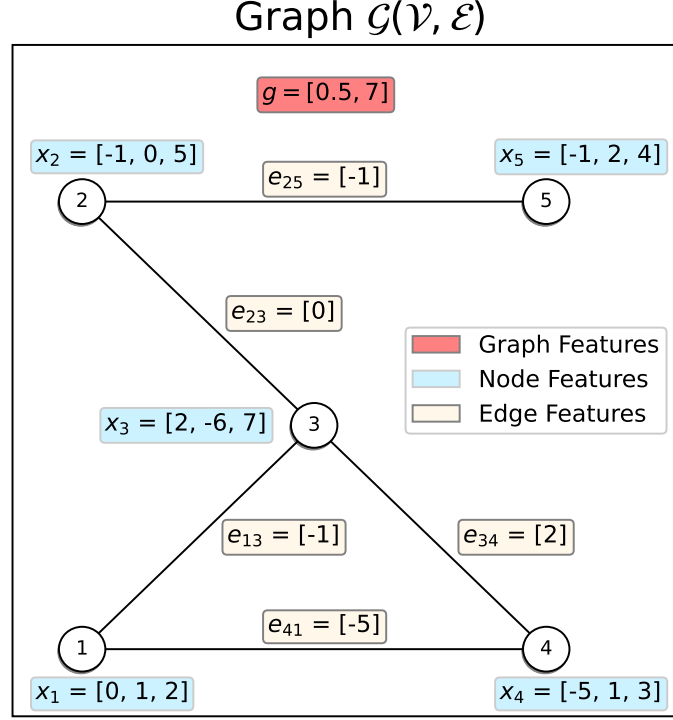


Figure 4.5: Example of an undirected graph, $\mathcal{G}(\mathcal{V}, \mathcal{E})$, with 5 nodes, each with a feature vector x_i . The edges between nodes i and j have feature vectors e_{ij} , and the graph-level feature vector is denoted g .

by a BatchNorm [85]—which normalises the input into each layer, stabilising training—and then a ReLU activation function [86], which enables the model to learn non-linear relationships. This is repeated for each neighbour node, and the results are averaged to give x_{MLP} . Additionally, inspired from the use in ResNet [87], a shortcut connection runs parallel to this with an output x_{skip} , and the final output of the EdgeConv block is the sum of these outputs with a ReLU activation function: $\text{ReLU}(x_{\text{MLP}} + x_{\text{skip}})$.

In a single EdgeConv layer, the same MLP is applied to each point on the graph, significantly reducing the number of parameters, whilst offering more efficient learning. Overall, the EdgeConv reads in a graph and outputs another graph with the same number of nodes, but with different feature vectors. By not changing the number of nodes, an arbitrary number of EdgeConv layers can be stacked on top of one another. This allows for hierarchical learning of features similar to normal convolutional networks, whereby the initial layers only access information from immediate neighbours, but later layers progressively incorporate information from the whole graph structure. Furthermore, the output of each EdgeConv layer can be viewed as a representation of the graph in a higher-dimensional latent space. By calculating the k -nearest neighbours in the latent space at the start of every EdgeConv layer, the connections between points can be dynamically learned. After successive EdgeConv layers, the new graph representation is reduced to a single vector using a Global Average Pooling layer, which takes the average of each node’s feature vector. This is passed through a final MLP which outputs a score between 0 and 1.

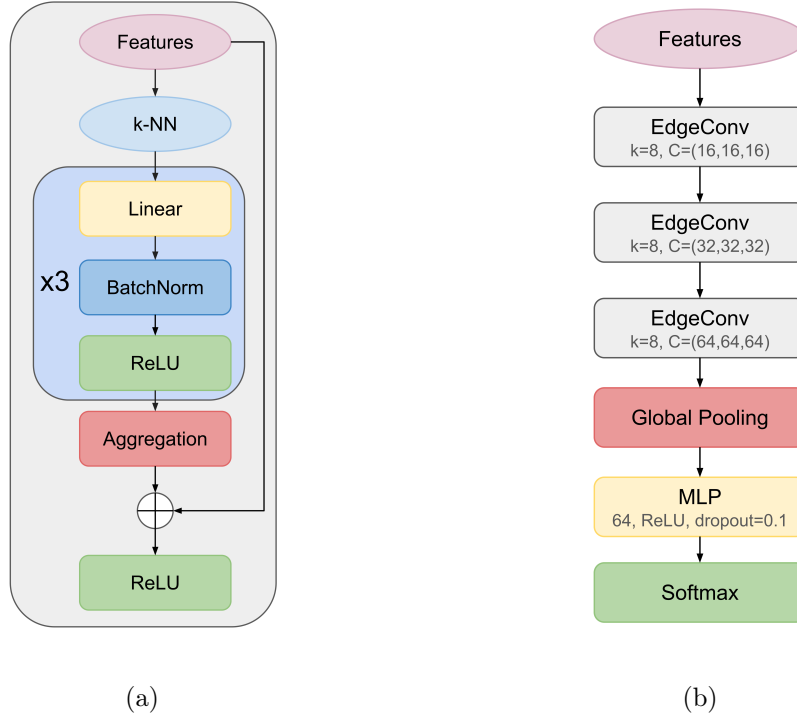


Figure 4.6: Architecture for the (left) EdgeConv layer and the (right) graph neural network for e/γ identification.

The full GNN architecture is shown in figure 4.6b, which is implemented using the PyTorch library [88]. This consists of three EdgeConv blocks with hidden units [(16, 16, 16), (32, 32, 32), (64, 64, 64)], each using $k=8$ nearest neighbours, and a final MLP layer with 64 hidden units. Performing a small gridsearch over the values of these hyperparameters did not lead to large changes in performance. To limit overfitting, the input into the final MLP also includes a dropout layer with a rate of 0.1 [89]. In total, the model comprises approximately 24,000 learnable parameters, representing a significant reduction compared to many modern networks, which often contain millions of parameters [87]. Before passing the graphs through the network, the input features are standardised such that they have a mean of 0 and a standard deviation of 1. Training is performed with a batch size of 150, limited by the available GPU memory. The model is optimised by minimising the binary cross-entropy loss using the Adam [90] optimiser with a learning rate of $1e-4$. Finally, the model that produces the lowest loss on the validation set is used as the final model.

4.3.3 BDT and GNN Performance

The output scores for the BDT model and the GNN are presented in figure 4.7, with a score of 0 being background-like and a score of 1 being signal-like. For both the BDT and GNN there is strong separation between the e/γ clusters and the pileup clusters. The performance of both models is evaluated on the test dataset using a Receiver Operating Characteristic (ROC) curve, where each point on the ROC curve shows the signal efficiency and the background rejection rate for a corresponding cut on the BDT score [91]. Additionally, the Area Under the ROC curve (AUC) is calculated, which gives a measure

of the classifier performance over the whole range of output scores. With perfect discrimination between signal and background, the AUC would be equal to 1. These are shown in figure 4.8, which is split into the low η and the high η regions. For both regions the GNN outperforms the BDT, most notably for the higher η region in which the GNN can match the performance of the BDT in the lower η region.

Using the output scores from the BDTs, baseline thresholds, known as *working points*, can be defined that correspond to a specific signal efficiency or background rejection rate. For the region $1.5 < |\eta| < 2.7$ a working point is defined for a signal efficiency of 97.5%, which corresponds to a background rejection of approximately 95%. For the region $2.7 < |\eta| < 3.0$ a tighter working point is used due to the larger contamination from pileup. This is defined as a signal efficiency of 90%, which corresponds to a background rejection of approximately 94%. With these thresholds, the BDT signal efficiency versus p_T and η is studied. These are shown in figure 4.9a for the low η region, high η region and for both regions combined. For both η regions the signal efficiency increases with electron p_T , with the low η region attaining almost 100% efficiency for $p_T > 50$ GeV, and both regions obtaining a signal efficiency exceeding 95% for $p_T > 60$ GeV. Furthermore, over the whole range of p_T , the low η region consistently has higher signal efficiency than the high η region. Finally, from figure 4.9c, the signal efficiency is generally seen to gradually decrease as a function of electron η , with exception in the first bin due to clusters being cut off at the HGCAL edge.

To compare the GNN to the BDT, working points are defined that have the same background rejection rate in the two η regions. Using these as the working points, the signal efficiency versus p_T and η is examined, as shown in figure 4.9b and 4.9d. At the same background rejection rate, the GNN significantly outperforms the BDT in all p_T and η bins. This is most significant for the $2.7 < |\eta| < 3.0$ region, with an increase in efficiency from nearly 60% to roughly 90% in the lowest p_T bin. Furthermore, the efficiency in both regions exceed 95% for a p_T above 30 GeV, which is 30 GeV lower than the BDT. Finally, the GNN obtains an almost entirely flat efficiency curve versus η , with significantly less degradation at the edges where the clusters are cut off.

Overall, the GNN performance suggests that it learns additional features that are not encapsulated by those used for training the BDT. This is possible due to the fact that the GNN operates on the low-level raw data itself with automatic feature extraction, rather than requiring manually-engineered feature extraction as with the BDT. In turn this leads to superior performance for the GNN, with significant improvements to the signal efficiency at a matching background rejection rate of the BDT, even with the relatively small size of the GNN model. Whilst, however, the implementation of a GNN at the L1T level is not currently possible due to computing constraints, the methodology is still applicable at the offline level or, potentially, at the HLT which already employs the GNN ParticleNet for b-tagging. Finally, as graph neural networks are capable of handling arbitrary shaped data — a feature often seen in particle physics — they are not limited to identification tasks, but can be also be applied to a broad range of other physics problems.

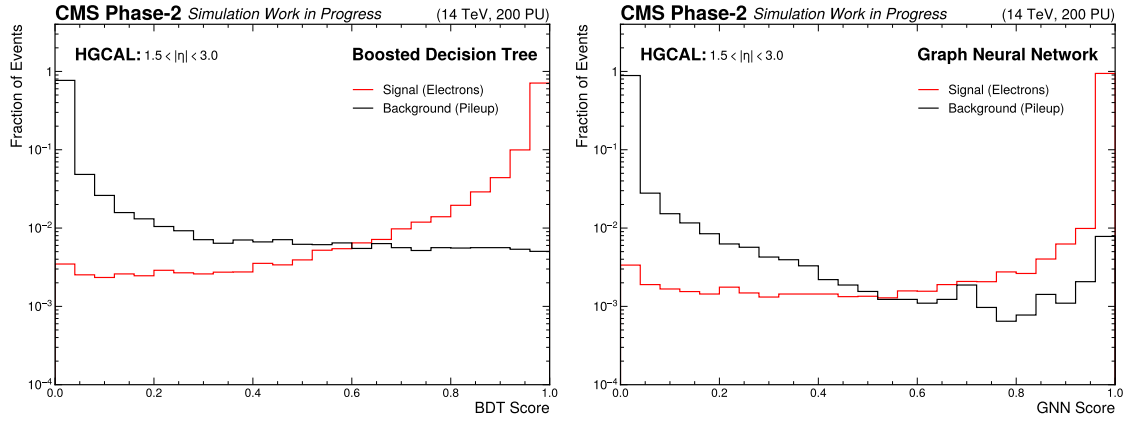


Figure 4.7: Distributions of the output scores for the (left) BDT and the (right) GNN for e/γ identification with the HGCAL L1T. The two BDTs trained for the regions $1.5 < |\eta| < 2.7$ and $2.7 < |\eta| < 3.0$ are combined.

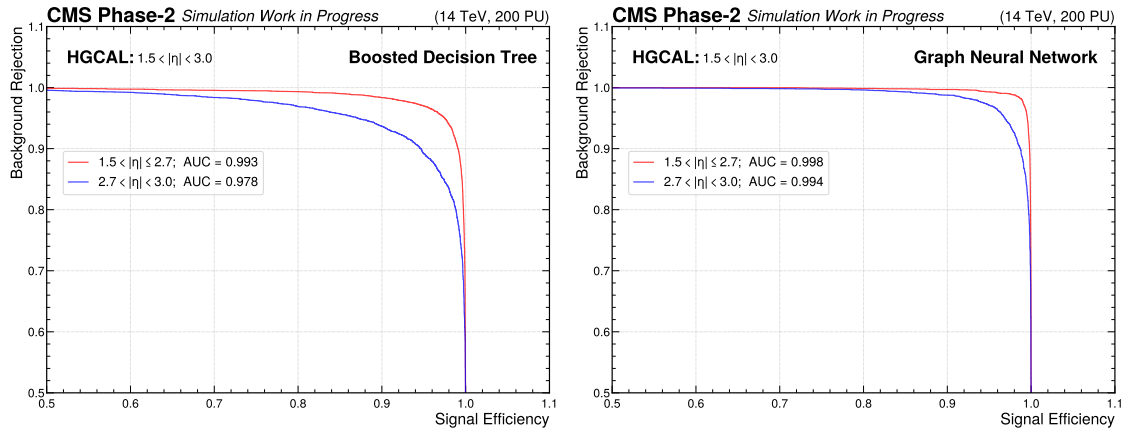


Figure 4.8: ROC curves with AUC score for the (left) BDT and (right) GNN for e/γ identification with the HGCAL L1T.

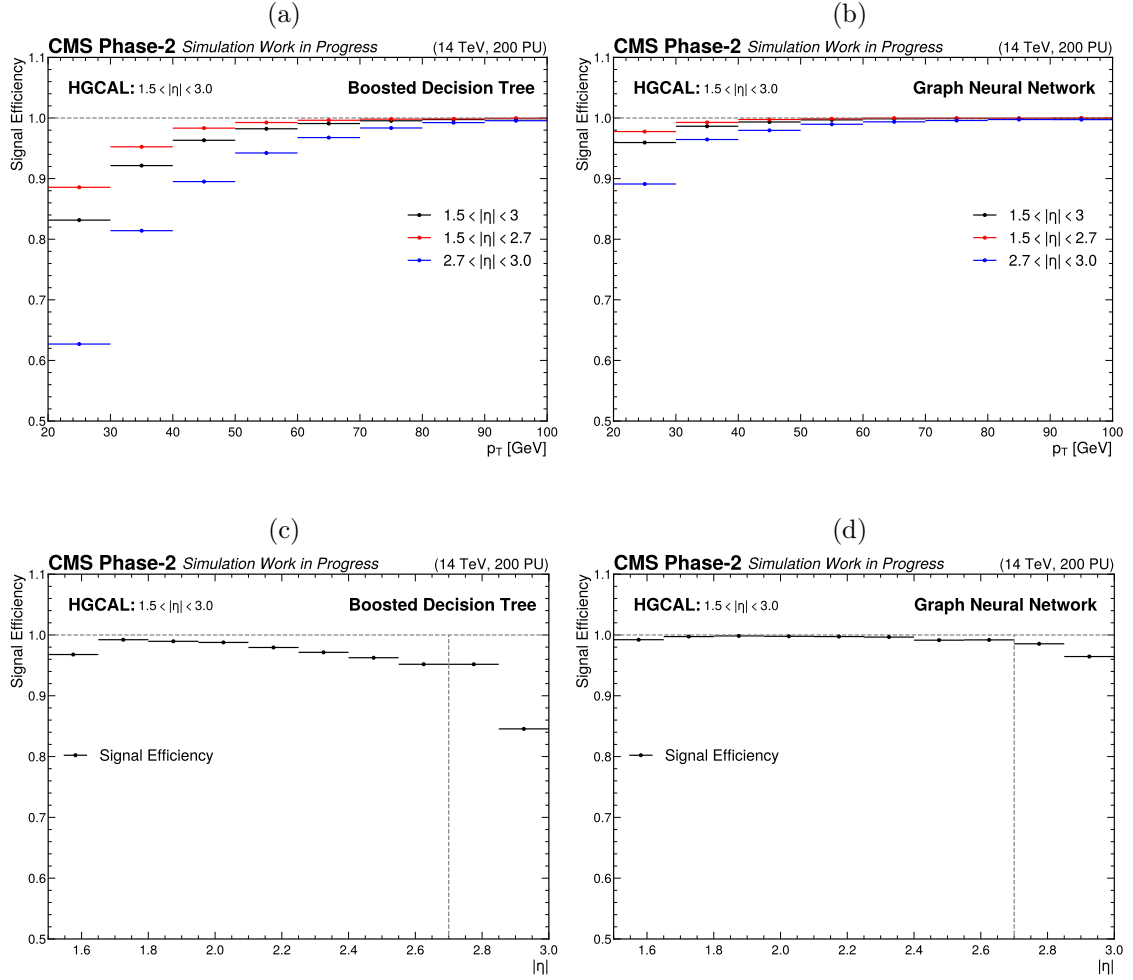


Figure 4.9: Signal efficiency versus p_T for the (top left) BDT and (top right) GNN, and versus η for the (bottom left) BDT and (bottom right) GNN. For the BDT, the working point is defined as a signal efficiency of 97.5% for $1.5 < |\eta| < 2.7$, and 90% for $2.7 < |\eta| < 3.0$. This is equivalent to a background rejection rate of approximately 95% for $1.5 < |\eta| < 2.7$, and 94% for $2.7 < |\eta| < 3.0$. The GNN working point is defined as having the same background rejection rate as the BDT.

Chapter 5

Search for Inert Scalars at CMS in the Dilepton Final State

Contents

5.1	Introduction	70
5.2	Samples	72
5.3	Base Selections	74
5.4	Event Selection	77
5.4.1	Input Features	78
5.4.2	Training and Performance	83
5.4.3	Binning Optimisation	89
5.4.4	pNN Interpolation	93
5.5	Statistical Model	97
5.6	Modelling Overview	98
5.7	Background Modelling	102
5.7.1	WW/tt Control Region	102
5.7.2	WZ Control Region	104
5.7.3	MisId Control Region	106
5.7.4	ZZ Control Region	113
5.7.5	Drell-Yan	116
5.8	MC Corrections and Uncertainties	121
5.9	Results	124

5.1 Introduction

This chapter presents a search for the production of scalars of the Inert Doublet Model (IDM) at the LHC using the CMS experiment. Despite the significant theoretical and phenomenological interest in this model, no direct search has been performed to this date. Therefore, the analysis presented here constitutes the first ever direct search for the IDM

scalars. As explained in section 2.3, generator-level studies suggest that the dilepton final state is the most sensitive channel, not only due to its clean signature, but also because of its relatively large cross-section compared to other IDM production modes [26]. Given this, this search is performed by studying events with the final state containing an opposite-sign same-flavour dilepton pair, $e^-e^+/\mu^-\mu^+$, accompanied by missing transverse energy from the inert scalars, H . However, due to the large Z boson background at the LHC, this search requires the invariant mass of the dilepton pair to correspond to the off-shell Z region: $12 < M_{\ell\ell} < 80$ GeV.

The dominant Feynman diagram for the dilepton final state is shown in 2.3(a), corresponding to the primary production mode: $q\bar{q} \rightarrow HH\ell^+\ell^-$. Crucially, this specific diagram depends on only the two neutral IDM masses, M_H and M_A , as the vertex factors are given solely by SM quantities. Whilst the charged scalar does enter through $q\bar{q} \rightarrow HH\ell^+\ell^-\nu_\ell\bar{\nu}_\ell$, as shown in figure 2.3(b), the smaller cross-section, and the weaker discrimination versus backgrounds, renders this negligible. As for the IDM coupling constants, the parameter λ_{345} enters through Higgs-related diagrams, such as in figure 2.3(c), however even for values of λ_{345} significantly larger than allowed from the constraints, this is found to have a negligible effect on the kinematics. λ_2 enters through the quartic coupling between IDM particles, which, at leading order, has no effect on the kinematics. Subsequently, the dilepton final state provides a highly sensitive signature to place limits on the masses of the neutral inert scalars, M_H and M_A , while being largely insensitive to choices of λ_2 , λ_{345} and $M_{H^\pm}$. Finally, as figure 2.3(a) is the dominant diagram, the kinematics of the dilepton signal is predominantly dependent on the mass splitting¹ $\Delta(M_A, M_H) = M_A - M_H$, with the cross-section driven by the absolute mass scale.

Standard Model processes can, however, mimic the signal process. These include processes with prompt leptons and real missing energy, for example diboson production $WW/ZZ \rightarrow \ell^+\ell^-\nu_\ell\bar{\nu}_\ell$ and $t\bar{t} \rightarrow b\bar{b}\ell^+\ell^-\nu_\ell\bar{\nu}_\ell$, where the neutrinos go undetected. As well as processes with real leptons, notably Drell-Yan $Z/\gamma^* \rightarrow \ell^-\ell^+ + \text{jets}$, with the mismeasurements of the jets leading to spurious missing energy. In addition, there are contributions from three-lepton events when one lepton falls out of acceptance, such as $WZ \rightarrow \ell^-\ell^+\ell'\nu_{\ell'}$. Finally, contributions from events with jets incorrectly reconstructed as leptons can also mimic the signal. This is predominantly from $W + \text{jets}$ and QCD di-jet events.

The following sections cover the main aspects of the IDM search, starting with the various data and simulated samples utilised for the analysis. This is followed by the definitions of different physics objects, and the set of base selections used to remove basic SM backgrounds. Next, the main signal extraction method is described, involving a Parametric Neural Network, that further removes SM backgrounds, and which is also used to define the signal region. Subsequently, the details of the control regions used to model the background sources are presented. Finally, the multiple systematic uncertainties are discussed, which are included in the statistical model when extracting the signal and deriving limits on the neutral IDM masses, M_H and M_A .

¹Note that, following convention, $M_H < M_A$

5.2 Samples

The search is performed using proton-proton collision data collected at the CMS experiment from 2016-2018 at a centre of mass energy of $\sqrt{s} = 13$ TeV, and in 2022 at $\sqrt{s} = 13.6$ TeV. In total, this corresponds to an integrated luminosity of 172.2 fb^{-1} , with 137.5 fb^{-1} collected in 2016-18, and 34.7 fb^{-1} in 2022.

In addition to data, simulated Monte-Carlo (MC) events are utilised (refer to section 3.4 for a detailed explanation of the simulation process). These not only model the different processes that contribute to a specific signature, but are also used for training the classifier to be used on data. To this end, simulations are used to model the parton-level interactions of the underlying physics events, this is then followed by PYTHIA 8 [92] to model the showering and hadronization of the outgoing particles. Next, this is mixed with simulated pile-up to match the conditions of the detector. Finally, for all simulated samples, the response of the CMS detector (section 3.2) is modelled using the GEANT4 package [66]. Using the simulated events, physics object are reconstructed using the same algorithms that are applied on data. This pipeline is also processed separately for each year, to account for the different conditions in each data-taking period.

Signal events are simulated at leading-order (LO) using MADGRAPH5_aMC@NLO [67]. Both the $q\bar{q} \rightarrow HH\ell^+\ell^-$ and $q\bar{q} \rightarrow HH\ell^+\ell^-\nu_\ell\bar{\nu}_\ell$ processes are simulated, with $\ell = e, \mu, \tau$. No constraints are placed on the internal propagators to ensure that all contributing diagrams, with the proper interferences, are considered. All final states containing τ particles are decayed using PYTHIA. Events are simulated with a dark matter mass, M_H , ranging from 60-180 GeV in steps of 10 GeV, and a mass splitting $\Delta(M_A, M_H) = M_A - M_H$ of 20-100 GeV in steps of 10 GeV. Additionally, extra samples are produced with a mass splitting of $\Delta(M_A, M_H) = 92$ GeV to cover the discontinuity at $\Delta(M_A, M_H) = M_Z = 91.2$ GeV from the Z resonance in the decay $A \rightarrow HZ$. The signal is generated with $\lambda_2 = 1$, however at leading order this has no effect on the kinematics. The Higgs-IDM coupling, λ_{345} , also has a negligible effect on the kinematics of this search, and so is set to a value of 10^{-6} , which is within the allowed values from direct detection constraints [26–31]. Finally, $M_{H^\pm}$ is set to its maximal value of $M_{H^\pm} = M_A + 50$ GeV, which is largely limited by the constraints from electroweak precision measurements. Again, for the parameter space under consideration, $M_{H^\pm}$, λ_2 and λ_{345} have negligible effects on the final limit, which is instead entirely driven by the neutral masses M_H and M_A . Two sets of signal events are simulated. The first are the Training Points, shown in figure 5.1 in blue, which are used for training the neural network and obtaining the final limit. Additional samples are created, shown in figure 5.1 in red, called the Validation Points, which are used to evaluate the interpolation performance and the associated uncertainties. Finally, throughout this chapter, the different points in the parameter space, for example $(M_H = 80, M_A = 130)$ GeV, are referred to as *mass points*.

Generator-level distributions of the invariant mass and missing energy are shown in figure 5.2 for a range of mass points. There are two main features of these distributions. First, the

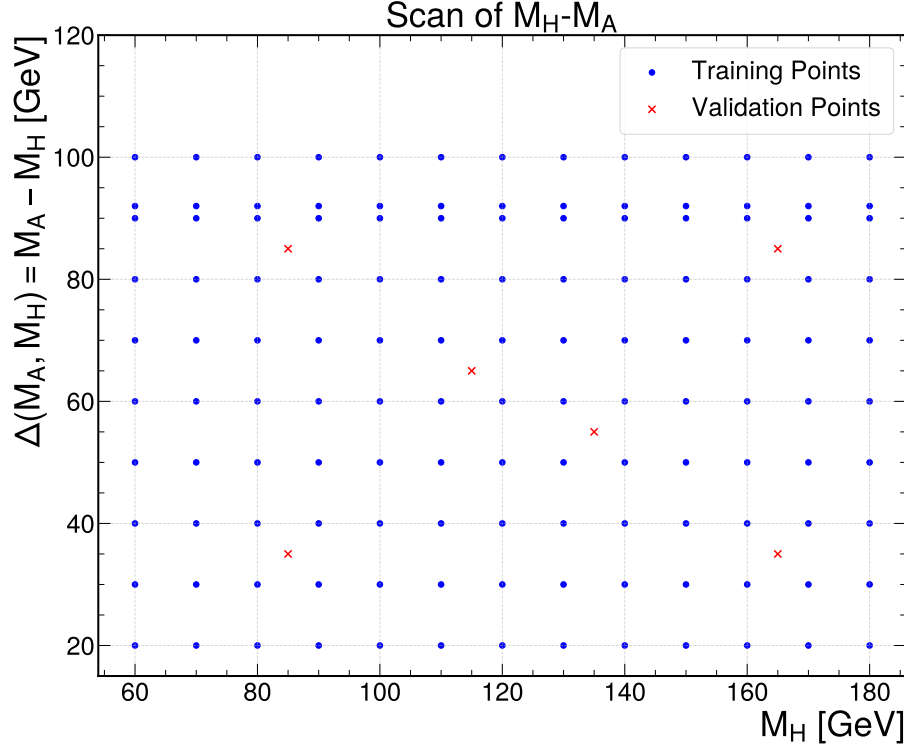


Figure 5.1: Simulated IDM points in $(M_H, \Delta(M_A, M_H) = M_A - M_H)$ plane. Blue points are used for training the network and calculating the final limits. Red points are used for validation of the interpolation. Additional points are simulated at $\Delta(M_A, M_H) = 92$ GeV due to the discontinuity at $\Delta(M_A, M_H) = M_Z = 91.2$ GeV from the Z boson becoming on-shell in the decay of $A \rightarrow HZ$.

invariant mass distribution has a maximum value² of $M_A - M_H$. This is due to the leading diagram, shown in figure 2.3(a), where the dileptons originate from the Z boson in the process $A \rightarrow ZH$. The mass splitting, therefore, dictates the kinematics of the Z , which in turn defines the kinematics of the dileptons. Consequently, this results in the kinematics being largely dominated by the mass-splitting $M_A - M_H$, with the cross-section driven by the absolute mass scale. Second, for the IDM phase space under investigation, despite the presence of two dark matter particles in the final state, the majority of mass points show little missing transverse energy compared to many previous dark matter searches [93, 94]. This is due to the fact that the two H particles are often produced back-to-back, leading to a cancellation.

Quark-initiated W^-W^+ and ZZ events are simulated at NLO in pQCD using POWHEG [95]. Diboson production from gluon-gluon fusion ($gg \rightarrow 2e2\nu, 2\mu2\nu, 2e2\mu, 2e2\tau, 2\mu2\tau, 4e, 4\mu, 4\tau$) is simulated with MCFM v7 [96]. Samples of quark-initiated WZ events are simulated using MADGRAPH5_aMC@NLO [67, 97] at NLO in pQCD.

The $t\bar{t}$ and single top quark background samples are produced at NLO in pQCD using POWHEG v2.0 and v1.0, respectively [68–70]. The $t\bar{t}$ cross section is calculated at NNLO in pQCD and includes resummation of next-to-next-to-leading logarithmic (NNLL) soft

²There are some events at higher values, but these are from sub-dominant diagrams.

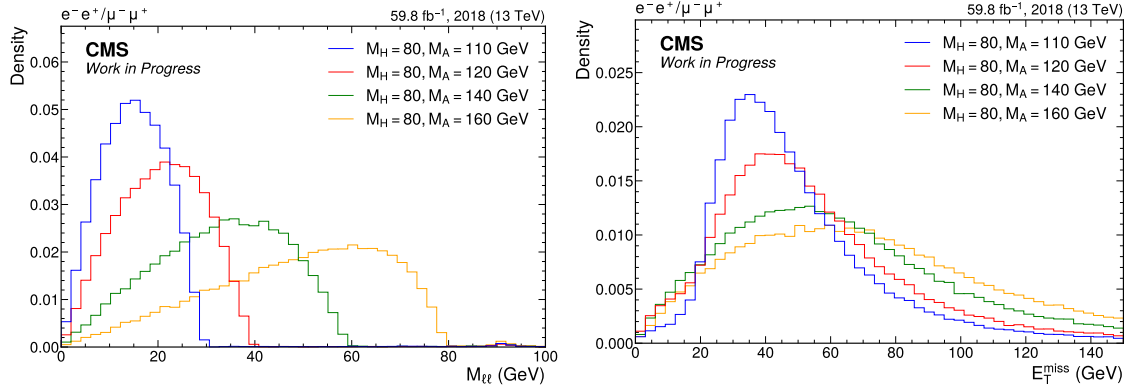


Figure 5.2: Generator-level plots of (left) invariant mass and (right) missing energy for a range of IDM signal samples. All histograms are normalised to unity.

gluon terms using the Top++v2.0 program [98]. Single top processes are also normalised to available NNLO calculations [99, 100]. Finally, top pair production in association with a vector boson, $t\bar{t}V$, and tri-boson events, VVV , are simulated to NLO in pQCD using MADGRAPH5_aMC@NLO.

Single boson events, including Drell-Yan $Z/\gamma^* + \text{jets}$ and $W + \text{jets}$, with a maximum of 4 jets, are simulated to LO accuracy with MADGRAPH5_aMC@NLO, and their cross-sections are scaled to the NNLO calculations [101, 102]. For jet matching, the MLM jet matching procedure [103] is used. Additionally, NLO Drell-Yan samples, $Z/\gamma^* + \text{jets}$ with a maximum of 3 jets using the FxFx jet merging scheme [104, 105], are used for deriving NLO/LO k-factors. This is presented in section 5.7.5.

5.3 Base Selections

Trigger

Data events are selected in multiple stages, starting with the trigger that follows the two-tier trigger strategy outlined in section 3.2.5. For this analysis, events are selected that pass the single-electron and single-muon triggers. These accept events that have at least one electron (muon) with a pseudorapidity of $|\eta| < 2.5$ ($|\eta| < 2.4$), passing the Tight ID, and having a p_T above a year-specific threshold. For electrons, the p_T thresholds are 27, 32, 32 and 30 GeV for 2016, 2017, 2018 and 2022, respectively. Whereas, for muons the p_T thresholds are 24, 27, 24, 27 GeV for 2016, 2017, 2018 and 2022, respectively. However, due to the finite resolution of the p_T , a drop in efficiency is observed around the trigger p_T threshold. Therefore, to ensure that events aren't selected that fall into this region, which is more sensitive to mismodelling, the leptons that fired the trigger are required to have a transverse momentum of $p_T = p_T^{\text{trigger}} + 2$ GeV. The selections applied after the trigger are much more stringent. As a result, the triggers have an efficiency for accepting events of interest greater than 99% compared to the full analysis selection.

Quantity	Electrons		Muons	
	Loose	Tight	Loose	Tight
p_T [GeV]	≥ 10	≥ 20	≥ 10	≥ 20
$ \eta $	≤ 2.5	≤ 2.5	≤ 2.4	≤ 2.4
Veto Transition	True	True	-	-
d_{xy} [cm]	≤ 0.3	≤ 0.3	≤ 0.3	≤ 0.3
d_z [cm]	≤ 0.2	≤ 0.2	≤ 0.2	≤ 0.2
Id	Loose	Tight	Loose	Tight
Isolation	-	-	Loose	Tight

Table 5.1: Electron and muon selections. Note that "Veto Transition" corresponds to vetoing electrons that fall within the transition region between the barrel and endcap of the ECAL: $1.44 < |\eta| < 1.57$. No additional isolation is included for electrons as this is included in the ID.

Object Selections

Objects within the event are reconstructed which are subsequently utilised to define additional selections. First, two working points (WP) are defined for electrons and muons: a Loose working point and a Tight working point. These are referred to as Loose and Tight leptons. The full selections are presented in table 5.1. A Loose electron is required to have $p_T > 10$ GeV, $|\eta| < 2.5$, and pass the Loose ID as outlined in section 3.3. For Tight electrons, these are required to pass the Loose WP, and have $p_T > 20$ GeV, whilst passing the Tight ID. A Loose muon is required to have $p_T > 10$ GeV, $|\eta| < 2.4$, and pass the Loose ID and Loose isolation. Subsequently, a Tight muon must pass the Loose working point, whilst having $p_T > 20$ GeV and both Tight ID and Tight isolation.

Jet candidates are formed using different reconstruction algorithms depending on the year. For 2016-18, all jets are reconstructed using the Charged Hadron Subtraction (CHS) method as described in section 3.3.4, whilst for 2022 this is upgraded to the PUPPI method. All jet candidates are required to have $p_T > 20$ GeV, $|\eta| < 2.4$ and have $\Delta R > 0.4$ with respect to all Loose/Tight electrons and muons. Furthermore, for CHS jets (2016-18) an additional pileup identification BDT is used to minimise the number of PU jets entering the analysis. Finally, to target SM backgrounds that include b-quarks, such as the decay of $t\bar{t}$, b-jet candidates are also identified. These are formed by selecting jets that pass the basic requirements above, as well as passing the Loose b-tagging working point in the DeepJet algorithm.

The missing energy of the event, $\vec{E}_T^{\text{miss}}$, is also utilised for this analysis. Following section 3.3.6, it is defined as the negative vectorial sum of all transverse momenta in an event, and as a consequence is dependent on the choice of jet reconstruction algorithm. For 2016-18, PF- $\vec{E}_T^{\text{miss}}$ is used, whilst for 2022, PUPPI- $\vec{E}_T^{\text{miss}}$ is used, leveraging the improved pileup rejection and $\vec{E}_T^{\text{miss}}$ resolution for the higher PU conditions seen in 2022.

Base Selections

With the physics objects defined, a set of kinematic selections are applied that have a high acceptance for all signal samples, whilst having a high rejection rate for reducible SM background sources. These are referred to as the *base selections*, and are presented in table 5.2. These are not intended to be the final selections, however, but serve only to remove the most distinguishable SM backgrounds. More extensive selections are applied at a later stage using a neural network.

The common final state for all signal points is a pair of prompt opposite-sign same-flavour leptons, $e^-e^+/\mu^-\mu^+$, which are required to pass the Tight working points. As no extra leptons are expected in the signal, any event with an additional Loose lepton is vetoed. The IDM signal contains two leptons recoiling off the dark matter particles, leading to the momentum of the dilepton system being approximately antiparallel to the missing energy³. Given this, a selection of $\Delta\phi(p_T^{\ell\ell}, \vec{E}_T^{\text{miss}}) \geq 1$ rad is required. This substantially reduces the contribution from processes that have missing energy only from jet mismeasurements, such as for $Z/\gamma^* + \text{jets}$, as in this case the missing energy is less correlated with the direction of the dilepton. Furthermore, the recoiling dilepton produces a larger $p_T^{\ell\ell}$, with the leptons being more collimated, leading to the selections $p_T^{\ell\ell} > 15$ GeV and $\cos(\Delta\phi(p_T^{l_1}, p_T^{l_2})) > 0.75$. Events with $Z/\gamma^* + \text{jets}$ are further reduced by requiring the invariant mass of the dilepton pair to correspond to the off-shell Z region, $12 \leq M_{\ell\ell} \leq 80$ GeV. As no b-jets are expected, events are rejected if they contain any b-tagged jets with $p_T > 20$ GeV and $|\eta| < 2.4$. Finally, since little hadronic activity is expected for the IDM signal, events with more than one jet with $p_T > 30$ GeV and $|\eta| < 2.4$ are rejected. Not only does this reduce $t\bar{t}$ events where the b-jets are produced centrally, but with neither jet being reconstructed as a b-jet, but also helps remove $Z/\gamma^* + \text{jets}$ events in the signal region, as these require hadronic activity to get a large $\vec{E}_T^{\text{miss}}$ signature (this is shown later, in figure 5.3, with DY, $Z/\gamma^* + \text{jets}$, generally having at least 1 jet after the base selections are applied).

Base Dilepton Selections	
At least 1 lepton with $p_T \geq p_T^{\text{trigger}} + 2$ GeV	
Opposite Sign Same Flavour	$e^-e^+/\mu^-\mu^+$
Lead/Sublead Working Points	Tight
Veto Lepton Working Point	Loose
Dilepton Invariant Mass	$12 \leq M \leq 80$ GeV
Dilepton $p_T^{\ell\ell}$	≥ 15 GeV
$\Delta\phi(p_T^{\ell\ell}, \vec{E}_T^{\text{miss}})$	≥ 1 rad
$\cos(\Delta\phi(p_T^{l_1}, p_T^{l_2}))$	≥ -0.75
No. Jets	$\leq 1, p_T \geq 30$ GeV, $ \eta \leq 2.4$
No. b Jets	$= 0, p_T \geq 20$ GeV, $ \eta \leq 2.4$

Table 5.2: Base selections to reduce SM backgrounds.

³In reality this is not perfectly balanced due to detector mismeasurements.

5.4 Event Selection

A neural network (NN) is employed to further remove SM background sources, and to maximise the sensitivity of the search. This has the benefit of being able to exploit the correlations between the kinematic features to improve the separation of signal and background. A difficulty with this analysis is that the NN architecture must consider the range of signal kinematics over the entire parameter space. On one extreme, a NN could be trained for each signal sample. However, this would lead to many networks needing to be trained and optimised, as well as ambiguity and possible inefficiencies when setting limits between the simulated points. On the other extreme, a single NN could be trained for the whole parameter space, but this would lead to a reduction in sensitivity as the differing kinematics of the signal samples are not considered. Instead, this thesis uses a *Parametric Neural Network* (pNN) [106]. A pNN has the benefit of needing only a single network, whilst still considering all possible signal kinematics. In addition, it provides smooth interpolation between the signal points that were simulated, offering consistently strong discrimination over the whole parameter space.

The main distinction between a standard NN and a pNN is the choice of input features. A standard NN is a function of the event features, $f(\vec{x})$, and is trained to separate signal from background without any reference to the parameters of the signal model. Conversely, a pNN extends the list of input features to include the signal parameters, $\vec{\theta}$, such that the network selections are dependent on the signal parameters, $f(\vec{x}, \vec{\theta})$. Therefore, the network is able to adapt its decisions based on the signal hypothesis being tested. Consequently, only a single network is required for the whole parameter space, as the network can naturally adapt its decisions based on which of the signal hypotheses is being tested. Moreover, given that pNNs represent smooth functions, it is possible to smoothly interpolate between the different signal hypotheses.

Training a pNN requires the event features, $\vec{x}$, the model parameters, $\vec{\theta}$, and the sample label, y . For the signal samples, $\vec{\theta}$ is just given by the value of M_H and M_A that the sample was simulated with. For example $\vec{\theta} = (M_H, M_A) = (80, 150)$ GeV. However, for the background MC there are no associated values for the parameters $\vec{\theta}$. Instead, these values are randomly selected from the set of parameters trained on, following Reference [107]. For example, if the network was trained on two signal samples with parameters $\{(80, 120), (80, 150)\}$, then the backgrounds are randomly assigned (80, 120) or (80, 150). This random assignment is repeated at the start of every epoch to ensure that the learned background distribution is independent of the model parameters.

The signal distributions are expected to vary smoothly with the model parameters $\vec{\theta}$. Correspondingly, the trained network $f(\vec{x}, \vec{\theta})$ has a smooth dependence on the signal parameters $\vec{\theta}$. As such, by learning the correlations between signal distributions and the input parameters, the network gains the ability to interpolate to intermediate points, even to those that were not included in the training of the network. When applying this to data and background MC — which carry no intrinsic values for $\vec{\theta}$ — this just means evaluating

$f(\vec{x}, \vec{\theta})$ at any value of $\vec{\theta}$ within the grid shown in figure 5.1. However, for signal samples — which do have intrinsic values for $\vec{\theta}$, which they were simulated at — this cannot be used. Instead, the signal distributions at intermediate points are calculated using spline interpolation. This is discussed in section 5.4.4.

To justify using the pNN architecture, the performance of the model must be comparable to other, simpler, methods—most notably, classifiers trained independently on individual mass points. In addition, strong performance is also required when using the model to interpolate between simulated mass points: a degradation in performance at intermediate points would render the model insufficient for setting limits over the whole parameter space. Furthermore, the reliability of the spline interpolation for the signal samples must also be evaluated to ensure consistent results.

The parametric neural network used in this thesis is covered in the following sections. First, the input features to the network are presented, which are found using an optimisation procedure using BDTs. Then, the architecture of the pNN is given along with its performance. Subsequently, the signal region is defined using the pNN output, and the optimisation of this is described. Finally, the performance of the pNN when interpolating between points in the parameter space is discussed.

5.4.1 Input Features

For maximum performance, the input features of the pNN must fully characterise the underlying distributions, with too few features potentially missing out on important information that can aid in separating signal from background. Conversely, including too many input features, some of which add no additional information, can lead to slower training times, and over-fitting to the training data, ultimately hindering the performance of the model. Given this, starting from an initial set of input features, an iterative process is performed to remove features that add little to the final performance.

The iterative process works as follows:

1. Using all the initial input features, train an XGBoost BDT using K-fold validation for a single mass point. K-fold validation is a technique in which the data is first split into K partitions. Then, training is done iteratively: for each iteration, one partition is used for validation, whilst the other K-1 partitions are used for training the model. This is repeated until each partition has been used once as the validation set. K-fold validation is utilised as it reduces the variance on the estimate of the performance.
2. Identify the least important input feature using the XGBoost "Gain" method and remove it from the list of features [81]. This method corresponds to the average improvement to the model's loss score for each feature across the whole tree. The larger the gain, the more important the feature is for classifying the samples.
3. Re-train the BDT with the new features.
4. Repeat steps (2)-(3) until a stopping criterion is met. For this thesis, the stopping

criterion is when the signal efficiency (at a background efficiency of 0.1%) has dropped below 1% compared to the signal efficiency when using all input features. The features that remain are taken as the final set of features for that mass point.

Due to the range of different signal distributions, however, some mass points may require certain features that other mass points do not. Therefore, to ensure that the required information for all points is included, this process is repeated for each point, and the final set of input features is the union of all individual features.

The initial set of features includes both low-level event features, such as lepton p_T , and high-level, derived features, such as the dilepton invariant mass, M_{ll} . The choice of high-level features used in the initial set is largely driven by the signal topology containing a dilepton with missing transverse energy. Due to this event signature, variables are included that have found use in similar searches, especially those that rely on discrimination using the correlation between the visible particles and missing transverse energy. All initial features are shown in table 5.3. Of particular importance are the JZB feature and the transverse mass features. The JZB feature — also referred to as the Jet-Z-Balance — measures the imbalance between the p_T of the dilepton system and of the hadronic recoil in $Z/\gamma^* + \text{jets}$ events [108]. This is given by $\text{JZB} = |\vec{E}_T^{\text{miss}} + \vec{p}_T^{\text{ll}}| - |\vec{p}_T^{\text{ll}}|$, where the first term is the hadronic recoil. For events without real $\vec{E}_T^{\text{miss}}$, such as $Z/\gamma^* + \text{jets}$, the $\vec{E}_T^{\text{miss}}$ comes from jet mismeasurements. In this case, $\vec{E}_T^{\text{miss}}$ is less correlated with the dilepton system, and therefore JZB is approximately symmetric around zero. Whereas, for the IDM signal and other SM processes with neutrinos, they are correlated, leading to a more negative, asymmetric, distribution.

Another important feature is the transverse mass:

$$M_T(\vec{p}_T^a, \vec{E}_T^{\text{miss}}) = \sqrt{2 * p_T^a * E_T^{\text{miss}} [1 - \cos(\Delta\phi(\vec{p}_T^a, \vec{E}_T^{\text{miss}}))]}, \quad (5.1)$$

where $\vec{p}_T^a$ is the 4-momentum for a single lepton, or dilepton system. Originally created for measuring the decay $W \rightarrow \ell\nu$, the transverse mass possesses useful qualities for measuring the W boson properties, such as having a maximum value at M_W for the W decay [109]. Whilst not the exact use-case here, it still provides useful discrimination between signal and SM backgrounds. In particular, it is effective against $Z/\gamma^* + \text{jets}$, where the missing energy is less correlated with the dilepton system. Similar to the JZB variable, the transverse mass encodes the correlation between the dilepton system and the missing energy, leading to larger values when they are antiparallel, as expected for the IDM signal. When $\vec{p}_T^a = \vec{p}_T^{\text{ll}}$ represents the dilepton system 4-momentum, this can be further generalised to include the dilepton invariant mass:

$$M_T^E(\vec{p}_T^{\text{ll}}, \vec{E}_T^{\text{miss}}) = \sqrt{2 * E_T^{\text{ll}} * E_T^{\text{miss}} [1 - \cos(\Delta\phi(\vec{p}_T^{\text{ll}}, \vec{E}_T^{\text{miss}}))]}, \quad (5.2)$$

where $E_T^{ll} = \sqrt{p_T^{ll^2} + M_{ll}^2}$.

The transverse mass has also been generalised to the situation where two particles are produced, each decaying to a visible particle and an invisible particle, this is known as the *stransverse mass* [110]. An example would be the cascade decay $pp \rightarrow X^+ X^- \rightarrow \ell^+ \ell^- \chi \chi$, where each hypothetical particle, X , decays to a lepton and a dark matter particle, $X^\pm \rightarrow \ell^\pm \chi$. The stransverse mass starts with the transverse mass defined for a massive invisible particle, χ :

$$m_T^2(\vec{p}_T^\alpha, \vec{p}_T^\chi; m_\chi) \equiv m_\chi^2 + 2 \left(p_T^\alpha E_T^\chi - \vec{p}_T^\alpha \cdot \vec{p}_T^{\chi 0} \right), \quad (5.3)$$

with $\vec{p}_T^\chi$ being the 4-momentum of the invisible particle with mass m_χ . However, as both invisible particles go undetected, the value for $\vec{p}_T^\chi$ of each particle is unknown, only the sum of their momenta is known: $\vec{E}_T^{miss} = \vec{p}_T^{\chi 1} + \vec{p}_T^{\chi 2}$. Therefore, instead of using equation 5.3, a minimisation is performed over all possible values of $(\vec{p}_T^{\chi 1}, \vec{p}_T^{\chi 2})$:

$$M_{T2}(\vec{p}_T^a, \vec{p}_T^b, \vec{E}_T^{miss}, m_\chi) = \min_{\vec{q}_T^{(1)} + \vec{q}_T^{(2)} = \vec{E}_T^{miss}} \left[\max \left\{ m_T(\vec{p}_T^a, \vec{q}_T^{(1)}, m_\chi), m_T(\vec{p}_T^b, \vec{q}_T^{(2)}, m_\chi) \right\} \right], \quad (5.4)$$

where "max" is the maximum of both m_T values. Consequently, M_{T2} provides a lower bound for the value of the transverse mass. Again, despite not being the exact use-case here, the stransverse mass is nevertheless a powerful variable for discriminating signal versus background. This is partly due to M_{T2} penalizing processes with no real missing energy, such as for $Z/\gamma^* + \text{jets}$, leading to smaller values for M_{T2} . Furthermore, even for processes that do contain real $\vec{E}_T^{miss}$ within a cascade decay, such as $W^- W^+$ and $t\bar{t}$, the invisible particles (neutrinos) are massless. Whereas, for the IDM, the invisible particles have mass M_H , shifting the M_{T2} distributions, and thus providing discrimination. To use M_{T2} purely for its discriminatory power, the variable is calculated under two assumptions: one with $m_\chi = 0$ GeV which is the SM-like hypothesis; and one with $m_\chi = 80$ GeV with is the IDM-like hypothesis. Both are used when finding the optimal variables for the pNN.

Example distributions after the base selections are shown in figure 5.3. The optimisation procedure is performed on each mass point with $K = 5$, and the process is stopped when the signal efficiency drops by more than 1% at a background efficiency of 0.1%. This threshold is chosen as it approximately corresponds to the most sensitive part of the search, where the signal-to-background ratio is highest. However, as this process is used to find the optimal features for a neural network (which can derive new features based on the inputs) and not a BDT (which cannot), the low-level features are never removed during this process. An example of this for a single point is shown in figure 5.4, which demonstrates that the

Table 5.3: Initial set of input features for the pNN before the iterative optimisation procedure.

4-Momentum Features	
Lead/Sublead Leptons	$p_T^\ell, \eta^\ell, \phi^\ell$
Dilepton System	$p_T^{\ell\ell}, \eta^{\ell\ell}, \phi^{\ell\ell}, M_{\ell\ell}$
Lead/Sublead jets	p_T^j, η^j, ϕ^j
Missing Energy	$\vec{E}_T^{\text{miss}}, \phi^{\text{miss}}$
Angular Variables	
Lepton-Lepton	$\cos(\Delta\phi(\ell_1, \ell_2)), \Delta R(\ell_1, \ell_2), \Delta\eta(\ell_1, \ell_2) $
Lepton-Missing Energy	$\Delta\phi(\ell_{1,2}, \vec{E}_T^{\text{miss}}), \Delta\phi(\ell\ell, \vec{E}_T^{\text{miss}})$
Jet-Lepton	$\Delta\phi(j, \ell\ell), \Delta R(j, \ell\ell), \Delta\phi(j, \ell_{1,2}), \Delta R(j, \ell_{1,2})$
Jet-MET	$\Delta\phi(j, \vec{E}_T^{\text{miss}})$
Momentum Sums	
S_T	$\sum_{\text{leptons}} p_T^i$
K_T	$\sum_{\text{leptons}} p_T^i + E_T^{\text{miss}}$
H_T	$\sum_{\text{leptons}} p_T^i + \sum_{\text{jets}} p_T^i + E_T^{\text{miss}}$
Momentum-Derived Features	
Ratios	$S_T/H_T, K_T/H_T, S_T/K_T$ $p_T^{l_1, l_2, \ell\ell, j_i} / (S_T, K_T, H_T), E_T^{\text{miss}} / (S_T, K_T, H_T)$
Balance Ratio	$(p_T^{\ell\ell} - E_{p_T}^{\text{miss}}) / p_T^{\ell\ell}$
JZB	$ \vec{E}_T^{\text{miss}} + \vec{p}_T^{\ell\ell} - \vec{p}_T^{\ell\ell} $
Transverse Masses	
$M_T(\vec{p}_T^a, \vec{E}_T^{\text{miss}})$	$\sqrt{2 * p_T^a * E_T^{\text{miss}} [1 - \cos(\Delta\phi(\vec{p}_T^a, \vec{E}_T^{\text{miss}}))]}$
$M_T^E(\vec{p}_T^{\ell\ell}, \vec{E}_T^{\text{miss}})$	$\sqrt{2 * E_T^{\ell\ell} * E_T^{\text{miss}} [1 - \cos(\Delta\phi(\vec{p}_T^{\ell\ell}, \vec{E}_T^{\text{miss}}))]}$ $\hookrightarrow E_T^{\ell\ell} = \sqrt{p_T^{\ell\ell 2} + M_{\ell\ell}^2}$
$M_{T2}(\vec{p}_T^a, \vec{p}_T^b, \vec{E}_T^{\text{miss}}, m_\chi = \{0, 80\} \text{GeV})$	$\min_{\vec{q}_T^{(1)} + \vec{q}_T^{(2)} = \vec{E}_T^{\text{miss}}} \left[\max \left\{ m_T(\vec{p}_T^a, \vec{q}_T^{(1)}), m_T(\vec{p}_T^b, \vec{q}_T^{(2)}) \right\} \right]$ $\hookrightarrow m_{T2}^2(\vec{p}_T^\alpha, \vec{p}_T^\chi; m_\chi) \equiv m_\chi^2 + 2 \left(p_T^\alpha E_T^\chi - \vec{p}_T^\alpha \cdot \vec{p}_T^{\chi_1^0} \right)$
Multiplicities	
Number of jets	N_j

Table 5.4: Final set of input features for the pNN after the iterative optimisation procedure.

4-Momentum Features	
Lead/Sublead Leptons	$p_T^\ell, \eta^\ell, \phi^\ell$
Dilepton System	$p_T^{\ell\ell}, \eta^{\ell\ell}, \phi^{\ell\ell}, M_{\ell\ell}$
Lead/Sublead jets	p_T^j, η^j, ϕ^j
Missing Energy	$\vec{E}_T^{\text{miss}}, \phi^{\text{miss}}$
Angular Variables	
Lepton-Lepton	$\Delta R(\ell_1, \ell_2)$
Jet-Lepton	$\Delta\phi(j, \ell\ell)$
Jet-MET	$\Delta\phi(j, \vec{E}_T^{\text{miss}})$
Momentum-Derived Features	
JZB	$ \vec{E}_T^{\text{miss}} + \vec{p}_T^{\ell\ell} - \vec{p}_T^{\ell\ell} $
Transverse Masses	
$M_T(\vec{p}_T^a, \vec{E}_T^{\text{miss}})$	$\sqrt{2 * p_T^a * E_T^{\text{miss}} [1 - \cos(\Delta\phi(\vec{p}_T^a, \vec{E}_T^{\text{miss}}))]}$
$M_T^E(\vec{p}_T^{\ell\ell}, \vec{E}_T^{\text{miss}})$	$\sqrt{2 * E_T^{\ell\ell} * E_T^{\text{miss}} [1 - \cos(\Delta\phi(\vec{p}_T^{\ell\ell}, \vec{E}_T^{\text{miss}}))]}$
	$\hookrightarrow E_T^{\ell\ell} = \sqrt{p_T^{\ell\ell^2} + M_{\ell\ell}^2}$
$M_{T2}(\vec{p}_T^a, \vec{p}_T^b, \vec{E}_T^{\text{miss}}, m_\chi = \{0, 80\}\text{GeV})$	$\min_{\vec{q}_T^{(1)} + \vec{q}_T^{(2)} = \vec{E}_T^{\text{miss}}} \left[\max \left\{ m_T(\vec{p}_T^a, \vec{q}_T^{(1)}), m_T(\vec{p}_T^b, \vec{q}_T^{(2)}) \right\} \right]$
	$\hookrightarrow m_{T2}^2(\vec{p}_T^\alpha, \vec{p}_T^\chi; m_\chi) \equiv m_\chi^2 + 2 \left(p_T^\alpha E_T^\chi - \vec{p}_T^\alpha \cdot \vec{p}_T^{\chi_1^0} \right)$

removal of many features have no impact on the final performance. Moreover, this also highlights the importance of the transverse mass features. Repeating this process for each mass point, and combining the individual features, leads to the final set of 26 training features displayed in table 5.4.

5.4.2 Training and Performance

Before training a model, the MC samples are split into three groups called training, validation and test sets, which correspond to 50%, 20% and 30% of the total sample, respectively. The training set is used for training the model, whilst the validation is used to validate the performance and prevent over-fitting during training. Finally, the test set is reserved for quantifying the final performance of the model, and to give an unbiased confirmation that there is negligible over-fitting.

The pNN is implemented with the PyTorch library [88] and consists of 5 feed-forward layers, each with 60 hidden units and an ELU activation function [111], corresponding to approximately 18,000 learnable parameters. The model architecture is determined through a grid search over the number of layers, hidden parameters, and activation functions, with the chosen model achieving the lowest loss on the validation set. The grid search parameters are shown in table 5.5. Before training, the train dataset is re-weighted such that the sum of the signal weights is equal to the sum of background weights. Furthermore, the signal samples are re-weighted such that the sum of each signal sample is equal, ensuring that each signal is treated equally without bias towards any individual sample. To promote stable and efficient training, the input features are standardised to have a mean of 0 and a standard deviation of 1: $x_{\text{scaled}} = (x - \mu)/\sigma$, where μ and σ are the feature mean and standard deviation, respectively.

The model is trained using the Adam optimiser [90] to minimise the loss of the weighted binary cross-entropy loss function:

$$\text{Weighted BCE} = -\frac{1}{N} \sum_{i=0}^N w_i [y_i \log(\text{pred}_i) + (1 - y_i) \log(1 - \text{pred}_i)], \quad (5.5)$$

where y_i is the label for event i , pred_i is the model prediction, and w_i is the event weight. This is performed for a maximum of 500 epochs using a batch size of 500, with a learning rate of 0.01 that is reduced by a factor of 0.2 if the validation loss does not drop after 25 epochs. If there is no improvement of the validation loss over 50 epochs, the training is stopped. The model that produces the lowest validation loss during training is chosen to be the final model.

During training, each batch is produced with the percentage of each background and signal process being approximately equal to the full training dataset. For example, if 10% of the full dataset were WW events, then approximately 10% of each batch would be WW events. This reduces random fluctuations during training, as each process is equally represented

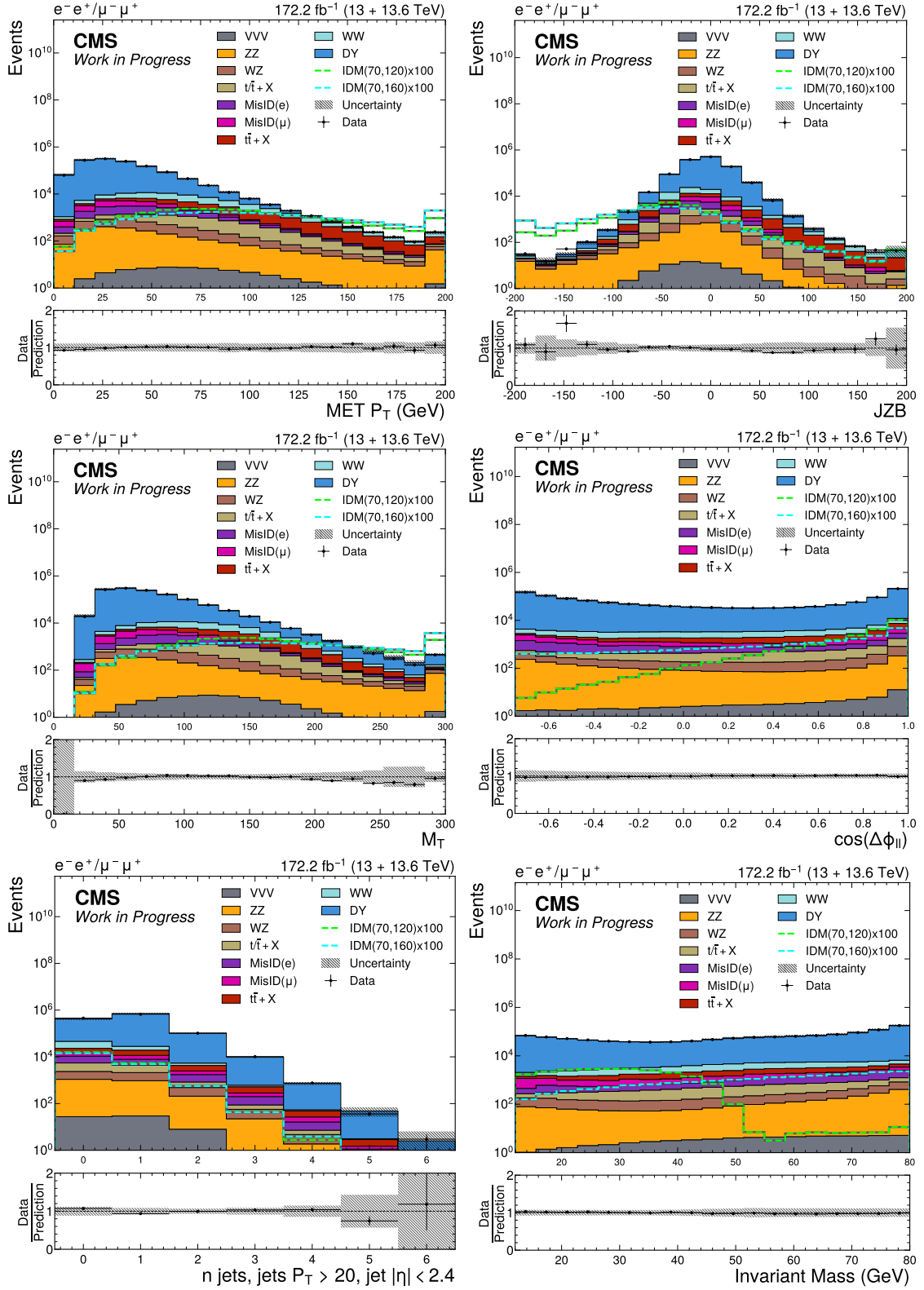


Figure 5.3: Distributions after the base selections described in table 5.2 for all years combined. Included are two representative IDM signal distributions which, for clarity, have their normalisations multiplied by 100. All distributions include contributions from under-flow and overflow events. All uncertainties are included in the uncertainty band.

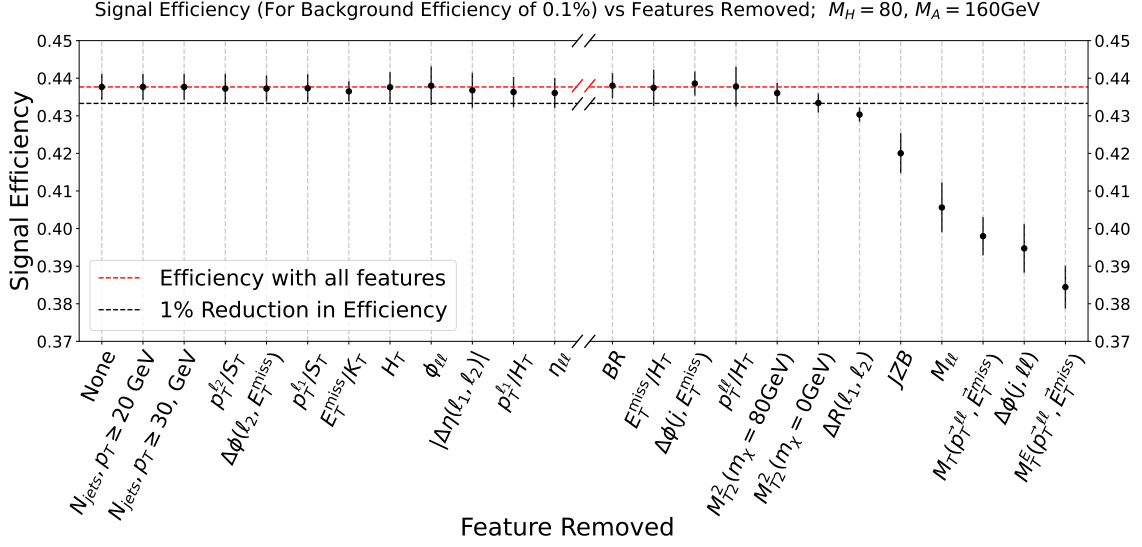


Figure 5.4: Example of the iterative method to find the optimal number of features for mass point $M_H = 80$, $M_A = 160$ GeV. Note that some middle features have been removed from the plot for clarity. All uncertainties are from the k=5 fold validation, with different folds resulting in a different final performance.

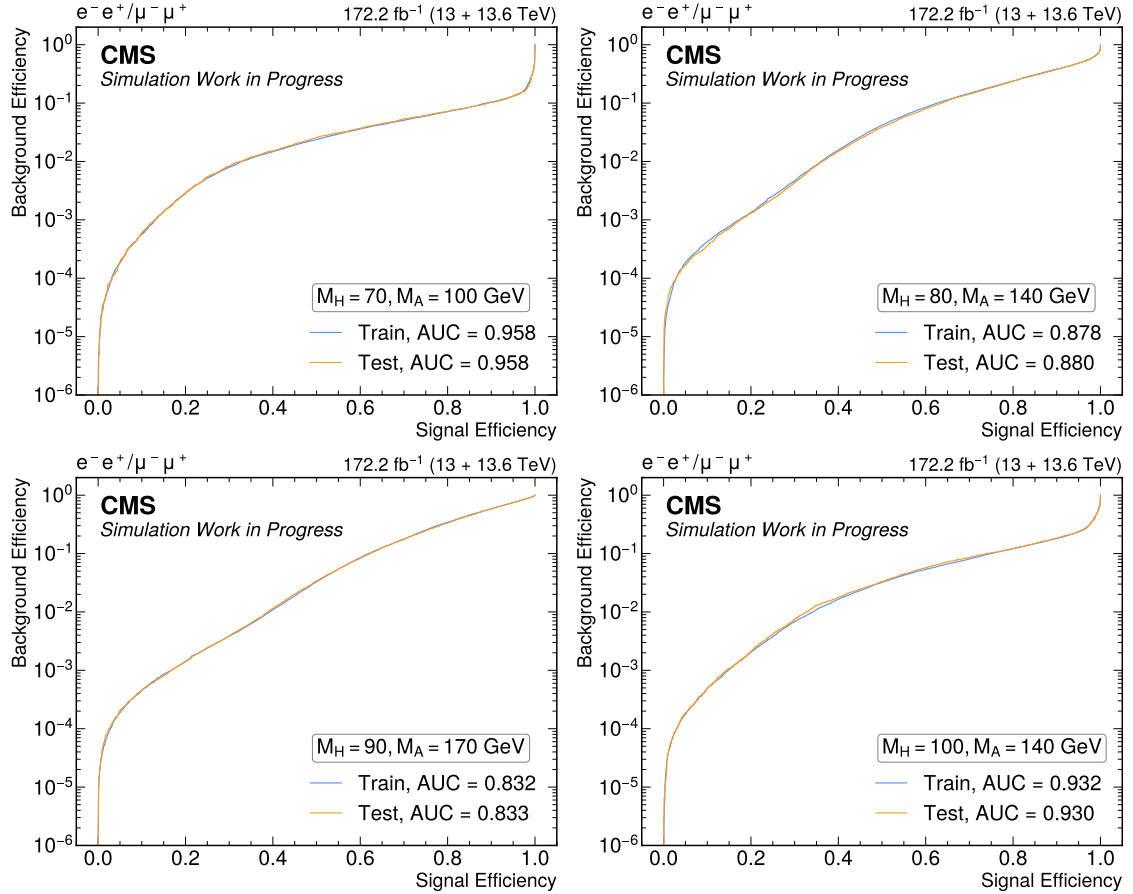


Figure 5.5: Example ROC curves and AUC scores for the training and test sets for a selection of mass points. All points show similar AUC scores for both the training and test sets, indicating negligible over-training.

Table 5.5: Value of the hyperparameters used in the grid search for the optimal model hyperparameters.

Number of layers	3, 4, 5, 6
Number of hidden units	30, 40, 50, 60, 70
Activation function	ReLU, ELU, GELU

within each batch. Finally, for each background sample in the batch, random values for the input model parameters, (M_H, M_A) , are chosen from the set of parameters the model was trained with: $\{(60, 80), (60, 90), (60, 100), \dots\}$.

The performance of the final model is evaluated using the ROC curve and the AUC scores, with comparisons made between the training and test set to determine any potential over-training. Example ROC curves are displayed in figure 5.5. For all points, there is strong separation between signal and background, with the majority achieving an $AUC > 0.9$. Moreover, there is negligible difference between the ROC curve and AUC scores for the training and test sets, indicating negligible over-training. Given this, the training, validation and test sets are combined when performing the final statistical tests of the analysis.

To ensure the pNN is not under-performing compared to networks trained on individual mass points, the pNN performance is compared to XGBoost BDTs trained on individual points. The results of this are shown in figure 5.6, which compares the AUC of the pNN versus the BDTs. Little difference is observed between the pNN and BDT, indicating no drop in performance from using a single pNN. This is true for the whole parameter space, with a maximum degradation in performance of less than 1%.

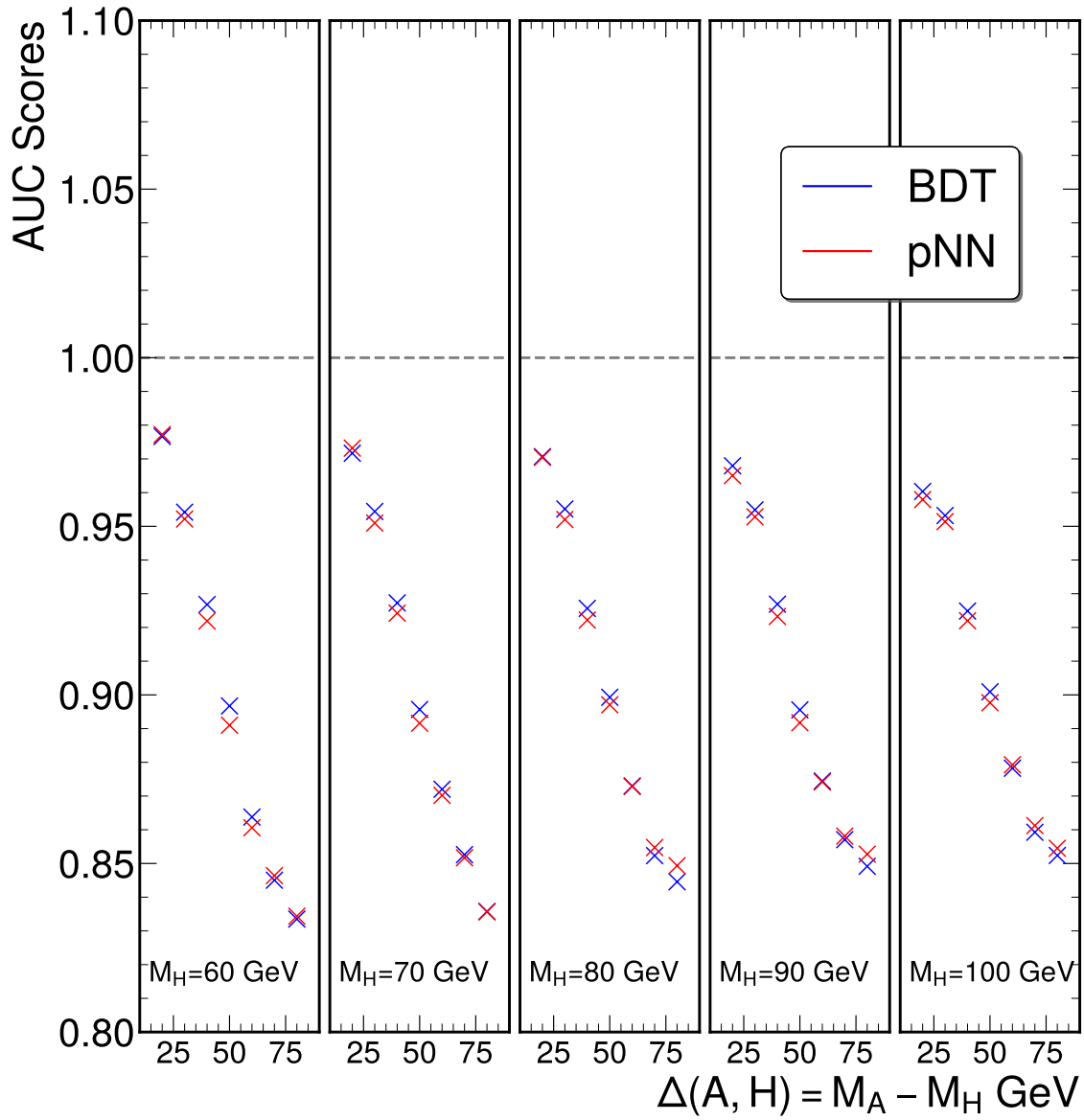


Figure 5.6: Comparison of the AUC between BDTs trained on individual mass points versus the pNN trained on all points simultaneously for $M_H = 60 - 100$ GeV. The AUC is calculated using the validation data set. A maximum drop in performance of approximately 1% is seen for the pNN compared to the BDTs.

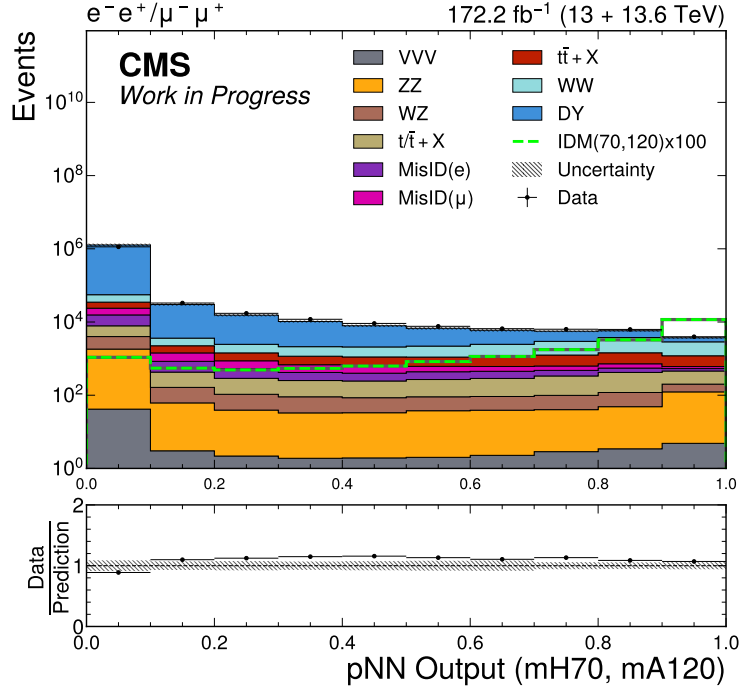


Figure 5.7: Distribution of the pNN output score for mass point $M_H = 70$, $M_A = 120$ GeV. Signal distributions are multiplied by 100 for clarity. All uncertainties are included in the uncertainty band.

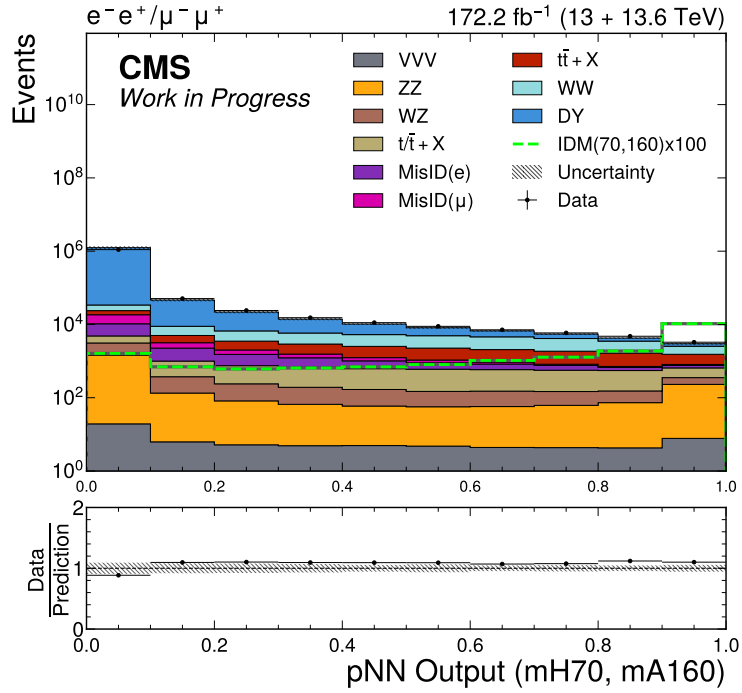


Figure 5.8: Distribution of the pNN output score for mass point $M_H = 80$, $M_A = 160$ GeV. Signal distributions are multiplied by 100 for clarity. All uncertainties are included in the uncertainty band.

5.4.3 Binning Optimisation

Using the pNN output, a set of bins are defined that maximise the sensitivity of the search. The optimisation of the binnings is performed separately for each of the two channels: e^-e^+ and $\mu^-\mu^+$. Ideally, each mass point would have bins that maximise its own sensitivity. However, this leads to ambiguity when interpolating the signal, as there are no well-defined bins for the intermediate mass points. As such, it is beneficial for all the mass points to share the same binning, permitted the loss of sensitivity is not too large. This is checked by first finding the optimal binning for each mass point, and then comparing to the sensitivity when optimising for all mass points.

To enhance the sensitivity, the choice of binning must also allow for shape discrimination between signal and background, exploiting the fact that the signal shape differs to the background shape, as illustrated in figure 5.7. In practice, this requires the finest binning boundaries possible, with the constraint that each bin has sufficient background statistics to avoid large unphysical discontinuities in the expected distributions.

The binning optimisation when considering individual mass points, or considering all points are simultaneously, goes as follows. For both methods the optimisation process starts with a bin boundary at a pNN score of 1 and scans towards lower pNN values, exploiting the fact that the signal pNN scores peak close to 1. Whilst scanning down, a potential bin is formed from the previous boundary and the new boundary, for example for a pNN score $\in [0.99, 1]$. If a certain criterion is met, then the bin is accepted and the process repeats, starting at the previous lowest boundary. Finally, this continues until the minimum bin boundary is at a pNN value of 0.9, which approximately corresponds to a minimum signal efficiency of 50%. When optimising for a single mass point, the procedure accepts a bin if the statistical uncertainty is less than 10%. Whereas, when optimising for all mass points simultaneously, the background statistical uncertainty is calculated for each of the mass points, and the bin is accepted if the statistical uncertainty is less than 10% for all points. Generally, these uncertainties will not be identical given the different background distributions for each point.

For a range of mass points, figure 5.9 shows the expected significance in the e^-e^+ channel for 2018 when using the binning optimised for each point, and the binning optimised for all points simultaneously. Little sensitivity is lost using the binning optimised for all points, with a maximum loss of approximately 1%. The reason for this is due to the fact that nearly all the sensitivity is from the largest-value pNN bin, which has relatively little change between the different mass points, as illustrated by figure 5.10. Note that the points with the smallest mass splitting were not included in this optimisation procedure because these had a large effect on the other points, whilst always having poor sensitivity.

The blinded signal region distributions using the optimised binning are shown in figures 5.11 and 5.12 for the $\mu^-\mu^+$ and e^-e^+ channels, respectively. The distributions correspond to the full luminosity of 2016-2022 combined. The finer binning in the $\mu^-\mu^+$ channel is predominantly due to the higher acceptance for muons than for electrons. Notably, for

all signal points, the signal distribution strongly peaks towards a pNN output of 1, with nearly all the sensitivity being in the last bin. Therefore, this suggests that the model is able to learn the shape of the signal and background, finding the area of phase space most consistent with the signal, whilst learning how this area of phase space changes depending on the signal hypothesis.

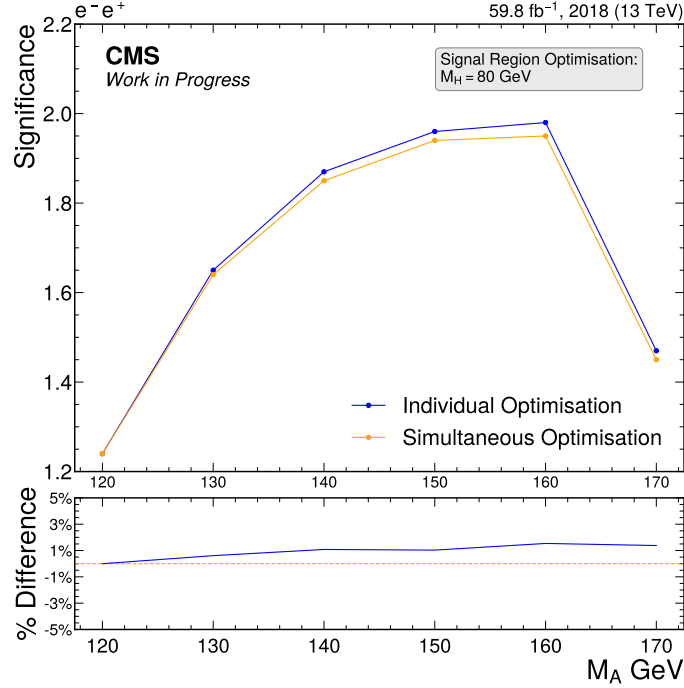


Figure 5.9: Comparison of the sensitivity for the e^-e^+ channel in 2018 when optimising for individual mass points, versus optimising for all points simultaneously. For simplicity, only masses with $M_H = 80$ GeV are presented. Overall, there is approximately a 1% drop in sensitivity when using the binning optimised for all points simultaneously.

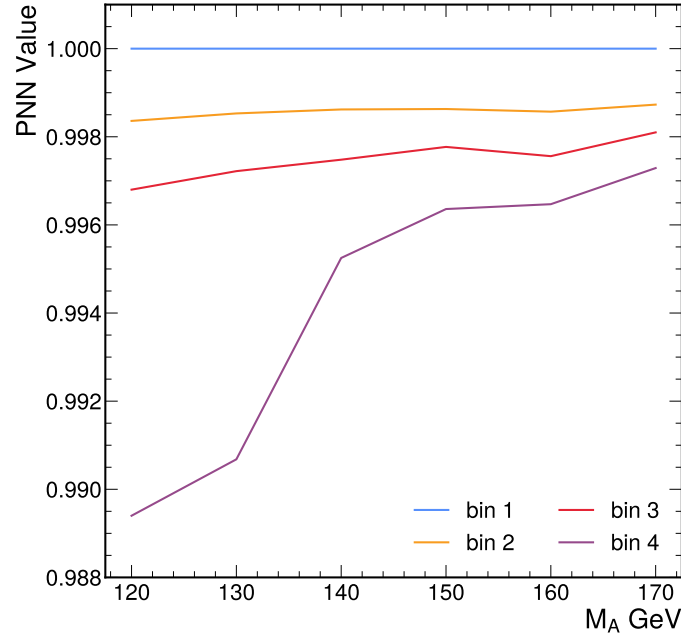


Figure 5.10: Comparison of the bin boundaries for the top 4 bins for $M_H = 80$ in the 2018 e^-e^+ channel when optimising each individual mass point. Notably, for the highest pNN bins, which are the most sensitive, the boundaries are similar across the parameter space.

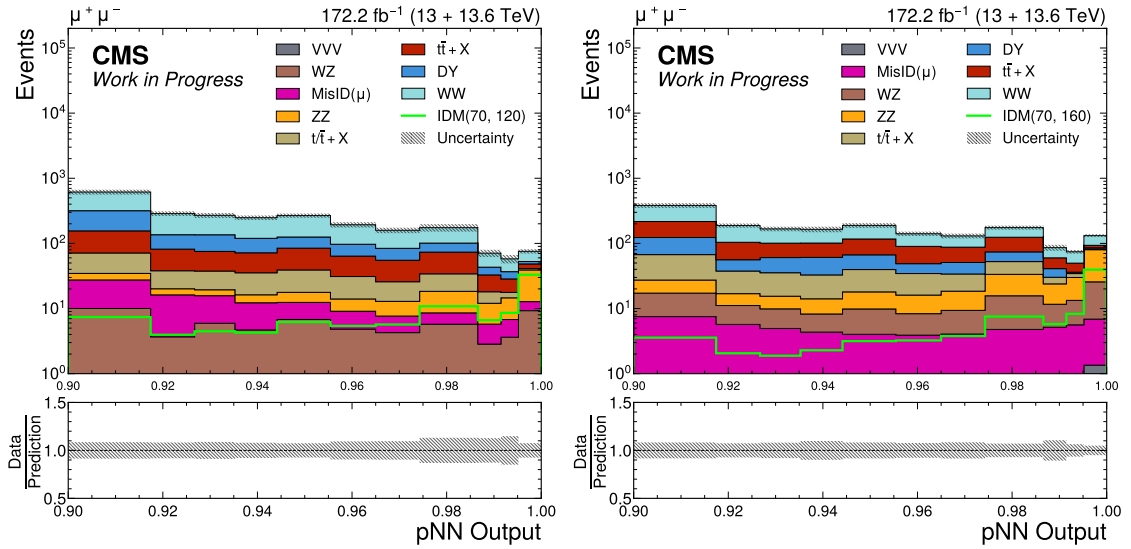


Figure 5.11: Blinded distributions of the signal region in the $\mu^-\mu^+$ channel for $(M_H, M_A) = (70, 120)$ GeV (left) and $(70, 160)$ GeV (right) for all years combined. The uncertainties shown include both statistical and systematic uncertainties.

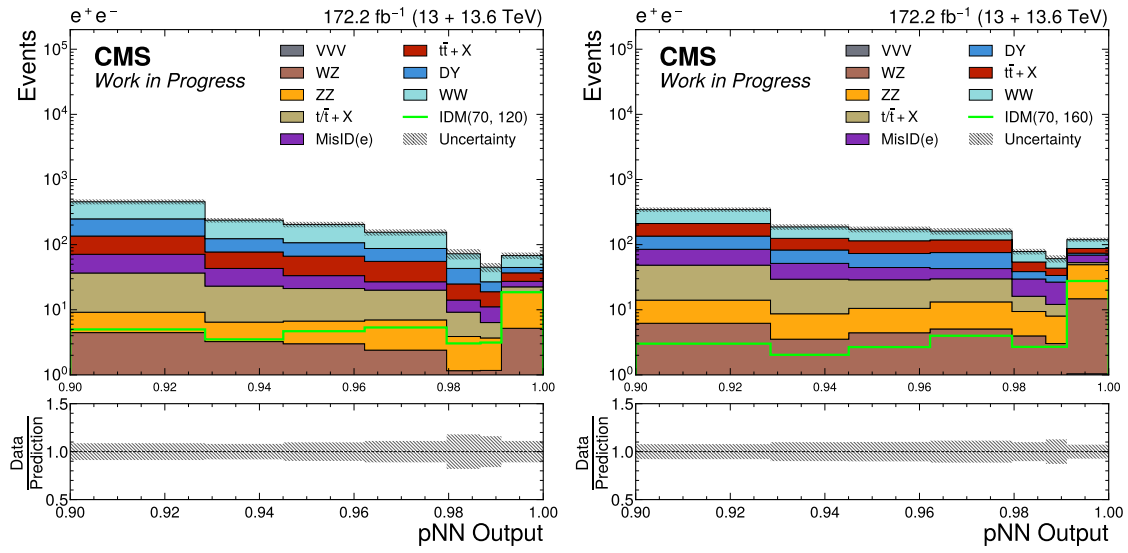


Figure 5.12: Blinded distributions of the signal region in the e^-e^+ channel for $(M_H, M_A)=(70, 120)$ GeV (left) and $(70, 160)$ GeV (right) for all years combined. The uncertainties shown include both statistical and systematic uncertainties.

5.4.4 pNN Interpolation

As previously mentioned, a notable benefit of pNNs is their ability to smoothly interpolate to intermediate points. Applied to data and background MC — which have no inherent IDM parameters — this is achieved by passing the datasets through the model with a choice of mass hypothesis, (M_H, M_A) . However, the model cannot be applied to signal samples in this manner, as they are simulated with specific IDM parameters. Instead, to find the signal distributions at intermediate points, spline interpolation is performed.

For this thesis, the spline interpolation is performed using radial basis functions, which are real-valued functions whose output depends only on distance, $\phi(\underline{x}) = \phi(||\underline{x}||)$ [112]. The aim is to approximate a function $F(\underline{x})$ for which we have a set of data points, $\underline{x}_i = \{x_1, x_2, \dots, x_N\}$, with values $y_i = \{y_1, y_2, \dots, y_N\}$. Using these data points, a linear combination of radial basis functions can be constructed with a function centred on every point:

$$S(\underline{x}) = \sum_{i=1}^N w_i \phi(\underline{x} - \underline{x}_i), \quad (5.6)$$

where w_i are the weights. The requirement $S(x_i) = y_i$ leads to the following matrix equation:

$$\underbrace{\begin{bmatrix} \phi(\underline{x}_1 - \underline{x}_1) & \phi(\underline{x}_1 - \underline{x}_2) & \cdots & \phi(\underline{x}_1 - \underline{x}_N) \\ \phi(\underline{x}_2 - \underline{x}_1) & \phi(\underline{x}_2 - \underline{x}_2) & \cdots & \phi(\underline{x}_2 - \underline{x}_N) \\ \vdots & \vdots & \ddots & \vdots \\ \phi(\underline{x}_N - \underline{x}_1) & \phi(\underline{x}_N - \underline{x}_2) & \cdots & \phi(\underline{x}_N - \underline{x}_N) \end{bmatrix}}_{\mathbf{\Phi}} \underbrace{\begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_N \end{bmatrix}}_{\mathbf{w}} = \underbrace{\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{bmatrix}}_{\mathbf{y}}, \quad (5.7)$$

from which the weights can be calculated using matrix inversion: $\mathbf{w} = \mathbf{\Phi}^{-1}\mathbf{y}$. Subsequently, $S(\underline{x})$ is an approximation for the underlying function $F(\underline{x})$ and can be used to interpolate between the initial data points. Common forms of the basis functions include cubic, $\phi(x) = (x/\epsilon)^3$, or Gaussian, e^{x^2/ϵ^2} , where ϵ is a shape hyperparameter. An example of this technique for estimating the function $F(x) = \sin(x)$ is shown in figure 5.13 for $\epsilon = 1.0$ and $\epsilon = 0.5$. For both values of ϵ , the spline prediction passes through all data points as required. However, figure 5.13 also illustrates that the spline interpolation between the data points differs significantly depending on ϵ . Therefore, to ensure robust interpolation when using splines, it is crucial that an appropriate value for the shape parameter, ϵ , is chosen. This can be determined with the following algorithm [112]:

1. Define a range of values for ϵ . For example, $\epsilon = [\epsilon_1, \epsilon_2, \dots, \epsilon_M] = [0.01, 0.02, \dots, 1.0]$.
2. Take a value for the shape parameter, ϵ_i .
3. For each data point, j , remove that point from the data set, and fit splines to the

rest of the data using ϵ_i , giving the function $S_j(x)$.

4. For each fitted spline, calculate $E_j = |S_j(x) - y_j|$, which is the error of the spline prediction (that was fit without point j) compared to the value of the point removed, y_j .
5. When the error for all data points has been calculated, take the mean error $\bar{E} = \frac{1}{N} \sum_j E_j$.
6. Repeat steps 2-5 with a different value of ϵ_i .
7. Take the value of ϵ_i that achieves the lowest $\bar{E}$.

The final value of ϵ is thus the value that, on average, had the best performance for interpolating when each of the data points were removed.

This method is employed when interpolating the signal distributions over the whole parameter space, using cubic radial basis functions. Due to the pNN binning being defined separately for each channel, the spline interpolation is performed separately for each case. Additionally, for each spline prediction, the bin-by-bin uncertainty is taken as the uncertainty from the nearest simulated point. Furthermore, to ensure that the interpolated predictions are correct, the Validation points from figure 5.1 are utilised. With these, the interpolated distributions are compared to the actual distributions, which are found by evaluating the pNN on the MC at each Validation point. An example is shown in figure 5.14 for both channels for 2016 with $M_H = 135, M_A = 190$ GeV, which shows that the spline predictions are compatible with the actual distributions from the simulations. This is true for all interpolations compared to the Validation points.

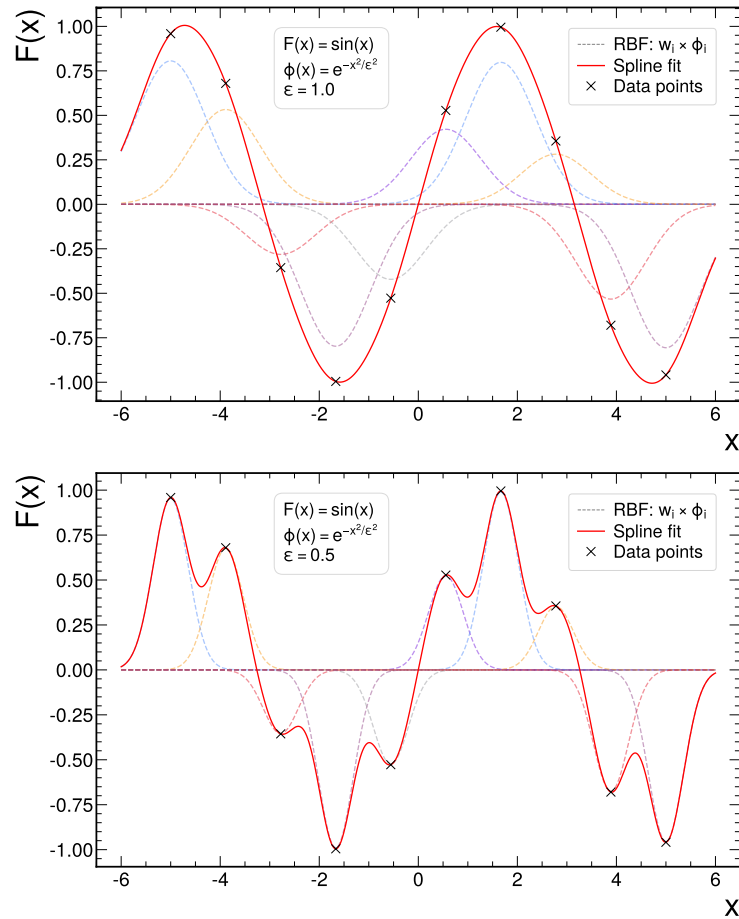


Figure 5.13: Example of spline interpolation of a sine wave, using Gaussian functions as the radial basis function for a shape parameter of (top) $\epsilon = 1.0$ and (bottom) $\epsilon = 0.5$. The full prediction from the spline interpolation is equal to the sum of all weighted basis functions (dotted lines). This demonstrates that whilst the spline prediction will pass through each point regardless of the shape parameter, ϵ , its ability to interpolate is dependent on ϵ .

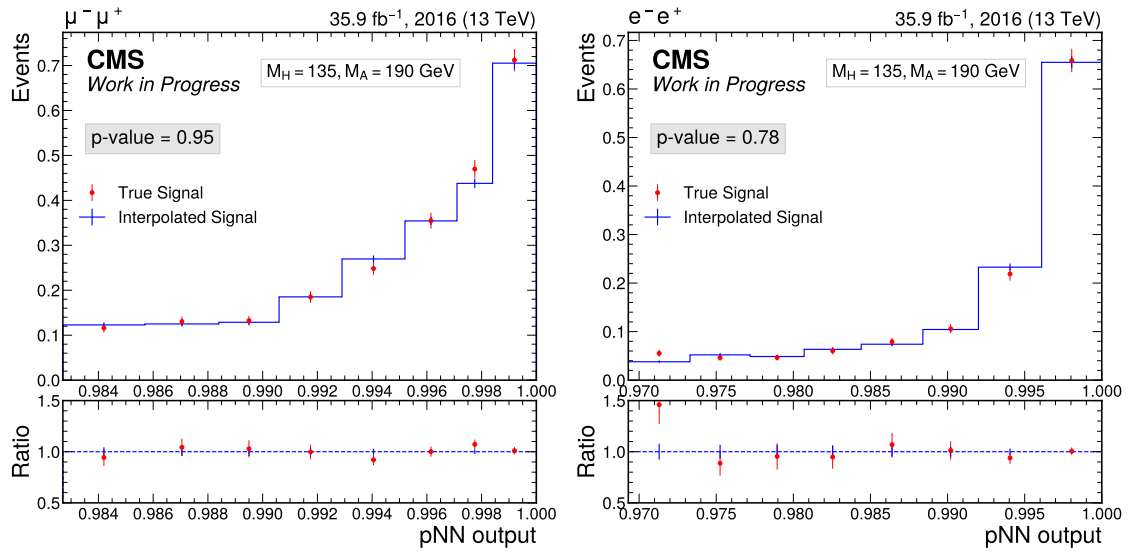


Figure 5.14: Comparisons between the interpolated signal (blue) and the actual signal (red). The interpolation is performed using cubic radial basis functions. The actual signal is found using the MC created at the validation points shown in figure 5.1, which are subsequently passed through the pNN. The uncertainties on the interpolated signal are calculated by using the same relative uncertainty from the nearest simulated point.

5.5 Statistical Model

The sensitivity of this analysis—and the reliability of the results—depends on precise determination of the SM backgrounds. The predictions given by simulation not only might be biased due to approximations in the matrix-element calculation and detector simulation, but can also be affected by large theoretical and experimental uncertainties. Consequently, to reduce the impact of these uncertainties, and to cover mismodelling differences, the largest background processes are constrained directly using data in background-enriched control regions (CRs). Before going into detail about each control region, which are outlined in the next section, this section outlines the statistical model used to extract the results.

The final results are extracted with a binned maximum likelihood fit performed simultaneously over all years and signal regions and control regions. The maximum likelihood takes the form

$$\mathcal{L}(\text{data}|\mu, \theta) = \prod_i^{N_i} \text{Poisson}\left(n_i \left| \mu \times s_i(\theta) + \sum_k^{N_k} b_{ik}(\theta) \right.\right) \cdot p(\hat{\theta}|\theta), \quad (5.8)$$

where i runs over all analysis categories and all bins for each category. Every bin follows the Poisson distribution with n_i observed events, and with s_i and b_i being the expected number of signal and background events, respectively. A scaling parameter, μ , is included to scale the signal to a desired hypothesis. $p(\hat{\theta}|\theta)$ represent the probability density functions for the nuisance parameters, θ , which are Gaussian distributed around an initial value, $\hat{\theta}$. Systematics that affect the shape of the distributions are also included, which are allowed to vary based on shape templates. For calculating distributions varied between the $\pm 1\sigma$ shape templates, a morphing function is used to interpolate between the different shapes [113].

The likelihood provides a basis for constructing a broad range of statistical tests, including estimating the value of a parameter or, most relevant for this thesis, hypothesis testing. For this analysis, the hypotheses are the Standard Model hypothesis, corresponding to $\mu = 0$, or the IDM hypothesis, corresponding to $\mu = 1$. The objective is to quantify whether the observed data is more consistent with the SM or with the IDM. This is achieved through the use of the *profiled likelihood ratio*:

$$\lambda(\mu) = \frac{\mathcal{L}(\text{data}|\mu, \hat{\hat{\theta}}(\mu))}{\mathcal{L}(\text{data}|\hat{\mu}, \hat{\theta})}, \quad (5.9)$$

where $\hat{\mu}$ and $\hat{\theta}$ are the values that maximise the likelihood, and $\hat{\hat{\theta}}$ is the value that maximises the likelihood for a given μ . For tests aimed at excluding parameter values — commonly referred to as *setting limits* — the following test statistic is used:

$$\tilde{t}_\mu = \begin{cases} \frac{\mathcal{L}(\text{data}|\mu, \hat{\theta}(\mu))}{\mathcal{L}(\text{data}|\hat{\mu}, \hat{\theta})} & 0 \leq \hat{\mu} \leq \mu, \\ \frac{\mathcal{L}(\text{data}|\mu, \hat{\theta}(\mu))}{\mathcal{L}(\text{data}|0, \hat{\theta}(0))} & \hat{\mu} < 0, \\ 0 & \mu \leq \hat{\mu}, \end{cases} \quad (5.10)$$

which follows the modified frequentist approach [114–116]. The case $\hat{\mu} < 0$ corresponds to an unphysical part of the parameter space, where the best-fit signal is negative. In this case, the denominator is limited to the background-only hypothesis, $\mathcal{L}(\text{data}|0, \hat{\theta}(0))$, which ensures that sensitivity is not artificially gained when $\hat{\mu}$ is negative. Furthermore, a fit of $\mu \leq \hat{\mu}$ corresponds to being more signal-like than the signal predicts. Consequently, it should not be used to reject the signal hypothesis and, therefore, it is set to a value of 0 and removed from the rejection region of the test.

For a given hypothesis, μ , it is possible to compute a p-value using the test statistic:

$$p_\mu = \int_{\tilde{t}_\mu^{\text{obs}}}^{\infty} f(\tilde{t}_\mu|\mu, \hat{\theta}(\mu)) d\tilde{t}_\mu, \quad (5.11)$$

where $\tilde{t}_\mu^{\text{obs}}$ is the observed value for the test statistic, and $f(\tilde{t}_\mu|\mu, \hat{\theta}(\mu))$ is the probability density function. From this, the quantity $\text{CL}_s(\mu)$ is defined as

$$\text{CL}_s(\mu) = \frac{p_\mu}{1 - p_0}, \quad (5.12)$$

where p_0 is the background-only (SM) hypothesis. A given signal hypothesis is then excluded at a confidence level of $1 - \alpha$ if

$$\text{CL}_s(\mu) < \alpha. \quad (5.13)$$

5.6 Modelling Overview

The composition of the background in the signal region depends on the mass point. Generally, however, the largest background is ZZ , which on average contributes approximately 35% to the most sensitive bin of the analysis. Next is W^-W^+ and $t\bar{t}$, which contribute 20% and 15%, respectively. WZ accounts for approximately 10%, and the remainder is split between MisId events and DY . At low mass splittings, for $M_A - M_H \lesssim 30$ GeV, DY starts to dominate. However, this corresponds to a region where the analysis sensitivity is intrinsically low due to poor discrimination of signal and background.

To ensure reliable results, it is necessary to validate the modelling of the main background sources. This is achieved using selections with comparable phase spaces with respect to the signal region, whilst being enriched in the background processes, and having little to no signal contamination. These are referred to as *control regions* (CR). The normalisations

of the dominant background processes are constrained through a simultaneous fit to all control regions, whilst their shape is taken from simulation. Shape-dependent systematic uncertainties are also included in the fit, allowing the shape of the distributions to vary. All sources of uncertainty are presented in section 5.8.

The following is an overview of the different control regions utilised for this thesis, with a schematic of how these connect to the signal region presented in figure 5.15.

$W^-W^+/t\bar{t}$ These enter through $W^-W^+/t\bar{t} \rightarrow \ell^-\ell^+ + X$. Two control regions are defined, both utilising the symmetry between $W^-W^+/t\bar{t} \rightarrow e^-e^+/\mu^-\mu^+ + X$ and $W^-W^+/t\bar{t} \rightarrow e^\pm\mu^\mp + X$. The first uses the same base selections as the signal region, except with opposite-sign opposite-flavour dilepton pairs, and requires all events to have $\text{pNN}(M_H, M_A) > 0.9$. The second control region, targetting the $t\bar{t}$ contribution, uses the same selections as the first, but requires at least 1 b-tagged jet. Using both control regions, the W^-W^+ and $t\bar{t}$ normalisations are constrained by fitting to the dilepton p_T spectrum.

ZZ ZZ enters the signal region from $ZZ \rightarrow 2\ell 2\nu$, with the CR utilising the symmetry with $ZZ \rightarrow 2\ell 2\ell'$. The CR is defined as a four lepton selection with two opposite-sign same-flavour (OSSF) dilepton pairs, $\ell^-\ell^+\ell'^-\ell'^+$. One of the dilepton pairs is taken as the signal dilepton, which is required to pass the signal region selections, whilst the other is used to emulate the neutrinos. Subsequently, events are required to pass $\text{pNN}(M_H, M_A) > 0.8$, after which the ZZ normalisation is constrained with a fit to the pNN output.

WZ WZ enters the signal region in two ways, both starting from the decay $W \rightarrow l\nu, Z \rightarrow ll$. The first, primary contribution, is when both signal leptons originate from the Z , and the lepton from the W falls out of acceptance. The second is when one signal lepton is from the Z and the other is from the W , and the second Z lepton falls out of acceptance. A control region is defined for each contribution, both utilising a three-lepton selection. The first requires an opposite-sign same-flavour lepton pair for the signal dilepton, and the second requires a same-sign same-flavour lepton pair for the signal dilepton. Subsequently, events are required to pass $\text{pNN}(M_H, M_A) > 0.9$, and the WZ normalisation is then constrained with a fit to the pNN output.

MisId MisId events are those where one or more of the leptons is from a jet that was misidentified as a lepton, primarily from W +jets and QCD di-jet events. Using a data-driven method [117, 118], the MisId estimate is found using a fake factor derived from data, and data samples where one or more leptons have failed the Tight working point. The MisId estimate is then constrained using same-sign control regions.

DY Drell-Yan enters the signal region from $Z/\gamma^* + \text{jets}$, where the jets are mismeasured leading to a fake $\vec{E}_T^{\text{miss}}$ signature. A side-band validation region is defined using the pNN output, from which an uncertainty on the Drell-Yan normalisation is derived. Furthermore, as LO Drell-Yan samples are used, NLO/LO k-factors are derived for

each year along with k-factor uncertainties to cover any miss-modelling in the MC distributions.

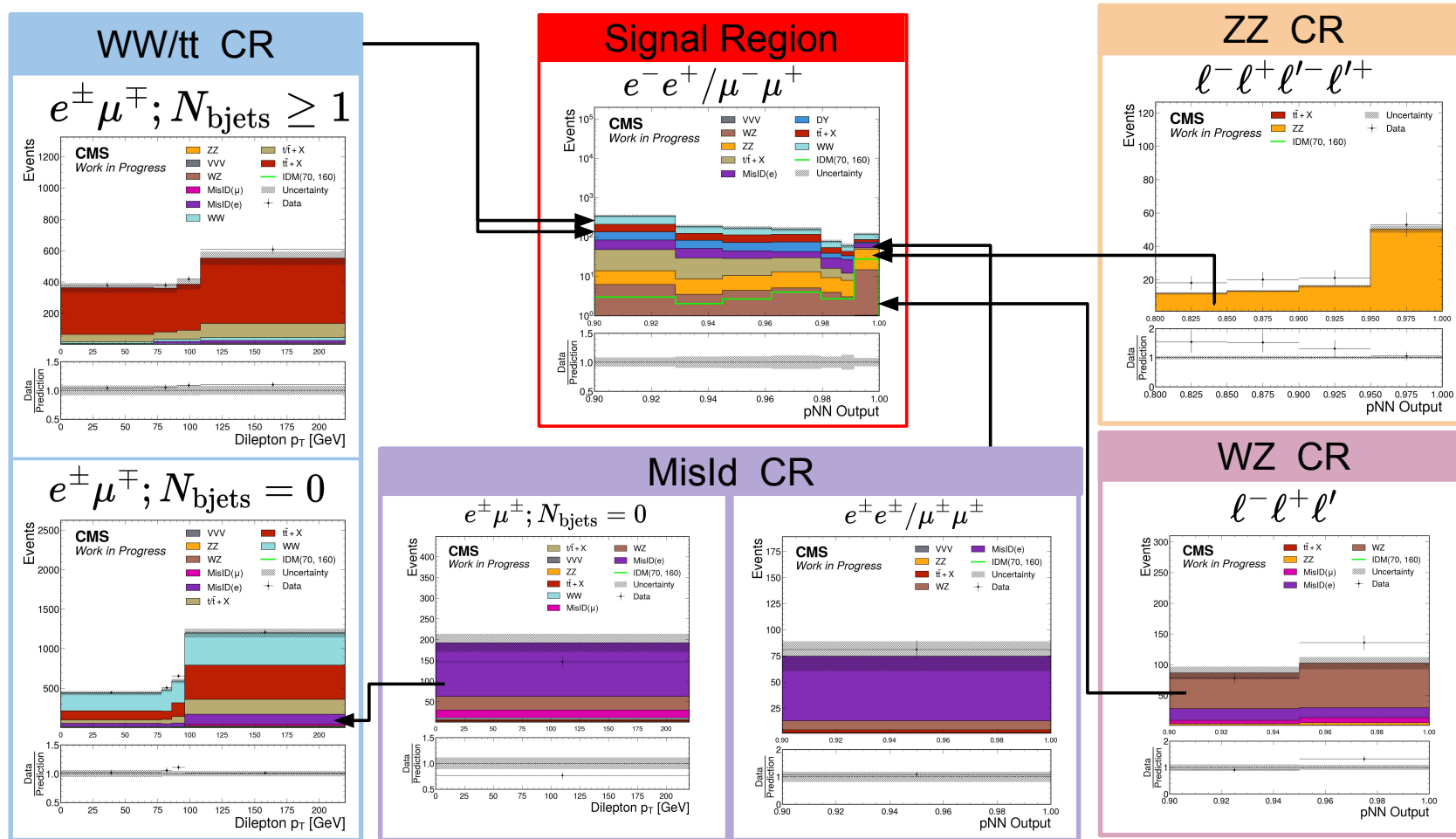


Figure 5.15: Overview of the control regions for the signal region (top centre). This includes the opposite-sign opposite-flavour control region with $N_{\text{bjets}} \geq 1$ (top left) and $N_{\text{bjets}} = 0$ (bottom left) for the $W^{-}W^{+}$ and $t\bar{t}$ processes, and the MisId control region (bottom centre) for the $N_{\text{bjets}} = 0$ $W^{-}W^{+}/t\bar{t}$ CR. Additionally, the $\ell^{-}\ell^{+}\ell'^{-}\ell'^{+}$ CR for the ZZ contribution (top right), and the $\ell^{-}\ell^{+}\ell'$ CR for the WZ process.

5.7 Background Modelling

5.7.1 $W^-W^+/t\bar{t}$ Control Region

The kinematic distributions for the decay $W^-W^+/t\bar{t} \rightarrow e^-e^+/\mu^-\mu^+ + X$ should be nearly identical to the decay $W^-W^+/t\bar{t} \rightarrow e^\pm\mu^\mp + X$, excluding reconstruction-specific factors such as the different triggering thresholds. As a consequence, to estimate how well the $W^-W^+/t\bar{t}$ processes are being modelled, a control region (CR) can be defined using the $e^\pm\mu^\mp$ state, as this significantly reduces other SM backgrounds such as $Z/\gamma^* + \text{jets}$, whilst also having little signal contamination. With this final state, two control regions are defined depending on the number of b-jets in the event. One CR selects events with 0 b-jets, which primarily targets W^-W^+ , and the other selects events with ≥ 1 b-jets, targetting $t\bar{t}$. By splitting into two CRs based on the number of b-jets, the constraints on both the W^-W^+ and $t\bar{t}$ normalisations are improved.

The first $W^-W^+/t\bar{t}$ CR starts with identical base-selections as in table 5.2, including requiring 0 b-jets, but targetting the $e^\pm\mu^\mp$ final state. To match the signal region phase space, all events are required to have $\text{pNN}(M_H, M_A) > 0.9$. It should be noted that by using the pNN output as part of the selection — which is dependent on the mass hypothesis (M_H, M_A) — then the control regions are also dependent on the mass hypothesis. Consequently, each individual mass point has a different $W^-W^+/t\bar{t}$ control region, thus ensuring that the CR is representative of the signal region for all points in the parameter space. Ideally, the $W^-W^+/t\bar{t}$ normalisations would be constrained with a fit to the pNN output directly, however the W^-W^+ and $t\bar{t}$ distributions are too similar, leading to unreliable, degenerate fits. Instead, the dilepton p_T is used as the observable in the fit, as this provides the necessary shape discrimination between W^-W^+ and $t\bar{t}$ to break the fit degeneracy and robustly constrain their normalisations. An example of this CR for two representative IDM mass points, $(M_H, M_A) = (70, 120)$ and $(70, 160)$ GeV, is shown in figure 5.16. The binning is determined using a sliding window technique from low p_T to higher p_T values, where a bin is accepted if the statistical uncertainty is below 5%. This ensures histogram bins that are fine-grain enough for separation between W^-W^+ and $t\bar{t}$, whilst limiting large discontinuities from limited MC statistics.

The second control region targets $t\bar{t}$ and has identical selections to the first, but with the requirement of ≥ 1 b-jets, and also fits to the dilepton p_T spectrum. Although the signal region requires no b-jets, whilst this CR does, they nevertheless cover a similar phase space. This is supported by generator-level studies, which show that over 80% of $t\bar{t}$ events in the signal region contain a b-jet emitted within the selection acceptance, but that is not reconstructed as a b-jet. An example of this CR is shown in figure 5.17, which employs the same binning technique as the first $W^-W^+/t\bar{t}$ CR. Utilising both control regions, the W^-W^+ and $t\bar{t}$ normalisations are constrained separately for each year, allowing for potential differences between years to be accounted for in the fit. There are multiple reasons why year-to-year differences are expected. First, each year has slightly different running conditions, with the most significant change being the pileup (refer to Figure 3.3).

Furthermore, due to detector degradation, each sub-detector unit will reconstruct slightly differently each year. Finally, even if the detector and conditions remained identical, the different trigger thresholds could also create year-to-year differences.

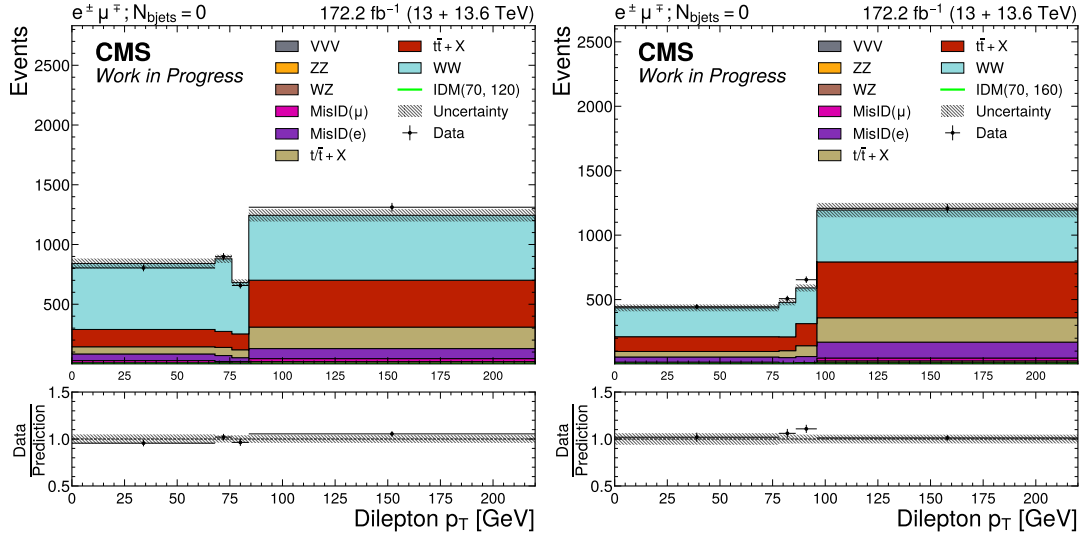


Figure 5.16: Distributions of the $W^-W^+/t\bar{t}$ control region with no b-jets for mass points $(M_H, M_A) = (70, 120)$ GeV (left) and $(70, 160)$ GeV (right). All uncertainties are included in the uncertainty band.

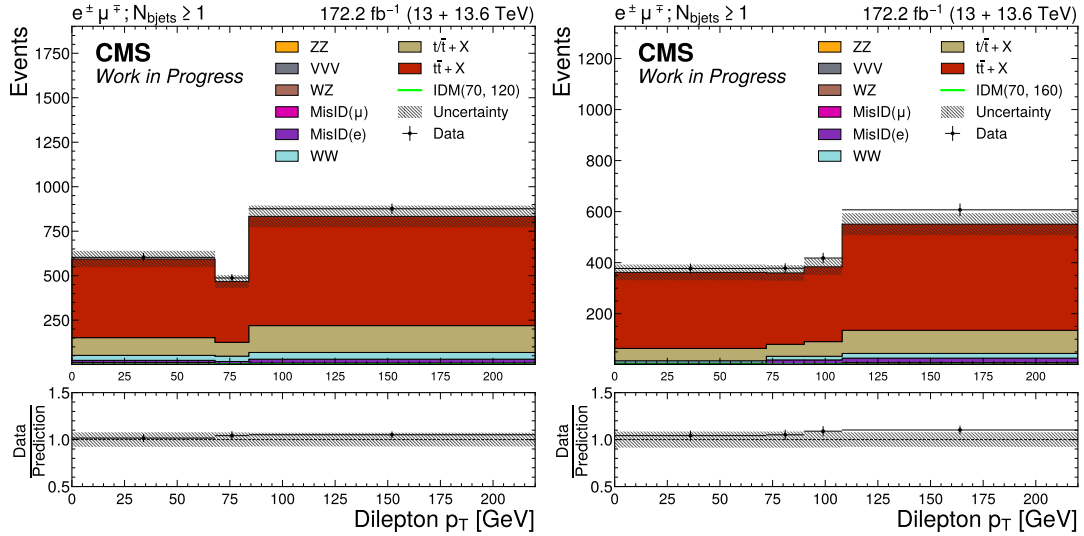


Figure 5.17: Distributions of the $W^-W^+/t\bar{t}$ control region with at least 1 b-jet for mass points $(M_H, M_A) = (70, 120)$ GeV (left) and $(70, 160)$ GeV (right). All uncertainties are included in the uncertainty band.

5.7.2 WZ Control Region

WZ events can enter the signal region through multiple decay pathways. The dominant contribution — accounting for over 80% of WZ events in the most sensitive bin — occurs when the two OSSF signal leptons originate from the Z boson, $Z \rightarrow \ell^- \ell^+$ ($\ell = e, \mu$), with the W boson decaying leptonically and its decay products evading detection. This typically proceeds via the decay chain $W \rightarrow \nu_\tau \tau (\rightarrow \ell \nu_\ell)$, where $\ell = e, \mu$, which, due to the presence of multiple neutrinos, produces a softer p_T spectrum for the ℓ . Notably, the parent τ is primarily produced within the detector acceptance. Alternatively, when the W boson decays to an electron or muon, a large proportion (approximately 40%) of these are still produced within the pseudorapidity acceptance of the analysis selections. As a result, the kinematic distributions for these events closely match the kinematics of $WZ \rightarrow \ell^- \ell^+ \ell' \nu_{\ell'}$, where the third lepton, $\ell' = e, \mu$, is reconstructed. This motivates the construction of a WZ control region using a three lepton event selection.

There is an additional, subdominant, contribution from when one signal lepton originates from the Z boson, the other from the W boson, and the remaining lepton from the Z falls out of acceptance. As in the previous decay chain, the majority of these events occur with all leptons produced within the acceptance of the analysis selections (approximately 65%), motivating the use of a three-lepton control region. Finally, few WZ events (<1%) enter the signal region where the W decays hadronically, predominantly due to the reduced missing energy signature of this decay as all missing energy would only arise from jet mismeasurements.

The first WZ control region targets events where both signal leptons come from the Z . This starts by selecting events with three leptons, two of which form an opposite-sign same-flavour dilepton pair that pass the Tight working points. The third lepton is also required to pass the Tight working point, chosen as such due to the large $Z/\gamma^* + \text{jets}$ background with a jet faking a lepton. Events are required to have the invariant mass of the three lepton system $M_{\ell\ell\ell} > 105$ GeV, which removes $Z + \gamma$ events where final stage radiation of a photon is incorrectly reconstructed as an electron. Furthermore, the invariant mass of all OSSF dilepton pairs must satisfy $M_{\ell-\ell^+} > 4$ GeV to match the generator level selections of the WZ MC samples. For events with three of the same lepton flavour (eee or $\mu\mu\mu$), there is ambiguity in which dilepton pairing to choose as the signal dilepton. In this case, the dilepton pair is chosen to be the one with invariant mass closest to the Z boson mass. After events with three leptons are selected, each event is also required to pass the base selections outlined in table 5.2, along with the requirement $\text{pNN}(M_H, M_A) > 0.9$. Subsequently, the pNN output is used as the observable in the fit. An example of this is shown in figure 5.18 for mass points $(M_H, M_A) = (70, 120)$ and $(70, 160)$ GeV. In general, an excess of data compared to MC is observed. However, previous WZ studies also noted comparable excesses compared to the NLO WZ samples [119]. Furthermore, the WZ normalisation has a relatively minor impact on the final sensitivity compared to other systematic uncertainties.

The second WZ control region targets the case where one signal lepton is from the Z

and the other is from the W . This requires the same selection as the first, but with a same-sign same-flavour dilepton pair. Likewise, the pNN output is used as the observable in the fit. An example of this control region is shown in figure 5.19. As in the first WZ control region, a similar excess of data events over MC prediction is observed. Utilising both control regions, the WZ normalisation is constrained separately for each year.

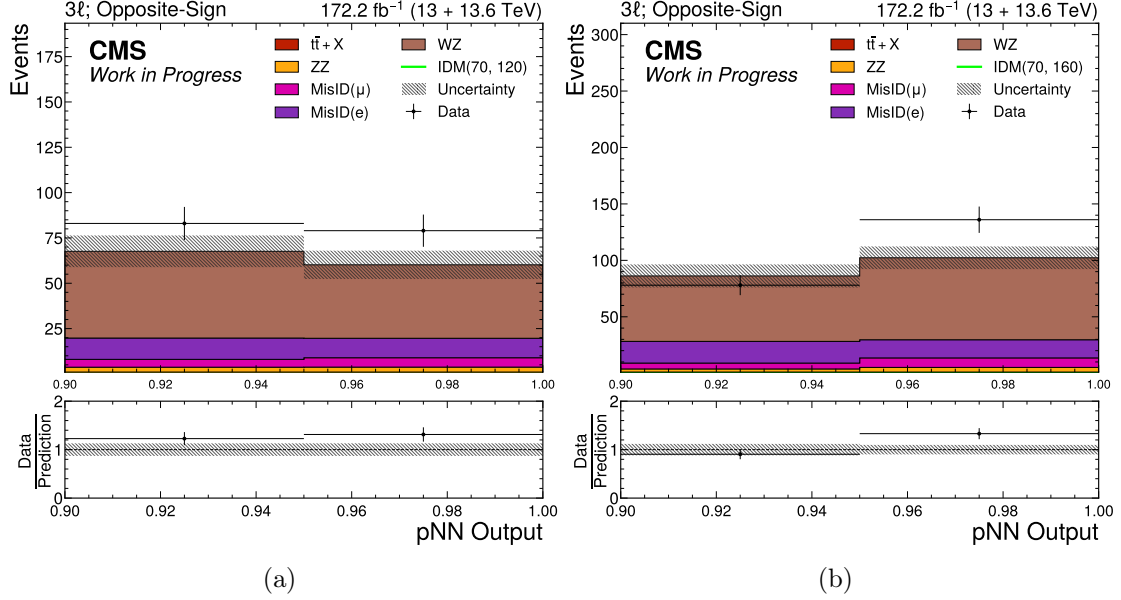


Figure 5.18: Distributions of the first, opposite-sign, WZ control region for mass points $(M_H, M_A) = (70, 120)$ GeV (left) and $(70, 160)$ GeV (right). All uncertainties are included in the uncertainty band.

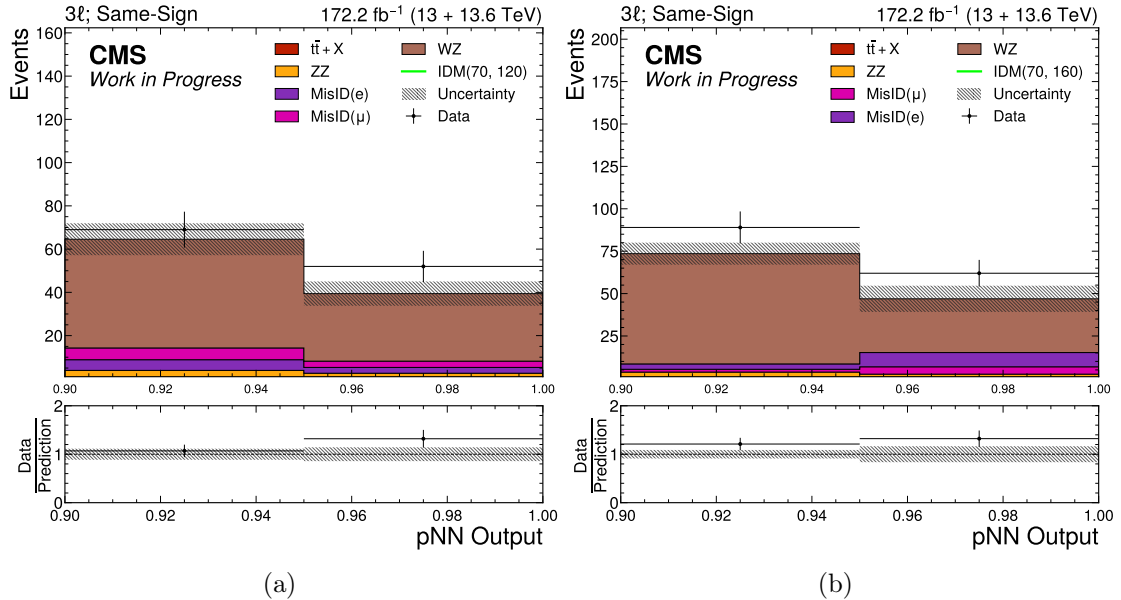


Figure 5.19: Distributions of the second, same-sign, WZ control region for mass points $(M_H, M_A) = (70, 120)$ GeV (left) and $(70, 160)$ GeV (right). All uncertainties are included in the uncertainty band.

5.7.3 MisId Control Region

Background SM events that include jets can enter the selections when one or more jets are falsely identified as an electron or a muon. Not only can these contribute to the signal regions, but they can also contribute to the control regions, predominantly the $W^-W^+/t\bar{t}$ control region with no b-jets, and the WZ control region. Examples of such processes include $W + \text{jets}$ events resulting in one prompt lepton from the W boson and a fake lepton from the jet, as well as QCD di-jet events where both jets fake leptons. However, events of this type are generally difficult to model using MC due to the poor modelling of the jet-to-lepton mis-reconstruction. Additionally, given the rarity of such reconstructions, large MC samples would be required which come at a significant computational cost. As a consequence, it is beneficial to instead use data-driven methods rather than relying solely on MC. The method used for this thesis is referred to as the MisId method [117, 118].

The MisId method works in two stages. First, a fake factor is derived for the probability that a jet, which has already been reconstructed as a Loose lepton, also passes the Tight working point:

$$\text{Fake Factor} = f = \frac{N(\text{Passes Tight ID})}{N(\text{Passes Loose ID})}, \quad (5.14)$$

where $N(X)$ is the number of leptons of type X .

Following this, the MisId contribution can be estimated using control samples of data. For example, for the signal region, one control sample requires a single lepton to fail the Tight ID, which is dominated by $W + \text{jets}$ events, whilst the other requires both leptons to fail the Tight ID, which is mainly QCD di-jet events. Using these samples, and the fake factors, each event is assigned a weight based on the probability that the event would pass the signal region selection. The final estimate is then given by the weighted data sample with the contamination from prompt events subtracted using MC.

Fake Rate

To measure the fake rate, a selection is required that is enriched in jets faking leptons, whilst having minimal contamination from SM sources containing prompt leptons. This is obtained using $Z/\gamma^* + \text{jets}$ events, where the third lepton is likely to be a jet fake. Specifically, events are selected with two opposite-sign same-flavour Tight leptons with an invariant mass compatible with the Z boson, $|M_{\ell-\ell^+} - M_Z| < 15 \text{ GeV}$, as well as an additional Loose lepton that is used as the probe for calculating the fake rate. To reduce the dominant SM background containing three prompt leptons, $WZ \rightarrow \ell^- \ell^+ \ell' \nu_{\ell'}$, every event is required to have $E_T^{\text{miss}} < 25 \text{ GeV}$, chosen as such due to WZ events having higher missing transverse energy from the neutrinos. Moreover, a selection of $M_{\ell-\ell^+} > 4 \text{ GeV}$ for all OSSF dilepton pairs is applied to ensure compatible phase space selections with the WZ MC. Finally, any event with a fourth Loose lepton is removed to reduce ZZ contamination.

The fake rate is measured in two bins of lepton pseudorapidity, $|\eta| \leq 1.479$ and $1.479 <$

$|\eta| < 2.5$ (2.4) for electron (muons) where $|\eta| = 1.479$ corresponds to the barrel-endcap boundary of the ECAL. This is further split into bins of lepton p_T , with $p_T = [20, 30, 50, 50+]$ GeV for electrons, and $p_T = [20, 30, 30+]$ GeV for muons. Fewer bins are used for muons due to the reduced number of events. Any remaining prompt contamination is subtracted using MC. The fake rate measurements are shown in figures 5.20-5.23, which are generally consistent across all years. It should be noted that, whilst the muon fake rates are higher than electron fake rates, this does not imply that jets are more likely to fake muons than electrons. The reason for this is that the fake rate is defined as passing the Tight ID, after already having passed the Loose ID, and is not the total rate for which jets fake electrons or muons.

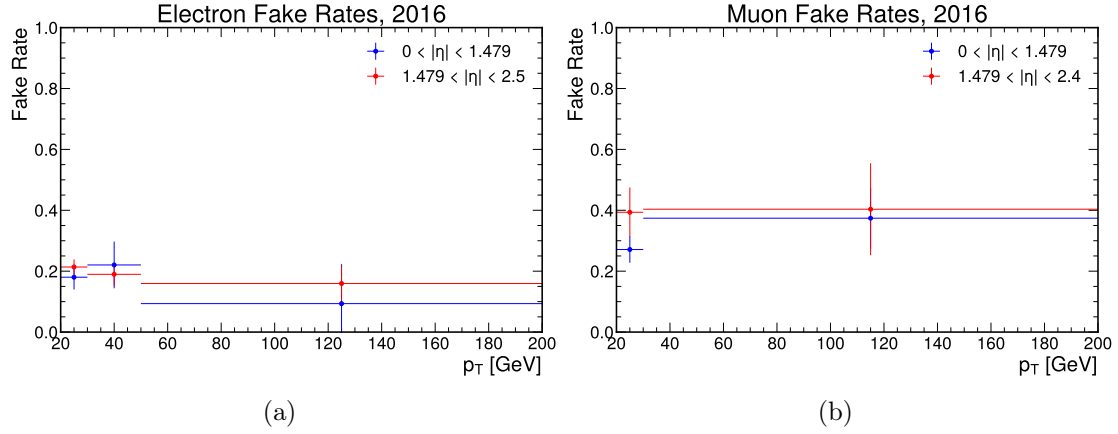


Figure 5.20: Fake rates for electrons (left) and muons (right) for 2016.

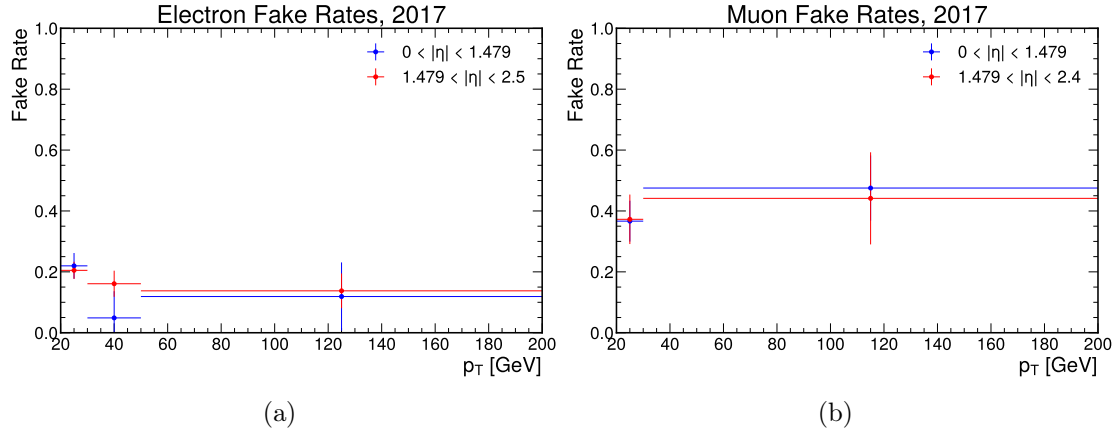


Figure 5.21: Fake rates for electrons (left) and muons (right) for 2017.

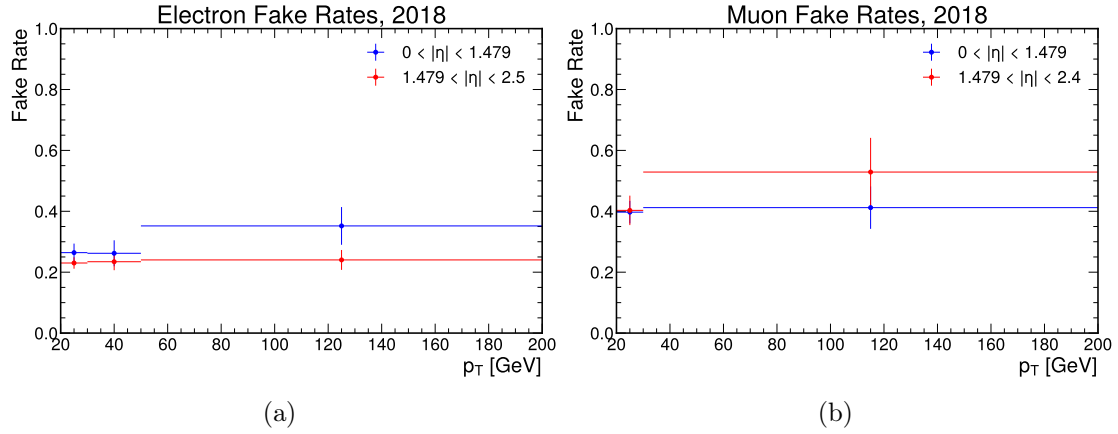


Figure 5.22: Fake rates for electrons (left) and muons (right) for 2018.

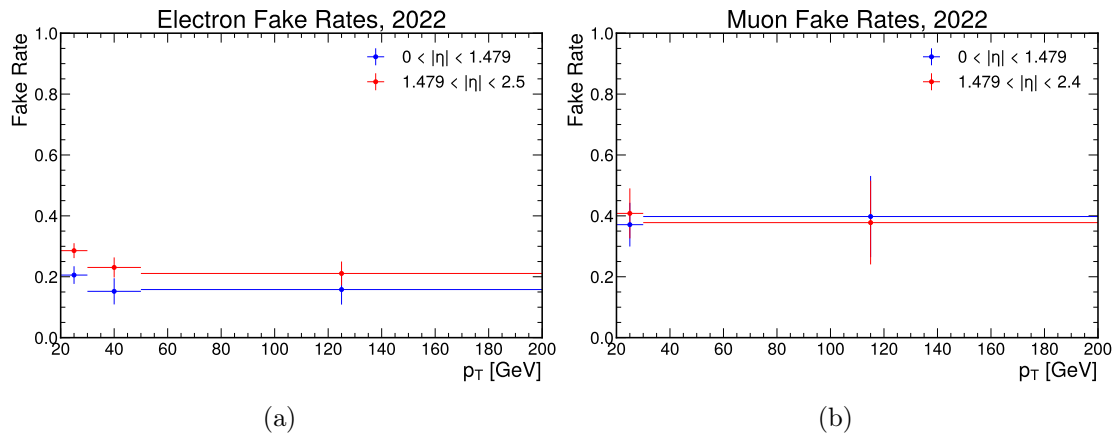


Figure 5.23: Fake rates for electrons (left) and muons (right) for 2022.

Calculating the MisId Estimate

To calculate the MisId contribution, data samples are selected where at least one lepton fails the Tight ID. For the case where the selection contains two Tight leptons, such as the signal region or $W^-W^+/\bar{t}t$ CR, the MisId samples are denoted as N_{TL} , N_{LT} and N_{LL} , where T denotes Tight and L denotes Loose but not Tight. Here, the objective is to use the samples with at least one Loose lepton to calculate the MisId contribution to the region of interest, denoted N_{TT} . However, the fake rate can only be used to get the probability that a jet that has already passed the Loose ID will pass the Tight ID, with probability f , or pass the Loose ID but not the Tight ID, with probability $1 - f$. Moreover, a jet fake can contribute to any of N_{TL} , N_{LT} or N_{LL} . Therefore, to calculate the MisId estimate, the number of Loose/Tight measured leptons (N_{TL} , N_{LT} and N_{LL}) must be related to the origin of the contribution: N_{FP} , N_{PF} and N_{FF} , where P are prompt leptons and F are leptons from fakes [117, 118]. Assuming all prompt leptons pass the Tight ID — which was checked in simulation to be the case in $> 95\%$ of events — these can be related to the previous quantities to find the quantity of interest, N_{TT} :

$$\begin{aligned}
 N_{TT} &= f_1 N_{FP} + f_2 N_{PF} + f_1 f_2 N_{FF} \\
 N_{TL} &= (1 - f_2) N_{PF} + f_1 (1 - f_2) N_{FF} \\
 N_{LT} &= (1 - f_1) N_{FP} + f_2 (1 - f_1) N_{FF} \\
 N_{LL} &= (1 - f_1)(1 - f_2) N_{FF},
 \end{aligned} \tag{5.15}$$

where f_i is the fake rate for lepton i . The MisId estimate is then calculated by solving for N_{TT} in terms of N_{TL} , N_{LT} and N_{LL} :

$$N_{TT} = \frac{f_1}{(1 - f_1)} N_{LT} + \frac{f_2}{(1 - f_2)} N_{TL} - \frac{f_1 f_2}{(1 - f_1)(1 - f_2)} N_{LL}. \tag{5.16}$$

The contamination from prompt leptons is subtracted from each of the yields N_{TL} , N_{LT} and N_{LL} using MC before evaluating N_{TT} . Furthermore, the uncertainty on the fake rate is propagated to an uncertainty on the MisId estimate.

For the three lepton case, as in the WZ control region, instead the related quantities are

$$\begin{aligned}
N_{TTT} &= f_1 N_{FPP} + f_2 N_{PFP} + f_3 N_{PPF} + f_1 f_2 N_{FFP} + f_1 f_3 N_{FPP} + f_2 f_3 N_{PPF} \\
&\quad + f_1 f_2 f_3 N_{FFF} \\
N_{TTL} &= (1 - f_3) N_{PPF} + f_1 (1 - f_3) N_{FPP} + f_2 (1 - f_3) N_{PFP} + f_1 f_2 (1 - f_3) N_{FFF} \\
N_{TLT} &= (1 - f_2) N_{PFP} + f_1 (1 - f_2) N_{FFP} + f_3 (1 - f_2) N_{PPF} + f_1 f_3 (1 - f_2) N_{FFF} \\
N_{LTT} &= (1 - f_1) N_{FPP} + f_2 (1 - f_1) N_{FFP} + f_3 (1 - f_1) N_{PFP} + f_2 f_3 (1 - f_1) N_{FFF} \quad (5.17) \\
N_{TLL} &= (1 - f_2)(1 - f_3) N_{PFP} + f_1 (1 - f_2)(1 - f_3) N_{FFF} \\
N_{LTL} &= (1 - f_1)(1 - f_3) N_{PFP} + f_2 (1 - f_1)(1 - f_3) N_{FFF} \\
N_{LLT} &= (1 - f_1)(1 - f_2) N_{FFP} + f_3 (1 - f_1)(1 - f_2) N_{FFF} \\
N_{LLL} &= (1 - f_1)(1 - f_2)(1 - f_3) N_{FFF}
\end{aligned}$$

Rearranging for N_{TTT} in terms of the other observables gives the following MisId estimate:

$$\begin{aligned}
N_{TTT} &= \frac{f_1}{(1 - f_1)} N_{LTT} + \frac{f_2}{(1 - f_2)} N_{TLT} + \frac{f_3}{(1 - f_3)} N_{TTL} \\
&\quad - \frac{f_1 f_2}{(1 - f_1)(1 - f_2)} N_{LLT} - \frac{f_1 f_3}{(1 - f_1)(1 - f_3)} N_{LTL} - \frac{f_2 f_3}{(1 - f_2)(1 - f_3)} N_{TLL} \quad (5.18) \\
&\quad + \frac{f_1 f_2 f_3}{(1 - f_1)(1 - f_2)(1 - f_3)} N_{LLL},
\end{aligned}$$

where, again, contamination from prompt leptons is subtracted off each of the yields N_{TTL} , N_{TLT} , N_{LTT} , N_{TLL} , N_{LTL} , N_{LLT} and N_{LLL} using MC.

MisId Control Region

To constrain the MisId estimate, same-sign control regions are defined, leveraging the fact that jets can be misidentified as leptons of either charge. For the MisId contribution to the signal regions, the control regions are constructed using identical selections but with a same-sign dilepton pair, $e^\pm e^\pm / \mu^\pm \mu^\pm$, and with the requirement $\text{pNN}(M_H, M_A) > 0.9$ to match the signal region phase space. The MisId estimate is then constrained by fitting to the pNN output in these control regions. To take into account that the electron fakes are modelled differently to muon fakes, the fit is performed separately for MisId events containing a Loose electron and for those containing a Loose muon. Examples of the MisId control regions are shown in figures 5.24 and 5.25 for the $e^\pm e^\pm$ and $\mu^\pm \mu^\pm$ channels, respectively. There is negligible contamination from the signal, as well as from genuine opposite-sign events where the sign of one lepton has been misidentified. For some MisId CRs, in particular the $\mu^\pm \mu^\pm$ channel, there is a large contribution from WZ which can produce genuine same-sign dilepton events. However, as the fit is performed to all control regions simultaneously, which includes the WZ control region, the fit remains stable and non-degenerate.

For the MisId estimate to the $W^-W^+/t\bar{t}$ control region with no b-jets⁴, events must pass identical selections but with a same-sign opposite-flavour dilepton pair, $e^\pm\mu^\pm$, and with the requirement $\text{pNN}(M_H, M_A) > 0.9$ to match the signal region phase space. Compared to the $e^\pm e^\pm/\mu^\pm\mu^\pm$ CRs, however, for the $W^-W^+/t\bar{t}$ CR an event can contain both a Loose electron and a Loose muon. In this case, because electron fakes are the dominant contribution, these are considered to be electron fake events in the fit. Note, however, events with two Loose leptons make up less than 3% of the total MisId contribution, as such the choice has little impact on the final fit. The pNN output is the observable used in the fit, with the electron-fake contribution and the muon-fake contribution constrained separately. Examples of this control region are shown in figure 5.26. The MisId estimate is constrained separately for each year, allowing for differences in the fake rates between years to be accounted for in the fit.

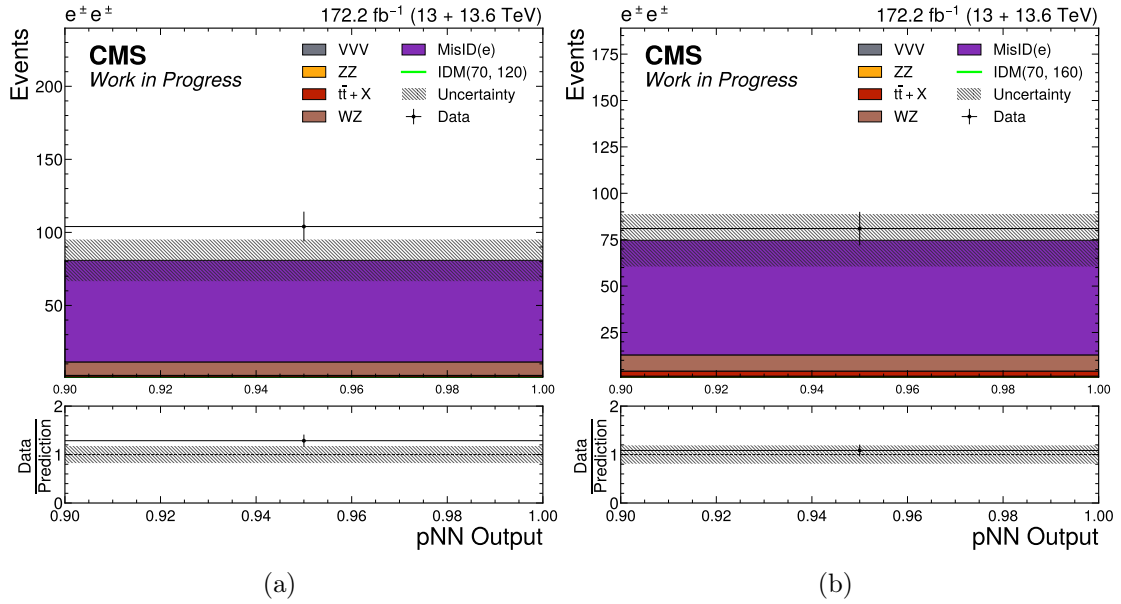


Figure 5.24: Distributions for the $e^\pm e^\pm$ MisId control region for mass points $(M_H, M_A) = (70, 120)$ GeV (left) and $(70, 160)$ GeV (right). All uncertainties are included in the uncertainty band.

⁴The $W^-W^+/t\bar{t}$ CR with ≥ 1 b-jets contains little MisId contribution and, therefore, has no associated MisId control region.

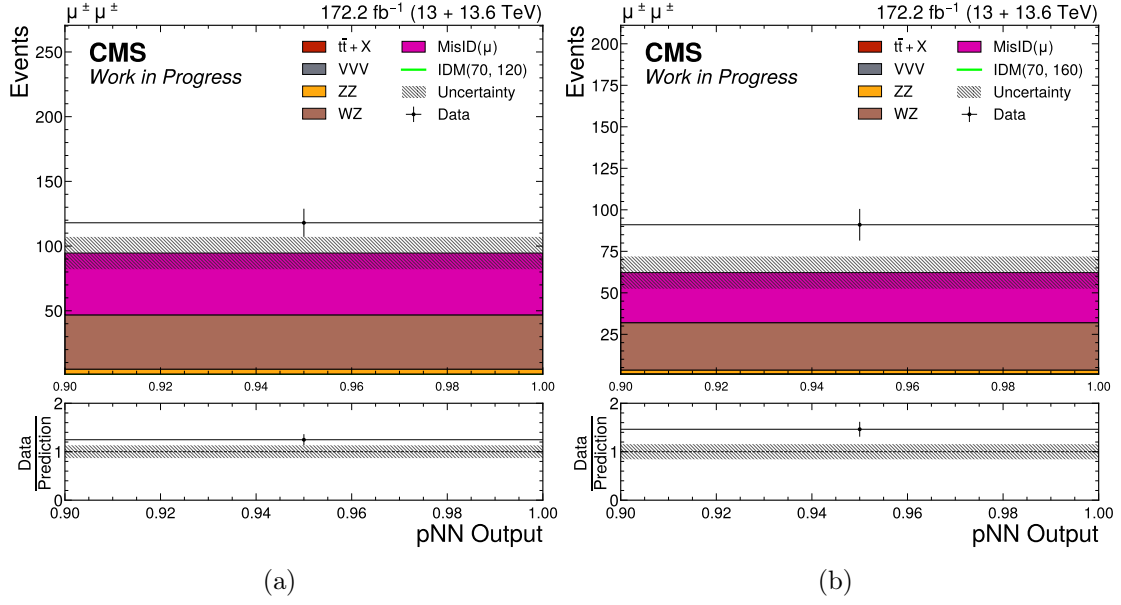


Figure 5.25: Distributions for the $\mu^{\pm}\mu^{\pm}$ MisId control region for mass points $(M_H, M_A)=(70, 120)$ GeV (left) and $(70, 160)$ GeV (right). All uncertainties are included in the uncertainty band.

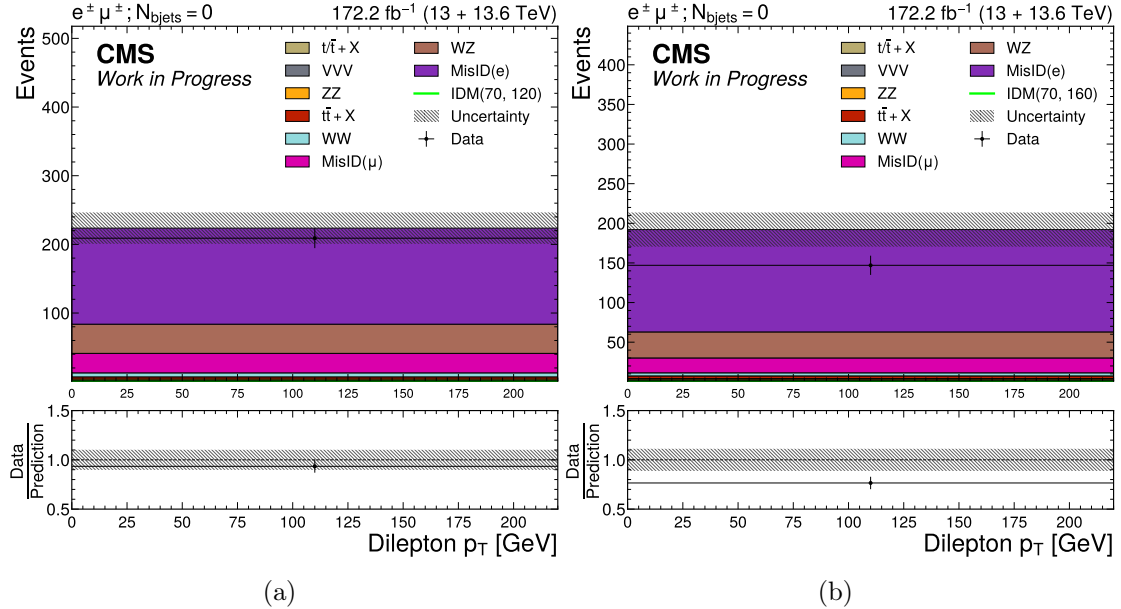


Figure 5.26: Distributions for the $e^{\pm}\mu^{\pm}$ MisId control region for mass points $(M_H, M_A)=(70, 120)$ GeV (left) and $(70, 160)$ GeV (right). All uncertainties are included in the uncertainty band.

5.7.4 ZZ Control Region

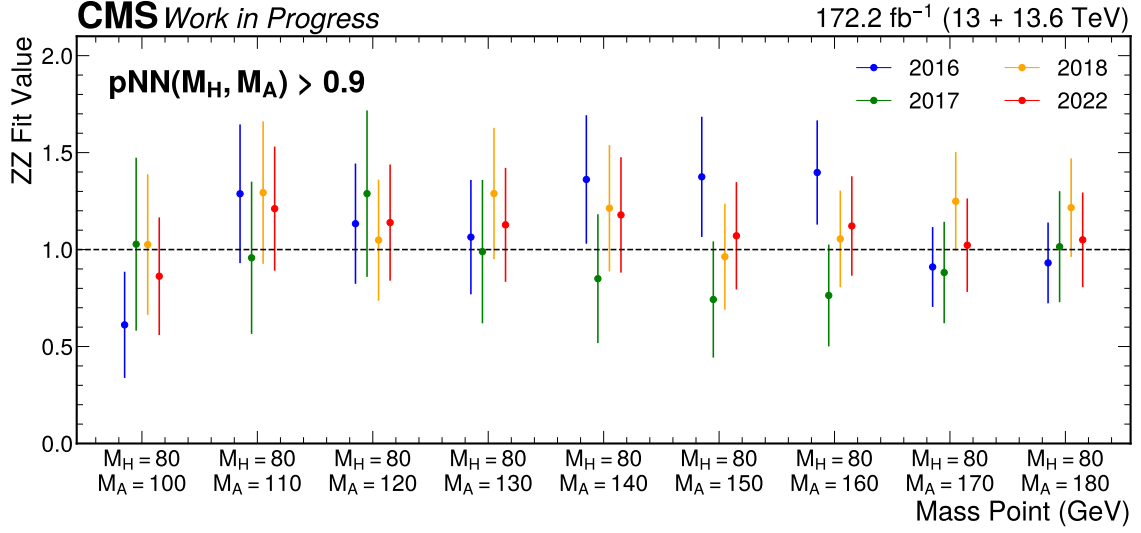
The decay $ZZ \rightarrow \ell^- \ell^+ \nu \nu$ constitutes the dominant background contribution in the most sensitive pNN bin, as the ZZ signature closely matches that of the IDM signal. In this ZZ decay, the OSSF dilepton pair passes the signal region selections, whilst the vector sum of the dineutrino system contributes to the missing transverse energy, $\vec{E}_T^{\text{miss}}$. This can be emulated with a four lepton selection that contains two OSSF dilepton pairs. In this case, one dilepton pair is taken to be the signal region leptons, whilst the other dilepton pair is removed from the event, and their dilepton momentum is added vectorially to $\vec{E}_T^{\text{miss}}$ to mimic the dineutrino signature.

The ZZ CR selection starts by requiring exactly four leptons, with two OSSF dilepton pairs. For events with 2 electrons and 2 muons there is no ambiguity between which leptons originated from which Z . However, for events with 4 electrons or 4 muons, there is. In this case, all possible dilepton pair combinations are constructed, and the combination that produces an invariant mass closest to the Z is chosen. For example, if the event contained $\ell_1^- \ell_2^+ \ell_3^- \ell_4^+$ then the possible dilepton combinations would be $\{(\ell_1^- \ell_2^+, \ell_3^- \ell_4^+), (\ell_1^- \ell_4^+, \ell_3^- \ell_2^+)\}$. If the corresponding invariant masses were $\{(30, 85), (49, 92)\}$ GeV, then the second combination ((49, 92) GeV) would be selected as this contains an invariant mass closest to the Z mass.

With the dilepton pairs chosen, at least one dilepton pair must pass the signal region selections outlined in table 5.2. The pair that passes the signal region selections is referred to as the *signal dilepton*, and the other pair is referred to as the *dineutrino candidate*. Given the high purity of ZZ events in the four lepton selection, and the low yield of the selection, instead of requiring both signal leptons to be Tight, only 2 Tight leptons are required over the whole event. This maintains the high purity of the selection whilst increasing the acceptance of events. If both dilepton pairs pass the signal region selections, then the signal dilepton pair is selected randomly. Furthermore, the invariant mass of the dineutrino candidates is required to be $|M_{\ell-\ell^+} - M_Z| < 10$ GeV to remove the case where they originate from a photon, which is not possible in the $2\ell 2\nu$ final state. Finally, the invariant mass of all OSSF dilepton pairs must satisfy $M_{\ell-\ell^+} > 5$ GeV to remove $Z + \gamma$ events where the photon pair-produces an OSSF dilepton pair.

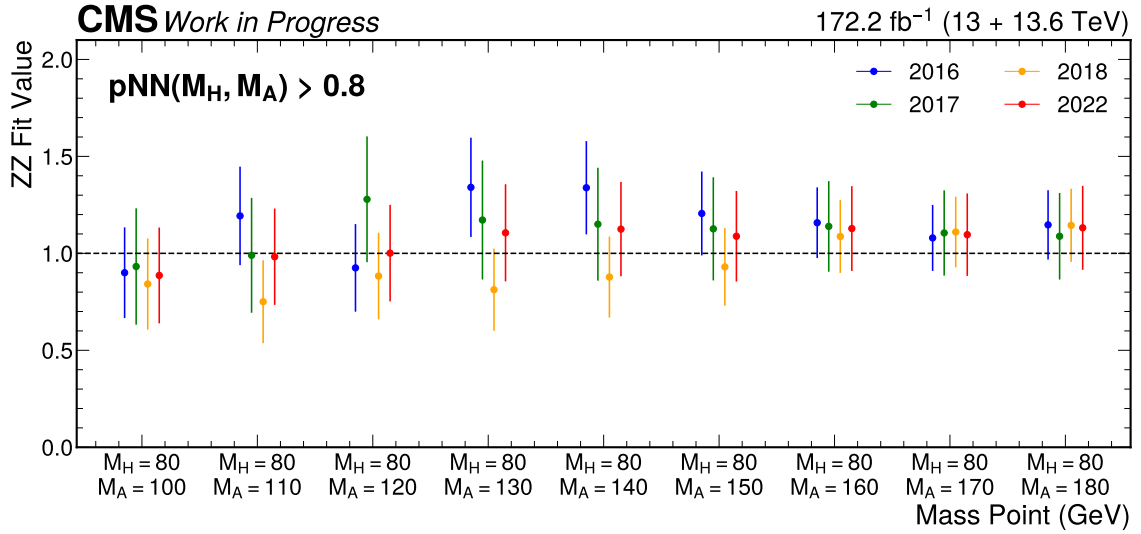
After events have been selected that pass the above criteria, the ZZ control region is defined based on a cut on the pNN output. However, using the same requirement as the signal region, $\text{pNN}(M_H, M_A) > 0.9$, resulted in low yields, leading to large uncertainties on the fits. This is shown in figure 5.27, which presents the post-fit ZZ normalisations and their uncertainties when fitting to a control region with $\text{pNN}(M_H, M_A) > 0.9$. To improve the ZZ yield in the CR, therefore, the selection is lowered to $\text{pNN}(M_H, M_A) > 0.8$. The post-fit normalisations for this selection are shown in figure 5.28. This increases the statistics, leading to reduced uncertainties on the ZZ normalisation, whilst still being compatible with a fit to $\text{pNN}(M_H, M_A) > 0.9$. Due to the shape of the ZZ distribution, which largely peaks at a value close to 0 and 1, loosening this requirement further does not increase the number of events significantly, and therefore provides only negligible improvement.

Examples of the ZZ control region are shown in figure 5.29.



(a)

Figure 5.27: Values of the ZZ normalisations after fitting to the ZZ control region for the selection $pNN(M_H, M_A) > 0.9$.



(a)

Figure 5.28: Values of the ZZ normalisations after fitting to the ZZ control region for the selection $pNN(M_H, M_A) > 0.8$.

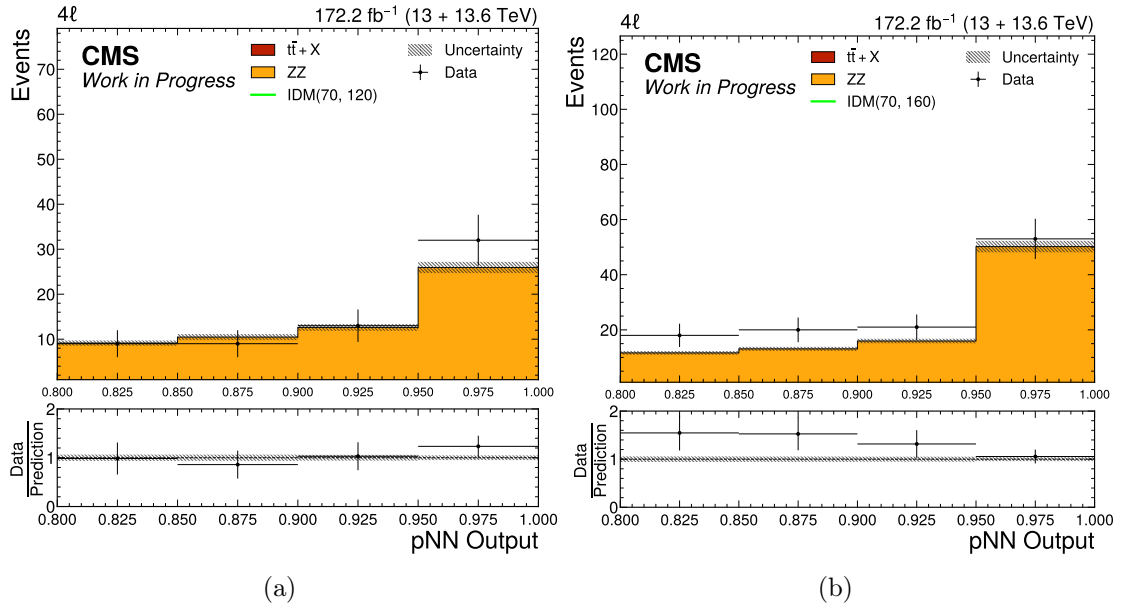


Figure 5.29: Distributions of the ZZ control region for mass points $(M_H, M_A) = (70, 120)$ GeV (left) and $(70, 160)$ GeV (right). All uncertainties are included in the uncertainty band.

5.7.5 Drell-Yan

$Z/\gamma^* + \text{jets}$ events can enter the signal region when the jets are mismeasured, leading to a fake $\vec{E}_T^{\text{miss}}$ signature. However, across most of the parameter space, they form a small background source in the most sensitive bins of the analysis. The largest contribution is in the low mass splitting region, $M_A - M_H \lesssim 40$ GeV, but this is still well below the W^-W^+ , $t\bar{t}$ and ZZ backgrounds; for larger mass splittings there is a negligible contribution. Despite this, as $Z/\gamma^* + \text{jets}$ does still enter into the lower pNN bins in the signal region, large mismodelling can negatively affect the fit when extracting the signal. Therefore, two methods are utilised to reduce any potential bias from $Z/\gamma^* + \text{jets}$ mismodelling.

The first method involves k-factor corrections to the Drell-Yan MC samples. Leading-order (LO) Drell-Yan samples are used for this thesis, primarily driven by the significant increase in statistics of the LO samples versus the Next-to-leading-order (NLO) samples. However, differences in the reconstruction-level kinematic distributions are observed, which motivates the use of kinematic-dependent k-factors to correct for the discrepancies. Examples of these discrepancies are shown in figure 5.30. The k-factors are implemented as a binned 2-dimensional re-weighting using the generator-level Z momentum, p_T^Z , and generator-level invariant mass, M_Z , with the following binning:

$$\begin{aligned} P_T^Z &= [0, 20, 40, 60, 80, 100, 100+] \text{ GeV} \\ M_Z &= [0, 50, 50+] \text{ GeV}. \end{aligned} \tag{5.19}$$

The k-factors are presented in figure 5.31. A similar trend is observed for all years, with the largest k-factors around 20% for low values of p_T^Z . The reconstruction-level distributions with the k-factors applied are shown in figure 5.32, displaying minimal discrepancy between the NLO and LO samples after the corrections. Furthermore, associated k-factor uncertainties are also derived. These are defined as the envelope of the uncorrected distribution (no k-factor), the corrected distribution (with k-factor), and the doubly-corrected distribution (k-factors squared). In the signal region, this roughly corresponds to a maximum uncertainty of 10%, however generally this has little impact on the final sensitivity.

The second method involves a validation region, using a side-band region of the pNN output. This follows identical selections to the signal region, but with the requirement $0.8 < \text{pNN}(M_H, M_A) < 0.9$, which has no overlap with the signal region and negligible signal contamination. Example distributions of this validation region are shown in figure 5.33. Whilst not entirely pure in $Z/\gamma^* + \text{jets}$ events for most of the parameter space, especially for higher mass splitting samples, it is still possible to perform a fit — including all other control regions — to get an estimate on the mismodelling of the $Z/\gamma^* + \text{jets}$ normalisation. Across the whole parameter space, the maximum value of the $Z/\gamma^* + \text{jets}$ fit is around 30% higher than nominal, with an uncertainty of approximately 25%. Therefore, as a very conservative estimate, an uncertainty of 75% is assigned to the Drell-Yan normalisation for all points in the parameter space. Notably, however, even with a large uncertainty on the $Z/\gamma^* + \text{jets}$ samples, the impact on the sensitivity remains minimal.

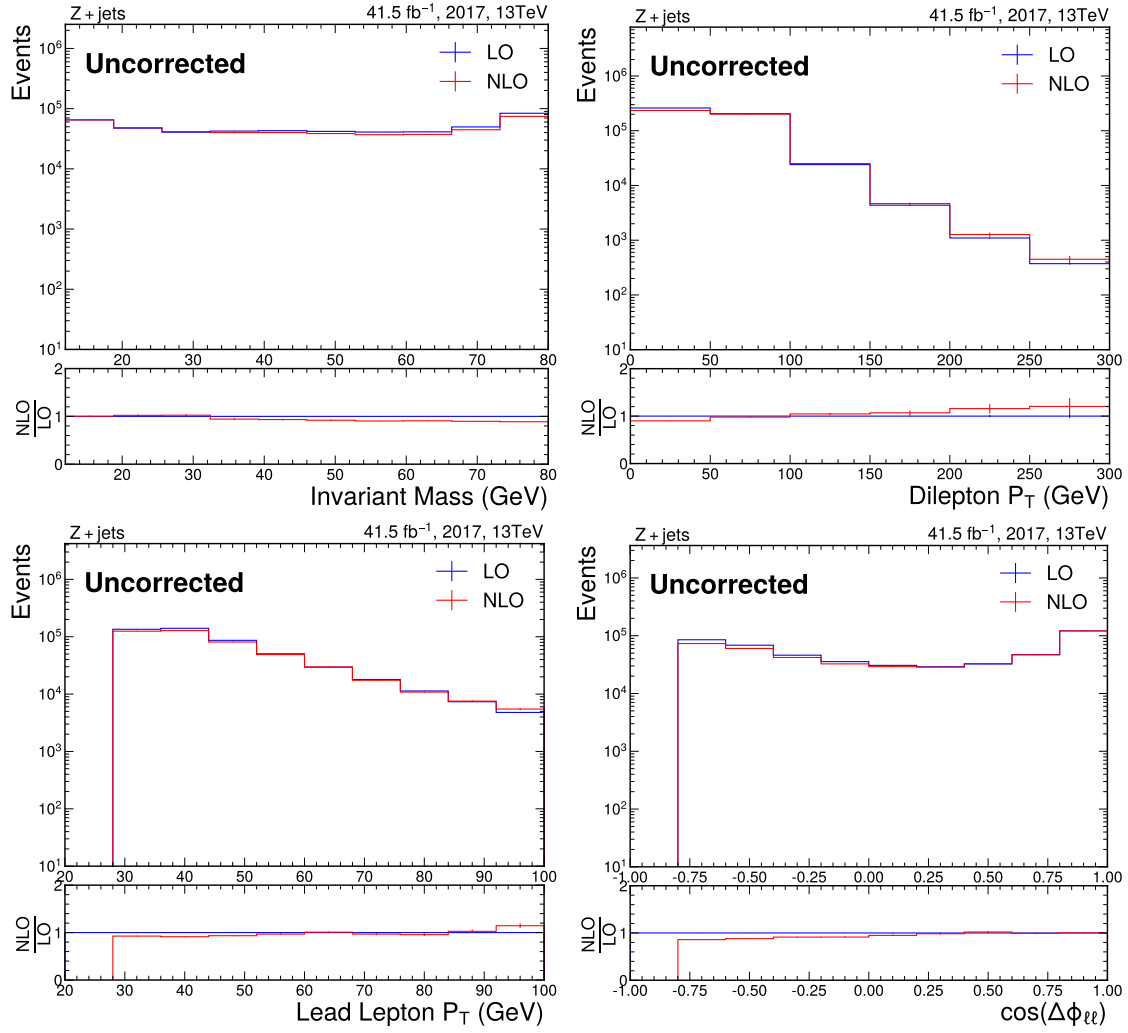
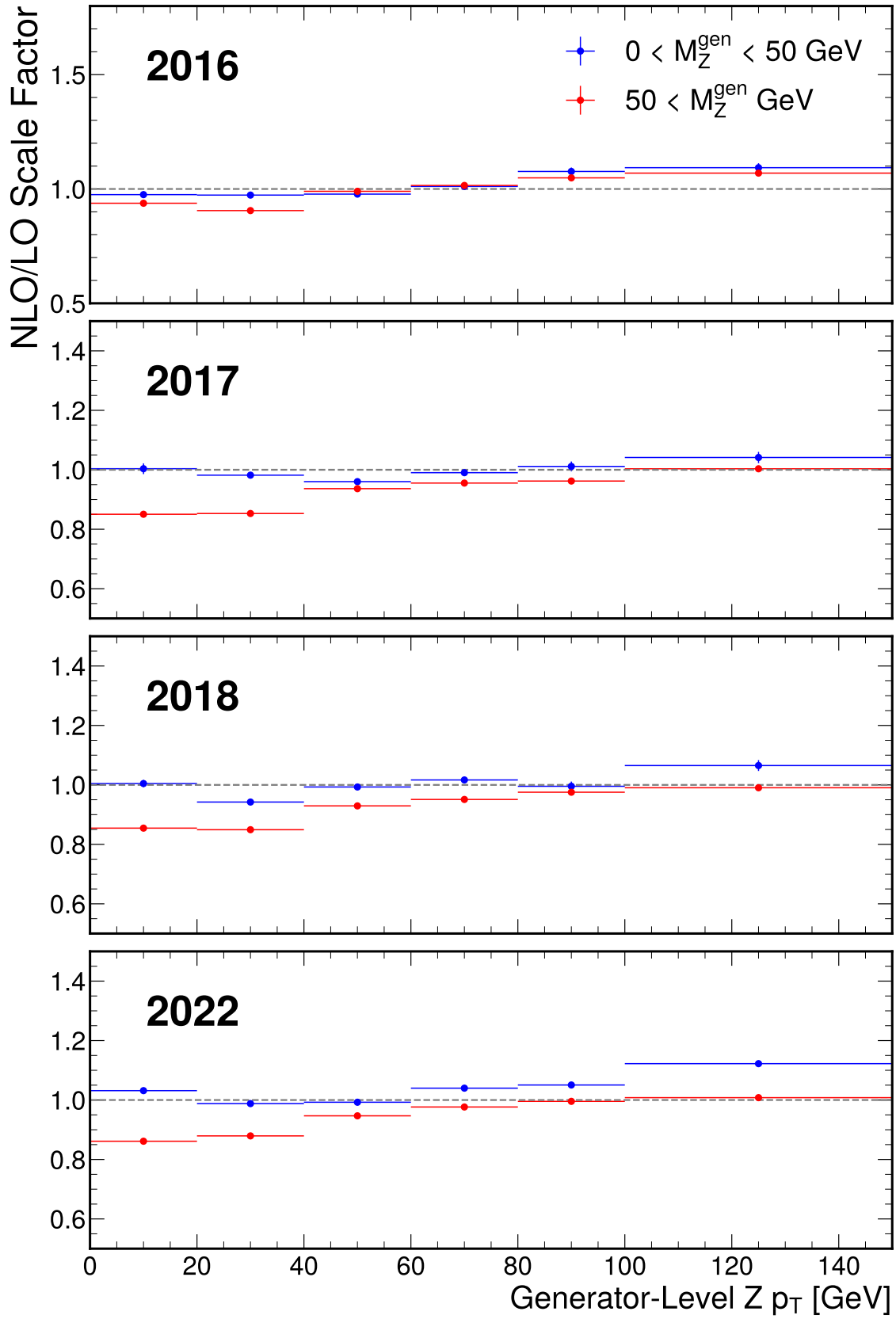


Figure 5.30: Distributions before the k-factor corrections of the dilepton invariant mass (top left), dilepton p_T (top right), lead lepton p_T (bottom left) and cosine of the ϕ angle between the leptons (bottom right) for the NLO and LO Drell-Yan samples for 2017. The ratio of the two distributions are shown in the bottom plots.



(a)

Figure 5.31: NLO/LO K-factors for all years, parameterised in terms of the generator-level $Z p_T$ and generator-level invariant mass, M_Z^{gen} . The last bin includes overflow.

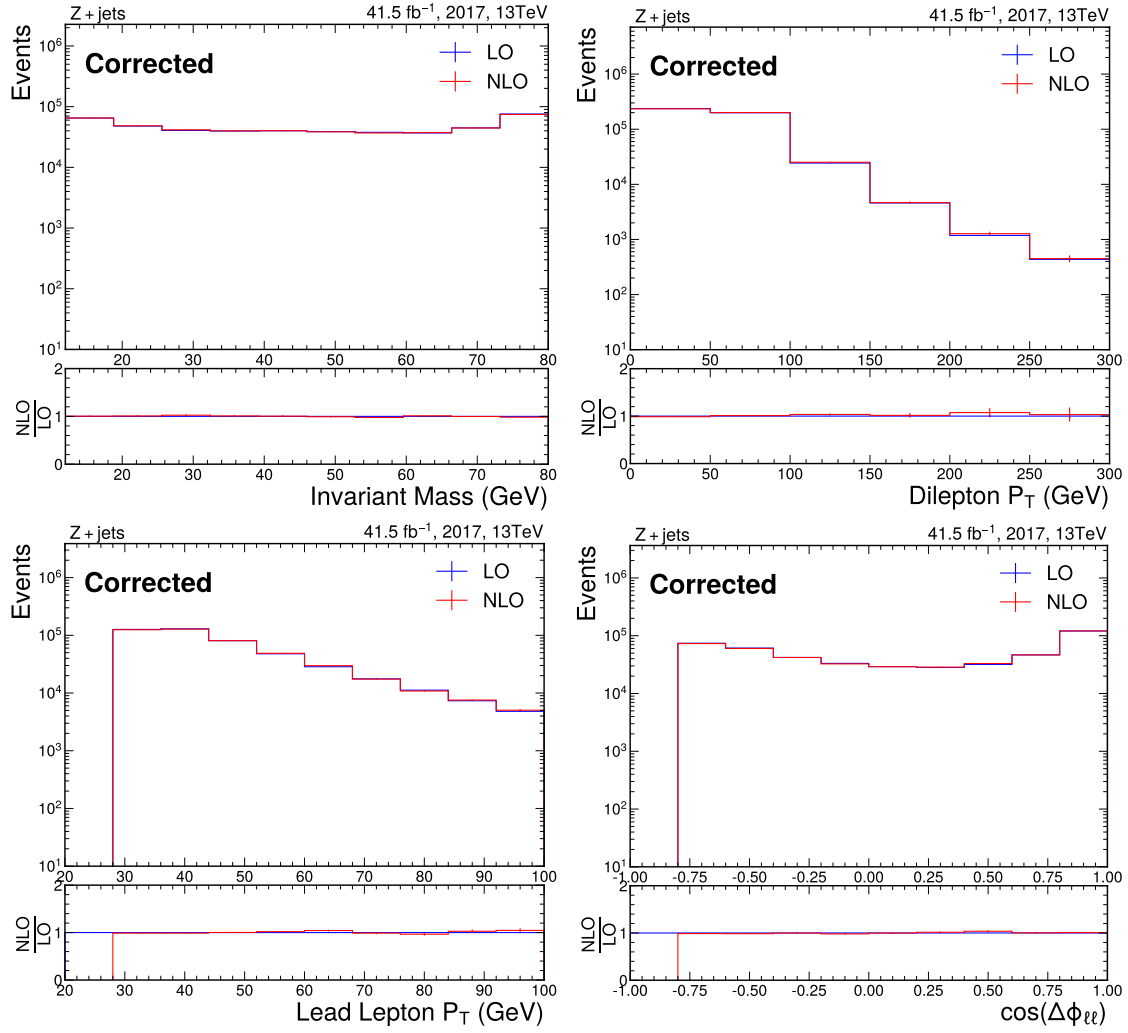


Figure 5.32: Distributions after the k-factor corrections of the dilepton invariant mass (top left), dilepton p_T (top right), lead lepton p_T (bottom left) and cosine of the ϕ angle between the leptons (bottom right) for the NLO and LO Drell-Yan samples for 2017. The ratio of the two distributions are shown in the bottom plots.

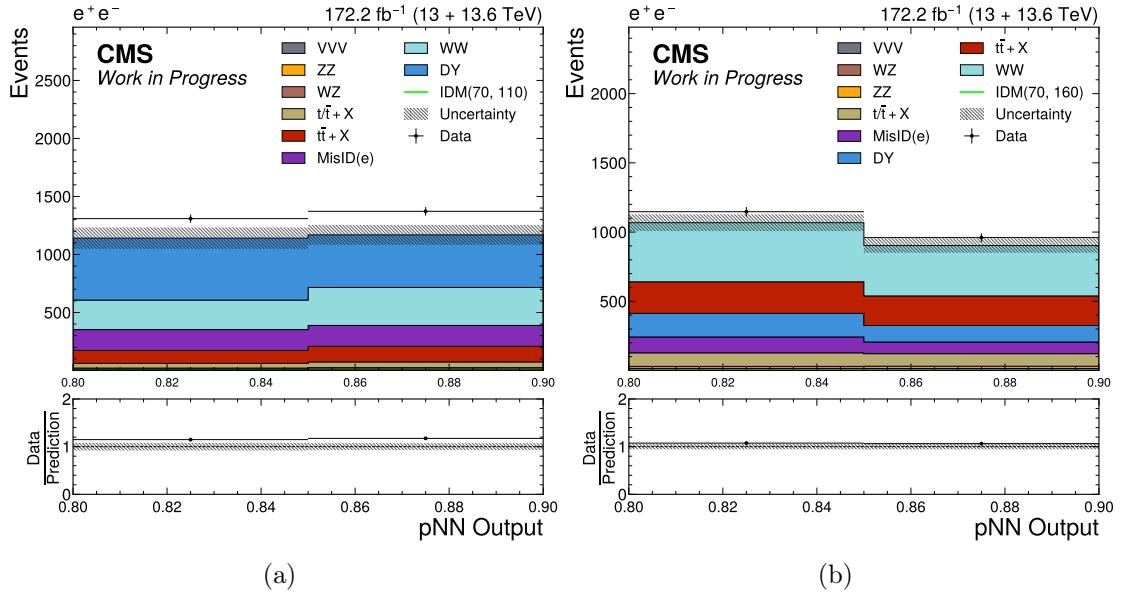


Figure 5.33: Distributions of the validation region for the $Z/\gamma^* + \text{jets}$ events for $(M_H, M_A) = (70, 110)$ GeV (left) and $(70, 160)$ GeV (right) for the e^-e^+ channel. This region is defined as the same selections as the signal region, but with $0.8 < \text{pNN}(M_H, M_A) < 0.9$.

5.8 MC Corrections and Uncertainties

Differences in the efficiencies, energy scales and energy resolutions between MC and data arise due to imperfect modelling in the simulations. This leads to discrepancies in the kinematic distributions which, as they are used as input into the pNN, can result in discrepancies in the pNN distributions. Corrections are, therefore, applied to MC to reduce this disparity. This section presents an overview of the corrections applied to MC, as well as the corresponding uncertainties. It should be noted that the stated uncertainties are calculated by finding the largest change in the individual bin yields in the signal regions. As such, these are specific to this analysis, and might not be representative for other analyses with different selections.

Lepton Corrections

Corrections are applied to leptons to account for mismodelling of the triggering, reconstruction, identification and isolation efficiencies [55]. An example of the electron reconstruction uncertainties on the pNN output for the $t\bar{t}$ process in the $e^\mp e^\pm$ final state is shown in figure 5.34(a). These are derived by the CMS Collaboration using data sets that are pure in $Z \rightarrow e^- e^+ / \mu^- \mu^+$ events. Subsequently, a tag-and-probe method is utilised, where one lepton is required to pass stringent selections, called the *tag lepton*, whilst the other lepton, called the *probe lepton*, is used for deriving the efficiency of a specified selection. By calculating the efficiencies in data and MC, corrections are derived to reduce the discrepancy. Uncertainties on these corrections are also computed, which are implemented as uncertainties to the shape of the distributions. In the signal region, for muons the uncertainties are of the order 0.5 – 1%, whereas for electrons they lie in the range 0.5 – 3%.

Jet Corrections

Corrections to the energy scale and energy resolution of jets are calculated by the CMS Collaboration to match data [57]. Within the barrel region, these are at the sub-percent level, rising to a few percent in the more forward regions. The uncertainties on these corrections are split into multiple sources, ranging from statistical uncertainties, to uncertainties in the QCD modelling. An example of the uncertainties on the pNN output from the jet energy scale uncertainties due to the PYTHIA fragmentation simulation is shown in figure 5.34(b) for the $t\bar{t}$ process in the $e^\mp e^\pm$ final state. Unlike the lepton efficiencies, however, the uncertainties on the jet energy scale and resolution can affect which events pass the selections, specifically the requirement of ≤ 1 jet with $p_T^j \geq 30$ GeV and $|\eta| < 2.4$. For example, if an event has two jets with $p_T=(33, 35)$ GeV and $\eta=(1, 2)$, then this would be rejected. After the energy corrections, however, this might be updated to $p_T=(28, 35)$ GeV, which would no longer be rejected by the jet veto selection. As a consequence, for every jet energy scale and resolution variation, there is a new set of MC which is passed through the whole analysis pipeline to give the expected change in the final distributions. Furthermore, for each set of events, the jet variations are propagated to the missing energy calculation. The magnitude of the uncertainties are dependent on the event topology,

but range from sub-percent to $\mathcal{O}(10\%)$. For Run 2 years (2016-18), there is an additional uncertainty on the jet pile-up identification, which runs at the sub-percent level.

Differences in the efficiencies for the b-tagging algorithm are also corrected by the CMS Collaboration. The uncertainties for which are decomposed depending on the flavour of jets, with separate uncertainties for the b-tagging of real b-jets or of non-b-jets. These are largely dependent on the specific process, with processes containing few b-jets, such as W^-W^+ and ZZ , having an uncertainty of less than a percent. As for processes commonly with b-jets, such as $t\bar{t}$ and single top events, these increase to approximately 0 – 3%.

Pileup Corrections

All MC samples include simulated pileup. The distributions of the simulated pileup does not, however, exactly match that of the data. Therefore, the CMS Collaboration derives corrections that re-weight MC events to match the distributions seen in data. The associated uncertainties are dependent on the process topology, but generally lie in the range 0 – 5%.

Pre-firing

In 2016 and 2017, due to a shift in the timing, certain energy deposits in the ECAL with $|\eta| > 2.5$ were being incorrectly associated with the previous bunch crossing [46]. Therefore, in some cases the L1T would incorrectly accept the previous bunch crossing when the current bunch crossing was the one of interest. This would thus lead to a drop in efficiency for events with high p_T electrons or jets. This is referred to as *L1T Pre-firing*. The magnitude of this drop in efficiency was calculated in data by the CMS Collaboration, and the corrections applied to MC. The exact uncertainty depends on the process, but averages out to approximately 1%.

Luminosity

Uncertainties in the luminosity calculation can affect the normalisations of processes that are not constrained by data, such as for signal or VVV processes. These are given by an uncertainty of 1%, 2%, 1.5% and 1.4% for 2016, 2017, 2018 and 2022, respectively, and additional uncertainties that are correlated between years that are of magnitude 0.6 – 2%.

Drell-Yan K-factor

As outlined in section 5.7.5, uncertainties for the Drell-Yan k-factor corrections are also derived, which affect the shape of the Drell-Yan distributions. Whilst the corrections before the pNN can be of the order 20%, in the signal region this falls into the range 0 – 10%. Additionally, an uncertainty of 75% on the Drell-Yan normalisation is implemented, which is derived using a side-band validation region.

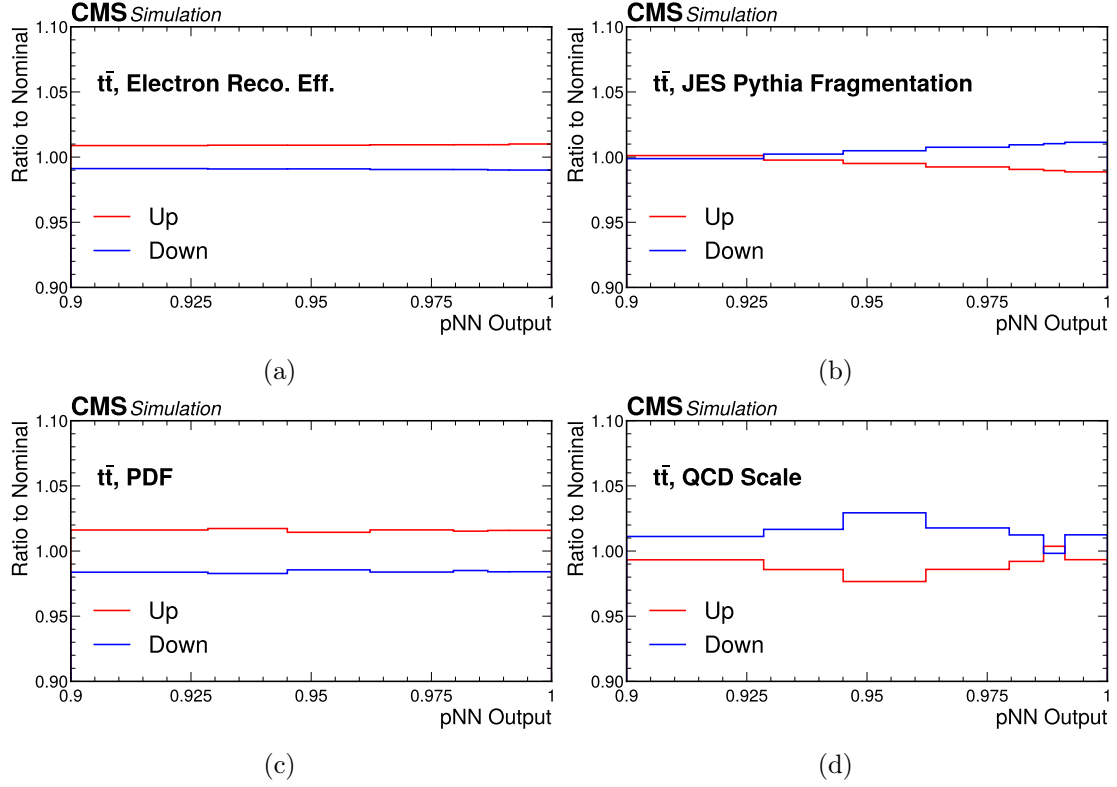


Figure 5.34: Example distributions of systematic uncertainties for $(M_H, M_A) = (70, 120)$ GeV for the e^-e^+ channels for $t\bar{t}$. All distributions are shown relative to the nominal distribution, with the up/down variations corresponding to the $\pm 1\sigma$ variations. Shown in the order: (a) electron reconstruction efficiency uncertainties; (b) PYTHIA fragmentation uncertainty to the jet energy scale; (c) PDF uncertainties; and (d) QCD scale uncertainties.

Theory Uncertainties

Uncertainties are also associated with the incorrect modelling of the underlying, generator-level, physics process. These include uncertainties in the modelling of the parton PDFs, as well as the uncertainties from finite-order simulations of QCD processes. An example of these uncertainties on the pNN output for the $t\bar{t}$ process in the $e^\mp e^\pm$ final state are shown in figure 5.34(c) and 5.34(d). These are dependent on the topology of the process, but are generally 0-3% for the PDF uncertainties, and 0-4% for the QCD uncertainties.

Statistical Uncertainties

Statistical uncertainties are associated with each process and each bin due to the finite number of MC events created. These uncertainties are included following the Barlow-Beeston method, which is outlined in References [113, 120]. Notably, as stated in section 5.4.4, the statistical uncertainties of the signal samples cover the uncertainties of the signal interpolation.

5.9 Results

The post-fit distributions for the signal region are presented in figures 5.35 and 5.36 for a range of mass points, with the stacked SM contributions obtained from the background-only fit. Generally, the observed data is found to be consistent with the background-only hypothesis ($\mu = 0$), with a maximum observed significance of 1.5σ . In addition, the entire set of post-fit distributions for the control regions for mass points $(M_H, M_A) = (70, 120)$ GeV and $(M_H, M_A) = (70, 160)$ GeV are presented in figures 5.37-5.40. It should be noted that, whilst these two example points both have $M_H = 70$ GeV, because the signal kinematics are almost entirely dominated by the mass splitting, $M_A - M_H$, these points are still representative of points with different M_H .

Overall, the SM predictions show good agreement with the data, with the simultaneous fit successfully correcting for the discrepancies observed in the pre-fit distributions. The normalisation factors from the fit for the dominant backgrounds are dependent on the year, but a rough outline of their magnitude and uncertainties are given in the following. Over the entire parameter space, the W^-W^+ normalisation factor is generally consistent with a value of 1.0 ± 0.1 . For $t\bar{t}$, an under-prediction is observed, leading to a consistent value of approximately 1.1 ± 0.1 . Whereas, the ZZ normalisation factor is dependent on the mass point, starting at approximately 0.9 ± 0.2 at $M_A - M_H = 20$ GeV, rising to roughly 1.1 ± 0.2 at $M_A - M_H = 90$ GeV. The largest excess is seen for WZ , which consistently has a factor of approximately 1.25 ± 0.25 . Finally, the normalisation factors for the MisId estimates are dependent on the channel, however these generally fall within the range 0.75-1.25 with an uncertainty of the order 30%.

Following the procedure described in the previous section, 95% CL upper limits are set on the two IDM parameters, M_H and M_A . These are derived with the Asymptotic approximation [116] using the COMBINE statistical analysis tool [121]. The expected and observed limits are presented in figure 5.41 in terms of M_H and the mass splitting $\Delta(M_A, M_H) = M_A - M_H$. The current limits based on a LEP recast and relic density constraints are overlaid on top. The observed limits are found to be consistent with the expected limits, lying within the 68% CL intervals shown by the dotted red lines. The hatched area indicates regions of parameter space excluded by the search, which shows a maximum exclusion for the dark matter mass of $M_H = 108$ GeV for a mass splitting of approximately $M_A - M_H = 80$ GeV. The excluded region marks a substantial improvement over previous IDM constraints, underlining the importance of a dedicated analysis that is optimised to the IDM signature.

Towards smaller mass splittings, however, the sensitivity tapers off. This is a result of limited acceptance as this region produces softer leptons, and is compounded by poorer discrimination against $Z/\gamma^* + \text{jets}$. Conversely, at a mass splitting above M_Z , the dilepton invariant mass becomes on-shell, leading to a sharp drop in acceptance due to the requirement $12 < M_{\ell\ell} < 80$ GeV. It should be noted, however, that this selection was chosen due to having poor discrimination against $Z/\gamma^* + \text{jets}$ for values of the invariant mass above this

range. Finally, the limit weakens towards larger values of M_H as a result of the decrease in the production cross-sections.

Whilst the exact impact of each systematic uncertainty on the limit is dependent on the mass point, for nearly all points in the parameter space the dominant source of uncertainty affecting the limit is the ZZ normalisation. This is exemplified in table 5.6, which shows the 10 most dominant uncertainties for $(M_H, M_A) = (70, 160)$ GeV, including their post-fit values, and the impact each systematic has on the signal strength μ when the systematic is varied up and down by 1σ . The large affect of the ZZ uncertainties is primarily driven by two factors. First, ZZ is the dominant irreducible background, having similar kinematics to the signal. Second, the limited statistics in the ZZ control region — stemming from the low acceptance of four-lepton events — leads to weak constraints on the ZZ normalisation that are directly propagated to the limit. All other systematic uncertainties are subleading in comparison, producing relatively small shifts in the expected sensitivity to exclusion. As a consequence, future improvements are expected, primarily from tighter constraints on the ZZ yield—for example, through increased statistics in the ZZ control region from the full Run 3 dataset.

Rank	Uncertainty source	Category	$\hat{\theta}$	Impact on μ
1	ZZ norm. (2018)	Background norm.	1.31 ± 0.22	± 0.07
2	ZZ norm. (2022)	Background norm.	1.29 ± 0.26	± 0.06
3	ZZ norm. (2017)	Background norm.	1.02 ± 0.22	± 0.06
4	ZZ norm. (2016)	Background norm.	1.32 ± 0.24	± 0.06
5	WW norm. (2017)	Background norm.	1.19 ± 0.12	± 0.03
6	QCD scale (diboson)	Theoretical	-0.05 ± 1	± 0.02
7	Electron reconstruction eff.	Experimental	-0.1 ± 1	± 0.02
8	WW norm. (2017)	Background norm.	1.11 ± 0.10	± 0.02
9	WW norm. (2017)	Background norm.	1.05 ± 0.10	± 0.02
10	Electron identification eff.	Experimental	-0.01 ± 1	± 0.01

Table 5.6: Dominant nuisance parameters affecting the signal strength extraction for $M_H = 70$ GeV and $M_A = 160$ GeV. The post-fit values $\hat{\theta}$ are with respect to their pre-fit values, and the impact corresponds to the change in the fitted signal strength μ when the nuisance parameter is shifted by $\pm 1\sigma$, with all other parameters profiled. The normalisation of a process is referred to as "norm.".

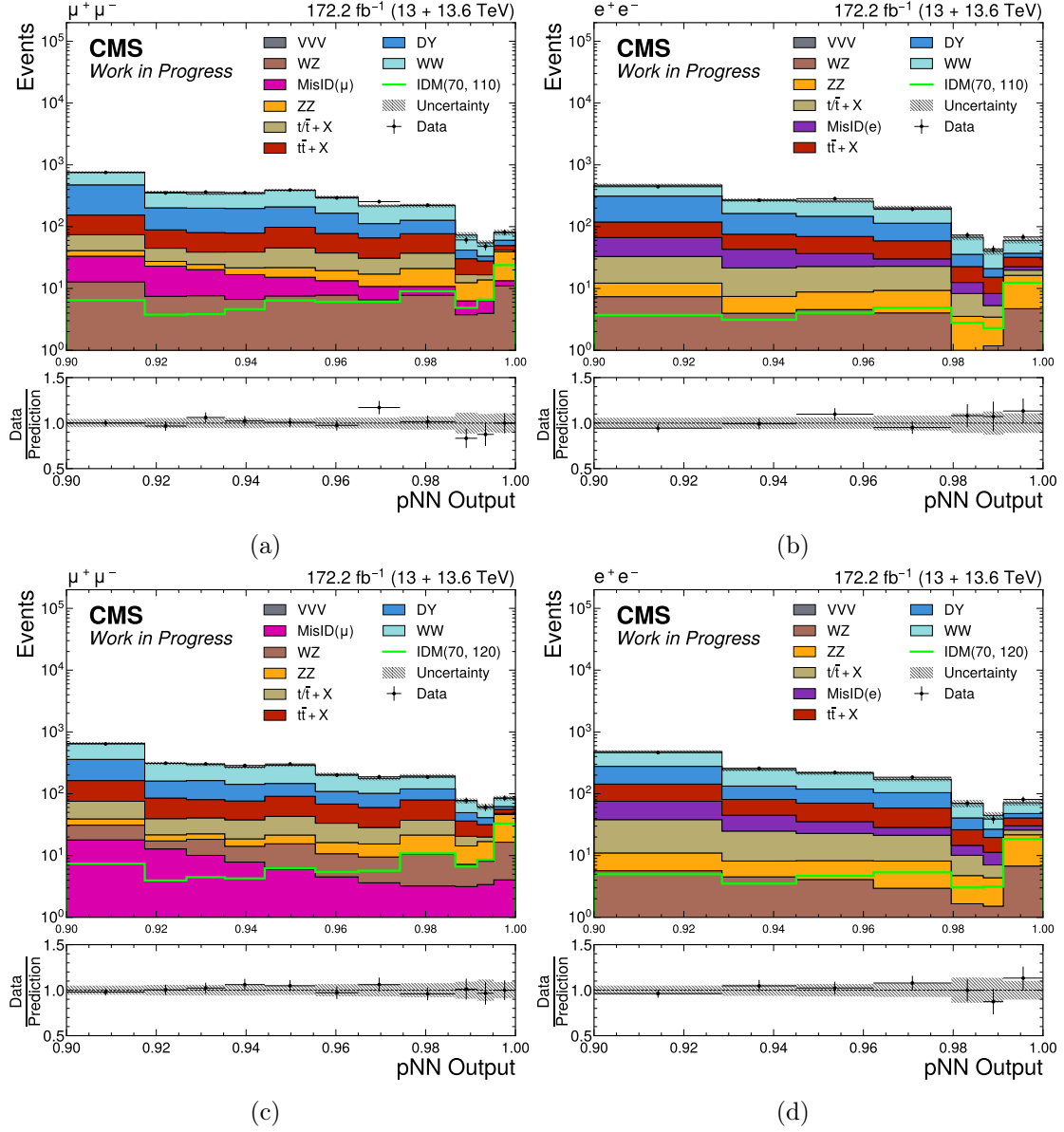


Figure 5.35: Post-fit signal region distributions for $(M_H, M_A) = (70, 110)$ GeV (top) and $(70, 120)$ GeV (bottom) for the $\mu^-\mu^+$ (left) and e^-e^+ (right) channels, respectively. The uncertainty shown includes both statistical and systematic uncertainties.

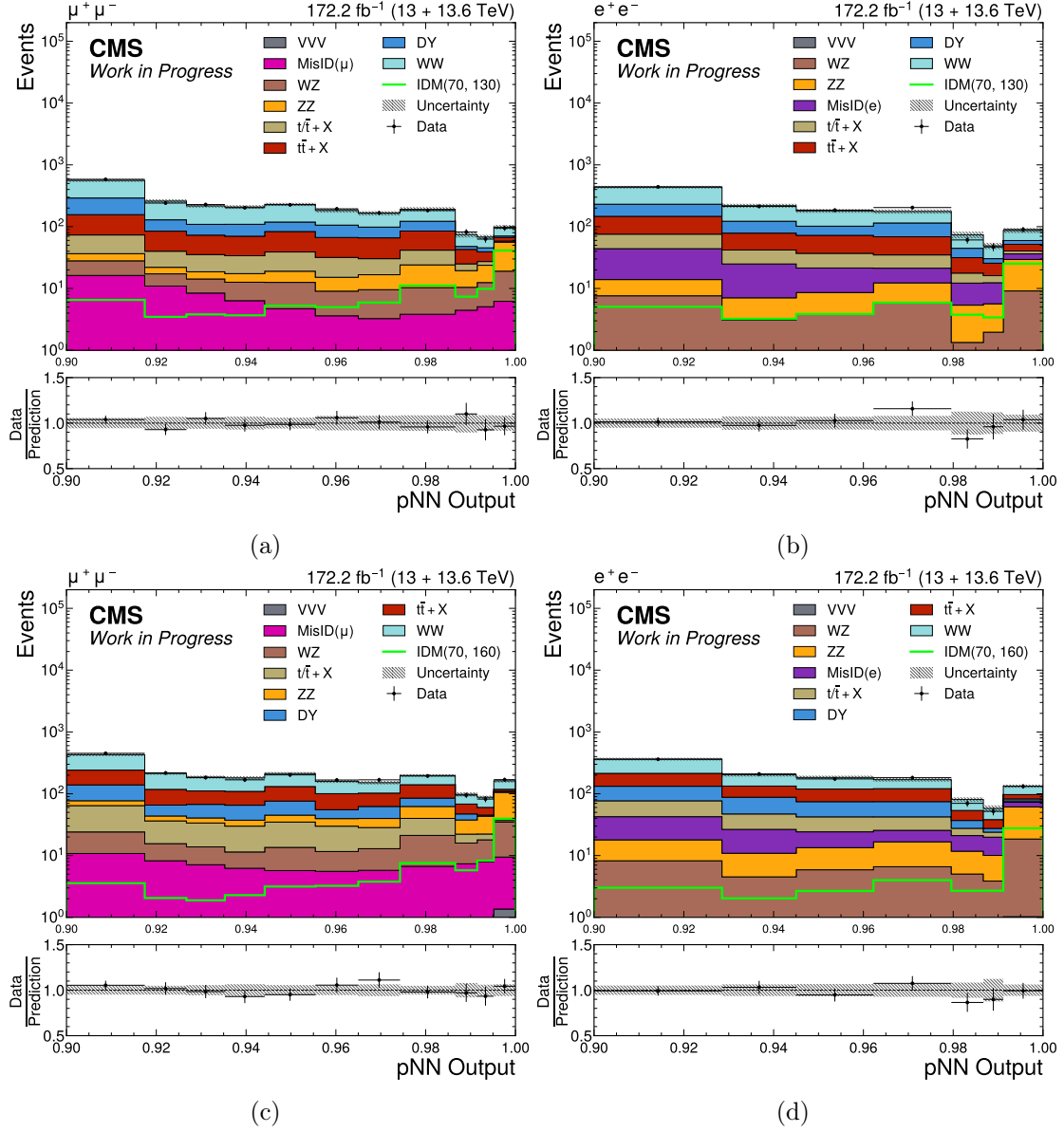


Figure 5.36: Post-fit signal region distributions for $(M_H, M_A) = (70, 130)$ GeV (top) and $(70, 160)$ GeV (bottom) for the $\mu^-\mu^+$ (left) and e^-e^+ (right) channels, respectively. The uncertainty shown includes both statistical and systematic uncertainties.

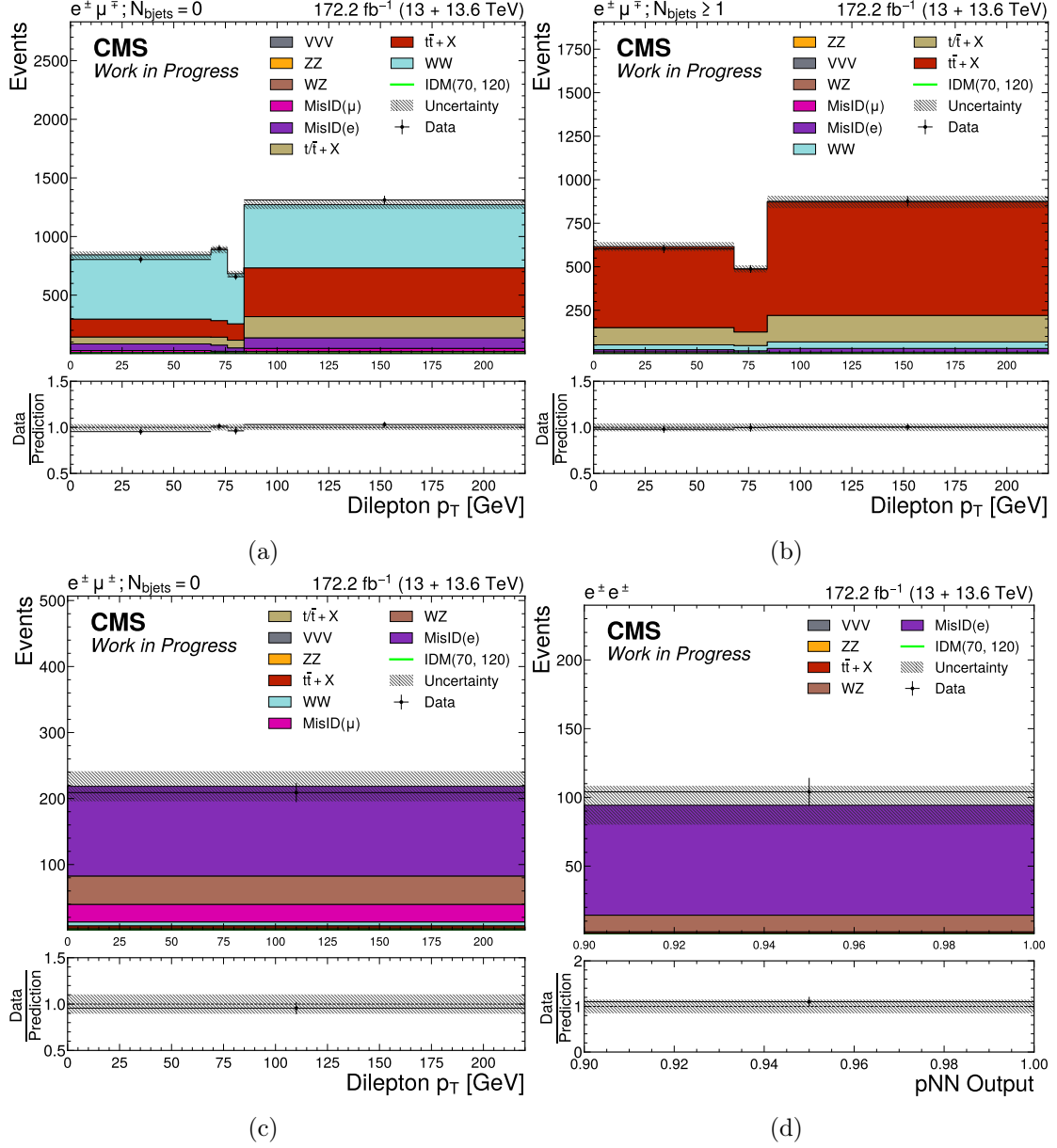


Figure 5.37: Post-fit distributions for $M_H = 70$ and $M_A = 120$ GeV. Control regions shown in the order: $W^- W^+ / t\bar{t}$ CR (top left), $t\bar{t}$ b-tagged CR (top right), $e^+ \mu^\pm$ MisId CR (bottom left) and $e^+ e^+$ CR (bottom right). The uncertainty shown includes both statistical and systematic uncertainties.

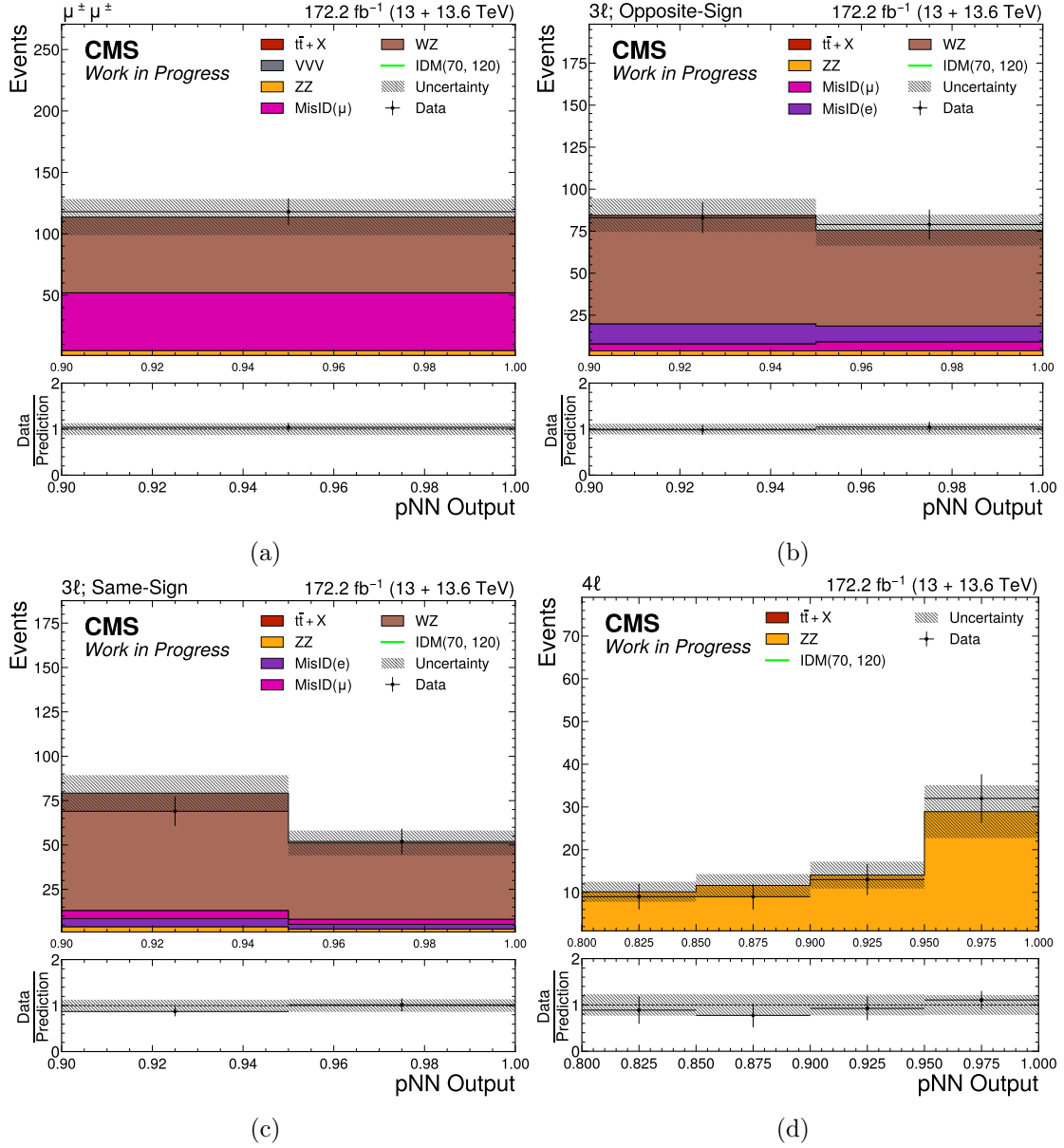


Figure 5.38: Post-fit distributions for $M_H = 70$ and $M_A = 120$ GeV. Shown in the order: $\mu^\pm \mu^\pm$ CR (top left), opposite-sign WZ CR (top right), same-sign WZ CR (bottom left) and ZZ CR (bottom right). The uncertainty shown includes both statistical and systematic uncertainties.

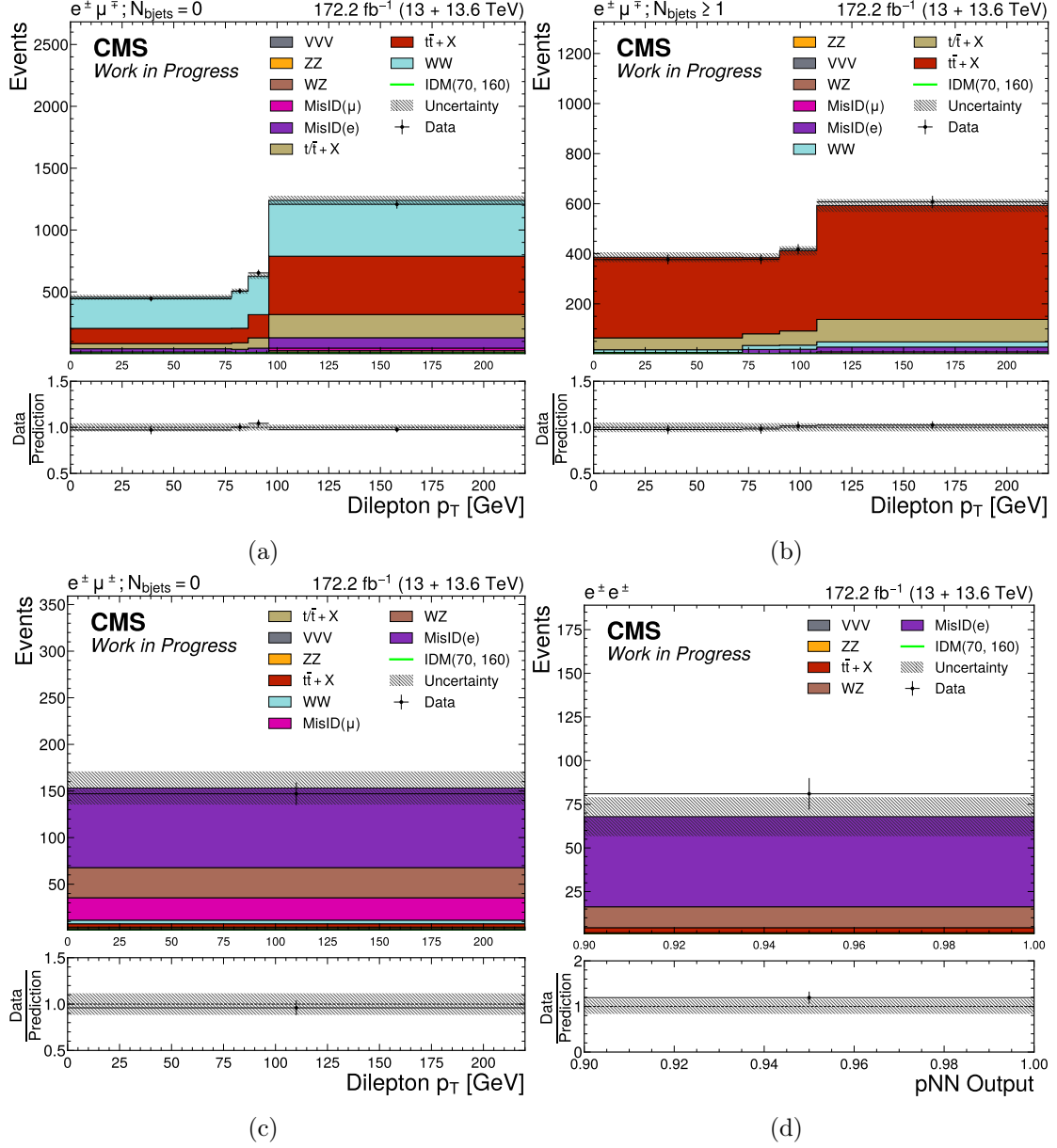


Figure 5.39: Post-fit distributions for $M_H = 70$ and $M_A = 160$ GeV. Control regions shown in the order: $W^- W^+ / t\bar{t}$ CR (top left), $t\bar{t}$ b-tagged CR (top right), $e^\pm \mu^\pm$ MisId CR (bottom left) and $e^\pm e^\pm$ CR (bottom right). The uncertainty shown includes both statistical and systematic uncertainties.

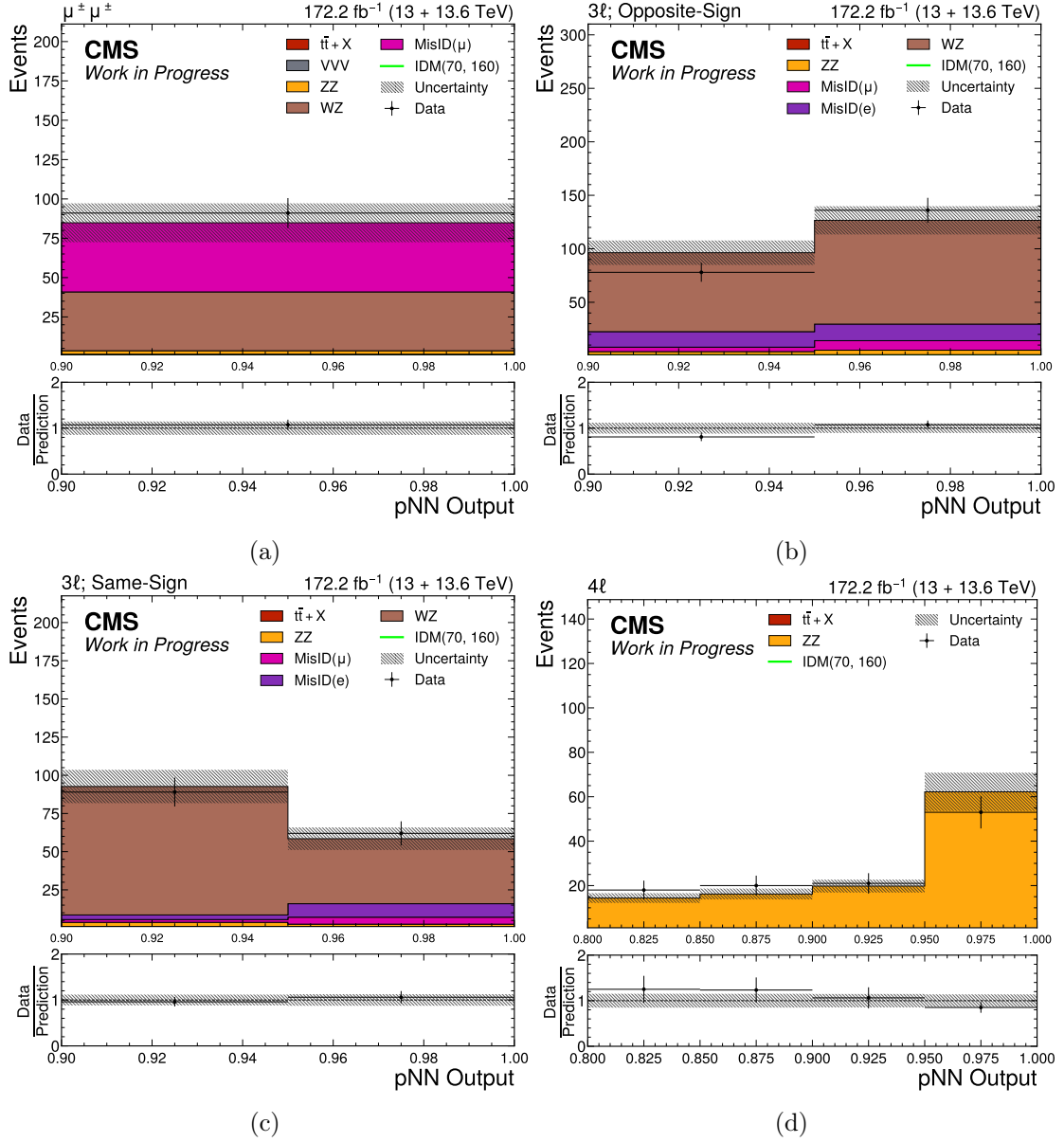


Figure 5.40: Post-fit distributions for $M_H = 70$ and $M_A = 160$ GeV. Shown in the order: $\mu^+\mu^-$ CR (top left), opposite-sign WZ CR (top right), same-sign WZ CR (bottom left) and ZZ CR (bottom right). The uncertainty shown includes both statistical and systematic uncertainties.

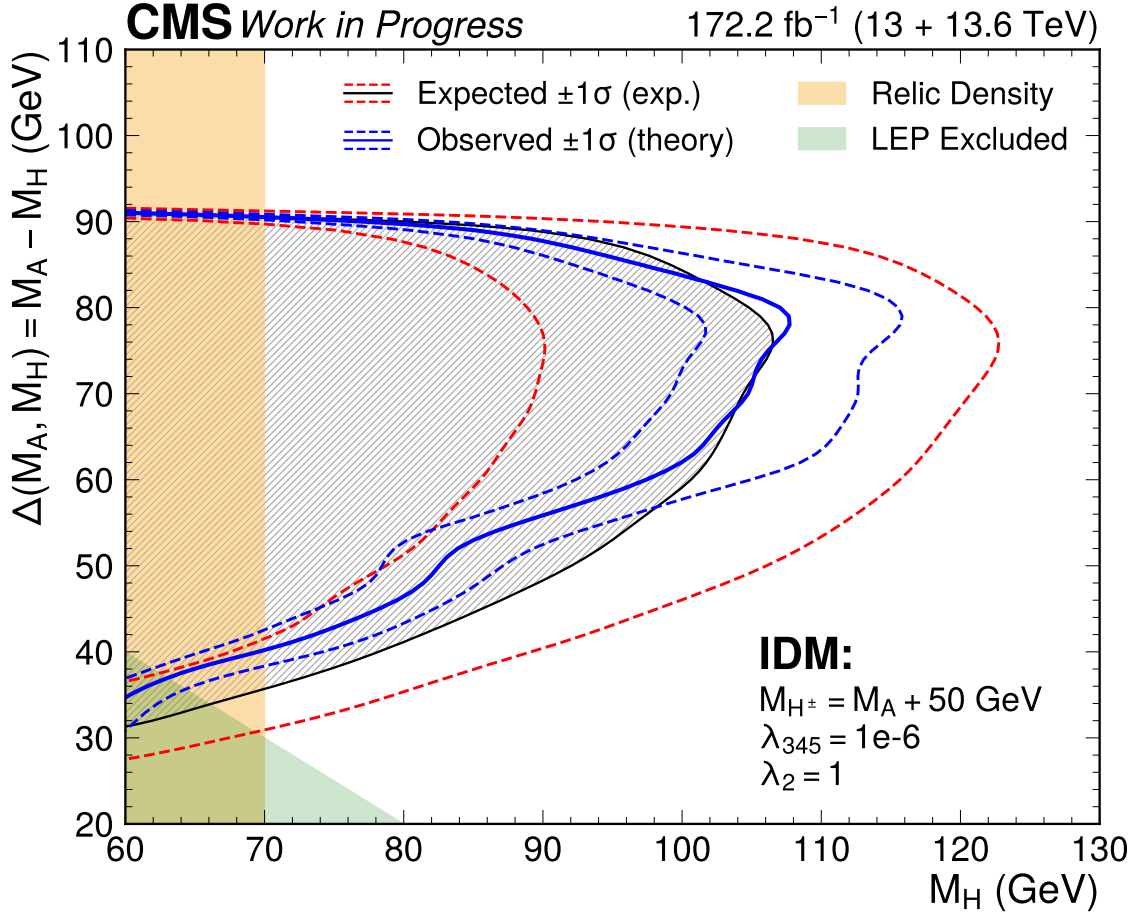


Figure 5.41: 95% CL exclusion limits in terms of the dark matter mass, M_H , and the mass splitting between the A and the H : $\Delta(M_A, M_H) = M_A - M_H$. The red dashed lines indicate the $\pm 1\sigma$ bands from experimental uncertainties, whilst the blue dashed lines indicate the $\pm 1\sigma$ bands from theory uncertainties on the signal samples. The exclusion limits from a LEP recast and relic density constraints are overlaid in green and yellow, respectively.

Chapter 6

Search for Inert Scalars at the FCC in the Dilepton Final State

Contents

6.1	Introduction	133
6.2	Samples	135
6.3	Object Selections	136
6.4	Event Selection	137
6.5	Expected Limits	142

6.1 Introduction

As the LHC transitions towards its final stage of operation, the High Luminosity LHC, the question remains of what experiments should follow. Whilst the successes of the LHC constitute a major achievement for high-energy particle physics — such as the discovery of the Higgs boson [8, 9]— it has yet to discover evidence of new physics, and leaves many important questions unanswered. The situation is made more complicated by the lack of evidence to guide the design of the next collider: possible signatures of new physics could lie in the extremely high-mass region over 100 TeV, or could be hiding in the low-mass, feebly-coupled region. The vast range of possible signatures thus requires a broad and flexible physics program to fully probe the possible parameter space. One such proposal for the next-generation of collider is the Future Circular Collider (FCC) [122–124].

The FCC is a proposed accelerator to be situated around the CERN complex, consisting of a ring with a circumference of approximately 91 km. This will operate in two primary stages, with the first stage, referred to as the *FCC-ee*, colliding electron-positron pairs at a maximum centre-of-mass energy of 365 GeV. Whilst the second stage, referred to as the *FCC-hh*, will involve proton-proton collisions at a centre-of-mass energy up to 100 TeV. The two-stage operation of the FCC is built upon the success and synergy between lower-energy precision experiments and higher-energy colliders, with precision measurements

historically playing an important role in particle discoveries. This is because predictions, such as a particle’s decay width, can be sensitive to higher order corrections involving heavier particles, even to those with far larger masses than are accessible at the centre-of-mass energy. As a result, discrepancies between predictions and measurements can indicate the presence of new physics with a mass reach far beyond the production capability of the collider itself, and with the reach extended from improvements in precision. Examples of such cases include the precision measurements of the W and Z bosons at LEP, which in turn predicted a range for the top quark mass before it was observed at the Tevatron collider [125–127]. The power of a precision experiment, therefore, extends beyond studying the SM itself, but also acts as a vital tool for particle discovery. The two-stage plan for the FCC leverages this synergy, extending both the precision and energy frontiers well beyond current experiments, whilst having the flexibility to cover a broad range of physics signatures. The focus of this chapter is on the first stage, the FCC-ee.

Colliding electron-positron pairs at a maximum centre-of-mass energy up to 365 GeV, the FCC-ee constitutes a significant increase over the previous high-energy lepton collider, the LEP experiment, which peaked at 209 GeV [125]. To enable precise measurements of different SM sectors, the FCC-ee will run in stages with differing centre-of-mass energies, eventually covering the full electroweak, Higgs and top sectors. This begins with a centre-of-mass energy around the Z pole at approximately 91 GeV, followed by the WW threshold at 160 GeV. Subsequently, this will be increased to the maximum Higgs production energy at 240 GeV, and finally the $t\bar{t}$ threshold at 365 GeV. Furthermore, the FCC intends to operate at luminosities multiple orders of magnitude above that of LEP, thus opening up the possibility to measure rarer SM processes that are currently inaccessible.

Despite the FCC-ee being designed primarily for precision measurements, it also provides multiple advantages for direct searches of new physics. Notably, lepton colliders operate with significantly fewer pileup events and a negligible QCD background, resulting in a cleaner environment when reconstructing physics objects. This is especially useful for processes involving low-mass resonances or soft leptons that would usually be hidden in the QCD background at the LHC. Furthermore, in contrast to hadron colliders, where the partons carry a fraction of the beam energy, the electrons have a well-defined initial state. Consequently, access to the full event 4-momentum provides superior performance in kinematic discrimination between signal and background, thus enhancing sensitivity. Finally, this is further enhanced by the high integrated luminosity expected at the FCC-ee, which enables searches for rarer, weakly coupled, BSM processes.

The varying conditions of the FCC-ee program, and the range of possible experimental searches, creates considerable challenges when designing a detector, requiring technological development beyond the current state-of-the-art. For one, an experiment is required to deal with the high luminosity and extreme event rate during the Z-pole measurements at $\sqrt{s} = 91$ GeV, forcing stringent requirements on the computing power to measure, store and process data. Furthermore, physics goals impose additional demands: many Higgs boson properties rely on ultra-precise measurements of lepton momentum [128]; excellent

flavour tagging is required to fully exploit $t\bar{t}$ events; fine calorimeter granularity is needed for effective particle flow reconstruction. Ultimately, meeting all demands within a single detector is likely not possible, however with four interaction points planned for the FCC-ee, a range of detector designs is already expected. As a baseline for studies on the physics performance at the FCC, all simulated samples are passed through a fast detector simulation of the IDEA detector [129, 130] using DELPHES 3.5.1pre05 [73].

This chapter presents a study on the sensitivity to the Inert Doublet Model at the FCC-ee at centre-of-mass energies of $\sqrt{s} = 240$ GeV and $\sqrt{s} = 365$ GeV. The analysis is an overview of the full study given in Reference [1], for which the signal extraction method — using a parametric neural network — and limit setting was implemented solely by the author of this thesis. Given that the analysis is based on simulations with simplified assumptions, extensive hyper optimisations and systematic studies are not performed. Instead, it is only meant to provide an estimate of the potential sensitivity of the FCC-ee to the IDM. This study is performed using the KEY4HEP framework [131] and analysed with the FCCAnalyses package [132].

This chapter is presented as follows. First, an overview of the Monte-Carlo (MC) samples is given. This is followed by an outline of the object selections and a basic set of base-selections, before moving onto the parametric neural network. Finally, the last section details the expected limits on the IDM parameters when the FCC-ee is running at different centre-of-mass energies.

6.2 Samples

Samples for e^-e^+ collisions are produced at centre-of-mass energies of $\sqrt{s} = 240$ GeV and $\sqrt{s} = 365$ GeV, corresponding to the Higgs and $t\bar{t}$ FCC-ee runs, respectively. For a full description of the simulation process, refer to section 3.4. The current expected values for the total integrated luminosity at $\sqrt{s} = 240$ GeV is 10.8 ab^{-1} , and for $\sqrt{s} = 365$ GeV is 2.7 ab^{-1} [133]. Lower collision energies are not simulated as these have limited production of IDM scalars for the non-excluded parts of the parameter space. For the simulation of the detector response, all samples are processed using DELPHES 3.5.1pre05 [73] with the IDEA detector setup [129].

The IDM signal samples are produced using the FCCAnalyses package [132] to leading-order using MADGRAPH5_aMC@NLO v2.8.1 [97] with hadronisation performed using Pythia v8.2 [92]. Both the $e^-e^+ \rightarrow HH\ell^+\ell^-$ and $e^-e^+ \rightarrow HH\ell^+\ell^-\nu_\ell\bar{\nu}_\ell$ processes are simulated, where $\ell = e, \mu, \tau$. No constraints are placed on the internal propagators to ensure that all interferences are considered. The quartic coupling¹ between IDM particles, λ_2 , has no effect on the signal distributions at leading-order and is therefore set to a constant value of 1. As in chapter 5, a grid over M_H and M_A is created, with $M_H = 70$ (80) GeV to $M_H = 115$ (180) GeV in steps of 5 GeV for $\sqrt{s} = 240$ (365) GeV. For each point, $\Delta(M_A, M_H) = M_A - M_H$ is scanned over from $\Delta(M_A, M_H) = 2$ GeV to the kinematic

¹For all couplings and relevant diagrams refer to section 2.3.

limit, $2M_H = \sqrt{s}$, in steps of 2 to 5 GeV.

For each point in the $M_H - \Delta(M_A, M_H)$ grid, three scenarios are simulated to cover any effects from λ_{345} (the Higgs-IDM coupling) and $M_{H^\pm}$. Scenario 1 simulates all IDM signal points with both λ_{345} and $M_{H^\pm}$ set to their minimum values: $\lambda_{345} = 0$ and $M_{H^\pm} = M_A$. Scenario 2 replaces λ_{345} with its maximum value, $\lambda_{345}^{\max}$, after all constraints are accounted for, which is approximately given by $\lambda_{345} = 0.0018 \times (M_H - 72\text{GeV})$ [26]. Finally, Scenario 3 replaces $M_{H^\pm}$ to its maximum allowed value, $M_{H^\pm}^{\max}$, given the model constraints.

In total 500,000 events are produced per point in the parameter grid, with a generator-level requirement of $p_T^\ell > 0.5$ GeV. Typically, the cross-section uncertainty for each point is below 0.1%. Generally, moving from Scenario 1 to Scenario 2, there is negligible change in the cross-sections. This is expected because λ_{345} affects only the Higgs-IDM coupling which is suppressed given that the phase space under consideration has $M_H > M_h/2$. The largest changes are seen in Scenario 3, but only for the $HH\ell^+\ell^-\nu_\ell\bar{\nu}_\ell$ final state where the cross-section is driven by the value of $M_{H^\pm}$.

Background samples are produced for the following processes: diboson production, W^-W^+ and ZZ ; inclusive lepton production of e^-e^+ , $\mu^-\mu^+$ and $\tau^-\tau^+$; Higgs production through ZH ; and, for $\sqrt{s} = 365$ GeV, $t\bar{t}$ production. For the inclusive e^-e^+ , due to the divergence from the t-channel scattering of $e^-e^+ \rightarrow e^-e^+$, a lower limit is placed on the invariant mass of the final state to $M_{\ell\ell} > 30$ GeV. Both the diboson and the $t\bar{t}$ samples are produced using PYTHIA v8.306 [72], whereas the remaining samples are produced with WHIZARD v3.0.3 [134] interfaced with PYTHIA 6.4.28 [71].

6.3 Object Selections

All objects are defined in Delphes using a right-handed coordinate system with the origin at the centre of the detector, the z-axis along the beam direction, the y-axis vertically upwards, and the x-axis towards the centre of the FCC. The azimuthal angle, ϕ , is defined as the angle to the x-axis in the x-y plane, and the polar angle, θ , is defined with respect to the z-axis in the y-z plane.

All reconstructed electrons, muons and photons are required to have $p > 5$ GeV. Jets are re-clustered after the removal of all reconstructed electron and muon candidates. The clustering uses the Durham algorithm which combines all candidates into exactly two jet objects, and which combines the four-vectors of all candidates to produce the jet four-vector [135]. It should be noted, however, that the number of jets required at the end of the clustering algorithm has no effect on the analysis as a veto is applied on any events with a jet. Finally, the missing energy is defined which includes the dark matter particles, H , as invisible.

To simulate the effect of detector resolution, the calorimeter energy of all objects is smeared following

$$\sigma_E = \sqrt{E^2 \times c^2 + E \times s^2 + n^2}, \quad (6.1)$$

where E is the energy, $c = 0.005$ (0.01) is a constant term, $s = 0.03$ (0.3) GeV $^{\frac{1}{2}}$ is a sampling term and $n = 0.002$ (0.05) GeV is a noise term for the electromagnetic (hadronic) parts of the object [130].

6.4 Event Selection

Base-Selections

To enhance the sensitivity of the search, a set of base-selections are applied to reduce the main Standard Model backgrounds, with the selections inspired by previous generator-level studies for the CLIC accelerator [36, 136]. Each event is required to have exactly two opposite-sign same-flavour leptons, $e^-e^+/\mu^-\mu^+$. All subsequent selections are shown in table 6.1. For $\sqrt{s} = 240$ GeV, all dilepton pairs are required to have an invariant mass of $M_{\ell\ell} < 120$ GeV and a dilepton momentum of $|p_z^{\ell\ell}| < 70$ GeV. Whereas, for $\sqrt{s} = 365$ GeV, this is replaced with $M_{\ell\ell} < -\frac{9}{14} \times |p_z^{\ell\ell}| + 200$ GeV and $|p_z^{\ell\ell}| < 140$ GeV. Both selections are shown in figure 6.1 with the SM backgrounds and a set of representative IDM points overlaid.

The backgrounds are further reduced by requiring $E_T^{\text{miss}} > 5$ GeV which removes approximately 90% of $e^-e^+ \rightarrow e^-e^+, \mu^-\mu^+, \tau^-\tau^+$ with negligible impact on the signal samples. Furthermore, as few additional objects are expected for the signal samples, all events are removed if they contain additional leptons, a photon with $E > 5$ GeV or a jet. Additionally, for $\sqrt{s} = 240$ GeV, events are required to have $p_T < 80(60)$ GeV for the leading (subleading) leptons. Whilst for $\sqrt{s} = 365$ GeV these are raised to $p_T < 140(80)$ GeV for the leading (subleading) leptons. Finally, as the dilepton system recoils off the IDM DM particles, all events are required to have $p^l/E^l > 0.1$.

The base selections reduce the ZZ and W^-W^+ backgrounds down to approximately 1.5% and 1%, respectively. Furthermore, for dilepton production, $ee \rightarrow e^-e^+, \mu^-\mu^+, \tau^-\tau^+$, the base selections have an acceptance efficiency of $< 1\%$. Whilst for the signal, the base selections have varying efficiency that is dependent on the signal topology. At the lowest mass-splitting region, for scenario S1, there is only a 2% (0.1%) efficiency for the production of HHll (HHllvv) due to the soft leptons evading acceptance. For higher mass splittings, this rises to approximately 65% and 6% for HHll and HHllvv, respectively. The expected number of events for each of the background sources, and a set of representative IDM points, are given in tables 6.2 and 6.3 for $\sqrt{s} = 240$ GeV and $\sqrt{s} = 365$ GeV, respectively. In addition, example kinematic distributions for both centre-of-mass energies are shown in figure 6.2 and 6.3. For points with $M_A - M_H < M_Z$, the invariant mass distribution peaks at around $M_A - M_H$, similar to the distributions in the CMS analysis in chapter 5. Whereas, for points with $M_A - M_H > M_Z$, the Z becomes on-shell, therefore the invariant mass distribution always peaks around M_Z .

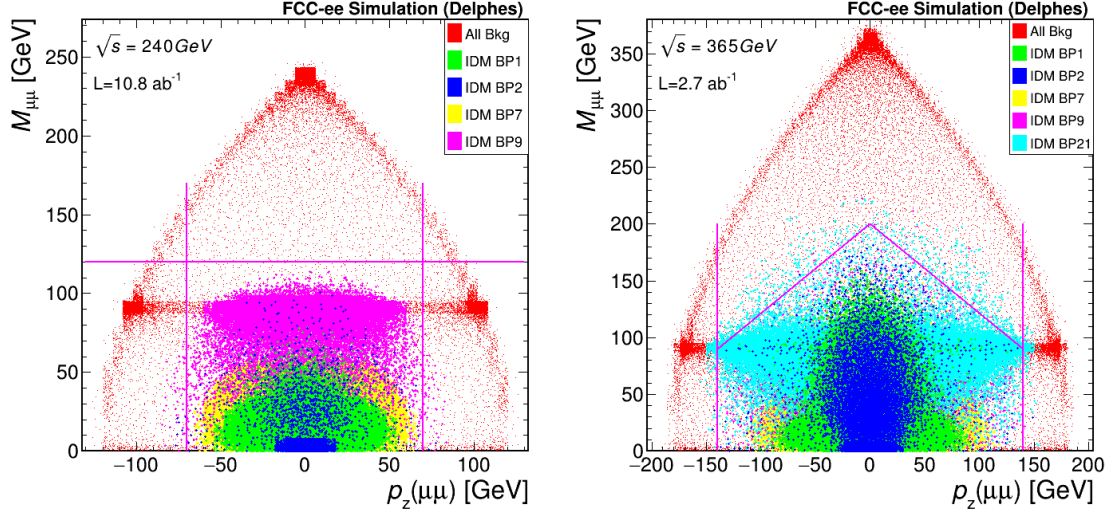


Figure 6.1: Dimuon momentum versus dimuon invariant mass for $\sqrt{s} = 240$ GeV (left) and $\sqrt{s} = 365$ GeV (right), with their corresponding base selections overlaid as the pink lines. All backgrounds are shown in red, and a set of representative IDM points are shown in green (BP1), blue (BP2), yellow (BP7) and pink (BP9) and cyan (BP21). The IDM parameters for the benchmarks points are described in Reference [36]. Figure taken from Reference [1].

Feature	240 GeV	365 GeV	Target Background
$ p_z^{\ell\ell} $ [GeV]	< 70	< 140	$ZZ, e^-e^+ \rightarrow \ell\ell$
$M_{\ell\ell}$ [GeV]	< 120	$< -\frac{9}{14} \times p_z^{\ell\ell} + 200$	$W^-W^+, e^-e^+ \rightarrow \ell\ell$
E_T^{miss} [GeV]	> 5	> 5	$ZZ, e^-e^+ \rightarrow \ell\ell$
$p_T^{l_1}$ [GeV]	< 80	< 140	$W^-W^+, ZZ, e^-e^+ \rightarrow \ell\ell$
$p_T^{l_2}$ [GeV]	< 60	< 80	$W^-W^+, ZZ, e^-e^+ \rightarrow \ell\ell$
$p^{\ell\ell}/E^{\ell\ell}$	> 0.1	> 0.1	$e^-e^+ \rightarrow \ell\ell$

Table 6.1: Base selections for $\sqrt{s} = 240$ GeV and $\sqrt{s} = 365$ GeV, with $\ell = e, \mu, \tau$.

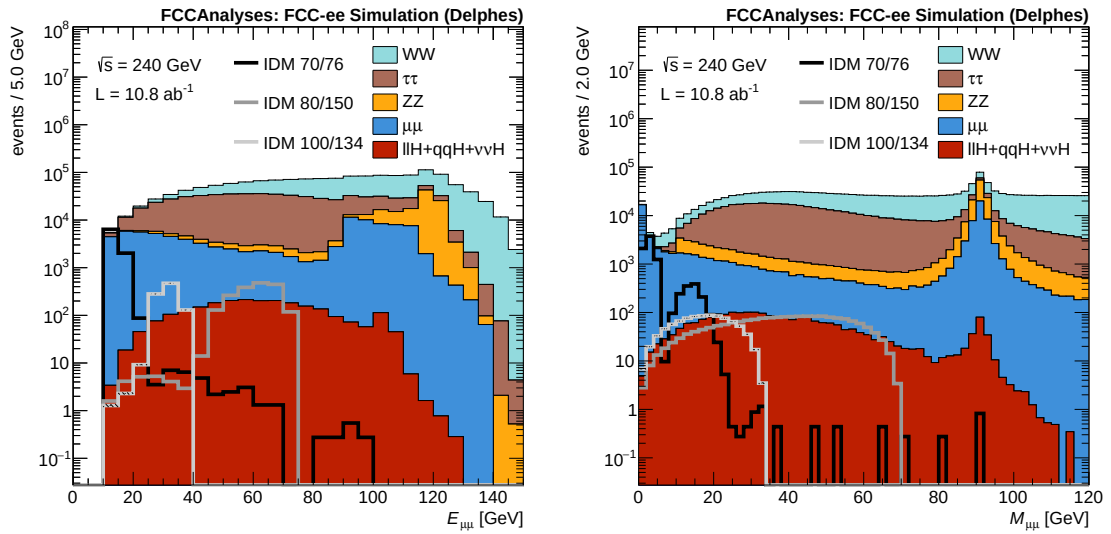


Figure 6.2: Distributions for the (left) dimuon energy, $E_{\mu\mu}$, and the (right) dimuon invariant mass, $M_{\mu\mu}$, for the SM backgrounds and three representative IDM mass points at $\sqrt{s} = 240$ GeV. The IDM points are given by IDM M_H/M_A with the masses in GeV.

Process	e^-e^+	$\mu^-\mu^+$
Background processes		
ZZ	$9.74 \times 10^4 \pm 1.60 \times 10^2$	$1.09 \times 10^5 \pm 1.69 \times 10^2$
W^+W^-	$8.02 \times 10^5 \pm 6.18 \times 10^2$	$8.72 \times 10^5 \pm 6.44 \times 10^2$
ee	$6.16 \times 10^5 \pm 8.05 \times 10^2$	0
$\mu\mu$	0	$9.80 \times 10^4 \pm 3.24 \times 10^2$
$\tau\tau$	$4.92 \times 10^5 \pm 6.88 \times 10^2$	$4.80 \times 10^5 \pm 6.80 \times 10^2$
$Z(ee)h$	$1.11 \times 10^2 \pm 2.68$	0
$Z(\mu\mu)h$	$6.09 \times 10^{-2} \pm 6.09 \times 10^{-2}$	$1.33 \times 10^2 \pm 2.85$
$Z(\nu\nu)h$	$2.15 \times 10^3 \pm 1.75 \times 10^1$	$2.25 \times 10^3 \pm 1.79 \times 10^1$
$Z(\tau\tau)h$	$1.62 \times 10^1 \pm 0.993$	$1.90 \times 10^1 \pm 1.08$
$Z(q\bar{q})h$	0	0
Representative signal mass points		
70–76 $HH\ell\ell$	$6.31 \times 10^3 \pm 4.16 \times 10^1$	$7.07 \times 10^3 \pm 4.41 \times 10^1$
70–76 $HH\ell\ell\nu\nu$	$1.24 \times 10^3 \pm 2.33 \times 10^1$	$1.45 \times 10^3 \pm 2.53 \times 10^1$
80–150 $HH\ell\ell$	$1.67 \times 10^3 \pm 4.55$	$1.86 \times 10^3 \pm 4.80$
80–150 $HH\ell\ell\nu\nu$	$3.06 \times 10^{-5} \pm 7.42 \times 10^{-6}$	$3.96 \times 10^{-5} \pm 8.44 \times 10^{-6}$
100–134 $HH\ell\ell$	$8.18 \times 10^2 \pm 2.25$	$8.94 \times 10^2 \pm 2.35$
100–134 $HH\ell\ell\nu\nu$	$1.14 \times 10^{-6} \pm 6.59 \times 10^{-7}$	$3.81 \times 10^{-7} \pm 3.81 \times 10^{-7}$

Table 6.2: Yields for all background samples and representative signal samples at $\sqrt{s} = 240$ GeV for ee and $\mu\mu$ final states.

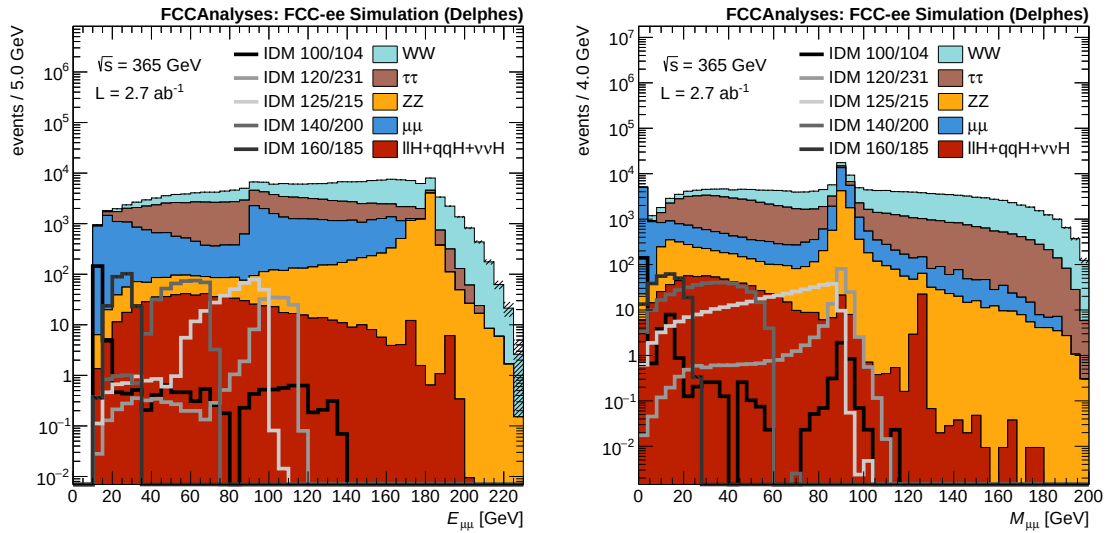


Figure 6.3: Distributions for the (left) dimuon energy, $E_{\mu\mu}$, and the (right) dimuon invariant mass, $M_{\mu\mu}$, for the SM backgrounds and three representative IDM mass points at $\sqrt{s} = 365$ GeV. The IDM points are given by IDM M_H/M_A with the masses in GeV.

Process	e^-e^+	$\mu^-\mu^+$
Background processes		
ZZ	$9.30 \times 10^3 \pm 3.75 \times 10^1$	$1.05 \times 10^4 \pm 3.98 \times 10^1$
W^+W^-	$8.75 \times 10^4 \pm 4.64 \times 10^2$	$9.32 \times 10^4 \pm 4.79 \times 10^2$
$t\bar{t}$	0	0
ee	$1.34 \times 10^5 \pm 4.30 \times 10^2$	0
$\mu\mu$	0	$2.86 \times 10^4 \pm 1.64 \times 10^2$
$\tau\tau$	$5.83 \times 10^4 \pm 1.57 \times 10^2$	$5.65 \times 10^4 \pm 1.55 \times 10^2$
$Z(ee)h$	$1.88 \times 10^1 \pm 0.613$	$9.98 \times 10^{-2} \pm 4.46 \times 10^{-2}$
$Z(\mu\mu)h$	$9.42 \times 10^{-3} \pm 9.42 \times 10^{-3}$	$1.74 \times 10^1 \pm 0.405$
$Z(\nu\nu)h$	$6.08 \times 10^2 \pm 6.34$	$6.72 \times 10^2 \pm 6.67$
$Z(\tau\tau)h$	$2.41 \times 10^0 \pm 0.157$	$2.11 \times 10^0 \pm 0.147$
$Z(q\bar{q})h$	0	0
Representative signal mass points		
100–104 $HH\ell\ell$	$1.09 \times 10^2 \pm 1.59$	$1.39 \times 10^2 \pm 1.80$
100–104 $HH\ell\ell\nu\nu$	$1.96 \times 10^1 \pm 1.00$	$1.96 \times 10^1 \pm 1.00$
120–231 $HH\ell\ell$	$1.25 \times 10^2 \pm 0.347$	$1.41 \times 10^2 \pm 0.368$
120–231 $HH\ell\ell\nu\nu$	$2.61 \times 10^{-4} \pm 7.59 \times 10^{-6}$	$2.07 \times 10^{-4} \pm 6.77 \times 10^{-6}$
125–215 $HH\ell\ell$	$3.14 \times 10^2 \pm 0.865$	$3.53 \times 10^2 \pm 0.918$
125–215 $HH\ell\ell\nu\nu$	$1.23 \times 10^{-2} \pm 1.14 \times 10^{-4}$	$9.92 \times 10^{-3} \pm 1.03 \times 10^{-4}$
140–200 $HH\ell\ell$	$3.42 \times 10^2 \pm 0.929$	$3.80 \times 10^2 \pm 0.979$
140–200 $HH\ell\ell\nu\nu$	$1.31 \times 10^{-2} \pm 1.21 \times 10^{-4}$	$1.06 \times 10^{-2} \pm 1.09 \times 10^{-4}$
160–185 $HH\ell\ell$	$2.19 \times 10^2 \pm 0.629$	$2.39 \times 10^2 \pm 0.656$
160–185 $HH\ell\ell\nu\nu$	$3.61 \times 10^{-3} \pm 4.81 \times 10^{-5}$	$2.83 \times 10^{-3} \pm 4.26 \times 10^{-5}$

Table 6.3: Yields for all background samples and representative signal samples at $\sqrt{s} = 365$ GeV for ee and $\mu\mu$ final states.

Parametric Neural Network

Following the base selections, a neural network is implemented to maximise the sensitivity of the search. However, given the variation in signal topologies, to optimally discriminate between signal and background over the whole parameter space, a Parametric Neural Network (pNN) is employed. By learning the correlations between the kinematic features and the signal hypotheses, a single network can not only apply the optimal selections based on the mass hypothesis, but can also naturally interpolate between the simulated mass points. The pNN utilised for this analysis follows the same construction and training methodology as in section 5.4.

The input features to the pNN include both low-level and high-level kinematic features, which are similar to those used in References [36, 136], as well as the two IDM masses² M_H and M_A . The following features are:

- Lead and sub-lead lepton features: $p_T^{l_{1,2}}, \eta^{l_{1,2}}, E^{l_{1,2}}, \text{charge}$.
- Cosine of azimuthal angle between the leptons, $\cos(\Delta\phi)$.
- The dilepton pair features: $E_{ll}, M_{ll}, p_T^{ll}, p_z^{ll}, \cos(\theta_{ll})$.
- The recoil mass assuming centre-of-mass energy of $\sqrt{s}$.
- The dilepton Lorentz boost p_{ll}/E_{ll} .
- Angle between l^- and the beam direction in the dilepton frame, $\cos(\theta^*)$.
- Angle between l^- and the dilepton boost direction in the dilepton frame, $\cos(\theta_R)$.

The pNN is implemented in PyTorch [88] and consists of four consecutive feed-forward layers containing 250 hidden units and a ReLU activation function, totalling approximately 200,000 learnable parameters. The last three layers are also supplemented with a drop out of 0.2. Finally, the output of the model is passed through a Sigmoid layer for binary classification. The model is optimised by reducing the weighted binary cross-entropy loss function, given in equation 5.5, using the Adam optimiser with a learning rate of 1e-4.

All samples are split into three datasets: training, validation and test, with fractions of the total dataset being 50%, 20% and 30%, respectively. The model is trained using the training dataset, with the validation dataset used to measure the model's performance during training. The final performance of the model, and the expected limits, are determined using the test dataset.

Before training, events are re-weighted such that the sum of the signal events equals the sum of all background events. Furthermore, to ensure that all signal samples are treated equally, the signal samples are re-weighted such that the sum of weights for each signal is equal. Moreover, all input features are standardised such that the mean of each feature is 0, and the standard deviation is 1.

²Only the two scalar masses are used because, for each scenario, the other parameters (λ_{345} , λ_2 and $M_{H^\pm}$) are kept fixed.

No distinction is made between di-electron and di-muon events, with a single network trained using both final states. Training is performed for a maximum of 100 epochs, with the training stopped if the validation loss does not decrease over 20 epochs. The final model is the model that achieved the lowest validation loss during the training procedure. This is repeated separately for each of the three sample scenarios. No overfitting was found between the training, validation and test datasets. Furthermore, the performance of the trained model is compatible with the performance of a BDT trained at each individual point.

The output distributions of the pNN that was trained on the Scenario 1 samples are shown in figures 6.4 and 6.5 for the $\sqrt{s} = 240$ and $\sqrt{s} = 365$, respectively. For both figures, the top row shows the full pNN output distribution for a low mass-splitting IDM sample and a higher mass-splitting sample. Whereas the bottom row shows the pNN output distribution above 0.9, matching the signal region that is defined in the following section. Notably, these figures highlight the large variation in signal topologies across the parameter space, with the lower mass-splitting signal points—such as $(M_H, M_A) = (70, 76)$ GeV—strongly peaking at a pNN output of 1. Conversely, higher mass-splitting points, such as $(M_H, M_A) = (80, 150)$ GeV, more closely resemble the WW process. Finally, to find the distributions at intermediate signal points that were not simulated, spline interpolation is utilised, following the same procedure as in section 5.4.4.

6.5 Expected Limits

To extract the limits, a maximum likelihood fit is performed using the CMS Combine package [121] using the asymptotic approximation [116]. The input to the fit is the pNN output above 0.9, which covers the maximum value of S/B for all signal mass points. Both the e^-e^+ and $\mu^-\mu^+$ final state distributions are fit simultaneously, with the only source of uncertainty being the statistical uncertainty of the samples. However, as no Monte-Carlo samples were produced for $e^-e^+ \rightarrow e^-e^+$ for $M_{ee} < 30$ GeV, the e^-e^+ final state is only included for $M_{\ell\ell} \geq 30$ GeV.

For $\sqrt{s} = 240$ GeV, the 95% CL exclusion limits are presented in figure 6.6(a) for each of the three scenarios in the $M_H - \Delta(M_A, M_H)$ plane. Both the Scenario 2 and Scenario 3 limits lie within the $\pm 1\sigma$ uncertainty bands for Scenario 1, indicating negligible effect from any kinematic differences between them. With 10.8 ab^{-1} of integrated luminosity, almost the entire parameter space is excluded up to $M_H = 110$ GeV, with an upper limit set by the on-shell kinematic limit shown by the dashed black line. This significantly extends the current constraints from LEP and from relic density observations that are overlaid in green and blue, respectively. Additionally, the $> 5\sigma$ discovery region is displayed in figure 6.6(b) for Scenario 1 for an integrated luminosity of 10.8 ab^{-1} . For the nominal FCC running scenario, the discovery reach extends up to approximately $M_H = 108$ GeV for a mass splitting of $M_A - M_H = 15$ GeV. Furthermore, the 5σ discovery contour at an integrated luminosity of 2 ab^{-1} is overlaid for comparison to an alternative linear lepton collider that operates at $\sqrt{s} = 250$ [137].

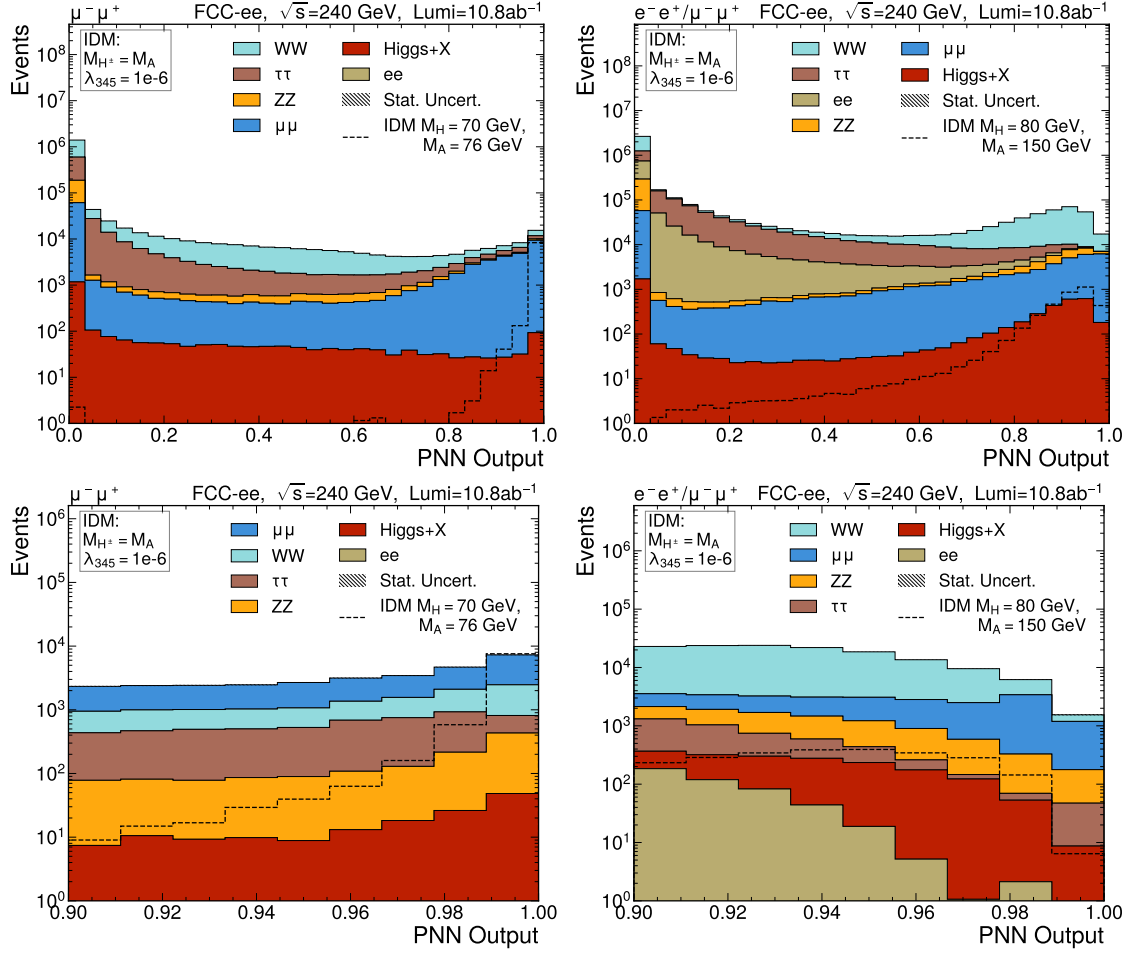


Figure 6.4: Distributions of the pNN output for two signal mass points at $\sqrt{s} = 240$ GeV. The top shows the full pNN output distributions whilst the bottom shows the signal region distributions. Note that for $M_{\ell\ell} \geq 30$ GeV, the e^-e^+ and the $\mu^-\mu^+$ channels are summed together, whilst for $M_{\ell\ell} < 30$ GeV only the $\mu^-\mu^+$ channel is used.

For $\sqrt{s} = 365$ GeV, the 95% CL exclusion limits and the $> 5\sigma$ discovery region are presented in figure 6.7 considering only the statistical uncertainties. Additionally, the 5σ discovery contour at a luminosity approximately a tenth of the nominal is overlaid for comparison to an alternative linear lepton collider which operates at $\sqrt{s} = 350$ GeV [137]. As in the $\sqrt{s} = 240$ GeV case, all scenarios lie within the $\pm 1\sigma$ uncertainty bands for Scenario 1. A decrease in the sensitivity is seen at a mass splitting of approximately $M_A - M_H = 90$ GeV due to the resonance around the Z boson mass. The 95% CL exclusion region indicates a maximal DM exclusion of roughly $M_H = 165$ GeV, whilst the discovery reach is shown to extend up to $M_H = 157$ GeV for a mass splitting of $M_A - M_H = 15$ GeV.

In comparison to the IDM limits presented in chapter 5 for the search using the CMS detector, the limits shown in figures 6.6 and 6.7 show improvements in two key areas. First, the limits set at the FCC-ee significantly extend down to the lower mass splitting region. This is a result of the relatively clean environment of the FCC-ee, which is free from the large QCD background of the LHC, enabling efficient reconstruction of low momentum

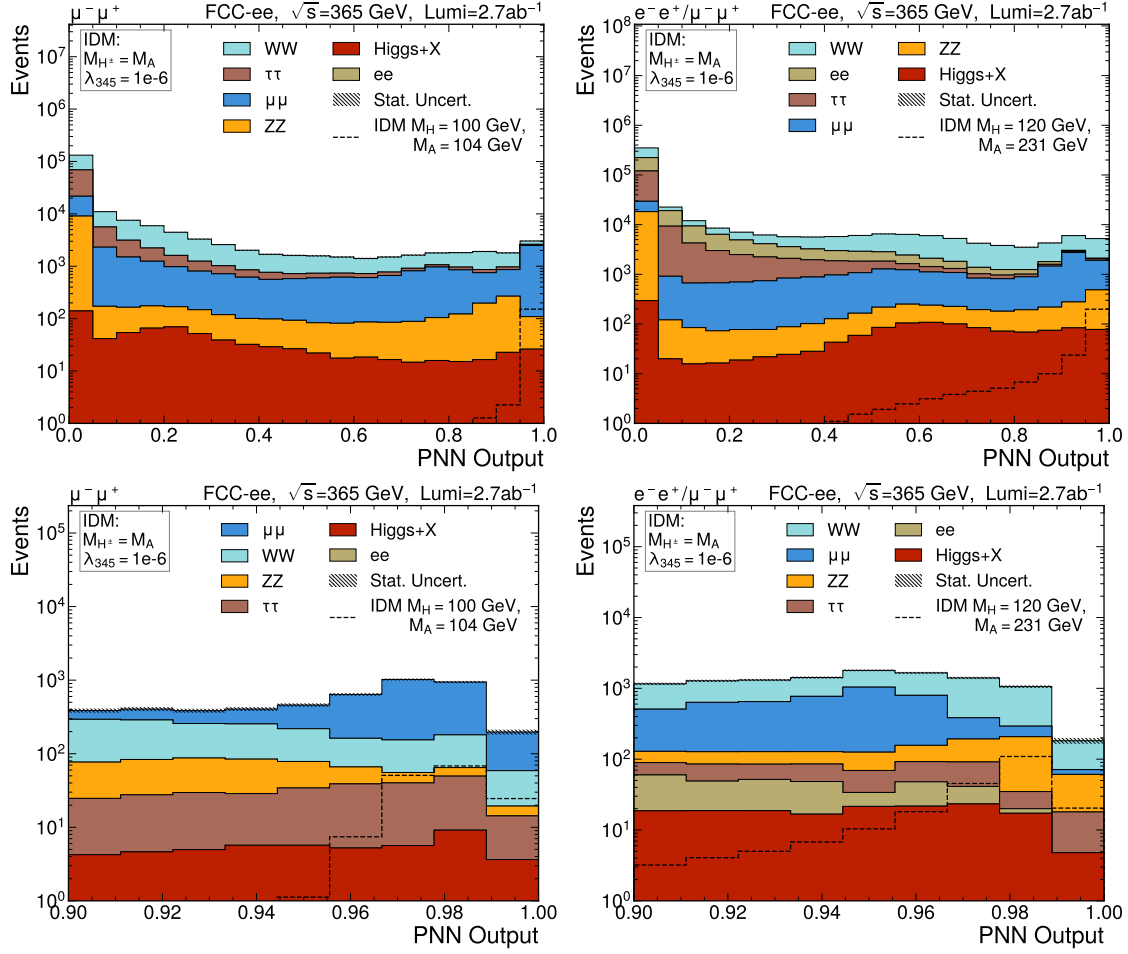


Figure 6.5: Distributions of the pNN output for two signal mass points at $\sqrt{s} = 365$ GeV. The top shows the full pNN output distributions whilst the bottom shows the signal region distributions. Note that for $M_{\ell\ell} \geq 30$ GeV, the e^-e^+ and the $\mu^-\mu^+$ channels are summed together, whilst for $M_{\ell\ell} < 30$ GeV only the $\mu^-\mu^+$ channel is used.

particles. As the lower mass splitting region generally has softer leptons, the FCC-ee, therefore, has a much higher acceptance for this area of the parameter space. Second, the FCC-ee is capable of extending the limits to mass splittings around the Z mass and beyond. This is primarily driven by the improved kinematic discrimination: the FCC-ee has a fixed initial state with the entire event 4-vector known, whereas at the LHC the initial state is unknown, with the partons carrying a fraction of the total energy. Furthermore, at LHC, the missing energy calculation is smeared by the large QCD background, significantly reducing discrimination power.

Despite only considering statistical uncertainties for this analysis, comparisons can be made to the CMS IDM analysis presented in chapter 5. First, given that the muon and electron uncertainties had little impact on the limits in chapter 5, these are not expected to dominate for the FCC, which is also likely to have reduced lepton uncertainties. In addition, the ZZ uncertainties have the largest impact in the CMS analysis, predominantly due to the limited number of events in the ZZ control region. However, given the high luminosity of the FCC, this is likely to be significantly reduced, thus reducing one of the largest sources

of uncertainty. Finally, with little-to-no hadronic activity expected for IDM events at the FCC, unlike the large hadronic pileup background of the LHC, jet uncertainties are also expected to be small.

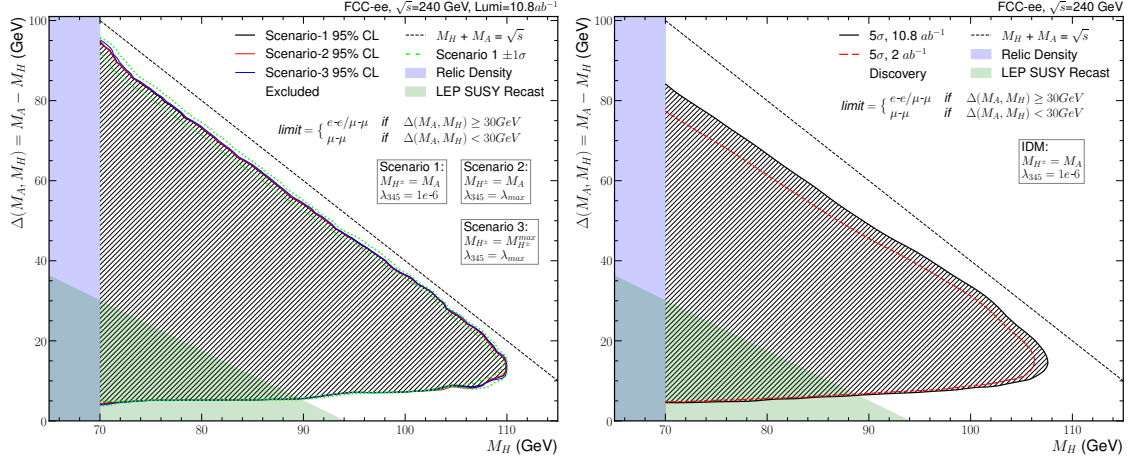


Figure 6.6: 95% CL exclusion limits (left) for all three scenarios and the contour for which there is $>5\sigma$ significance (right) at $\sqrt{s} = 240$ GeV. Additionally, the significance contour for a reduced integrated luminosity of 2 ab^{-1} is presented.

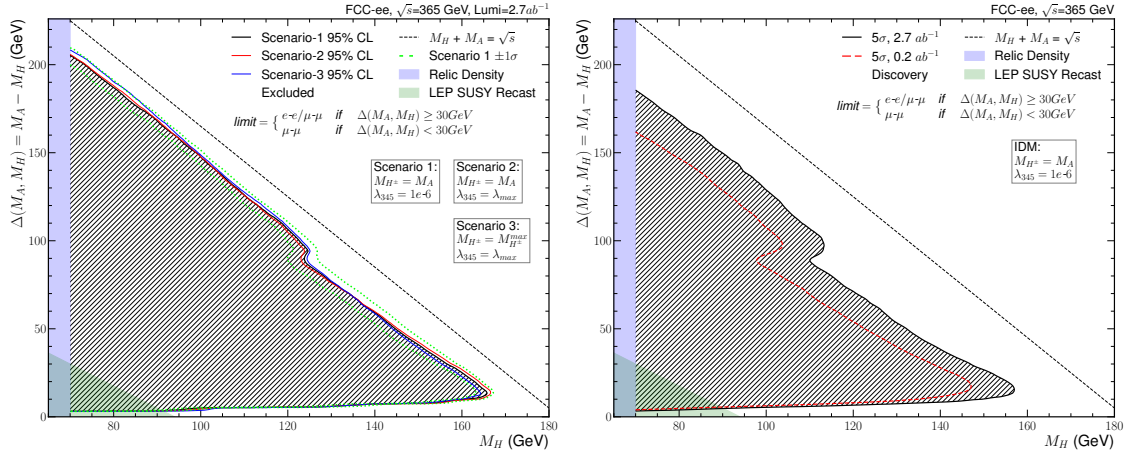


Figure 6.7: 95% CL exclusion limits (left) for all three scenarios and the contour for which there is $>5\sigma$ significance (right) at $\sqrt{s} = 365$ GeV. Additionally, the significance contour for a reduced integrated luminosity of 0.2 ab^{-1} is presented.

Chapter 7

Conclusion

The Standard Model of particle physics is the most successful theory of Nature, precisely describing the fundamental constituents of matter and their interactions via the weak, strong and electromagnetic forces. Nevertheless, major problems are apparent. These include, but are not limited to, the following issues: the SM contains no description of gravity; it cannot describe neutrino oscillations; it offers no solution to the matter-antimatter asymmetry of the universe; and contains no viable candidate for dark matter. Most relevant to this thesis is the Standard Model's lack of a dark matter particle. Despite decades of cosmological evidence for its existence, its true nature remains a mystery. As a result of this tension, beyond the Standard Model theories are proposed to rectify this issue. One such theory, outlined in this thesis, is the Inert Doublet Model (IDM). By introducing a second Higgs doublet, and an additional unbroken $\mathbb{Z}_2$ symmetry, the IDM predicts four new scalar particles: two neutral particles, H and A ; and two charged particles $H^\pm$. Due to the $\mathbb{Z}_2$ symmetry, the lightest of these particles, H , remains stable. Moreover, the new scalars have direct couplings only to the SM gauge bosons, resulting in limited interactions with the matter particles. Consequently, H presents an interesting, viable dark matter candidate. Despite the phenomenological interest in this model, no direct search had ever been performed, with only loose constraints on the IDM parameters.

Chapter 5 presented the first ever direct search for the Inert Doublet Model, using events containing an opposite-sign same-flavour dilepton pair. As the dilepton final state is mediated by SM gauge bosons, for which the couplings involve only SM factors, the sensitivity of this channel is predominantly dependent on the two neutral masses M_H and M_A , with negligible dependence on other IDM parameters. The analysis was performed using proton-proton collision data collected by the CMS Experiment at $\sqrt{s} = 13$ TeV from 2016-2018, and at $\sqrt{s} = 13.6$ TeV in 2022. In total, this corresponds to an integrated luminosity of 172.2 fb^{-1} . After events passed a base selection, that removed the most reducible SM backgrounds, a parametric neural network (pNN) was utilised. The pNN was capable of maximising the sensitivity of the search by exploiting the correlations between different kinematic features, whilst requiring only a single network for the entire parameter space. Subsequently, the pNN output was used to define the signal region, along with control

regions that have comparable phase spaces, whilst being enriched in different SM background sources. All analysis categories were utilised in a binned maximum likelihood fit to extract the signal, performed simultaneously across all years. No significant excess above the Standard Model expectation was observed, with a maximum local significance of 1.5σ . Additionally, 95% CL upper limits were set on the neutral IDM scalars, which observed a maximum exclusion for $M_H = 108$ GeV at a mass splitting of approximately $M_A - M_H = 80$ GeV. The results showed significant improvements upon previous IDM constraints, highlighting the importance of dedicated analyses optimised for the IDM signature.

The final chapter presented a study on the physics potential of the Future Circular Collider (FCC). The FCC is a proposed next-generation collider that will operate in two stages, motivated by the synergies between previous low-energy precision experiments and high-energy colliders. In the first stage, referred to as the FCC-ee, the FCC will collide electron-positron pairs up to a maximum centre-of-mass energy of $\sqrt{s} = 365$ GeV. This is followed by the FCC-hh, colliding proton-proton pairs at a centre-of-mass energy up to $\sqrt{s} = 100$ TeV. This chapter focussed on the FCC-ee, specifically the physics reach of such a collider for searches of the Inert Doublet Model. Using solely simulations and simplified systematics, a search in the opposite-sign same-flavour dilepton final state was presented, utilising a similar methodology to the CMS IDM search presented in chapter 5. The signal was extracted using a binned maximum likelihood fit, and 95% expected upper limits and $> 5\sigma$ regions were presented for the FCC-ee operating at $\sqrt{s} = 240$ GeV and $\sqrt{s} = 365$ GeV. For both scenarios, at the nominal integrated luminosity, nearly the entire kinematically-allowed parameter space is expected to be excluded. Notably, these results were shown to be complimentary to the exclusion limits presented in chapter 5, with significant improvements in both the very low mass splitting and the very high mass splitting regions of the parameter space.

The search for the nature of dark matter continues. However, looking forward, many interesting analyses are possible with regard to the IDM. For searches in the dilepton channel, the increased statistics of the full Run 3 data set presents an exciting opportunity to push the mass reach further than current limits. This is amplified by improved statistics in control regions—notably the ZZ control region—enabling tighter constraints on the dominant backgrounds. The strategies and methodologies developed during Run 3 will naturally be applicable during the High-Luminosity LHC, which will benefit from a near ten-fold increase in data. Beyond the LHC, the FCC-ee collider would open the door to regions inaccessible to a hadron collider, pushing the boundaries in the very low and very high mass splitting parameter space. Complimentary final states, such as those containing trileptons, and their statistical combination with the dilepton channel can further tighten constraints. Ultimately, the combined program of the LHC, HL-LHC, and FCC presents unprecedented prospects for a comprehensive test of the Inert Doublet Model.

Bibliography

- [1] Anubha Bal et al. “Search for additional scalar bosons within the Inert Doublet Model in a final state with two leptons at the FCC-ee”. In: *The European Physical Journal C* 85.11 (Feb. 2025), p. 891. ISSN: 1434-6052. DOI: <https://doi.org/10.1140/epjc/s10052-025-14622-y>. (Visited on 09/16/2025).
- [2] Tomio KOBAYASHI. “Experimental verification of the standard model of particle physics”. In: *Proceedings of the Japan Academy, Series B* 97.5 (2021), pp. 211–235. DOI: [10.2183/pjab.97.013](https://doi.org/10.2183/pjab.97.013).
- [3] Sheldon L. Glashow. “Partial-symmetries of weak interactions”. In: *Nuclear Physics* 22.4 (1961), pp. 579–588. ISSN: 0029-5582. DOI: [https://doi.org/10.1016/0029-5582\(61\)90469-2](https://doi.org/10.1016/0029-5582(61)90469-2). URL: <https://www.sciencedirect.com/science/article/pii/0029558261904692>.
- [4] Steven Weinberg. “A Model of Leptons”. In: *Phys. Rev. Lett.* 19 (21 Nov. 1967), pp. 1264–1266. DOI: [10.1103/PhysRevLett.19.1264](https://doi.org/10.1103/PhysRevLett.19.1264). URL: <https://link.aps.org/doi/10.1103/PhysRevLett.19.1264>.
- [5] Peter Ware Higgs. “Broken symmetries, massless particles and gauge fields”. In: *Phys. Lett.* 12 (1964), pp. 132–133. DOI: [10.1016/0031-9163\(64\)91136-9](https://doi.org/10.1016/0031-9163(64)91136-9). URL: <https://cds.cern.ch/record/641590>.
- [6] F. Englert and R. Brout. “Broken Symmetry and the Mass of Gauge Vector Mesons”. In: *Phys. Rev. Lett.* 13 (9 Aug. 1964), pp. 321–323. DOI: [10.1103/PhysRevLett.13.321](https://doi.org/10.1103/PhysRevLett.13.321). URL: <https://link.aps.org/doi/10.1103/PhysRevLett.13.321>.
- [7] X. Fan et al. “Measurement of the Electron Magnetic Moment”. In: *Physical Review Letters* 130.7 (Feb. 2023). ISSN: 1079-7114. DOI: [10.1103/physrevlett.130.071801](https://doi.org/10.1103/physrevlett.130.071801). URL: <http://dx.doi.org/10.1103/PhysRevLett.130.071801>.
- [8] CMS Collaboration. “Observation of a new boson at a mass of 125 GeV with the CMS experiment at the LHC”. In: *Physics Letters B* 716.1 (Sept. 2012), pp. 30–61. ISSN: 0370-2693. DOI: [10.1016/j.physletb.2012.08.021](https://doi.org/10.1016/j.physletb.2012.08.021). URL: <http://dx.doi.org/10.1016/j.physletb.2012.08.021>.
- [9] ATLAS Collaboration. “Observation of a new particle in the search for the Standard Model Higgs boson with the ATLAS detector at the LHC”. In: *Physics Letters B* 716.1 (Sept. 2012), pp. 1–29. ISSN: 0370-2693. DOI: [10.1016/j.physletb.2012.08.020](https://doi.org/10.1016/j.physletb.2012.08.020). URL: <http://dx.doi.org/10.1016/j.physletb.2012.08.020>.

- [10] Gianfranco Bertone and Dan Hooper. “History of dark matter”. In: *Reviews of Modern Physics* 90.4 (Oct. 2018). ISSN: 1539-0756. DOI: [10.1103/revmodphys.90.045002](https://doi.org/10.1103/revmodphys.90.045002). URL: <http://dx.doi.org/10.1103/RevModPhys.90.045002>.
- [11] Matts Roos. “Dark Matter: The evidence from astronomy, astrophysics and cosmology”. In: (2010). arXiv: [1001.0316](https://arxiv.org/abs/1001.0316) [astro-ph.CO]. URL: <https://arxiv.org/abs/1001.0316>.
- [12] Y. Fukuda et al. “Evidence for Oscillation of Atmospheric Neutrinos”. In: *Physical Review Letters* 81.8 (Aug. 1998), pp. 1562–1567. ISSN: 1079-7114. DOI: [10.1103/physrevlett.81.1562](https://doi.org/10.1103/physrevlett.81.1562). URL: <http://dx.doi.org/10.1103/PhysRevLett.81.1562>.
- [13] Laurent Canetti, Marco Drewes, and Mikhail Shaposhnikov. “Matter and antimatter in the universe”. In: *New Journal of Physics* 14.9 (Sept. 2012), p. 095012. ISSN: 1367-2630. DOI: [10.1088/1367-2630/14/9/095012](https://doi.org/10.1088/1367-2630/14/9/095012). URL: <http://dx.doi.org/10.1088/1367-2630/14/9/095012>.
- [14] E. Noether. “Invariante Variationsprobleme”. ger. In: *Nachrichten von der Gesellschaft der Wissenschaften zu Göttingen, Mathematisch-Physikalische Klasse* 1918 (1918), pp. 235–257. URL: <http://eudml.org/doc/59024>.
- [15] G. S. Guralnik, C. R. Hagen, and T. W. B. Kibble. “Global Conservation Laws and Massless Particles”. In: *Phys. Rev. Lett.* 13 (20 Nov. 1964), pp. 585–587. DOI: [10.1103/PhysRevLett.13.585](https://doi.org/10.1103/PhysRevLett.13.585). URL: <https://link.aps.org/doi/10.1103/PhysRevLett.13.585>.
- [16] Particle Data Group, P A Zyla, and Barnett et al. “Review of Particle Physics”. In: *Progress of Theoretical and Experimental Physics* 2020.8 (Aug. 2020), p. 083C01. ISSN: 2050-3911. DOI: [10.1093/ptep/ptaa104](https://doi.org/10.1093/ptep/ptaa104). eprint: https://academic.oup.com/ptep/article-pdf/2020/8/083C01/34673740/rpp2020-vol2-2015-2092_18.pdf. URL: <https://doi.org/10.1093/ptep/ptaa104>.
- [17] CMS Collaboration. “Measurement of the Higgs boson mass and width using the four-lepton final state in proton-proton collisions at $\sqrt{s} = 13$ TeV”. In: *Physical Review D* 111.9 (May 2025). ISSN: 2470-0029. DOI: [10.1103/physrevd.111.092014](https://doi.org/10.1103/physrevd.111.092014). URL: <http://dx.doi.org/10.1103/PhysRevD.111.092014>.
- [18] Elliot Leader and Enrico Predazzi. “An Introduction to Gauge Theories and Modern Particle Physics”. In: *Cambridge Monographs on Particle Physics, Nuclear Physics and Cosmology* (1996).
- [19] Michael E. Peskin and Daniel V. Schroeder. “An Introduction to quantum field theory”. In: (1995). DOI: [10.1201/9780429503559](https://doi.org/10.1201/9780429503559).
- [20] Planck Collaboration. “Planck 2013 results. I. Overview of products and scientific results”. In: *A&A* 571 (2014), A1. DOI: [10.1051/0004-6361/201321529](https://doi.org/10.1051/0004-6361/201321529). URL: <https://doi.org/10.1051/0004-6361/201321529>.
- [21] F. Zwicky. “Die Rotverschiebung von extragalaktischen Nebeln”. In: *Helv. Phys. Acta* 6 (1933), pp. 110–127. DOI: [10.1007/s10714-008-0707-4](https://doi.org/10.1007/s10714-008-0707-4).
- [22] Guido D’Amico, Marc Kamionkowski, and Kris Sigurdson. “Dark Matter Astrophysics”. In: (2009). arXiv: [0907.1912](https://arxiv.org/abs/0907.1912) [astro-ph.CO]. URL: <https://arxiv.org/abs/0907.1912>.
- [23] K. G. Begeman, A. H. Broeils, and R. H. Sanders. “Extended rotation curves of spiral galaxies: dark haloes and modified dynamics”. In: *Monthly Notices of the Royal Astronomical Society* 249.3 (Apr. 1991), pp. 523–537. ISSN: 0035-8711. DOI: [10.1093/mnras/249.3.523](https://doi.org/10.1093/mnras/249.3.523).

- eprint: <https://academic.oup.com/mnras/article-pdf/249/3/523/18160929/mnras249-0523.pdf>. URL: <https://doi.org/10.1093/mnras/249.3.523>.
- [24] Douglas Clowe et al. “A Direct Empirical Proof of the Existence of Dark Matter*”. In: *The Astrophysical Journal* 648.2 (Aug. 2006), p. L109. DOI: [10.1086/508162](https://doi.org/10.1086/508162). URL: <https://dx.doi.org/10.1086/508162>.
- [25] Nilendra G. Deshpande and Ernest Ma. “Pattern of symmetry breaking with two Higgs doublets”. In: *Phys. Rev. D* 18 (7 Oct. 1978), pp. 2574–2576. DOI: [10.1103/PhysRevD.18.2574](https://doi.org/10.1103/PhysRevD.18.2574). URL: <https://link.aps.org/doi/10.1103/PhysRevD.18.2574>.
- [26] Jan Kalinowski et al. “IDM Benchmarks for the LHC and Future Colliders”. In: *Symmetry* 13.6 (June 2021), p. 991. ISSN: 2073-8994. DOI: [10.3390/sym13060991](https://doi.org/10.3390/sym13060991). URL: <http://dx.doi.org/10.3390/sym13060991>.
- [27] Agnieszka Ilnicka, Maria Krawczyk, and Tania Robens. “Inert doublet model in light of LHC Run I and astrophysical data”. In: *Physical Review D* 93.5 (Mar. 2016). ISSN: 2470-0029. DOI: [10.1103/physrevd.93.055026](https://doi.org/10.1103/physrevd.93.055026). URL: <http://dx.doi.org/10.1103/PhysRevD.93.055026>.
- [28] Daniel Dercks and Tania Robens. “Constraining the Inert Doublet Model using Vector Boson Fusion”. In: *The European Physical Journal C* 79.11 (Nov. 2019). ISSN: 1434-6052. DOI: [10.1140/epjc/s10052-019-7436-6](https://doi.org/10.1140/epjc/s10052-019-7436-6). URL: <http://dx.doi.org/10.1140/epjc/s10052-019-7436-6>.
- [29] Agnieszka Ilnicka, Tania Robens, and Tim Stefaniak. “Constraining extended scalar sectors at the LHC and beyond”. In: *Modern Physics Letters A* 33.10n11 (Apr. 2018), p. 1830007. ISSN: 1793-6632. DOI: [10.1142/s0217732318300070](https://doi.org/10.1142/s0217732318300070). URL: <http://dx.doi.org/10.1142/S0217732318300070>.
- [30] Jan Kalinowski et al. “Benchmarking the Inert Doublet Model for ee colliders”. In: *Journal of High Energy Physics* 2018.12 (Dec. 2018). ISSN: 1029-8479. DOI: [10.1007/jhep12\(2018\)081](https://doi.org/10.1007/jhep12(2018)081). URL: [http://dx.doi.org/10.1007/JHEP12\(2018\)081](http://dx.doi.org/10.1007/JHEP12(2018)081).
- [31] Johannes Braathen et al. “Probing the Inert Doublet Model via vector-boson fusion at a muon collider”. In: *Journal of High Energy Physics* 2025.5 (May 2025). ISSN: 1029-8479. DOI: [10.1007/jhep05\(2025\)055](https://doi.org/10.1007/jhep05(2025)055). URL: [http://dx.doi.org/10.1007/JHEP05\(2025\)055](http://dx.doi.org/10.1007/JHEP05(2025)055).
- [32] Ethan Dolle et al. “Dilepton Signals in the Inert Doublet Model”. In: *Physical Review D* 81.3 (Feb. 2010), p. 035003. DOI: [10.1103/PhysRevD.81.035003](https://doi.org/10.1103/PhysRevD.81.035003). (Visited on 07/11/2023).
- [33] Xinyu Miao, Shufang Su, and Brooks Thomas. “Trilepton Signals in the Inert Doublet Model”. In: *Physical Review D* 82.3 (Aug. 2010), p. 035009. ISSN: 1550-7998, 1550-2368. DOI: [10.1103/PhysRevD.82.035009](https://doi.org/10.1103/PhysRevD.82.035009). arXiv: [1005.0090 \[hep-ph\]](https://arxiv.org/abs/1005.0090). (Visited on 07/13/2023).
- [34] Amitava Datta et al. “Exploring Collider Signatures of the Inert Higgs Doublet Model”. In: *Physical Review D* 95.1 (Jan. 2017), p. 015017. ISSN: 2470-0010, 2470-0029. DOI: [10.1103/PhysRevD.95.015017](https://doi.org/10.1103/PhysRevD.95.015017). arXiv: [1610.00648 \[hep-ph\]](https://arxiv.org/abs/1610.00648). (Visited on 07/12/2023).
- [35] A. Belyaev et al. “Advancing LHC probes of dark matter from the inert two-Higgs-doublet model with the monojet signal”. In: *Physical Review D* 99.1 (Jan. 2019). ISSN: 2470-0029. DOI: [10.1103/physrevd.99.015011](https://doi.org/10.1103/physrevd.99.015011). URL: <http://dx.doi.org/10.1103/PhysRevD.99.015011>.

- [36] Jan Kalinowski et al. “Exploring Inert Scalars at CLIC”. In: *Journal of High Energy Physics* 2019.7 (July 2019), p. 53. ISSN: 1029-8479. DOI: [10.1007/JHEP07\(2019\)053](https://doi.org/10.1007/JHEP07(2019)053). arXiv: [1811.06952](https://arxiv.org/abs/1811.06952) [hep-ex, physics:hep-ph]. (Visited on 07/14/2023).
- [37] Lyndon Evans and Philip Bryant. “LHC Machine”. In: *Journal of Instrumentation* 3.08 (Aug. 2008), S08001. DOI: [10.1088/1748-0221/3/08/S08001](https://doi.org/10.1088/1748-0221/3/08/S08001). URL: <https://dx.doi.org/10.1088/1748-0221/3/08/S08001>.
- [38] ALICE Collaboration. “The ALICE experiment at the CERN LHC”. In: *JINST* 3 (2008), S08002. DOI: [10.1088/1748-0221/3/08/S08002](https://doi.org/10.1088/1748-0221/3/08/S08002).
- [39] ATLAS Collaboration. “The ATLAS Experiment at the CERN Large Hadron Collider”. In: *Journal of Instrumentation* 3.08 (Aug. 2008), S08003. DOI: [10.1088/1748-0221/3/08/S08003](https://doi.org/10.1088/1748-0221/3/08/S08003). URL: <https://dx.doi.org/10.1088/1748-0221/3/08/S08003>.
- [40] CMS Collaboration. “The CMS Experiment at the CERN LHC”. In: *Journal of Instrumentation* 3.08 (Aug. 2008), S08004. ISSN: 1748-0221. DOI: [10.1088/1748-0221/3/08/S08004](https://doi.org/10.1088/1748-0221/3/08/S08004). URL: <https://dx.doi.org/10.1088/1748-0221/3/08/S08004> (visited on 02/08/2025).
- [41] LHCb Collaboration. “The LHCb Detector at the LHC”. In: *Journal of Instrumentation* 3.08 (Aug. 2008), S08005. DOI: [10.1088/1748-0221/3/08/S08005](https://doi.org/10.1088/1748-0221/3/08/S08005). URL: <https://dx.doi.org/10.1088/1748-0221/3/08/S08005>.
- [42] CMS Collaboration. “CMS Luminosity - Public Results”. In: (2025). URL: <https://twiki.cern.ch/twiki/bin/view/CMSPublic/LumiPublicResults> (visited on 08/31/2025).
- [43] CMS Collaboration. “CMS Technical Design Report for the Pixel Detector Upgrade”. In: (2012). URL: <https://cds.cern.ch/record/1481838> (visited on 02/08/2025).
- [44] CMS Collaboration. “The CMS Barrel Calorimeter Response to Particle Beams from 2 to 350 GeV/c”. In: *The European Physical Journal C* 60.3 (Apr. 1, 2009), pp. 359–373. ISSN: 1434-6052. DOI: [10.1140/epjc/s10052-009-0959-5](https://doi.org/10.1140/epjc/s10052-009-0959-5). URL: <https://doi.org/10.1140/epjc/s10052-009-0959-5> (visited on 03/01/2025).
- [45] CMS Collaboration. “Performance of the CMS muon detector and muon reconstruction with proton-proton collisions at $\sqrt{s} = 13$ TeV”. In: *Journal of Instrumentation* 13.06 (June 2018), P06015–P06015. ISSN: 1748-0221. DOI: [10.1088/1748-0221/13/06/p06015](https://doi.org/10.1088/1748-0221/13/06/p06015). URL: <http://dx.doi.org/10.1088/1748-0221/13/06/P06015>.
- [46] CMS Collaboration. “Performance of the CMS Level-1 trigger in proton-proton collisions at $\sqrt{s} = 13$ TeV”. In: *Journal of Instrumentation* 15.10 (Oct. 2020), P10017–P10017. ISSN: 1748-0221. DOI: [10.1088/1748-0221/15/10/p10017](https://doi.org/10.1088/1748-0221/15/10/p10017). URL: <http://dx.doi.org/10.1088/1748-0221/15/10/P10017>.
- [47] CMS Collaboration. “Development of the CMS detector for the CERN LHC Run 3”. In: *Journal of Instrumentation* 19.05 (May 2024), P05064. ISSN: 1748-0221. DOI: [10.1088/1748-0221/19/05/p05064](https://doi.org/10.1088/1748-0221/19/05/p05064). URL: <http://dx.doi.org/10.1088/1748-0221/19/05/P05064>.
- [48] CMS Collaboration. “Particle-flow reconstruction and global event description with the CMS detector”. In: *Journal of Instrumentation* 12.10 (Oct. 2017), P10003–P10003. ISSN: 1748-0221. DOI: [10.1088/1748-0221/12/10/p10003](https://doi.org/10.1088/1748-0221/12/10/p10003). URL: <http://dx.doi.org/10.1088/1748-0221/12/10/P10003>.

- [49] R. Frühwirth. “Application of Kalman filtering to track and vertex fitting”. In: *Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment* 262.2 (1987), pp. 444–450. ISSN: 0168-9002. DOI: [https://doi.org/10.1016/0168-9002\(87\)90887-4](https://doi.org/10.1016/0168-9002(87)90887-4). URL: <https://www.sciencedirect.com/science/article/pii/0168900287908874>.
- [50] W Adam et al. “Reconstruction of electrons with the Gaussian-sum filter in the CMS tracker at the LHC”. In: *Journal of Physics G: Nuclear and Particle Physics* 31.9 (July 2005), N9–N20. ISSN: 1361-6471. DOI: [10.1088/0954-3899/31/9/n01](https://doi.org/10.1088/0954-3899/31/9/n01). URL: <http://dx.doi.org/10.1088/0954-3899/31/9/N01>.
- [51] CMS Collaboration. “Search for neutral Higgs bosons decaying to tau pairs in pp collisions at $\sqrt{s} = 7$ TeV”. In: *Physics Letters B* 713.2 (June 2012), pp. 68–90. ISSN: 0370-2693. DOI: <https://doi.org/10.1016/j.physletb.2012.05.028>. URL: <https://www.sciencedirect.com/science/article/pii/S0370269312005564>.
- [52] CMS Collaboration. “Performance of CMS muon reconstruction in pp collision events at $\sqrt{s} = 7$ TeV”. In: *Journal of Instrumentation* 7.10 (Oct. 2012), P10002–P10002. ISSN: 1748-0221. DOI: [10.1088/1748-0221/7/10/p10002](https://doi.org/10.1088/1748-0221/7/10/p10002). URL: <http://dx.doi.org/10.1088/1748-0221/7/10/P10002>.
- [53] A. Bodek et al. “Extracting muon momentum scale corrections for hadron collider experiments”. In: *The European Physical Journal C* 72.10 (Oct. 2012). ISSN: 1434-6052. DOI: [10.1140/epjc/s10052-012-2194-8](https://doi.org/10.1140/epjc/s10052-012-2194-8). URL: <http://dx.doi.org/10.1140/epjc/s10052-012-2194-8>.
- [54] CMS Collaboration. “Performance of electron reconstruction and selection with the CMS detector in proton-proton collisions at $\sqrt{s} = 8$ TeV”. In: *Journal of Instrumentation* 10.06 (June 2015), P06005–P06005. ISSN: 1748-0221. DOI: [10.1088/1748-0221/10/06/p06005](https://doi.org/10.1088/1748-0221/10/06/p06005). URL: <http://dx.doi.org/10.1088/1748-0221/10/06/P06005>.
- [55] CMS Collaboration. “Electron and photon reconstruction and identification with the CMS experiment at the CERN LHC”. In: *Journal of Instrumentation* 16.05 (May 2021), P05014. ISSN: 1748-0221. DOI: [10.1088/1748-0221/16/05/p05014](https://doi.org/10.1088/1748-0221/16/05/p05014). URL: <http://dx.doi.org/10.1088/1748-0221/16/05/P05014>.
- [56] Matteo Cacciari, Gavin P Salam, and Gregory Soyez. “The anti-ktjet clustering algorithm”. In: *Journal of High Energy Physics* 2008.04 (Apr. 2008), pp. 063–063. ISSN: 1029-8479. DOI: [10.1088/1126-6708/2008/04/063](https://doi.org/10.1088/1126-6708/2008/04/063). URL: <http://dx.doi.org/10.1088/1126-6708/2008/04/063>.
- [57] CMS Collaboration. *Jet algorithms performance in 13 TeV data*. Tech. rep. Geneva: CERN, 2017. URL: <https://cds.cern.ch/record/2256875>.
- [58] Daniele Bertolini et al. “Pileup per particle identification”. In: *Journal of High Energy Physics* 2014.10 (Oct. 2014). ISSN: 1029-8479. DOI: [10.1007/jhep10\(2014\)059](https://doi.org/10.1007/jhep10(2014)059). URL: [http://dx.doi.org/10.1007/JHEP10\(2014\)059](http://dx.doi.org/10.1007/JHEP10(2014)059).
- [59] CMS Collaboration. “Pileup mitigation at CMS in 13 TeV data”. In: *Journal of Instrumentation* 15.09 (Sept. 2020), P09018–P09018. ISSN: 1748-0221. DOI: [10.1088/1748-0221/15/09/p09018](https://doi.org/10.1088/1748-0221/15/09/p09018). URL: <http://dx.doi.org/10.1088/1748-0221/15/09/P09018>.

- [60] Biplob Bhattacharjee et al. “Associated jet and subjet rates in light-quark and gluon jet discrimination”. In: *Journal of High Energy Physics* 2015.4 (Apr. 2015). ISSN: 1029-8479. DOI: [10.1007/jhep04\(2015\)131](https://doi.org/10.1007/jhep04(2015)131). URL: [http://dx.doi.org/10.1007/JHEP04\(2015\)131](http://dx.doi.org/10.1007/JHEP04(2015)131).
- [61] E. Bols et al. “Jet flavour classification using DeepJet”. In: *Journal of Instrumentation* 15.12 (Dec. 2020), P12012–P12012. ISSN: 1748-0221. DOI: [10.1088/1748-0221/15/12/p12012](https://doi.org/10.1088/1748-0221/15/12/p12012). URL: <http://dx.doi.org/10.1088/1748-0221/15/12/P12012>.
- [62] CMS Collaboration. “Performance of missing energy reconstruction in 13 TeV pp collision data using the CMS detector”. In: (2016). URL: <https://cds.cern.ch/record/2205284>.
- [63] Johann Brehmer and Kyle Cranmer. *Simulation-based inference methods for particle physics*. 2020. arXiv: [2010.06439](https://arxiv.org/abs/2010.06439) [hep-ph]. URL: <https://arxiv.org/abs/2010.06439>.
- [64] Jon Butterworth et al. “PDF4LHC recommendations for LHC Run II”. In: *Journal of Physics G: Nuclear and Particle Physics* 43.2 (Jan. 2016), p. 023001. ISSN: 1361-6471. DOI: [10.1088/0954-3899/43/2/023001](https://doi.org/10.1088/0954-3899/43/2/023001). URL: <http://dx.doi.org/10.1088/0954-3899/43/2/023001>.
- [65] V. Khachatryan et al. “Event generator tunes obtained from underlying event and multi-parton scattering measurements”. In: *The European Physical Journal C* 76.3 (Mar. 2016). ISSN: 1434-6052. DOI: [10.1140/epjc/s10052-016-3988-x](https://doi.org/10.1140/epjc/s10052-016-3988-x). URL: <http://dx.doi.org/10.1140/epjc/s10052-016-3988-x>.
- [66] S. Agostinelli et al. “Geant4—a simulation toolkit”. In: *Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment* 506.3 (2003), pp. 250–303. ISSN: 0168-9002. DOI: [https://doi.org/10.1016/S0168-9002\(03\)01368-8](https://doi.org/10.1016/S0168-9002(03)01368-8). URL: <https://www.sciencedirect.com/science/article/pii/S0168900203013688>.
- [67] Rikkert Frederix and Stefano Frixione. “Merging meets matching in MC@NLO”. In: *Journal of High Energy Physics* 2012.12 (Dec. 2012). ISSN: 1029-8479. DOI: [10.1007/jhep12\(2012\)061](https://doi.org/10.1007/jhep12(2012)061). URL: [http://dx.doi.org/10.1007/JHEP12\(2012\)061](http://dx.doi.org/10.1007/JHEP12(2012)061).
- [68] John M. Campbell et al. “Top-pair production and decay at NLO matched with parton showers”. In: *JHEP* 04 (2015), p. 114. DOI: [10.1007/JHEP04\(2015\)114](https://doi.org/10.1007/JHEP04(2015)114). arXiv: [1412.1828](https://arxiv.org/abs/1412.1828) [hep-ph].
- [69] Simone Alioli et al. “NLO single-top production matched with shower in POWHEG: s - and t -channel contributions”. In: *JHEP* 09 (2009), p. 111. DOI: [10.1007/JHEP02\(2010\)011](https://doi.org/10.1007/JHEP02(2010)011). arXiv: [0907.4076](https://arxiv.org/abs/0907.4076) [hep-ph].
- [70] Emanuele Re. “Single-top Wt -channel production matched with parton showers using the POWHEG method”. In: *Eur. Phys. J. C* 71 (2011), p. 1547. DOI: [10.1140/epjc/s10052-011-1547-z](https://doi.org/10.1140/epjc/s10052-011-1547-z). arXiv: [1009.2450](https://arxiv.org/abs/1009.2450) [hep-ph].
- [71] Torbjörn Sjöstrand, Stephen Mrenna, and Peter Skands. “PYTHIA 6.4 physics and manual”. In: *Journal of High Energy Physics* 2006.05 (May 2006), pp. 026–026. ISSN: 1029-8479. DOI: [10.1088/1126-6708/2006/05/026](https://doi.org/10.1088/1126-6708/2006/05/026). URL: <http://dx.doi.org/10.1088/1126-6708/2006/05/026>.
- [72] Christian Bierlich et al. “A comprehensive guide to the physics and usage of PYTHIA 8.3”. In: (2022). arXiv: [2203.11601](https://arxiv.org/abs/2203.11601) [hep-ph]. URL: <https://arxiv.org/abs/2203.11601>.

- [73] J. de Favereau et al. “DELPHES 3: a modular framework for fast simulation of a generic collider experiment”. In: *Journal of High Energy Physics* 2014.2 (Feb. 2014). ISSN: 1029-8479. DOI: [10.1007/jhep02\(2014\)057](https://doi.org/10.1007/jhep02(2014)057). URL: [http://dx.doi.org/10.1007/JHEP02\(2014\)057](http://dx.doi.org/10.1007/JHEP02(2014)057).
- [74] Stefan Höche. *Introduction to parton-shower event generators*. 2015. arXiv: [1411.4085](https://arxiv.org/abs/1411.4085) [hep-ph]. URL: <https://arxiv.org/abs/1411.4085>.
- [75] O. Aberle et al. “High-Luminosity Large Hadron Collider (HL-LHC): Technical design report”. In: CERN Yellow Reports: Monographs (2020). DOI: [10.23731/CYRM-2020-0010](https://doi.org/10.23731/CYRM-2020-0010). URL: <https://cds.cern.ch/record/2749422>.
- [76] Gabriella Pasztor. “The Phase-2 Upgrade of the CMS Detector”. In: (Mar. 2023), p. 045. DOI: [10.22323/1.422.0045](https://doi.org/10.22323/1.422.0045).
- [77] CMS Collaboration. “The Phase-2 Upgrade of the CMS Level-1 Trigger”. In: (2020). Final version. URL: <https://cds.cern.ch/record/2714892>.
- [78] Alves, Bruno. “Cluster reconstruction in the HGCal at the Level 1 trigger”. In: *EPJ Web of Conf.* 295 (2024), p. 02006. DOI: [10.1051/epjconf/202429502006](https://doi.org/10.1051/epjconf/202429502006). URL: <https://doi.org/10.1051/epjconf/202429502006>.
- [79] “The Phase-2 Upgrade of the CMS Endcap Calorimeter”. In: (2017). DOI: [10.17181/CERN.IV8M.1JY2](https://doi.org/10.17181/CERN.IV8M.1JY2). URL: <https://cds.cern.ch/record/2293646>.
- [80] CMS Collaboration. “Technical proposal for a MIP timing detector in the CMS experiment Phase 2 upgrade”. In: (2017). DOI: [10.17181/CERN.2RSJ.UE8W](https://doi.org/10.17181/CERN.2RSJ.UE8W). URL: <https://cds.cern.ch/record/2296612>.
- [81] Tianqi Chen and Carlos Guestrin. “XGBoost: A Scalable Tree Boosting System”. In: KDD ’16 (Aug. 2016), pp. 785–794. DOI: [10.1145/2939672.2939785](https://doi.org/10.1145/2939672.2939785). URL: <http://dx.doi.org/10.1145/2939672.2939785>.
- [82] Jerome Friedman, Trevor Hastie, and Robert Tibshirani. “Additive Logistic Regression: A Statistical View of Boosting”. In: *The Annals of Statistics* 28 (Apr. 2000), pp. 337–407. DOI: [10.1214/aos/1016218223](https://doi.org/10.1214/aos/1016218223).
- [83] Huilin Qu and Loukas Gouskos. “Jet tagging via particle clouds”. In: *Physical Review D* 101.5 (Mar. 2020). ISSN: 2470-0029. DOI: [10.1103/physrevd.101.056019](https://doi.org/10.1103/physrevd.101.056019). URL: <http://dx.doi.org/10.1103/PhysRevD.101.056019>.
- [84] Yue Wang et al. “Dynamic Graph CNN for Learning on Point Clouds”. In: (2019). arXiv: [1801.07829](https://arxiv.org/abs/1801.07829) [cs.CV]. URL: <https://arxiv.org/abs/1801.07829>.
- [85] Sergey Ioffe and Christian Szegedy. *Batch Normalization: Accelerating Deep Network Training by Reducing Internal Covariate Shift*. 2015. arXiv: [1502.03167](https://arxiv.org/abs/1502.03167) [cs.LG]. URL: <https://arxiv.org/abs/1502.03167>.
- [86] Vladimír Kunc and Jiří Kléma. *Three Decades of Activations: A Comprehensive Survey of 400 Activation Functions for Neural Networks*. 2024. arXiv: [2402.09092](https://arxiv.org/abs/2402.09092) [cs.LG]. URL: <https://arxiv.org/abs/2402.09092>.
- [87] Kaiming He et al. “Deep Residual Learning for Image Recognition”. In: (2016), pp. 770–778. DOI: [10.1109/CVPR.2016.90](https://doi.org/10.1109/CVPR.2016.90).

- [88] Adam Paszke et al. “PyTorch: An Imperative Style, High-Performance Deep Learning Library”. In: (2019). arXiv: [1912.01703](https://arxiv.org/abs/1912.01703) [cs.LG]. URL: <https://arxiv.org/abs/1912.01703>.
- [89] Nitish Srivastava et al. “Dropout: A Simple Way to Prevent Neural Networks from Overfitting”. In: *Journal of Machine Learning Research* 15 (June 2014), pp. 1929–1958.
- [90] Diederik P. Kingma and Jimmy Ba. “Adam: A Method for Stochastic Optimization”. In: (2017). arXiv: [1412.6980](https://arxiv.org/abs/1412.6980) [cs.LG]. URL: <https://arxiv.org/abs/1412.6980>.
- [91] Andrew P. Bradley. “The use of the area under the ROC curve in the evaluation of machine learning algorithms”. In: *Pattern Recognition* 30.7 (1997), pp. 1145–1159. ISSN: 0031-3203. DOI: [https://doi.org/10.1016/S0031-3203\(96\)00142-2](https://doi.org/10.1016/S0031-3203(96)00142-2). URL: <https://www.sciencedirect.com/science/article/pii/S0031320396001422>.
- [92] Torbjörn Sjöstrand et al. “An introduction to PYTHIA 8.2”. In: *Computer Physics Communications* 191 (June 2015), pp. 159–177. ISSN: 0010-4655. DOI: [10.1016/j.cpc.2015.01.024](https://doi.org/10.1016/j.cpc.2015.01.024). URL: <http://dx.doi.org/10.1016/j.cpc.2015.01.024>.
- [93] CMS Collaboration. “Search for supersymmetry in final states with two oppositely charged same-flavor leptons and missing transverse momentum in proton-proton collisions at $\sqrt{s} = 13$ TeV”. In: *Journal of High Energy Physics* 2021.4 (Apr. 2021). ISSN: 1029-8479. DOI: [10.1007/jhep04\(2021\)123](https://doi.org/10.1007/jhep04(2021)123). URL: [http://dx.doi.org/10.1007/JHEP04\(2021\)123](http://dx.doi.org/10.1007/JHEP04(2021)123).
- [94] CMS Collaboration. “Search for dark matter produced in association with a leptonically decaying Z boson in proton-proton collisions at $\sqrt{s} = 13$ TeV”. In: *The European Physical Journal C* 81.1 (Jan. 2021). ISSN: 1434-6052. DOI: [10.1140/epjc/s10052-020-08739-5](https://doi.org/10.1140/epjc/s10052-020-08739-5). URL: <http://dx.doi.org/10.1140/epjc/s10052-020-08739-5>.
- [95] Tom Melia et al. “ W^+W^- , WZ and ZZ production in the POWHEG BOX”. In: *JHEP* 11 (2011), p. 078. DOI: [10.1007/JHEP11\(2011\)078](https://doi.org/10.1007/JHEP11(2011)078). arXiv: [1107.5051](https://arxiv.org/abs/1107.5051) [hep-ph].
- [96] John Campbell and Tobias Neumann. “Precision phenomenology with MCFM”. In: *Journal of High Energy Physics* 2019.12 (Dec. 2019). ISSN: 1029-8479. DOI: [10.1007/jhep12\(2019\)034](https://doi.org/10.1007/jhep12(2019)034). URL: [http://dx.doi.org/10.1007/JHEP12\(2019\)034](http://dx.doi.org/10.1007/JHEP12(2019)034).
- [97] J. Alwall et al. “The automated computation of tree-level and next-to-leading order differential cross sections, and their matching to parton shower simulations”. In: *Journal of High Energy Physics* 2014.7 (July 2014). ISSN: 1029-8479. DOI: [10.1007/jhep07\(2014\)079](https://doi.org/10.1007/jhep07(2014)079). URL: [http://dx.doi.org/10.1007/JHEP07\(2014\)079](http://dx.doi.org/10.1007/JHEP07(2014)079).
- [98] Michał Czakon and Alexander Mitov. “Top++: A program for the calculation of the top-pair cross-section at hadron colliders”. In: *Computer Physics Communications* 185.11 (Nov. 2014), pp. 2930–2938. ISSN: 0010-4655. DOI: [10.1016/j.cpc.2014.06.021](https://doi.org/10.1016/j.cpc.2014.06.021). URL: <http://dx.doi.org/10.1016/j.cpc.2014.06.021>.
- [99] John Campbell, Tobias Neumann, and Zack Sullivan. “Single-top-quark production in the t -channel at NNLO”. In: *JHEP* 02 (2021), p. 040. DOI: [10.1007/JHEP02\(2021\)040](https://doi.org/10.1007/JHEP02(2021)040). arXiv: [2012.01574](https://arxiv.org/abs/2012.01574) [hep-ph].
- [100] Nikolaos Kidonakis and Nodoka Yamanaka. “Higher-order corrections for tW production at high-energy hadron colliders”. In: *JHEP* 05 (2021), p. 278. DOI: [10.1007/JHEP05\(2021\)278](https://doi.org/10.1007/JHEP05(2021)278). arXiv: [2102.11300](https://arxiv.org/abs/2102.11300) [hep-ph].

- [101] Kirill Melnikov and Frank Petriello. “Electroweak gauge boson production at hadron colliders through $O(\alpha_s^2)$ ”. In: *Physical Review D* 74.11 (Dec. 2006). ISSN: 1550-2368. DOI: [10.1103/PhysRevD.74.114017](https://doi.org/10.1103/PhysRevD.74.114017). URL: <http://dx.doi.org/10.1103/PhysRevD.74.114017>.
- [102] John M. Campbell, R. Keith Ellis, and Ciaran Williams. “Vector boson pair production at the LHC”. In: *Journal of High Energy Physics* 2011.7 (July 2011). ISSN: 1029-8479. DOI: [10.1007/jhep07\(2011\)018](https://doi.org/10.1007/jhep07(2011)018). URL: [http://dx.doi.org/10.1007/JHEP07\(2011\)018](http://dx.doi.org/10.1007/JHEP07(2011)018).
- [103] J. Alwall et al. “Comparative study of various algorithms for the merging of parton showers and matrix elements in hadronic collisions”. In: *Eur. Phys. J. C* 53 (2008), p. 473. DOI: [10.1140/epjc/s10052-007-0490-5](https://doi.org/10.1140/epjc/s10052-007-0490-5). arXiv: [0706.2569](https://arxiv.org/abs/0706.2569) [hep-ph].
- [104] J. Alwall et al. “The automated computation of tree-level and next-to-leading order differential cross sections, and their matching to parton shower simulations”. In: *JHEP* 07 (2014), p. 079. DOI: [10.1007/JHEP07\(2014\)079](https://doi.org/10.1007/JHEP07(2014)079). arXiv: [1405.0301](https://arxiv.org/abs/1405.0301) [hep-ph].
- [105] Rikkert Frederix and Stefano Frixione. “Merging meets matching in MC@NLO”. In: *JHEP* 12 (2012), p. 061. DOI: [10.1007/JHEP12\(2012\)061](https://doi.org/10.1007/JHEP12(2012)061). arXiv: [1209.6215](https://arxiv.org/abs/1209.6215) [hep-ph].
- [106] Pierre Baldi et al. “Parameterized neural networks for high-energy physics”. In: *The European Physical Journal C* 76.5 (Apr. 2016). ISSN: 1434-6052. DOI: [10.1140/epjc/s10052-016-4099-4](https://doi.org/10.1140/epjc/s10052-016-4099-4). URL: <http://dx.doi.org/10.1140/epjc/s10052-016-4099-4>.
- [107] Luca Anzalone, Tommaso Diotalevi, and Daniele Bonacorsi. “Improving parametric neural networks for high-energy physics (and beyond)”. In: *Machine Learning: Science and Technology* 3.3 (Sept. 2022), p. 035017. ISSN: 2632-2153. DOI: [10.1088/2632-2153/ac917c](https://doi.org/10.1088/2632-2153/ac917c). URL: <http://dx.doi.org/10.1088/2632-2153/ac917c>.
- [108] CMS Collaboration. “Search for physics beyond the standard model in events with a Z boson, jets, and missing transverse energy in pp collisions at $\sqrt{s}=7$ TeV”. In: *Physics Letters B* 716.2 (2012), pp. 260–284. ISSN: 0370-2693. DOI: <https://doi.org/10.1016/j.physletb.2012.08.026>. URL: <https://www.sciencedirect.com/science/article/pii/S0370269312008635>.
- [109] Triparno Bandyopadhyay et al. “A twisted tale of the transverse-mass tail”. In: *Journal of High Energy Physics* 2023.8 (Aug. 2023). ISSN: 1029-8479. DOI: [10.1007/jhep08\(2023\)135](https://doi.org/10.1007/jhep08(2023)135). URL: [http://dx.doi.org/10.1007/JHEP08\(2023\)135](http://dx.doi.org/10.1007/JHEP08(2023)135).
- [110] Alan Barr, Christopher Lester, and Phil Stephens. “A variable for measuring masses at hadron colliders when missing energy is expected; mT2: the truth behind the glamour”. In: *Journal of Physics G: Nuclear and Particle Physics* 29.10 (Sept. 2003), pp. 2343–2363. ISSN: 1361-6471. DOI: [10.1088/0954-3899/29/10/304](https://doi.org/10.1088/0954-3899/29/10/304). URL: <http://dx.doi.org/10.1088/0954-3899/29/10/304>.
- [111] Djork-Arné Clevert, Thomas Unterthiner, and Sepp Hochreiter. *Fast and Accurate Deep Network Learning by Exponential Linear Units (ELUs)*. 2016. arXiv: [1511.07289](https://arxiv.org/abs/1511.07289) [cs.LG]. URL: <https://arxiv.org/abs/1511.07289>.
- [112] Shmuel Rippa. “An algorithm for selecting a good parameter c in radial basis function interpolation”. In: *Advances in Computational Mathematics* 11 (Nov. 1999), pp. 193–210. DOI: [10.1023/A:1018975909870](https://doi.org/10.1023/A:1018975909870).
- [113] J. S. Conway. “Incorporating Nuisance Parameters in Likelihoods for Multisource Spectra”. In: (2011). arXiv: [1103.0354](https://arxiv.org/abs/1103.0354) [physics.data-an]. URL: <https://arxiv.org/abs/1103.0354>.

- [114] A L Read. “Presentation of search results”. In: *J. Phys. G* 28.10 (2002), pp. 2693–704. DOI: [10.1088/0954-3899/28/10/313](https://doi.org/10.1088/0954-3899/28/10/313). URL: <https://cds.cern.ch/record/722145>.
- [115] Thomas Junk. “Confidence level computation for combining searches with small statistics”. In: *Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment* 434.2–3 (Sept. 1999), pp. 435–443. ISSN: 0168-9002. DOI: [10.1016/S0168-9002\(99\)00498-2](https://doi.org/10.1016/S0168-9002(99)00498-2). URL: [http://dx.doi.org/10.1016/S0168-9002\(99\)00498-2](http://dx.doi.org/10.1016/S0168-9002(99)00498-2).
- [116] Glen Cowan et al. “Asymptotic formulae for likelihood-based tests of new physics”. In: *The European Physical Journal C* 71.2 (Feb. 2011). ISSN: 1434-6052. DOI: [10.1140/epjc/s10052-011-1554-0](https://doi.org/10.1140/epjc/s10052-011-1554-0). URL: <http://dx.doi.org/10.1140/epjc/s10052-011-1554-0>.
- [117] CMS Collaboration. “Search for physics beyond the standard model in dilepton mass spectra in proton-proton collisions at $\sqrt{s} = 8$ TeV”. In: *Journal of High Energy Physics* 2015.4 (Apr. 2015). ISSN: 1029-8479. DOI: [10.1007/jhep04\(2015\)025](https://doi.org/10.1007/jhep04(2015)025). URL: [http://dx.doi.org/10.1007/JHEP04\(2015\)025](http://dx.doi.org/10.1007/JHEP04(2015)025).
- [118] “Search for t-channel scalar and vector leptoquark exchange in the high mass dimuon and dielectron spectrum in proton-proton collisions at $\sqrt{s} = 13$ TeV”. In: (2024). URL: <https://cds.cern.ch/record/2905397>.
- [119] CMS Collaboration. “Measurement of the inclusive and differential WZ production cross sections, polarization angles, and triple gauge couplings in pp collisions at $\sqrt{s} = 13$ TeV”. In: *Journal of High Energy Physics* 2022.7 (July 2022). ISSN: 1029-8479. DOI: [10.1007/JHEP07\(2022\)032](https://doi.org/10.1007/JHEP07(2022)032). URL: [http://dx.doi.org/10.1007/JHEP07\(2022\)032](http://dx.doi.org/10.1007/JHEP07(2022)032).
- [120] Roger Barlow and Christine Beeston. “Fitting using finite Monte Carlo samples”. In: *Computer Physics Communications* 77.2 (1993), pp. 219–228. ISSN: 0010-4655. DOI: [https://doi.org/10.1016/0010-4655\(93\)90005-W](https://doi.org/10.1016/0010-4655(93)90005-W). URL: <https://www.sciencedirect.com/science/article/pii/001046559390005W>.
- [121] CMS Collaboration. “The CMS Statistical Analysis and Combination Tool: Combine”. In: *Computing and Software for Big Science* 8.1 (Nov. 2024). ISSN: 2510-2044. DOI: [10.1007/s41781-024-00121-4](https://doi.org/10.1007/s41781-024-00121-4). URL: <http://dx.doi.org/10.1007/s41781-024-00121-4>.
- [122] Michael Benedikt et al. “FCC-ee: The Lepton Collider”. In: 2 (2019). DOI: [10.1140/epjst/e2019-900045-4](https://doi.org/10.1140/epjst/e2019-900045-4). URL: <https://cds.cern.ch/record/2651299>.
- [123] I. Agapov et al. “Future Circular Lepton Collider FCC-ee: Overview and Status”. In: (2022). arXiv: [2203.08310](https://arxiv.org/abs/2203.08310) [physics.acc-ph]. URL: <https://arxiv.org/abs/2203.08310>.
- [124] Alain Blondel and Patrick Janot. “FCC-ee overview: new opportunities create new challenges”. In: (2021). arXiv: [2106.13885](https://arxiv.org/abs/2106.13885) [hep-ex]. URL: <https://arxiv.org/abs/2106.13885>.
- [125] R W Assmann, M Lamont, and S Myers. “A Brief History of the LEP Collider”. In: *Nucl. Phys. B, Proc. Suppl.* 109 (2002), pp. 17–31. DOI: [10.1016/S0920-5632\(02\)90005-8](https://doi.org/10.1016/S0920-5632(02)90005-8). URL: <https://cds.cern.ch/record/549223>.
- [126] In: *Physics Reports* 427.5–6 (May 2006), pp. 257–454. ISSN: 0370-1573. DOI: [10.1016/j.physrep.2005.12.006](https://doi.org/10.1016/j.physrep.2005.12.006). URL: <http://dx.doi.org/10.1016/j.physrep.2005.12.006>.
- [127] CDF Collaboration. “Observation of Top Quark Production in $\bar{p}p$ Collisions with the Collider Detector at Fermilab”. In: *Phys. Rev. Lett.* 74 (14 1995), pp. 2626–2631. DOI: [10.1016/0370-2693\(95\)00100-0](https://doi.org/10.1016/0370-2693(95)00100-0).

- 1103/PhysRevLett.74.2626. URL: <https://link.aps.org/doi/10.1103/PhysRevLett.74.2626>.
- [128] Paolo Azzurri et al. “A special Higgs challenge: measuring the mass and production cross section with ultimate precision at FCC-ee”. In: *The European Physical Journal Plus* 137.1 (Dec. 2021). ISSN: 2190-5444. DOI: [10.1140/epjp/s13360-021-02202-4](https://doi.org/10.1140/epjp/s13360-021-02202-4). URL: <http://dx.doi.org/10.1140/epjp/s13360-021-02202-4>.
- [129] Giovanni F. Tassielli. “A proposal of a drift chamber for the IDEA experiment for a future e^+e^- collider”. In: *PoS ICHEP2020* (2021), p. 877. DOI: [10.22323/1.390.0877](https://doi.org/10.22323/1.390.0877).
- [130] The IDEA Study Group. “The IDEA detector concept for FCC-ee”. In: (2025). arXiv: [2502.21223](https://arxiv.org/abs/2502.21223) [physics.ins-det]. URL: <https://arxiv.org/abs/2502.21223>.
- [131] Gerardo Ganis, Clément Helsens, and Valentin Völkl. “Key4hep, a framework for future HEP experiments and its use in FCC”. In: (2021). arXiv: [2111.09874](https://arxiv.org/abs/2111.09874) [hep-ex]. URL: <https://arxiv.org/abs/2111.09874>.
- [132] FCC Collaboration. “FCCAnalyses”. In: (2025). URL: <https://github.com/HEP-FCC/FCCAnalyses> (visited on 08/31/2025).
- [133] Patrick Janot et al. “Integrated Luminosities and Sequence of Events for the FCC Feasibility Study Report”. In: (Dec. 2024). DOI: [10.17181/yr3v6-dgh16](https://doi.org/10.17181/yr3v6-dgh16). URL: <https://doi.org/10.17181/yr3v6-dgh16>.
- [134] Wolfgang Kilian, Thorsten Ohl, and Jürgen Reuter. “WHIZARD—simulating multi-particle processes at LHC and ILC”. In: *The European Physical Journal C* 71.9 (Sept. 2011). ISSN: 1434-6052. DOI: [10.1140/epjc/s10052-011-1742-y](https://doi.org/10.1140/epjc/s10052-011-1742-y). URL: <http://dx.doi.org/10.1140/epjc/s10052-011-1742-y>.
- [135] Matteo Cacciari, Gavin P. Salam, and Gregory Soyez. “FastJet user manual: (for version 3.0.2)”. In: *The European Physical Journal C* 72.3 (Mar. 2012). ISSN: 1434-6052. DOI: [10.1140/epjc/s10052-012-1896-2](https://doi.org/10.1140/epjc/s10052-012-1896-2). URL: <http://dx.doi.org/10.1140/epjc/s10052-012-1896-2>.
- [136] Aleksander Filip Zarnecki et al. “Searching Inert Scalars at Future e^+e^- Colliders”. In: (2020). arXiv: [2002.11716](https://arxiv.org/abs/2002.11716) [hep-ph]. URL: <https://arxiv.org/abs/2002.11716>.
- [137] J. Altmann et al. “ECFA Higgs, Electroweak, and Top Factory Study”. In: 5/2025 (2025). DOI: [10.23731/CYRM-2025-005](https://doi.org/10.23731/CYRM-2025-005). URL: <https://doi.org/10.23731/CYRM-2025-005>.